

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

(1a)

- write F as $/$, G as $\backslash$
- render $a \leq b$ as $\boxed{\frac{a}{b}}$
- **rewrite** $X \leq XF$ as $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj



now we have $\wedge/ = /$ and $\vee\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

$F \dashv G$	F 's type	monad FG	$1 \sqsubseteq FG$	$GF \sqsubseteq 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleleft$	$A \rightarrow A \otimes A$	$\triangleleft \triangleleft = 1$	$1 \sqsubseteq \triangleleft \triangleleft$	$\triangleleft \triangleleft \sqsubseteq 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleleft \triangleleft = \triangleleft$	$\triangleleft \triangleleft \triangleleft = \triangleleft$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \sqsubseteq \multimap \multimap$	$\multimap \multimap \sqsubseteq 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^\circ \circ = R$	$R^\circ \circ = R$	$R^\circ \circ = R$	$(R \cup S)^\circ = S^\circ \cup R^\circ$	$(R \cap S)^\circ = R^\circ \cap S^\circ$	$R^\circ \circ = R$	$R^\circ \circ = R$
$\circ \dashv \triangleleft \dashv \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \dashv \multimap \dashv \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \sqsubseteq R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cap S)$	$R \sqsubseteq R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \sqsubseteq R$				
$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \sqsubseteq (RS)/S$	$(T/S) S \sqsubseteq T$	$(S \cup T)R = SR \cup TR$	$(R \cap R')/S = (R/S) \cap (R'/S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \sqsubseteq S \setminus (SR)$	$S(S \setminus T) \sqsubseteq T$	$R(S \cup T) = RS \cup RT$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \sqsubseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \sqsubseteq Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \sqsubseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \sqsubseteq A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \sqsubseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \sqsubseteq A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$(S \cup T)R = SR \cup TR$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^\circ = Xg^\circ \cap Yg^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \dashv \cdot f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$R(S \cup T) = RS \cup RT$	$f(X \cap Y) = fX \cap fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \dashv \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow EA$	$\exists : EB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, EB) \rightarrow (A \rightarrow B)$	1	$\frac{f\exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$
$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, EB)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f\exists}{\exists} = f$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$	$\frac{\frac{f\exists}{\exists} \exists}{\exists} = \frac{f\exists}{\exists}$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
 \begin{array}{ccccc}
 \boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
 \text{f a map} & & & & \\
 \xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
 \xRightarrow{\text{indirect equality}} & & & & & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
 \end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = fR/S$$

(1.2b)

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\leftarrow \triangleleft$	Δ
$\cdot S$	$R/S_1 \ S_2=R/S_2/S_1$	$S \setminus (R/W) = (S \setminus R)/W$	$R/(Sg) = (Rg^\circ)/S$	$\text{Map}(f) \implies f(R/S) = fR/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(RnR')/S = (R/S) \cap (R'/S)$
$S \cdot$	$S \setminus (R/W) = (S \setminus R)/W$	$S_1 \ S_2 \setminus R = S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ / S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (RnR') = (S \setminus R) \cap (S \setminus R')$
$\cdot f$	$\text{Map}(f) \implies R/fS = (R/S)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(RS)^\circ = S^\circ R^\circ$			$(RS)^\circ = S^\circ R^\circ$		$\text{Simple}(g) \implies (XnY)g^\circ = Xg^\circ nYg^\circ$
$f^\circ \cdot$	$\text{Map}(f) \implies f(R/S) = fR/S$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(RS)^\circ = S^\circ R^\circ$		$(RS)^\circ = S^\circ R^\circ$		$f(T_1 n T_2) = fT_1 n fT_2$
Rn	—	—	—	—	$R \Rightarrow S \Rightarrow T = RnS \Rightarrow T$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow SnT = (R \Rightarrow S) \cap (R \Rightarrow T)$
$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ / T^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv \circ)$ both bends: the $^\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$
$\leftarrow \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv \circ)$ both bends: the $^\circ$ section's display	—	—
u	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$Rn(S \cup T) = RnS \cup RnT$	$(R \cup S)^\circ = S^\circ \cup R^\circ$		
$\emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$\emptyset R = \emptyset$	$R \emptyset = \emptyset$	$Rn \emptyset = \emptyset$	$\emptyset^\circ = \emptyset$		

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $\text{Map}(f) \implies R/fS = (R/S)f^\circ$ — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = \text{EA}$ they are the image triple:

(1.3b)

operator	acts by	name
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = E(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R / T = \{(a, a) : \forall b. aRb\}$.

(1.3c)

$$\mathcal{D} \dashv \cdot T \dashv 1 \cap \cdot / T$$

Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — Tf still reaches only $\text{im } f$, and $\forall_f f \emptyset = B \setminus \text{im } f$ is still not $\emptyset$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^\circ = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

(1.2b)

The rows that chain forever say nothing by it: $\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $\circ \dashv \circ \dashv \circ$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

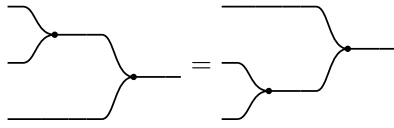
§2. Relations

(2a)

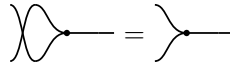
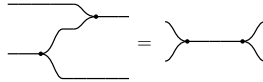
$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ^{-1}$, and

$C: (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

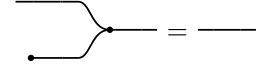
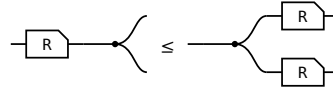
(2b)



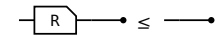
$\triangleright$ associative



$\triangleright$ commutative



$\circ$ is its unit



Frobenius, one half — the other is its $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$

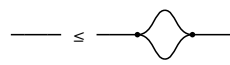
$R \circ \leq \circ$

§2.1. $\forall$ object A , $(A, \triangleleft, \circ) \dashv (A, \triangleright, \circ)$

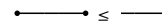
(2.1a)



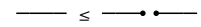
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



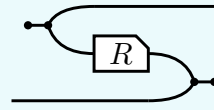
$1 \leq \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



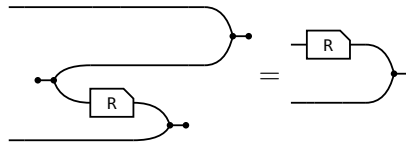
$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow \mathbf{a} \otimes \mathbf{a}$ opens a pair of wires out of nothing and $\triangleright \circ : \mathbf{b} \otimes \mathbf{b} \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\circ^\circ$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



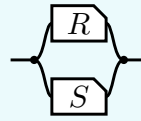
(3.1a)

R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ \triangleleft$ bends the $\mathbf{a}$ strand down and the $\triangleright \circ$ brings it back up, while the $\mathbf{b}$ strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

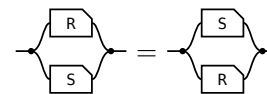
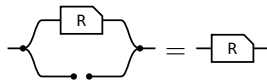
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \langle \downarrow (R \otimes S) \rangle$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



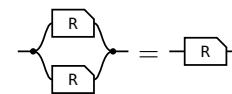
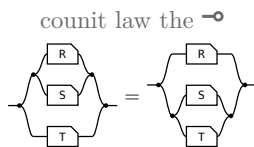
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\top = \text{copy} \circ \text{merge}$ (the paper's Lemma 4.11).

(4b)



unit: one half of $\top$ per end — the merge's unit law absorbs the copy , the copy's counit law the merge **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

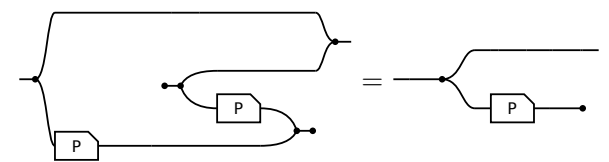
semi-distributivity, and what supplies it	picture
$R \cap (S \cap T) \leq R \cap S \cap R \cap T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = F R \cap F S$.	

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq 1 \cap R R^\circ$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

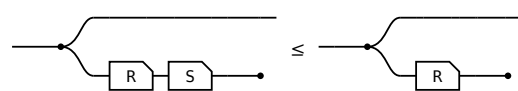
(5c)

$\text{dom}(R) \triangleq 1 \cap R R^\circ$
$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X R$
$\text{dom}(RS) \sqsubseteq \text{dom}(R)$
$\text{dom}(R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = 1$
$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \sqsubseteq 1$
$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(RS)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(fg)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(fg)$
$\text{Entire}(RS) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \sqsubseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

Equality is S entire, which is the same picture read as $\text{dom}(R) = 1 \iff R \text{ entire}$.

§6. Maps

(6a)

surjective $\exists R \circ R, \text{ that is, } R \circ \text{entire.}$	single valued $R \circ R \subseteq \text{id}$
entire $\exists R \circ R \circ R. \text{ With single valued, a map.}$	injective $R \circ R \subseteq \text{id}$

(6b)

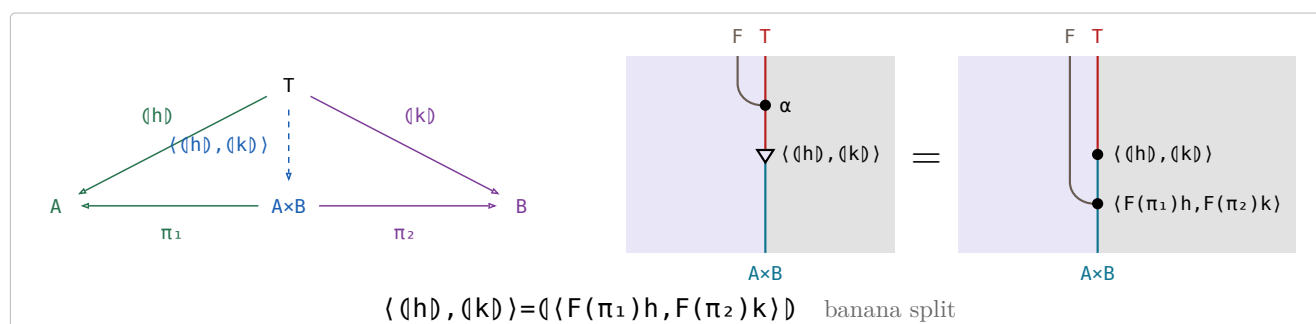
law	picture
$\text{Simple}(F) \Rightarrow F(R \circ S) = F \circ R \circ F \circ S$ $F \text{ single valued}$	

§7. Reduce in $\mathcal{S}et$

(7a)

definition	type	note
$\text{listr } A ::= \text{nil} \mid \text{cons}(A, \text{listr } A)$	$A \rightarrow A$	The cons-lists over A, the datatype every row below folds.
$\text{sum} = ([\text{zero}, \text{plus}])$	$[A] \rightarrow A$	$\text{plus}(a, b) = a + b.$
$\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$	$[A] \rightarrow \text{Nat}$	π_2 drops the head and keeps the count of the tail, succ adds one for the head.
$\text{average} \triangleq (\text{sum}, \text{length}) \text{div}$	$[\text{Nat}] \rightarrow \text{Nat}$	$\text{div}(m, n) = m/n$, with $\text{div}(0, 0) = 0$ so average is total. Traverses the list twice.
banana-split law $\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$	$T \rightarrow A \times B$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to $\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1)h, F(\pi_2)k \rangle$	$FT \rightarrow A \times B$	All that $\triangleq$ reduce leaves to check: the fork satisfies the defining equation.
the instance $\langle \text{sum}, \text{length} \rangle = ([\text{zeros}, \text{pluss}])$	$[\text{Nat}] \rightarrow \text{Nat} \times \text{Nat}$	$\text{pluss}(a, (b, n)) = (a + b, n + 1)$, so average runs in one pass.
$\text{preds } n = [n, n-1, \dots, 1]$	$\text{Nat} \rightarrow [\text{Nat}]$	apply the earlier special case to write preds as $\langle k \rangle \pi_1$.

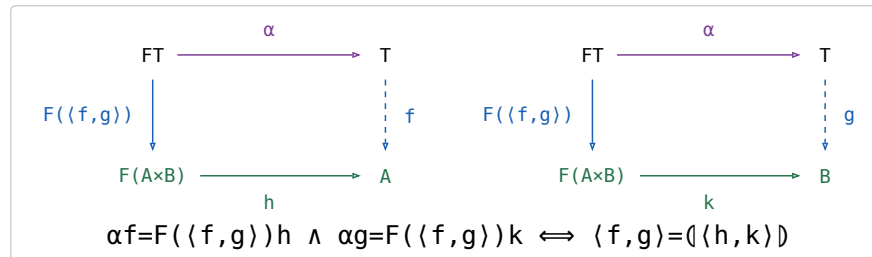
(7b)



§7.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ creates them: the product of $h : FA \rightarrow A$ and $k : FB \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$.
- π_1 and π_2 are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (7a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in mutual recursion the two arrows leave $F(A \times B)$, not FA and FB , so each sees BOTH components. The product is the case where they factor as $F(\pi_1)h$ and $F(\pi_2)k$. What it says: **any algebra on a product carrier is folded by a fork**.

(7.1a)



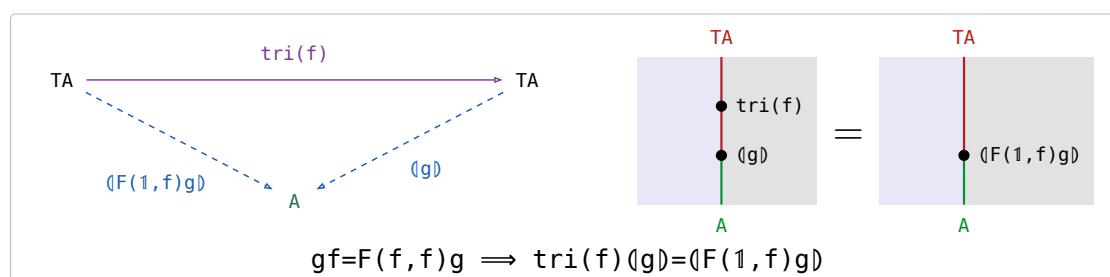
§7.2. Ruby triangles

(7.2a)

stage	definition
informally	$\text{tri}(f)[a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f(a_1), \dots, f^i(a_i), \dots, f^n(a_n)]$
for cons-lists	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$
with the base functor named	$F(A, B) = 1 + A \times B$, $\alpha = [\text{nil}, \text{cons}]$, so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$
abstractly	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : TA \rightarrow TA$.

(7.2b)



Horner's rule is Horner's rule, generalised from cons-lists to any initial type. Fusion reduces it to $F(1, T(f)) \alpha(g) = F(1, (g)) F(1, f) g$.

§7.3. Depth of a tree

(7.3a)

the datatype	$\text{tree } A ::= \text{tip}(A) \mid \text{bin}(\text{tree } A, \text{tree } A)$, base functor $F(A,B)=A+B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$
the fold	$\llbracket [g, h] \rrbracket$ is the unique f with $f(\text{tip}(a)) = g(a)$ and $f(\text{bin}(x, y)) = h(f(x), f(y))$
$\text{tree}(f)$	$\text{tree}(f) = \llbracket F(f, 1) \rrbracket_{[\text{tip}, \text{bin}]}$; pointwise $\text{tree}(f)(\text{tip}(a)) = \text{tip}(f(a))$ and $\text{tree}(f)(\text{bin}(x, y)) = \text{bin}(\text{tree}(f)(x), \text{tree}(f)(y))$
max	$\text{max} = \llbracket [1, \text{bmax}] \rrbracket$, where $\text{bmax}(a, b)$ is the larger of a and b
depths	$\text{depths} = \text{tree}(\text{zero}) \text{ tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor
depth	$\text{depth} = \text{depths } \text{max}$

§8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images: $S(y) \subseteq R(x)$. \ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

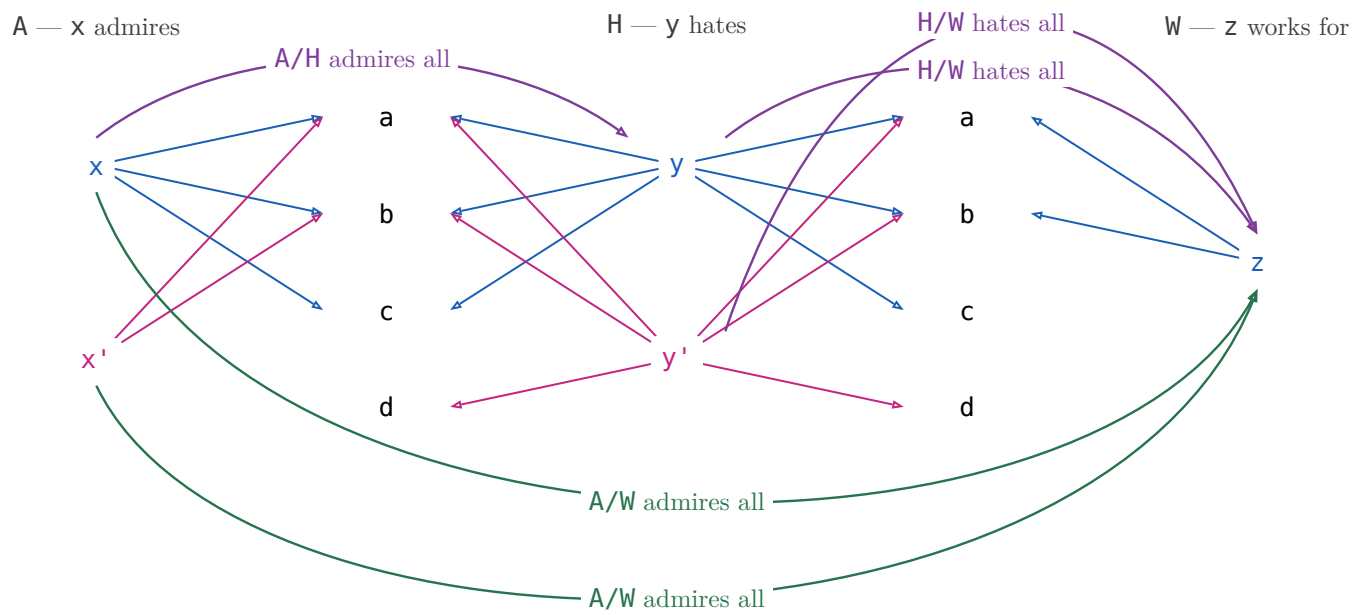
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \subseteq R/W$

(8.1a)



(8.1b)

$x (A/H) y$ — x admires everyone y hates

$y (H/W) z$ — y hates everyone z works for

$x (A/W) z$ — x admires everyone z works for

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \subseteq R/W$.

$X \in R/S \iff XS \in R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \in S \setminus R \iff SX \in R$ The mirror — divide on the left when x comes first.	
$(R/S)S \in R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
$S \setminus (S \setminus R) \in R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f \ (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \in R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
$1 \in R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R)(R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \in R/R$ above it is an equality.	
$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: η , υ , $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§9. $\frac{R}{S}$

(9a)

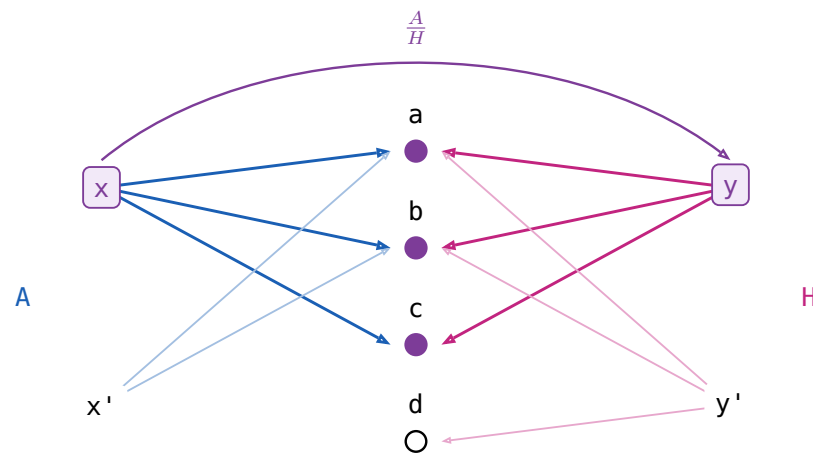
$R/\textcircled{S} \triangleq (R/S) \cap (S/R)^\circ$ what R sends x to and nothing else is what S sends y to. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(9b)

$x (A/H) y$ — x admires everyone y hates
 $x ((H/A)^\circ) y$ — x admires only people y hates
 $x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(9c)



(9d)

law	the reading
$X \in R/\textcircled{S} \iff X S \in R \wedge X^\circ R \in S$ X sits inside the matching exactly when it carries S back inside R and, reversed, carries R back inside S .	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(R/\textcircled{S})^\circ = S/\textcircled{R}$ reversing the matching swaps the two relations.	Matching is symmetric.
$(R/\textcircled{S})(S/\textcircled{W}) \in R/\textcircled{W}$ a match from R to S followed by one from S to W is a match from R to W .	And transitive.
$(R/\textcircled{S}) S \in R$ following the matching by S lands inside R .	$(\exists y. x(\frac{R}{S})y \wedge y S p) \rightarrow x R p$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$
$(R/\textcircled{S}) R = R$ matching R against itself and then following R gives R back.	$(\exists y. x(\frac{R}{R})y \wedge y R p) \iff x R p$ x and y admire the same people $y=x$ always qualifies ($1 \in R \circ R$ below)
$1 \in R/\textcircled{S} R$ every x matches itself.	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(R/\textcircled{S})(R/\textcircled{S}) = R/\textcircled{S} R$ matching twice matches no more pairs than matching once.	So the relation admires the same people is an equivalence relation.
$\text{Symmetric}(X) \implies X \in R/\textcircled{S} R \iff X R \in R$ a symmetric X sits inside the matching exactly when following it by R adds nothing to R .	The largest symmetric arrow that leaves R alone.
$\text{simplePart}(R) \triangleq R/\textcircled{S} 1$ matching R against the identity keeps the x whose image is a single point.	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{dom}(R/\textcircled{S}) = 1 \cap (R/S)(S/R)$ the x that match something are those the two one-sided divisions take round the loop back to x .	Its domain is the domain of simplicity of R .

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \exists_R &= \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

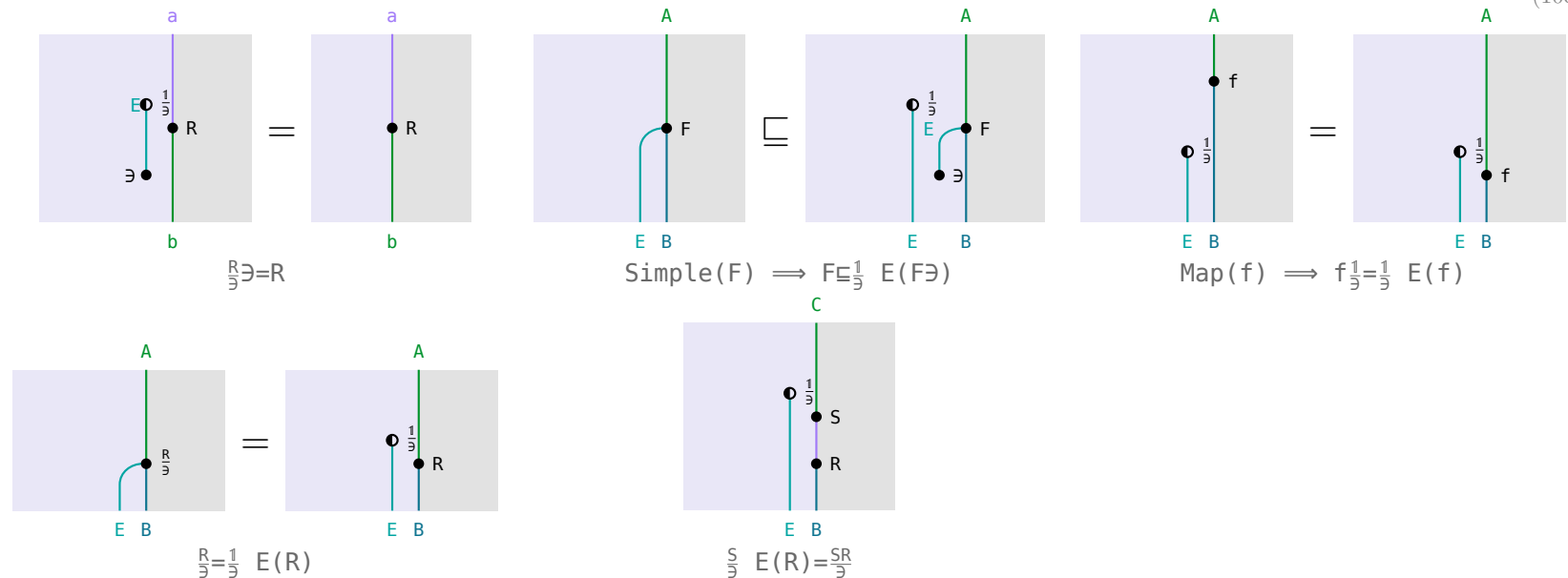
$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : EB \rightarrow B$, dropping the subscript.

$$\exists \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

law	the reading
1 $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow EB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\text{Map}(\frac{R}{\exists})$	
5 $\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6 $\text{Simple}(F) \Rightarrow F \subseteq \frac{F\exists}{\exists}$	A partial choice of sets is inside the total one.
7 the singleton map is monic: $\frac{1}{\exists}$ $(\frac{1}{\exists})^\circ \subseteq \mathbb{1}$ two points with the same one-person set are the same point.	The one-person set.
8 $\exists \exists = \mathbb{1}$	Make the set of a set, then read it back one level down.
9 fusion: $\text{Map}(f) \Rightarrow \frac{fR}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10 $\text{Map}(f) \wedge \mathbb{1} \square = \exists \square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11 $R\square = \exists \square \Rightarrow R/_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : EA \rightarrow EB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §10.3
15 $\exists \triangleq \exists/\exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
16 $\subseteq = \supseteq^\circ$	$xs \subseteq ys \iff ys \supseteq xs$

(10c)



§10.3. $\mathcal{A} \cong \text{Kleisli}(E)$

(10.3a)

$$E(R) \triangleq \frac{R}{\exists}, \quad \frac{1}{\exists} : A \rightarrow EA, \quad \text{union} = E(\exists) : E(EA) \rightarrow EA$$

$$f \diamond g \triangleq fE(g)\text{union}, \quad \text{for } f : A \rightarrow EB \text{ and } g : B \rightarrow EC$$

the monad is on $\text{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{1}{\exists}$, union and $fE(g)\text{union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xlongequal{\text{definition of } \diamond, \text{ union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right)E(\exists)} \xlongequal{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right)} \\ & \xlongequal{(10b), \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xlongequal{(10b), \text{absorption}} \boxed{\frac{SR}{\exists}} \end{aligned}$$

the first three steps only unfold $\diamond$ — E is a functor, so $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\right)E(\exists) = E(R)$ and the union is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of $\frac{1}{\exists}$.

1 goes to $\frac{1}{\exists}$, the Kleisli identity, by definition — so $\frac{1}{\exists}$ is an isomorphism of categories $\mathcal{A} \cong \text{Kleisli}(E)$, and §10.1's $i \dashv E$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.

§11. Relator

(11a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

(11b)

the statement

$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$.

Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.

$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.

Two relators agreeing on maps are equal.

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$

$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$

The **power relator** P — $x \sqsubseteq P(R) y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§11.1. Fork $\langle R, S \rangle$

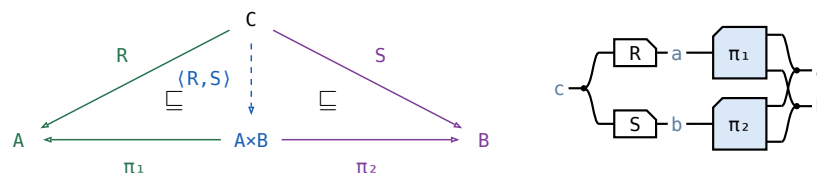
(11.1a)

The **fork** of $R : \mathcal{C} \rightarrow \mathcal{A}$ and $S : \mathcal{C} \rightarrow \mathcal{B}$ is $\langle R, S \rangle \triangleq R\pi_1 \cap S\pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

(11.1b)

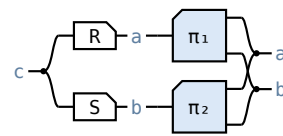
$$\langle R, S \rangle \pi_1 = \text{dom}(S)R \quad \langle R, S \rangle \pi_2 = \text{dom}(R)S$$

(11.1c)



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In Rel , $c \sqsubseteq \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

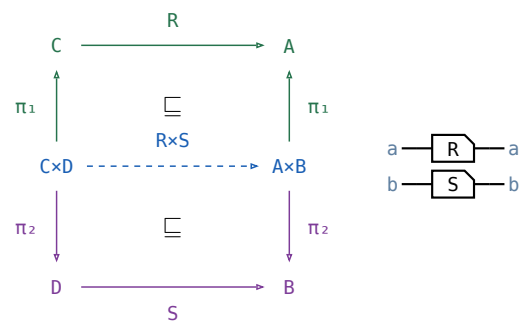
No $\circ$ survives the translation. $\pi_1 = 1 \otimes \circ$ discards the second component, so $\pi_1 \circ = 1 \otimes \circ$ **creates** one out of nothing, and n is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



$\circ \triangleright = 1$ on each half

§11.1.1. Relational product $R \times S$

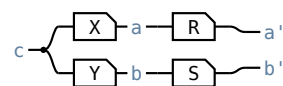
$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.



Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In Rel , $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§11.1.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



the law	picture
1 $R \times S \triangleq (\pi_1 R, \pi_2 S)$	
2 $\langle X, Y \rangle (R \times S) = (XR, YS)$ both sides are the same strokes — absorption. Its $\sqsubseteq$ half is that half at $S := \mathbb{1}$ and at $R := \mathbb{1}$ — stages of their proof of this row; the corollary is the $(R \times S)$ $(U \times V) = (RU) \times (SV)$ it yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$.	
3 $\langle R, S \rangle \pi_1 = \text{dom}(S) R$ (11.1b)	
4 $\langle R, S \rangle \pi_2 = \text{dom}(R) S$ (11.1b)	
5 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$	
6 $\langle R, S \rangle^\circ \langle U, V \rangle \sqsubseteq (R^\circ U) \times (S^\circ V)$	
7 $\text{Map}(f) \implies f \langle R, S \rangle = \langle fR, fS \rangle$ f a map; it fails for an arbitrary arrow;	
8 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$; only $\sqsubseteq$ for an arbitrary R, S	
9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$ reading $\times$ as the relational product, does Rel have exponentials?	

§11.2. Coproduct $[R, S] : A+B \rightarrow C$

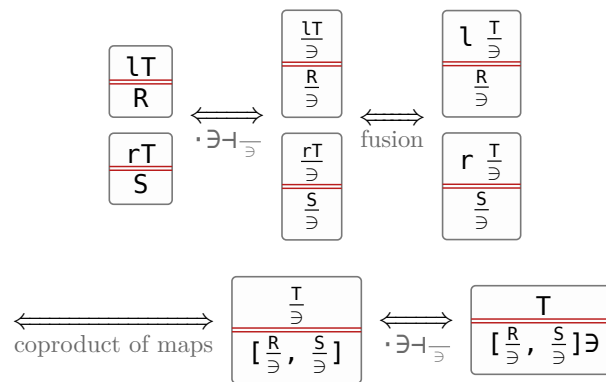
(11.2a)

the statement	picture
$[R, S] \triangleq \mathfrak{l} \circ R \cup r \circ S$ The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
$R + S \triangleq [R\mathfrak{l}, S\mathfrak{r}]$	
$\mathfrak{l}[R, S] = R, r[R, S] = S$ $\mathfrak{l}T = R \wedge rT = S \implies T = [R, S]$	
$\mathfrak{l}\mathfrak{l}^\circ = 1, r\mathfrak{r}^\circ = 1$	
$\mathfrak{l}r^\circ = \emptyset, r\mathfrak{l}^\circ = \emptyset$	
$\mathfrak{l}^\circ \mathfrak{l} \cup r^\circ r = 1$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$	

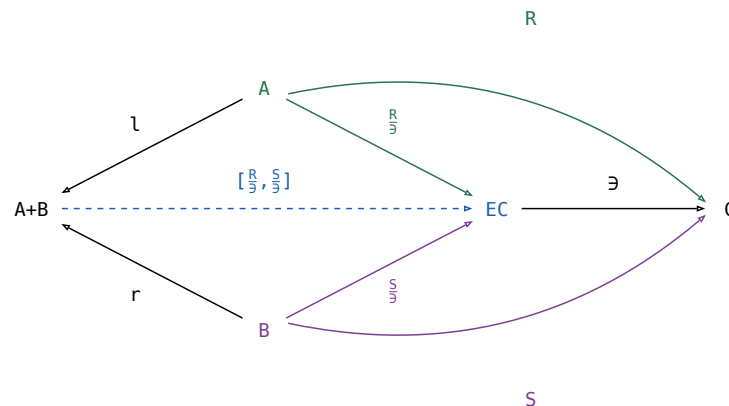
§11.2.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free: $\mathfrak{l}, r$ were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow EC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(11.2.1a)



(11.2.1b)



Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists = l \circ R \cup r \circ S$.

(11.2.1c)

the law	picture
$[R, S] \triangleq l \circ R \cup r \circ S$ (11.2a)'s first row	
$R + S \triangleq [Rl, Sr]$	
$[U, V] \circ [R, S] = U \circ R \cup V \circ S$ (11.2a)'s last row;	
$[1, 0] = l \circ, [0, 1] = r \circ$ $[l, r] = 1$ which is (5.9)	
prove (5.11), and say why duality does not carry it over from the product law	
$(R + S) \cap [P, Q] [U, V] \circ =$ $(R \cap P U \circ) + (S \cap Q V \circ)$ $[P, Q] [U, V] \circ$ is a full 2×2 of composites; $R + S$ is diagonal, so the meet cuts the two off-diagonal branches	

§11.3. The power relator $P(R)$

(11.3a)

For $R : A \rightarrow B$,

$$P(R) \triangleq (\epsilon \setminus R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

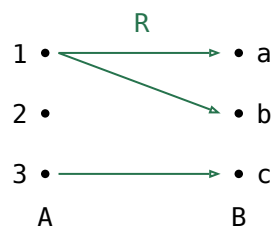
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(11.3b)

	in Rel	in words	$\text{im}(xs)$	partner
$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	every y with some $xs \ R \ y$ is in ys	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{every } y \text{ with some } xs \ R \ y$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall x. x \ R \ \text{some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$	—	every x and every y has a partner

(11.3c)



xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

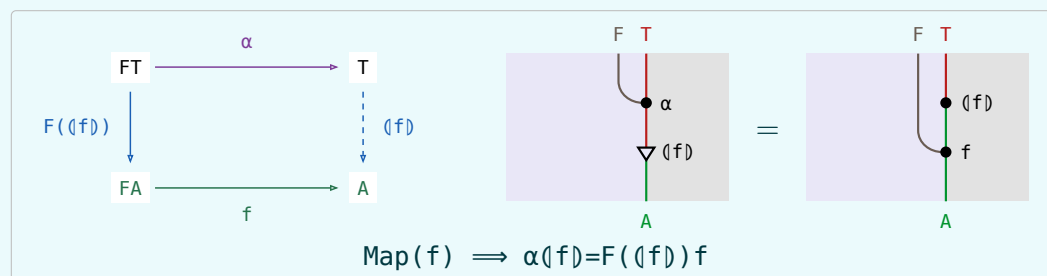
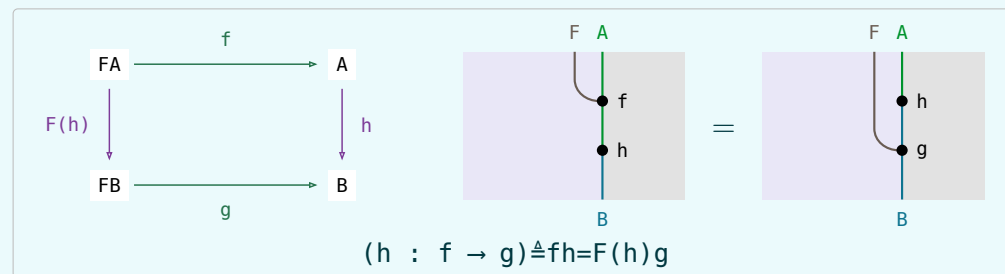
(11.3d)

law	the reading
$X \sqsubseteq P(R) \iff X \sqsubseteq \exists R \text{ and } X^\circ \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(RS) = P(R)P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§11.4. Initial algebra

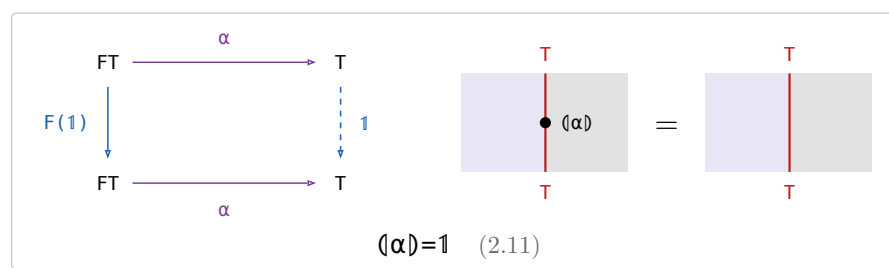
(11.4a)

An **F-algebra** is a map $f : F(A) \rightarrow A$; A is its **carrier**. An **F-homomorphism** from $f : F(A) \rightarrow A$ to $g : F(B) \rightarrow B$ is a map $h : A \rightarrow B$ with $f \circ h = F(h) \circ g$. The **initial algebra** $\alpha : F(T) \rightarrow T$ is the F-algebra with exactly one F-homomorphism $\langle f \rangle : T \rightarrow A$ to every F-algebra f .



§11.4.1. Reflection

(11.4.1a)

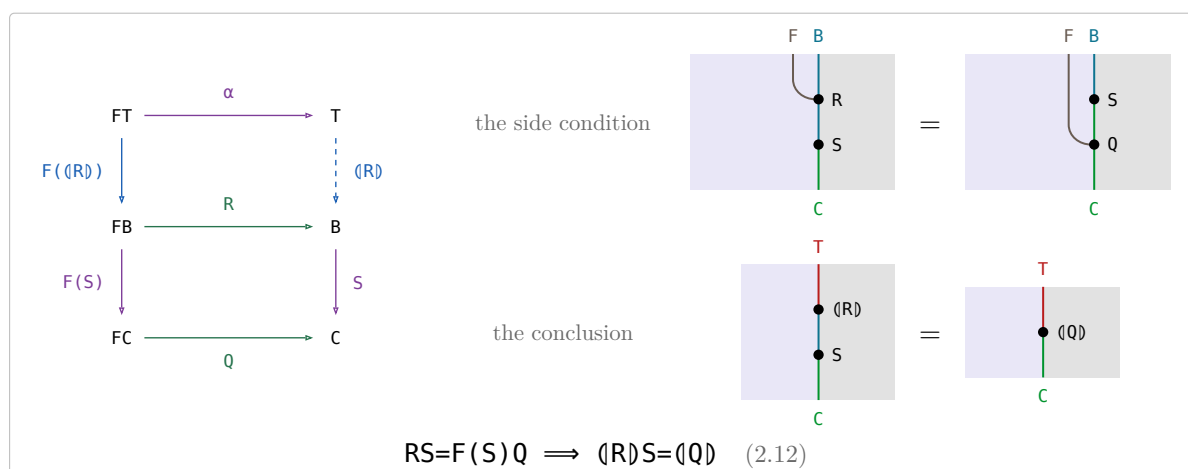


Taking a value apart with α and putting it straight back is doing nothing.

§11.4.2. Fusion: $(R:FB)$ absorb $S:R \rightarrow Q$ (in $\text{Alg}(F)$) and becomes $\langle Q \rangle$

When $S : B \rightarrow C$ is an F-homomorphism from $R : F(B) \rightarrow B$ to $Q : F(C) \rightarrow C$, folding with R and then applying S is folding with Q .

(11.4.2a)



§11.5. Type relator

(11.5a)

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha)$$

(11.5b)

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way.
functor	$T(1) = 1$ and $T(R)T(S) = T(RS)$	Acting by the identity changes nothing, and two actions in a row are one action.
type functor fusion	$T(f)(h) = (F(f, 1)h)$	A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because F is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$.
naturality of α	$\alpha T(f) = F(f, T(f)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$\text{PreservesRecip}(F) \implies T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

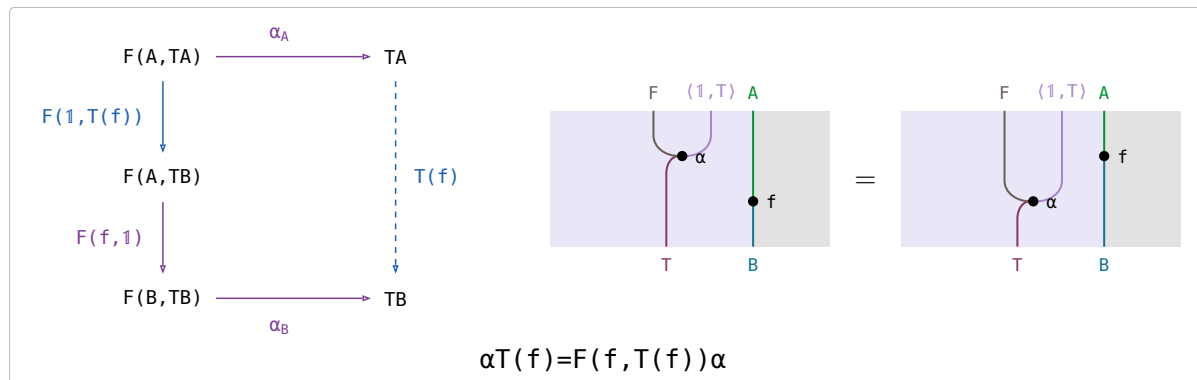
§11.5.1. Type functor

(11.5.1a)

Let F be a bifunctor taking both the parameter A and the recursive position TA , with an initial algebra $\alpha_A : F(A, TA) \rightarrow TA$ for every object A . Then T is a functor, acting on a map $f : A \rightarrow B$ by

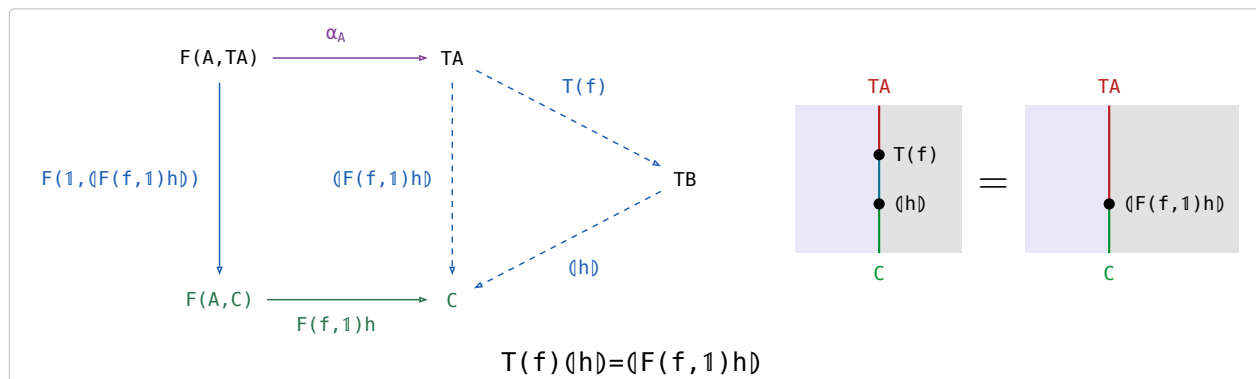
$$T(R) \triangleq (F(R, 1) \alpha)$$

(11.5.1b)



- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ is a bifunctor and a wire is a unary functor, so the two arguments are packed first: $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$ sends A to (A, TA) , and $F(\langle 1, T \rangle(A))$ is $F(A, TA)$.
- The picture is three wires — F , $\langle 1, T \rangle$, and the object — and the region between the first two is $\mathcal{A} \times \mathcal{A}$.
- α is then an ordinary natural transformation $F \circ \langle 1, T \rangle \Rightarrow T$: its bead eats the F and $\langle 1, T \rangle$ wires, and the T wire is born under it.
- $F(f, T(f))$ costs no notation. It is the bead f on the object wire with $\langle 1, T \rangle$ and F running past: $\langle 1, T \rangle$ is what turns f into the pair $(f, T(f))$, and F is what applies it.
- The law is the naturality of α , which is exactly the freedom to slide that f bead past it.

(11.5.1c)



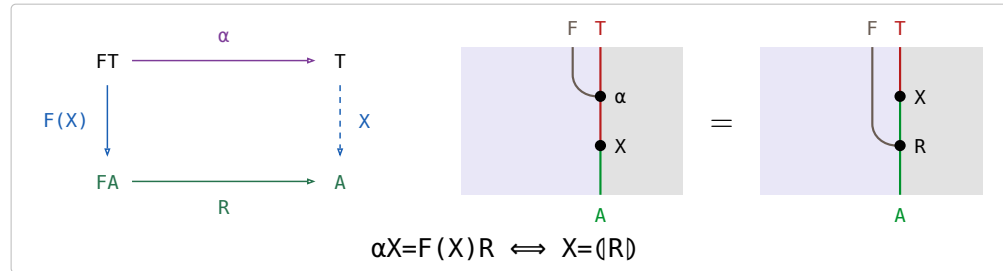
§11.6. Reduce

(11.6a)

let F be a relator and has **initial algebra** $\alpha : F(T) \rightarrow T$ in the subcategory of functions. α is also initial in the allegory:

§11.6.1. The defining equation

(11.6.1a)



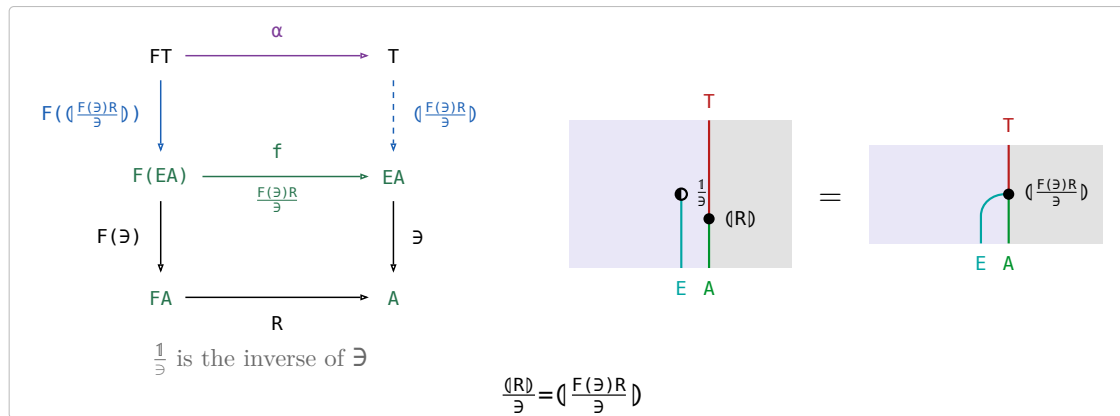
An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and A .

(11.6.1b)

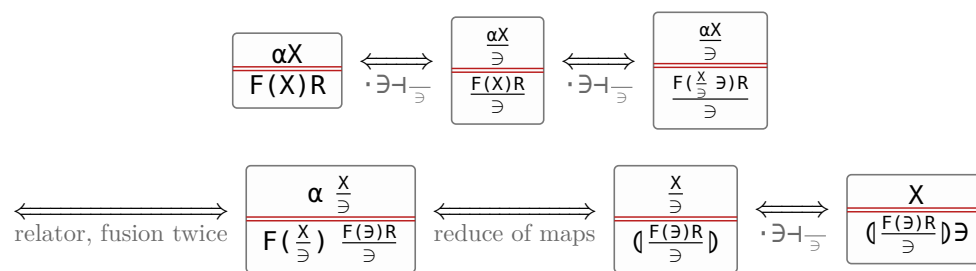
$\langle [c, f] \rangle = \text{reduce}(c, f)$		
catamorphism: arrows of the form $\langle f \rangle$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
part	Nat	[A]
datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
the fold, pointwise	$\langle [c, f] \rangle(\text{zero}) = c$ $\langle [c, f] \rangle(\text{succ}(n)) = f(\langle [c, f] \rangle(n))$	$\langle [c, f] \rangle(\text{nil}) = c$ $\langle [c, f] \rangle(\text{cons}(a, x)) = f(a, \langle [c, f] \rangle(x))$

§11.6.2. $\langle R \rangle = \langle \frac{F(\exists)R}{\exists} \rangle \exists$

(11.6.2a)



(11.6.2b)



the last two rows at $X := \langle R \rangle$:

$\alpha X = F(X)R \iff X = \langle R \rangle$ (11.6.2c)		
a relation X out of $\mathbf{T}$ satisfies the fold equation of the relation R exactly when it is $\langle R \rangle$, the fold of the map $\frac{F(\exists)R}{\exists}$ followed by $\exists$		
formula — each equation of the chain		reason
	$\alpha X = F(X)R$	
	$\frac{\alpha X}{\exists} = \frac{F(X)R}{\exists}$	$\frac{\exists}{\exists}$ is an isomorphism (10b) — $\text{frac}(X, \exists)\exists = X$
	$\frac{\alpha X}{\exists} = \frac{F(\frac{X}{\exists})R}{\exists}$	cancellation, backwards (10b) — $X = \text{frac}(X, \exists)\exists$
	$\alpha \frac{X}{\exists} = F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}$	relator; fusion, backwards, twice $\triangleq$ relator — $F(RS) = F(R)F(S)$; (10b)'s fusion at the maps α and $F(\frac{X}{\exists})$
	$\frac{X}{\exists} = \langle \frac{F(\exists)R}{\exists} \rangle$	the fold of a map algebra (11.4a) — the unique map satisfying its defining equation
	$X = \langle \frac{F(\exists)R}{\exists} \rangle \exists$	cancellation (10b) — $\text{frac}(X, \exists)\exists = X$

§11.6.3. $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$

(11.6.3a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

(11.6.3b)

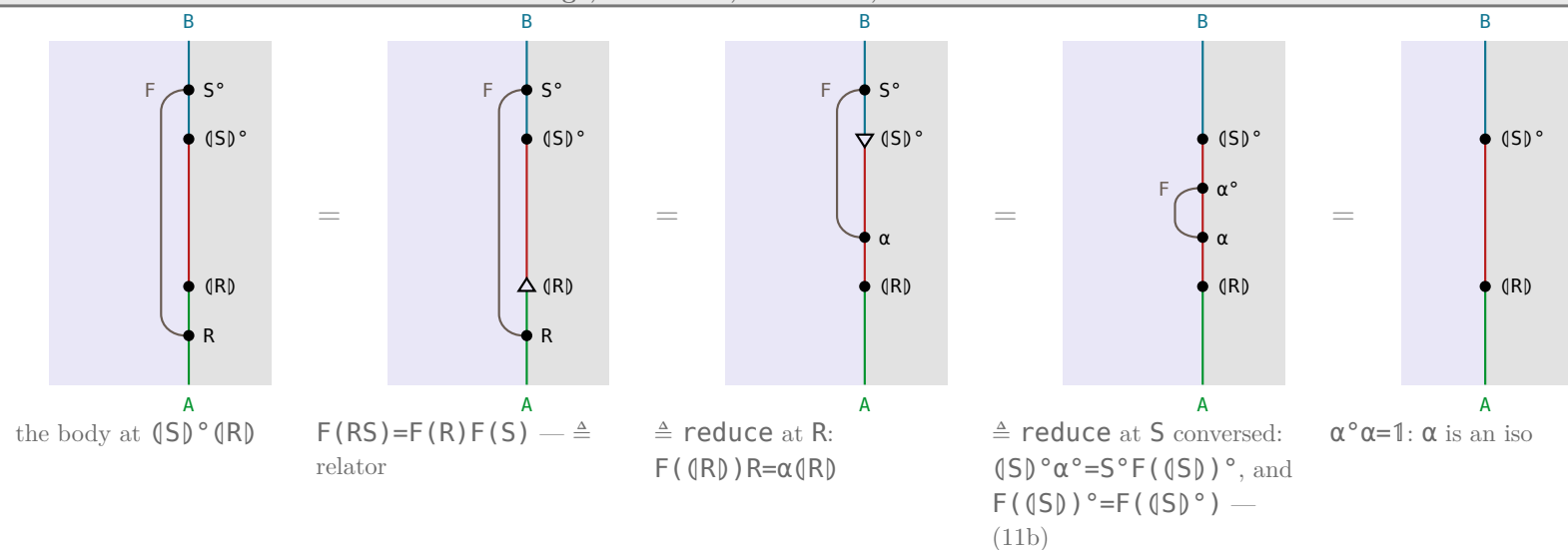
the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §11.6.4 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski: the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

§11.6.4. $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$

$$\text{PreservesRecip}(F) \Rightarrow S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$$

(11.6.4a)

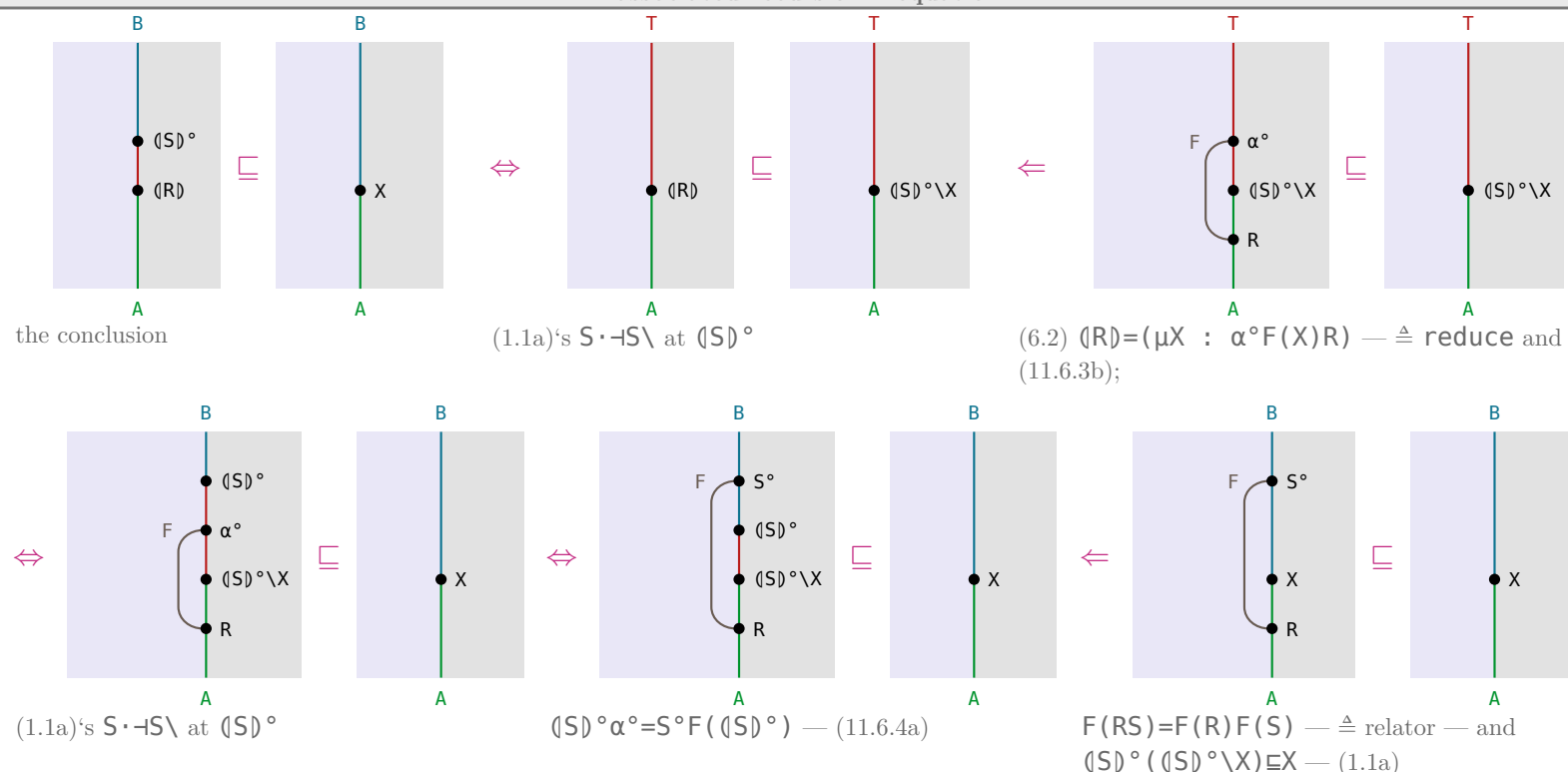
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((S)^\circ (R))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



$$\text{PreservesRecip}(F) \wedge S^\circ F(X) R \sqsubseteq X \Rightarrow (S)^\circ (R) \sqsubseteq X$$

(11.6.4b)

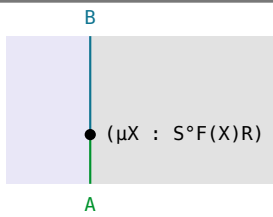
hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(S)^\circ (R)$ is included in X if X satisfies the associated recursion inequation



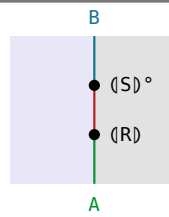
$$\text{PreservesRecip}(F) \implies \langle S \rangle^\circ \langle R \rangle = (\mu X : S^\circ F(X) R)$$

(11.6.4c)

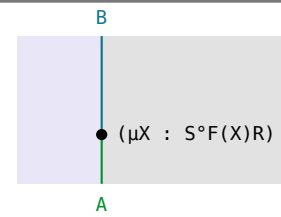
hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



$$\triangleq \mu \text{ at } \varphi(X) := S^\circ F(X) R$$

 $\sqsubseteq$


(11.6.3b)'s $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$
 at $Y := \langle S \rangle^\circ \langle R \rangle$, whose
 $S^\circ F(\langle S \rangle^\circ \langle R \rangle) R = \langle S \rangle^\circ \langle R \rangle$ is (11.6.4a)

 $\sqsubseteq$


(11.6.4b) at $X := (\mu X : S^\circ F(X) R)$, whose
 $S^\circ F(\mu X : S^\circ F(X) R) R \sqsubseteq (\mu X : S^\circ F(X) R)$ is
 (11.6.3b)'s $\varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$;

§12. Combinatorial functions

(12a)

definition	type	note
$[A] ::= \text{nil} \mid \text{cons}(A, [A])$	$\mathcal{A} \rightarrow \mathcal{A}$	The list type, under the short name it keeps.
$\text{list}(R) = ([\text{nil}, (R \times 1) \text{cons}])$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
$\text{subseq} = ([\text{nil}, \text{cons} \circ \pi_2])$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
$\text{prefix} = ([\text{nil}, \text{nil} \circ \text{cons}])$ $\text{prefix} = \text{cat} \circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc} \circ \pi_1$.
$\text{suffix} = \text{cat} \circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons} \circ \pi_2$; as a reduce it needs snoc -lists.
$\text{segment} = \text{suffix} \text{ prefix}$	$[A] \rightarrow [A]$	A contiguous stretch of xs : a suffix, then a prefix of that.
$\text{partition} = \text{concat}^\circ$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^* \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
$\text{concat} = ([\text{nil}, \text{cat}])$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\supseteq}$, listing the prefixes by increasing length.
tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\supseteq}$ by decreasing length — the opposite order.
$\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\supseteq} \text{est}(R^\circ)$	$[A] \rightarrow [A]$	The longest subsequence of xs whose every element passes p . $\text{est}(R^\circ)$ is $\triangleq \text{est}$
$R \triangleq \text{length} \leq \text{length}^\circ$	$[A] \rightarrow [A]$	The preorder filter and takewhile maximise over: the longer list wins. $\geq \triangleq \leq^\circ$
$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\supseteq} \text{est}(R^\circ)$	$[A] \rightarrow [A]$	The same with prefix for subseq : the longest prefix whose every element passes p .
$\text{mss} \triangleq \frac{\text{segment sum}}{\supseteq} \text{est}(\geq)$	$[A] \rightarrow A$	Maximum segment sum. $\text{segment} = \text{suffix prefix}$ splits it into $\frac{\text{prefix sum}}{\supseteq} \text{est}(\geq)$ on each suffix.

§12.1. $\frac{\text{subseq}}{\supseteq} = ([\text{nil} \frac{1}{\supseteq}, (\frac{1 \times \supseteq}{\supseteq} \text{E}(\text{cons}), \pi_2) \text{cup}])$

(12.1a)

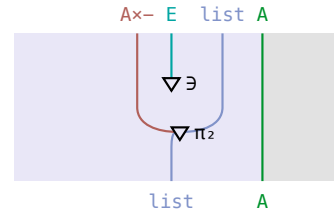
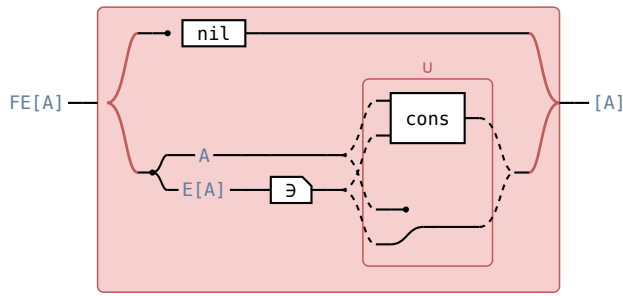
$\text{cup} \triangleq \frac{\pi_1 \supseteq \cup \pi_2 \supseteq}{\supseteq} : EA \times EA \rightarrow EA$, so $\frac{R \cup S}{\supseteq} = (\frac{R}{\supseteq}, \frac{S}{\supseteq}) \text{cup}$.

$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}] \quad (12.1b)$$

the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (11.6.2b) at **subseq** = $\langle [\text{nil}, \text{cons } u \pi_2] \rangle$, $\triangleq$ **subseq**.

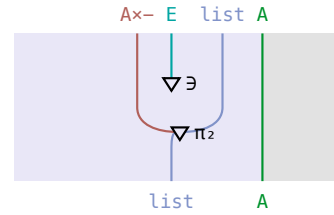
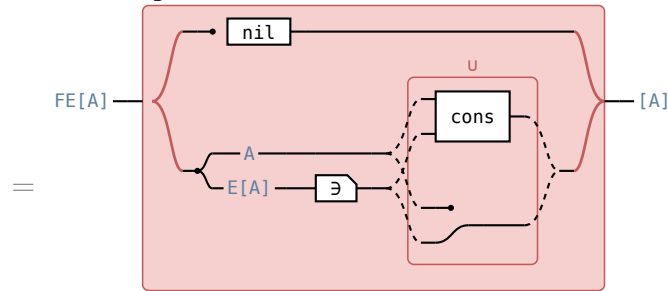
circuit — the fork is $F([A]) = 1 + A \times [A]$: **nil** above, the pair below

Hinze–Marsden



$$\frac{F(\exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

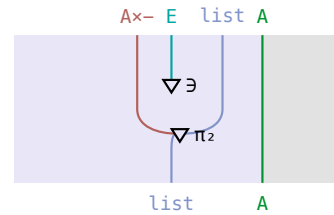
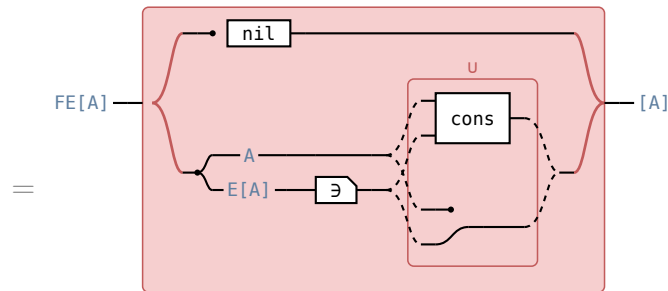
the π_2 operand of $\text{cons } u \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists)\pi_2$



$$\frac{(1+1 \times \exists)[\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X) = 1 + A \times X \triangleq \text{subseq}$$

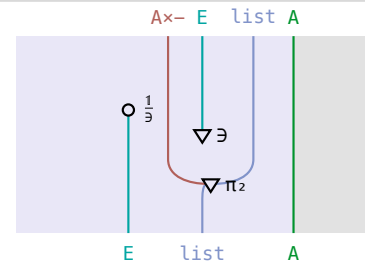
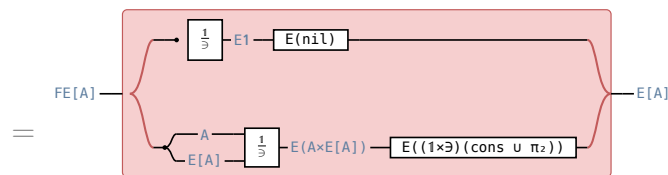
the same operand under $1 + 1 \times \exists$, whose $1 \times \exists$ summand it sits in



$$\frac{[\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]}{\exists}$$

$$R + S \triangleq [Rl, Sr], l[R, S] = R, r[R, S] = S \quad (11.2a)$$

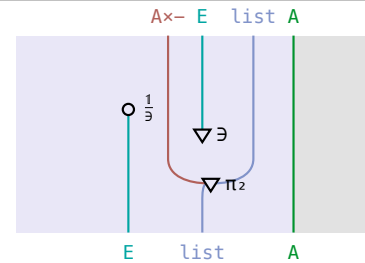
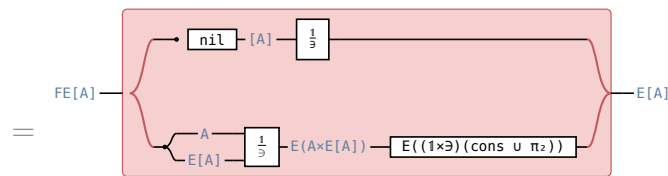
the π_2 operand of the second arm $(1 \times \exists)(\text{cons } u \pi_2)$



$$[\frac{\text{nil}}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]$$

$$(11.2.1a) \text{ at } T := [\text{nil}, (1 \times \exists)(\text{cons } u \pi_2)]$$

the π_2 operand under its $1 \times \exists$, the arm (12.1c)'s square rewrites, $(1 \times \exists)\pi_2 = \pi_2 \exists$

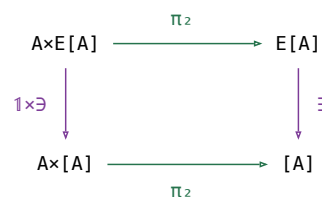


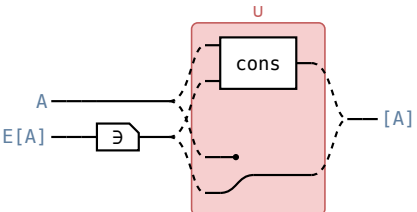
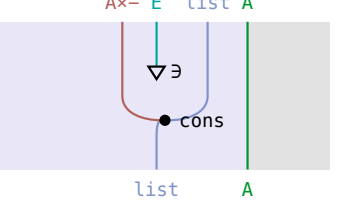
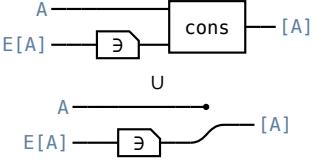
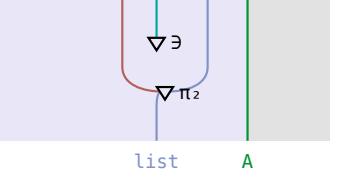
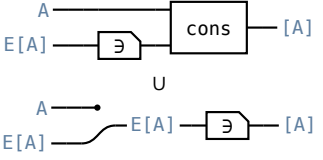
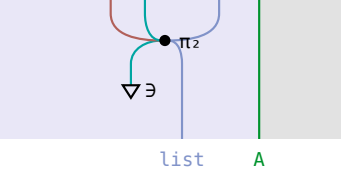
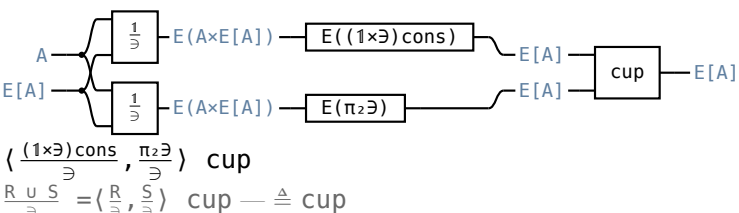
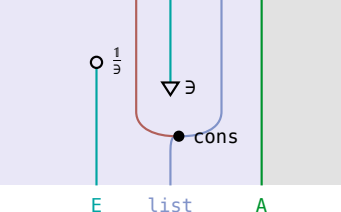
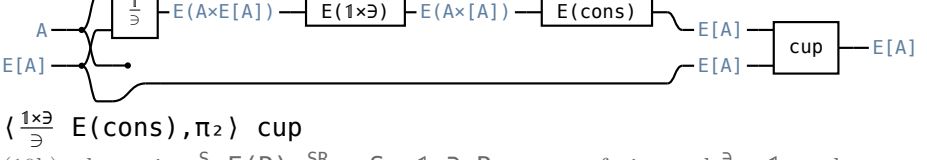
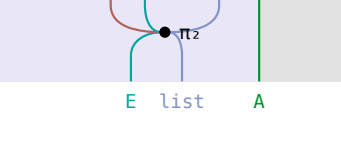
$$[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \pi_2)}{\exists}]$$

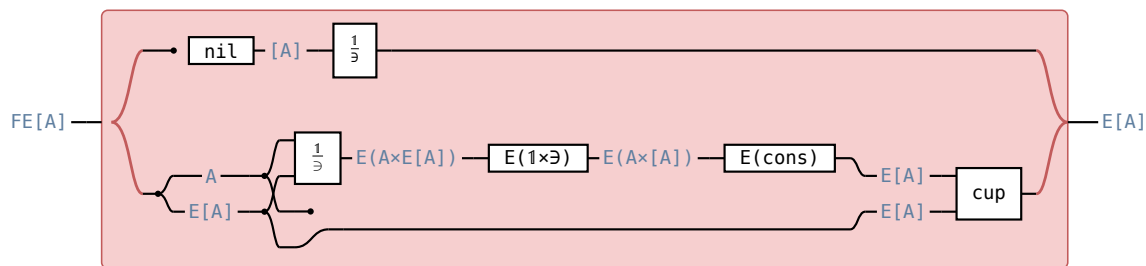
$$(10b), \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$

the same operand; the two rows differ only in the **nil** arm

(12.1c)



$\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists} = \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ power transpose of join: the power transpose of the join of two relations is $\left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$, where cup is the function that returns the union of two sets	(12.1d)
circuit	Hinze–Marsden
 $\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists}$ (12.1b)'s second branch	 the cons operand of $\text{cons} \cup \pi_2$
 $\frac{(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2}{\exists}$ $T(X_1 \cup X_2) = TX_1 \cup TX_2$ — (1.2b)	 the π_2 operand of $(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2$
 $\frac{(1 \times \exists)\pi_2}{\exists} = \pi_2 \exists$ — (11.1.1b) at π_2, an equality because $\mathbb{1}$ is entire	 the π_2 operand of $(1 \times \exists)\text{cons} \cup \pi_2 \exists$
 $\frac{(1 \times \exists)\text{cons}, \pi_2 \exists}{\exists} \text{cup}$ $\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \triangleq \text{cup}$	 the cons operand under its $1\% \exists$
 $\left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ (10b), absorption $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$ at $S := \mathbb{1} \times \exists$, $R := \text{cons}$; fusion and $\frac{\exists}{\exists} = \mathbb{1}$ on the π_2 operand	 the π_2 operand, bare π_2



$$\left[\frac{\text{nil}}{\exists}, \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes $P\text{cons}$; cons is a map, and there $P(\text{cons}) = E(\text{cons})$ — (11.3d).

(12.1e)

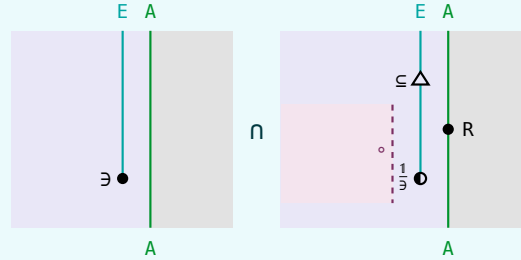
§13. Optimisation Problems

§13.1. $\text{est}(R) \triangleq \exists n(\epsilon \setminus R^\circ)$

(13.1a)

For $R : A \rightarrow A$, $\text{est}(R) \triangleq \exists n(\epsilon \setminus R^\circ) : EA \rightarrow A$ $X \in \epsilon \setminus R^\circ \iff EX \in R^\circ$, and EX runs $y \rightarrow xS \rightarrow x$, member first, so its pair (y, x) is in R° exactly when $x R y$; at $R \triangleq \leq$ that is $x \leq y$ for every $y \in xS$, the least — $\epsilon \setminus \leq$ would give $y \leq x$, the greatest.

$\text{est}(R) = \exists n \subseteq (\frac{R}{S})^\circ$ x is the $\text{est}(R)$ of xS when $x \in xS$ and $xS \subseteq \Lambda(R)(x)$, the set of every y with $x R y$ — (10.2a) at $Z \triangleq R^\circ$



$xS (\text{est}(R)) x \iff x \in xS \wedge (\forall y \in xS. x R y)$ the same predicate under the same letter, $\min R$, so $\text{est}(R) = \min(R^\circ)$ once R is an arrow;

$xS (\text{est}(R)) x \iff (x \text{ in } xS)$ and all $x R \setminus : xS$ in q

$\text{est}(R) = \exists n \text{ all } R^\circ$ all $R \triangleq \epsilon \setminus R$, q 's all; the chains below keep it written $\epsilon \setminus$

$E(R) \triangleq \frac{\exists R}{S} : EA \rightarrow EB$, $xS E(R) yS \iff yS = \{y \mid \exists x \in xS. x R y\}$ the image of xS , (10b)

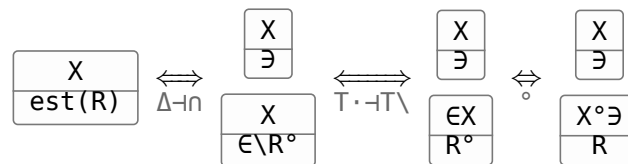
$P(R) : EA \rightarrow EB$, $xS P(R) yS \iff (\forall x \in xS. \exists y \in yS. x R y) \wedge (\forall y \in yS. \exists x \in xS. x R y)$ every x and every y has a partner, (11.3b)

(13.1b)

the law	what it says
$X \subseteq \text{est}(R) \iff X \subseteq \exists$ and $X^\circ \subseteq R$	in the set, and below every element of it
$\frac{1}{S} (\epsilon \setminus R) = R$	bounding a singleton is bounding its element
$\frac{S}{S} (\epsilon \setminus R) = S^\circ \setminus R$	bound S 's image without building the set
$\text{union} \triangleq \frac{\exists \exists}{S} : E(EA) \rightarrow EA$	flattens a set of sets
$\text{union} (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)$	bound a union by bounding each member set
$\frac{1}{S} \text{est}(R) = 1 \cap R^\circ$	a singleton's minimum is its element, where R is reflexive $\frac{S}{S} \text{est}(R)$ at $S := 1$
$\frac{S}{S} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)$	an S -value that points to every S -value
$\frac{S}{S} \text{est}(R \cap S^\circ S) = \frac{S}{S} \text{est}(R)$	only R between values S gives one argument counts — context
$E(S) \text{est}(R) = (\exists S) \cap ((\exists S)^\circ \setminus R^\circ)$	the same for the image of a set $\frac{S}{S} \text{est}(R)$ at $S := \exists S$
$\text{Map}(f) \implies P(f) \text{est}(R) = \text{est}(f R f^\circ) f$	shunt a function through a minimum
$P(S) \text{est}(R) = (\exists S) \cap (\epsilon \setminus (S R^\circ))$ R reflexive	fusion with the power relator $\exists$ is the only proof here that tabulates
$P(S) \text{est}(R) \subseteq (\exists S) \cap (\epsilon \setminus (S R^\circ))$	the half of the row above that costs nothing
$P(\text{est}(R)) \text{est}(R) \subseteq \text{union est}(R)$ R transitive	a minimum in each set, then a minimum of those
$RR \subseteq R \implies P(\text{est}(R)) \text{est}(R) = P(\text{dom}(\text{est}(R))) \text{union est}(R)$	the same as an equality, once empty sets are dropped

§13.1.1. $X \subseteq \text{est}(R) \iff X \subseteq \exists$ and $X^\circ \subseteq R$

(13.1.1a)



§13.1.2. $\frac{1}{S} (\epsilon \setminus R) = R$

(13.1.2a)

$$\begin{array}{c} X \\ \hline \frac{1}{\exists} (\epsilon \backslash R) \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \frac{1}{\exists}^\circ X \\ \hline \epsilon \backslash R \end{array} \xleftrightarrow{T^\circ \cdot \neg T^\circ} \begin{array}{c} \epsilon \frac{1}{\exists}^\circ X \\ \hline R \end{array} \xleftrightarrow{\circ} \begin{array}{c} (\frac{1}{\exists} \exists)^\circ X \\ \hline R \end{array} \xleftrightarrow{i \cdot \neg E} \begin{array}{c} X \\ \hline R \end{array}$$

§13.1.3. $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$

$\frac{S}{\exists}$ gathers the S -image of a point into one set and $\epsilon \backslash R$ asks that every member of that set be R -related to the target, so the set cancels and $S^\circ \backslash R$ asks it of the S -image directly.

(13.1.3a)

$$\begin{array}{c} X \\ \hline \frac{S}{\exists} (\epsilon \backslash R) \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \frac{S}{\exists}^\circ X \\ \hline \epsilon \backslash R \end{array} \xleftrightarrow{T^\circ \cdot \neg T^\circ} \begin{array}{c} \epsilon \frac{S}{\exists}^\circ X \\ \hline R \end{array} \xleftrightarrow{\circ} \begin{array}{c} (\frac{S}{\exists} \exists)^\circ X \\ \hline R \end{array} \xleftrightarrow{\cdot \exists \neg \frac{S}{\exists}} \begin{array}{c} S^\circ X \\ \hline R \end{array} \xleftrightarrow{T^\circ \cdot \neg T^\circ} \begin{array}{c} X \\ \hline S^\circ \backslash R \end{array}$$

§13.1.4. $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\text{union} R)$

(13.1.4a)

$$\begin{array}{c} X \\ \hline \text{union} (\epsilon \backslash R) \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \text{union}^\circ X \\ \hline \epsilon \backslash R \end{array} \xleftrightarrow{T^\circ \cdot \neg T^\circ} \begin{array}{c} \epsilon \text{union}^\circ X \\ \hline R \end{array} \xleftrightarrow{\circ} \begin{array}{c} (\text{union} \exists)^\circ X \\ \hline R \end{array} \xleftrightarrow{\cdot \exists \neg \frac{S}{\exists}} \begin{array}{c} (\exists \exists)^\circ X \\ \hline R \end{array} \xleftrightarrow{\circ, T^\circ \cdot \neg T^\circ} \begin{array}{c} X \\ \hline \epsilon \backslash (\text{union} R) \end{array}$$

§13.1.5. $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \backslash R^\circ)$

(13.1.5a)

$$\begin{array}{c} X \\ \hline \frac{S}{\exists} \text{est}(R) \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \frac{S}{\exists}^\circ X \\ \hline \text{est}(R) \end{array} \xleftrightarrow{\Delta \neg \cap} \begin{array}{c} \frac{S}{\exists}^\circ X \\ \hline \exists \end{array} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} X \\ \hline \frac{S}{\exists} \exists \end{array} \xleftrightarrow{\cdot \exists \neg \frac{S}{\exists}, (13.1.3a)} \begin{array}{c} X \\ S \end{array} \xleftrightarrow{\Delta \neg \cap} \begin{array}{c} X \\ \hline S \cap (S^\circ \backslash R^\circ) \end{array}$$

§13.1.6. $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(R \cap S^\circ S)$

(13.1.6a)

$$\begin{array}{c} X \\ \hline \frac{S}{\exists} \text{est}(R \cap S^\circ S) \end{array} \xleftrightarrow{(13.1.5a), \circ} \begin{array}{c} X \\ \hline S \cap (S^\circ \backslash (R^\circ \cap S^\circ S)) \end{array} \xleftrightarrow{\Delta \neg \cap, T^\circ \cdot \neg T^\circ} \begin{array}{c} X \\ S \end{array} \xleftrightarrow{\Delta \neg \cap, S^\circ \cdot \text{monotone}} \begin{array}{c} X \\ S \end{array} \xleftrightarrow{T^\circ \cdot \neg T^\circ, \Delta \neg \cap} \begin{array}{c} X \\ \hline S \cap (S^\circ \backslash R^\circ) \end{array} \xleftrightarrow{(13.1.5a)} \begin{array}{c} X \\ \hline \frac{S}{\exists} \text{est}(R) \end{array}$$

§13.1.7. $P(f) \text{est}(R) = \text{est}(f R f^\circ) f$

(13.1.7a)

$$\begin{array}{c} P(f) \text{est}(R) \\ \hline f \text{ a map} \end{array} \xleftrightarrow{P=E \text{ on maps, (13.1.5a)}} \begin{array}{c} (\exists f) \cap ((\exists f)^\circ \backslash R^\circ) \end{array} \xleftrightarrow{\circ, T^\circ \cdot \neg T^\circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} (\exists f) \cap (\epsilon \backslash (f R^\circ)) \end{array} \xleftrightarrow{\text{modular law, } f \text{ simple}} \begin{array}{c} (\exists \cap ((\epsilon \backslash (f R^\circ)) f^\circ)) f \end{array} \xleftrightarrow{\cdot f \neg f^\circ, \circ, \text{est}} \begin{array}{c} \text{est}(f R f^\circ) f \end{array}$$

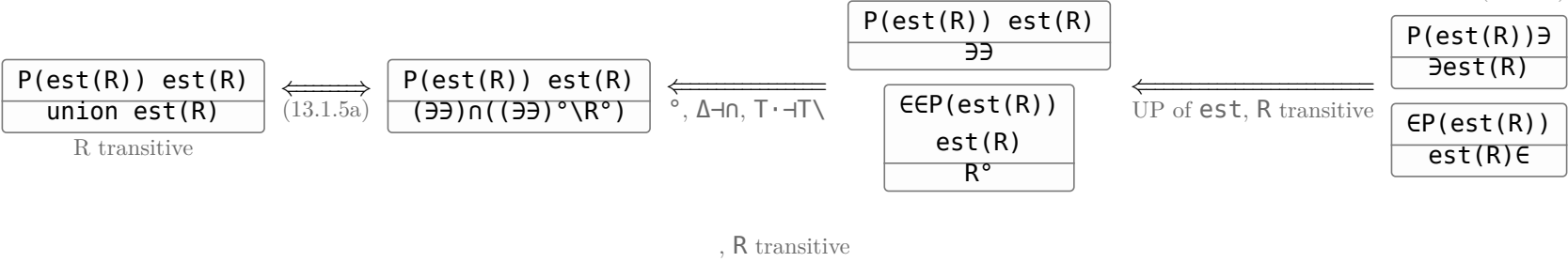
§13.1.8. $P(S) \text{est}(R) \subseteq (\exists S) \cap (\epsilon \backslash (S R^\circ))$

(13.1.8a)

$$\begin{array}{c} P(S) \text{est}(R) \\ \hline (\exists S) \cap (\epsilon \backslash (S R^\circ)) \end{array} \xleftrightarrow{\Delta \neg \cap, T^\circ \cdot \neg T^\circ} \begin{array}{c} P(S) \text{est}(R) \\ \hline \exists S \end{array} \xleftrightarrow{\text{UP of est}} \begin{array}{c} P(S) \exists \\ \hline \exists S \end{array} \xleftrightarrow{\text{UP of est}} \begin{array}{c} \epsilon P(S) \text{est}(R) \\ \hline S R^\circ \end{array} \xleftrightarrow{\text{UP of est}} \begin{array}{c} \epsilon P(S) \\ \hline S \epsilon \end{array}$$

§13.1.9. $P(\text{est}(R)) \text{est}(R) \subseteq \text{union est}(R)$

(13.1.9a)



§13.2. Lax natural transformations (LaT)

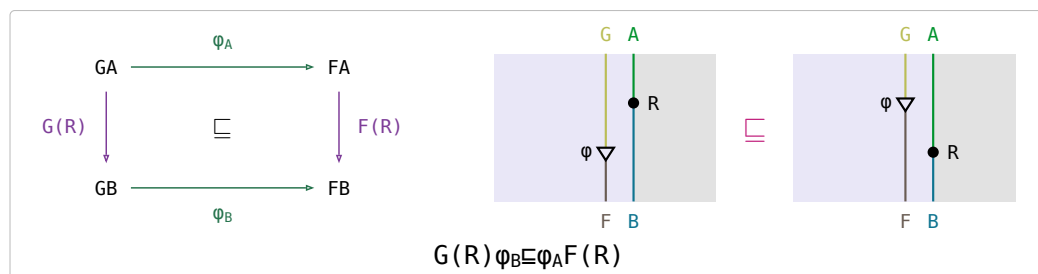
(13.2a)

For relators $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and components $\varphi_A : GA \rightarrow FA$, φ is **lax at** $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ — both sides $GA \rightarrow FB$, the left through the component at B , the right through the one at A .

$\varphi : G \Rightarrow F$ is a **lax natural transformation** (LaT) when it is lax at **every** R . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality $G(f)\varphi = \varphi F(f)$: laxness is about relations only.

(13.2b)



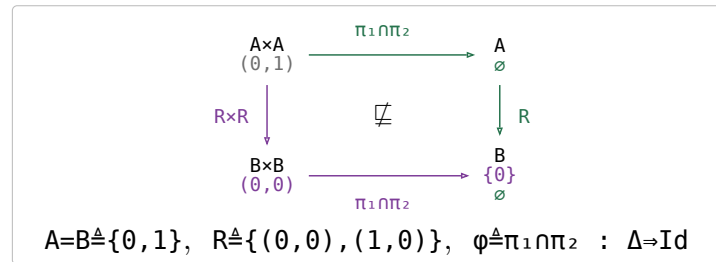
closed under	commutative diagram	string diagram
composition $\psi\phi$	$H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\phi_B \sqsubseteq \phi_A F(R)$ give $H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)$	
horizontal composition $\chi\phi$	$\phi : G \Rightarrow F$ with $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and $\chi : L \Rightarrow K$ with $L, K : \mathcal{D} \rightarrow \mathcal{E}$ give $\chi \circ \phi : L \circ G \Rightarrow K \circ F$, the family $A \mapsto \chi_{GA} K(\phi_A)$ $L(G(R))\chi_{GB} \sqsubseteq \chi_{GA} K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))$ is K applied to ϕ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\phi) : K \circ G \Rightarrow K \circ F$, K composed on the outside; $\phi := \mathbb{1}_G$ gives $\chi G : L \circ G \Rightarrow K \circ G$, G composed on the inside two candidates, $\chi_{GA} K(\phi_A)$ and $L(\phi_A)\chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
union $\phi \cup \psi$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)$	
a relator K $K(\phi)$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ gives $K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))$	
product $\phi \times \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi \times \psi : G \times G' \Rightarrow F \times F'$ $(R \times S)(U \times V) = (RU) \times (SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle \phi, \psi \rangle = \Delta(\phi \times \psi)$, and its ONE cost is the lax copy law $R \triangleleft \sqsubseteq \triangleleft (R \times R)$ — (2b)	
coproduct $\phi + \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi + \psi : G + G' \Rightarrow F + F'$ $(R + S)(U + V) = (RU) + (SV)$ and monotonicity in both slots; the co-fork is the derived case $[\phi, \psi] = (\phi + \psi) \nabla$, and ∇ costs nothing the step it would need is $(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)$, the wrong direction of (4c) a counterexample: (13.2.1a)	
meet — fails $\phi \cap \psi$		

Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A (XY) \sqsubseteq (X'Y') R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$, quantified over R ; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference

§13.2.1. meet is not closed in LaT

counterexample — $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$ lax natural does NOT give $\varphi \cap \psi$ lax natural

(13.2.1a)



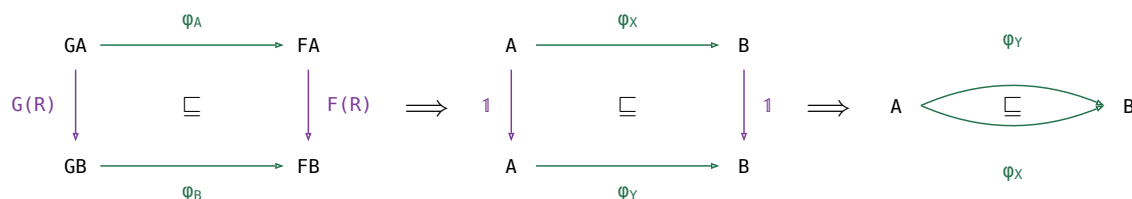
§13.2.2. Two stacked towers

(13.2.2a)

	lower — inside ONE allegory	upper — between allegories
0-cell	an object A	an allegory $\mathcal{C}$ — a whole lower tower
1-cell	a relation $R : A \rightarrow B$	a relator $F : \mathcal{C} \rightarrow \mathcal{D}$
2-cell	$R \sqsubseteq S$	a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$
order	$R \sqsubseteq S$	$\varphi \sqsubseteq \psi$ componentwise
union	$R \cup S$	$\varphi \cup \psi$, survives (13.2c)
product	$R \times S \triangleq \times$	$\varphi \times \psi$, survives; a TENSOR, not a categorical product; SYMMETRIC (11.1b) (13.2c)
coproduct	$R + S$	$\varphi + \psi$, survives; a BIPRODUCT (13.2c)
meet	$R \cap S$	$\varphi \cap \psi$ componentwise, fails (13.2.1a)
converse	R°	φ° , fails, oplax: $\varphi_A^\circ \mathbf{G}(R) \sqsubseteq \mathbf{F}(R) \varphi_B^\circ$
zero object	z with $1 \ z = 0$	the constant relator at such a z , initial and terminal

§13.2.3. RES is a special case of LaT

(13.2.3a)



$\mathbf{G} \triangleq \text{const } A, \mathbf{F} \triangleq \text{const } B$ — the relators $X \rightarrow A$ and $X \rightarrow B$, each sending every arrow to $1 \triangleq \text{lax}$
 with $\mathcal{C}$ two objects and one non-identity arrow $X \rightarrow Y$, a LaT $\text{const } A \Rightarrow \text{const } B$ is the pair $\varphi_X, \varphi_Y : A \rightarrow B$ with $\varphi_Y \sqsubseteq \varphi_X$ — the datum RES

not every LaT is constant: $\exists : \mathbf{P} \Rightarrow \text{Id}$ (11.3d) $\pi_2 : (\mathbf{A} \times -) \Rightarrow \text{Id}$ (12.1c) $\triangleleft : \text{Id} \Rightarrow \Delta$ (2b)

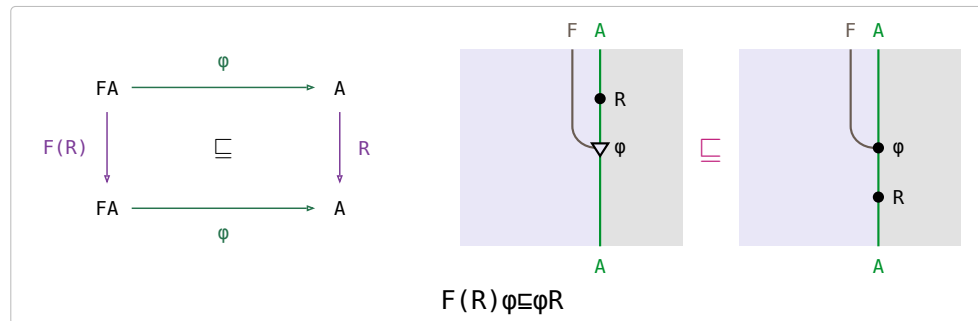
§13.3. Monotonic algebras

(13.3a)

An F -algebra $\varphi : FA \rightarrow A$ is **monotonic on** $R : A \rightarrow A$ when it is lax at R at $G := F$, $F := \text{Id}$: $F(R) \varphi \sqsubseteq \varphi R$. R is an **endorelation**, so $B = A$ and the two components φ_A, φ_B are the one arrow φ — an algebra, not a family.

For a map $f : FA \rightarrow A$ that is $f \circ F(R) f \in R$ (1.1a)'s $f \circ \cdot \dashv f \cdot$ at $X := F(R) f$, $Y := R$, equivalently $F(R) \sqsubseteq f R f^\circ \cdot f \dashv \cdot f^\circ$ then $f^\circ \cdot \dashv f \cdot, .$ $(\leq \times \leq) + \sqsubseteq + \leq$ — addition on Nat is monotonic on $\leq$, which at the point level reads $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$.

(13.3b)



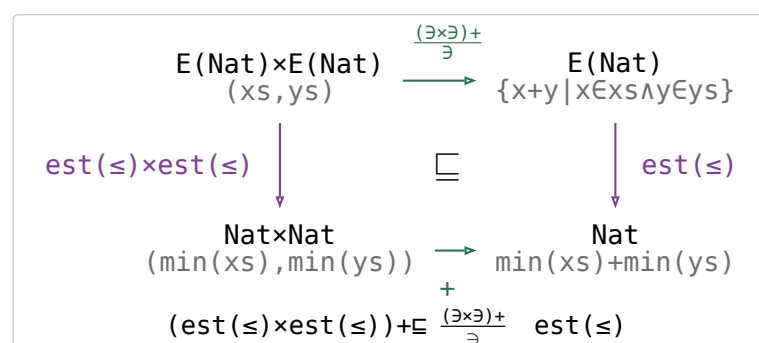
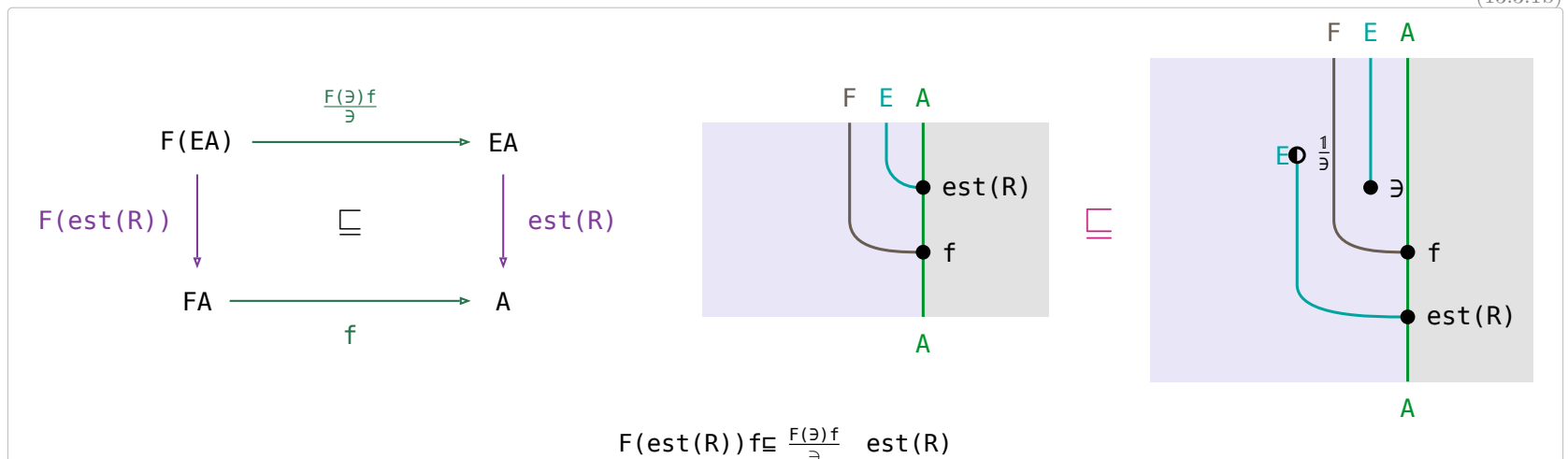
§13.3.1. Function f is monotonic on R iff it distributes over R

(13.3.1a)

$f : FA \rightarrow A$ **distributes over** R if $F(\text{est}(R)) f \sqsubseteq \frac{F(\exists)f}{\exists} \text{est}(R)$.

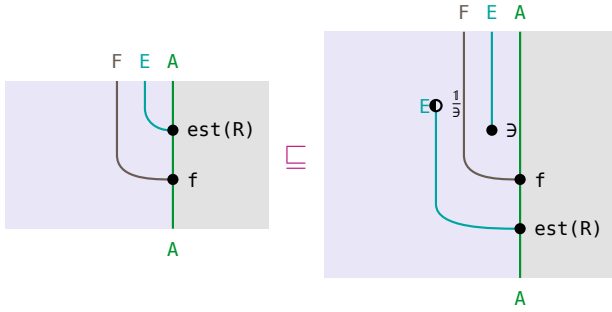
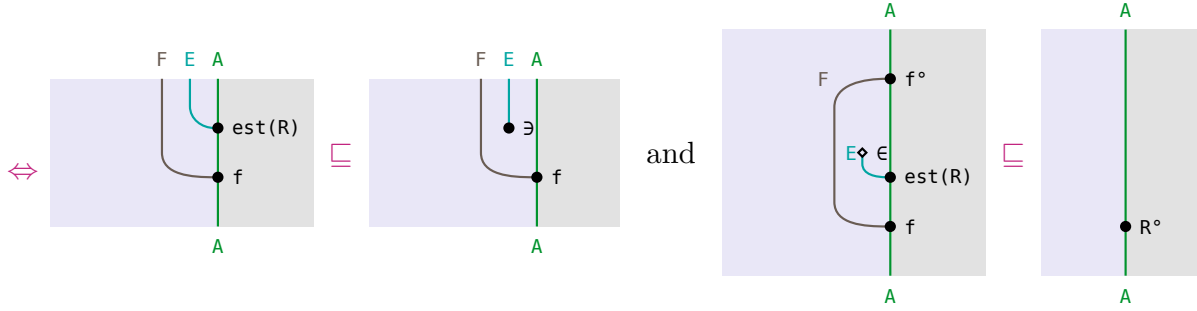
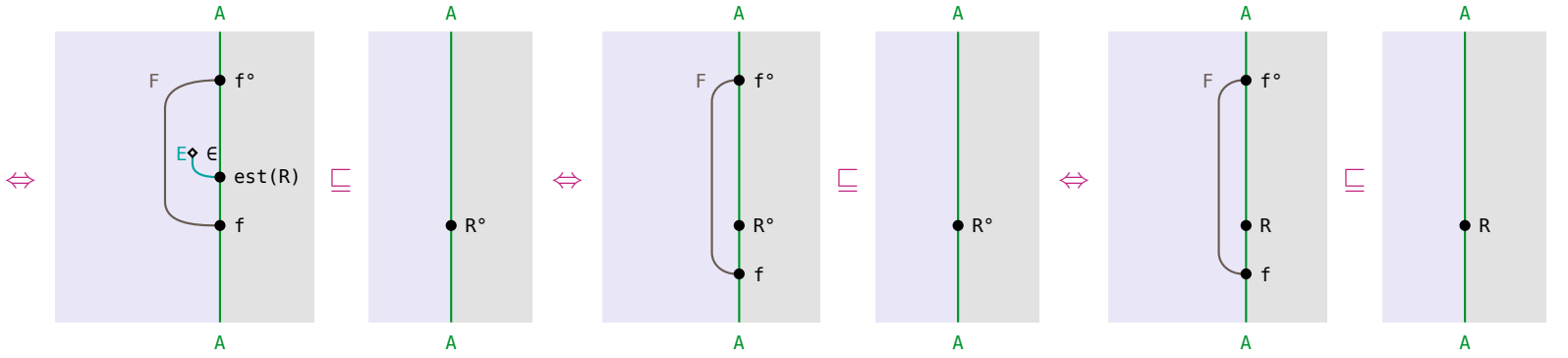
$+$ distributes over $\leq$, at the point level $\min(xs) + \min(ys)$ is a least element of $\{x + y \mid x \in xs \wedge y \in ys\}$ for xs, ys non-empty and $\min \triangleq \text{est}(\leq)$.

(13.3.1b)



$$f^\circ F(R) f \sqsubseteq R \iff F(\text{est}(R)) f \sqsubseteq \frac{F(\exists) f}{\exists} \text{est}(R)$$

(13.3.1c)

 function f is monotonic over R if and only if it distributes over R ; f a map, R reflexive

 f distributes over R — $\triangleq$ distributes — the fraction bent as (10.1a)

 (13.1.5a) splits the bound in two, (8.1c) moving $(F(\exists) f)^\circ$ across

 $\text{est}(R) \sqsubseteq \exists$ — $\triangleq$ est — so the first conjunct drops

 $(F(\exists) f)^\circ = f^\circ F(\epsilon)$ — $\triangleq$ $^\circ$ — and ϵ
 $\text{est}(R) = R^\circ$ — $\triangleq$ est , R reflexive

 both sides conversed — $F(R^\circ)^\circ = F(R)$, (11b)

(13.3.2a)

circuit	Hinze–Marsden
$\langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq \langle \frac{\langle S \rangle}{\sup} \text{ est}(R) \rangle$ the conclusion: one minimum kept at each step is below every result collected and one minimum taken at the end	
$\langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq \langle S \rangle$ and $\langle S \rangle^\circ \langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq R^\circ$ (13.1.5a) at $\langle S \rangle$; $\langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq \langle S \rangle^\circ \setminus R^\circ \iff \langle S \rangle^\circ \langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq R^\circ$ — (8.1c)	
$\frac{S}{\sup} \text{ est}(R) \sqsubseteq S$ the left conjunct; (13.1.5a), and $\langle - \rangle$ is monotone	
$A \text{---} \boxed{S} \text{---} \boxed{F(R^\circ)} \text{---} FA \text{---} \boxed{\frac{1}{\sup}} \text{---} EFA \text{---} \boxed{E(S)} \text{---} EA \text{---} \boxed{\text{est}(R)} \text{---} A \sqsubseteq A \text{---} \boxed{R} \text{---} A$ the right conjunct; $\langle S \rangle^\circ \langle \frac{S}{\sup} \text{ est}(R) \rangle$ is the least X with $X = S^\circ F(X)$ ($\frac{S}{\sup} \text{ est}(R)$) hylomorphism theorem — so Knaster–Tarski leaves this one inequation $\frac{S}{\sup} = \frac{1}{\sup} E(S)$ — (10.1a)	
$A \text{---} \boxed{R} \text{---} \boxed{S} \text{---} FA \text{---} \boxed{\frac{1}{\sup}} \text{---} EFA \text{---} \boxed{E(S)} \text{---} EA \text{---} \boxed{\text{est}(R)} \text{---} A$ $S^\circ F(R^\circ) \sqsubseteq R^\circ S^\circ$ — $\triangleq$ monotonic at S, conversed; $F(R)^\circ = F(R^\circ)$ — (11b)	
$A \text{---} \boxed{R} \text{---} \boxed{R} \text{---} A$ $S^\circ \langle \frac{S}{\sup} \text{ est}(R) \rangle \sqsubseteq S^\circ (S^\circ \setminus R^\circ) \sqsubseteq R^\circ$ — (13.1.5a), (8.1c)	
$A \text{---} \boxed{R} \text{---} A$ R transitive	

§13.3.3. $\text{takewhile}(p) = ([\text{nil}, (\lambda x. p \rightarrow \text{cons}, \rightarrow \text{nil})])$

(13.3.3a)

name	definition	type	example	in words
F	$FX = 1 + A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$		nothing, or a head and a tail
nil, cons	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$ $\triangleq \text{subseq}$	$1 \rightarrow [A],$ $A \times [A] \rightarrow [A]$	$\text{cons}(3, [1, 2]) = [3, 1, 2]$	the empty list; a head onto a tail
α	$[\text{nil}, \text{cons}]$	$F[A] \rightarrow [A]$		both constructors as one map
p	a coreflexive	$A \rightarrow A$	$p \triangleq \text{even} \rightarrow 2 \rightarrow p \rightarrow 2$, and $3 \notin \text{dom}(p)$	$\{(a, a) \mid a \text{ passes the test}\}$
R	$\text{length} \leq \text{length}^\circ$, a preorder	$[A] \rightarrow [A]$	$[1] R [1, 2]$	$xs R ys \iff \text{length}(xs) \leq \text{length}(ys)$
$\rightarrow \text{nil}$	the constant nil — the second nil of prefix	$A \times [A] \rightarrow [A]$	$(\rightarrow \text{nil})(3, [1, 2]) = \text{nil}$	drop the pair, return nil
prefix	$([\text{nil}, \rightarrow \text{nil} \cup \text{cons}]) \triangleq \text{subseq}$	$[A] \rightarrow [A]$	$[3, 1, 2] \text{ prefix } [3, 1]$	$xs \text{ prefix } ys \iff \exists zs. xs = ys \# zs$ — at each cons, stop or keep the head
S	$[\text{nil}, \rightarrow \text{nil} \cup (p \times 1) \text{ cons}]$	$F[A] \rightarrow [A]$	$(4, [2]) S [4, 2]$, and $(3, [2]) S \text{ nil only}$	prefix's algebra with one extra p — stop, or keep a head that passes p
$(g \rightarrow X, Y)$	X where g is defined and Y where it is not	$A \rightarrow B$		the test picks the branch

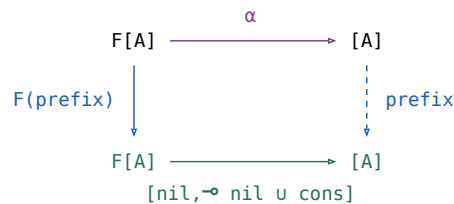
$\text{nil} R = \tau$, $\text{nil} R^\circ = \text{nil}$ nil is the shortest list — below every list, and above only itself, so it loses every $\text{est}(R^\circ)$

prefix = $([\text{nil}, \rightarrow \text{nil} \cup \text{cons}])$

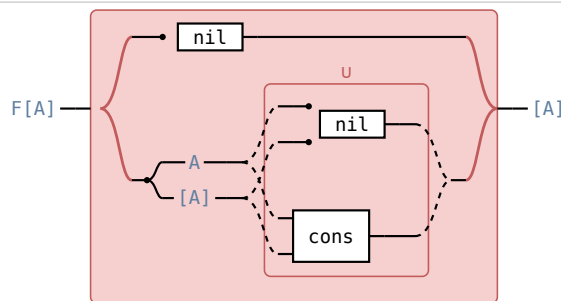
(13.3.3b)

the fold whose algebra, at each **cons**, stops with **nil** or keeps the head: $xs \text{ prefix } ys \iff \exists zs. xs = ys \# zs$

the defining square $\alpha \text{ prefix} = F(\text{prefix})[\text{nil}, \rightarrow \text{nil} \cup \text{cons}]$ — build the list and then take a prefix, or take a prefix of the tail and then rebuild with the algebra



$F = 1 + A \times -$, so $F(\text{prefix}) = 1 + 1 \times \text{prefix}$: the head passes, the fold recurses on the tail alone



the algebra $[\text{nil}, \rightarrow \text{nil} \cup \text{cons}] : F([A]) \rightarrow [A]$ as a circuit: **nil** on the **1** branch; on a pair (a, ys') two outputs, **nil** and $\text{cons}(a, ys')$

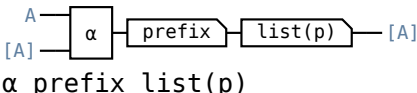
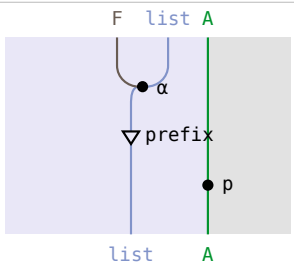
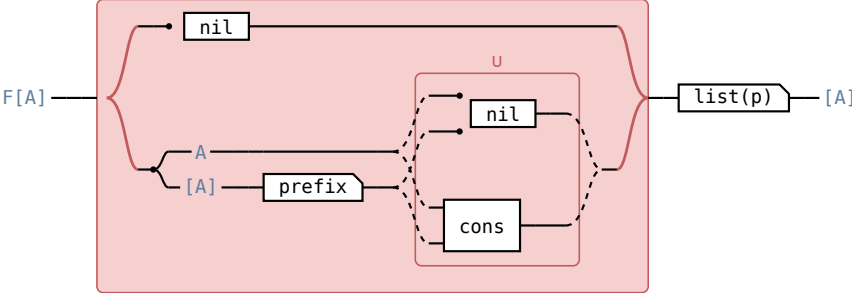
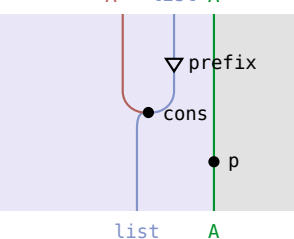
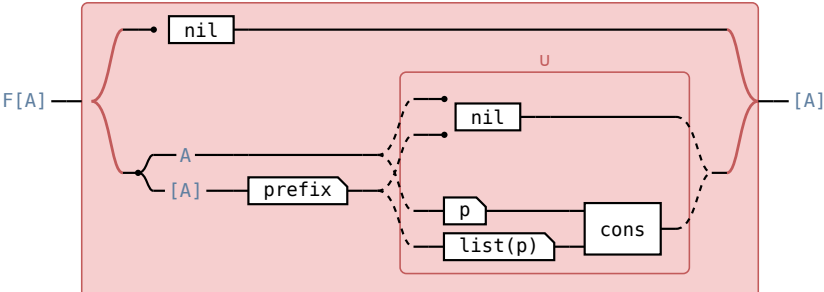
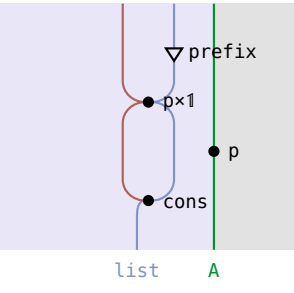
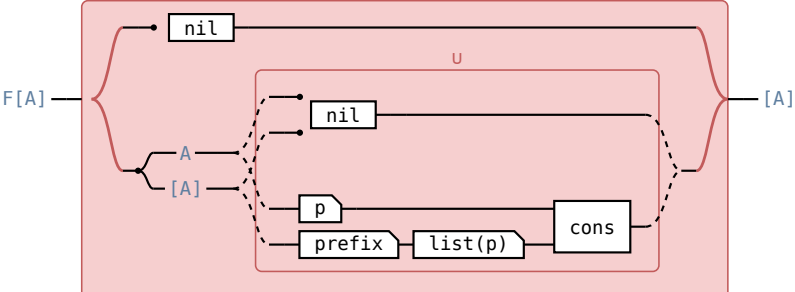
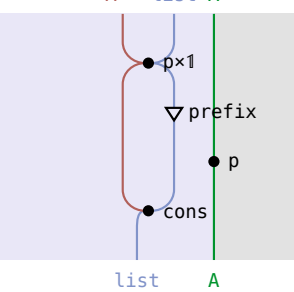
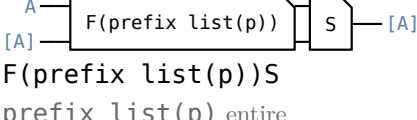
Hinze–Marsden



the **cons** operand of $\rightarrow \text{nil} \cup \text{cons}$; $\rightarrow \text{nil}$ makes a constant and draws nothing

$\text{nil prefix } ys \iff ys = \text{nil}$
 $\text{cons}(a, xs') \text{ prefix } ys \iff ys = \text{nil} \vee \exists ys'. xs' \text{ prefix } ys' \wedge ys = \text{cons}(a, ys')$

a right fold: α° peels, $F(\text{prefix})$ recurses, the algebra runs on the way back; $\text{init}^* (\triangleq \text{subseq})$ is the tail-recursive form
 lax natural only, $\text{list}(R) \text{ prefix} \sqsubseteq \text{prefix}$
 $\text{list}(R): \text{list}(R)$ first needs an R -image of every element, **prefix** first may have dropped the ones without

$\alpha \text{prefix list}(p) = F(\text{prefix list}(p))S$ (13.3.3c)	
building the list and then keeping a p -passing prefix of it is keeping one of the tail first, and then building with S this same diagram is subseq 's: algebra $[\text{nil}, \pi_2 \cup \text{cons}]$, type $[A] \rightarrow [A]$	
circuit — the fork is $F([A]) = 1 + A \times [A]$: nil above, the pair below	Hinze–Marsden
 $\alpha \text{prefix list}(p)$	 the cons branch alone, without $1 +$ or $\multimap \text{nil}$
 $F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup \text{cons}] \text{list}(p)$ defining equation	 the cons operand of $\multimap \text{nil} \cup$
 $F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup (p \times \text{list}(p)) \text{cons}]$ $\text{list}(p)$ through cons	 the $(p \times \text{list}(p))$ cons operand of $\multimap \text{nil} \cup$ $(p \times \text{list}(p))$ cons
 $[\text{nil}, \multimap \text{nil} \cup (p \times (\text{prefix list}(p))) \text{cons}]$ relator, prefix entire	
 $F(\text{prefix list}(p))S$ $\text{prefix list}(p)$ entire	

$\triangleq$ **reduce** reads that off as $\text{prefix list}(p) = \{S\}$. fusion cannot: $\text{list}(p)$ is not entire, $(1 \times \text{list}(p)) \multimap \text{nil} \sqsubseteq \multimap \text{nil}$, and no algebra meets the side condition. ,

$(1 \times R^\circ)(\multimap \text{nil} \cup (p \times 1)\text{cons}) \sqsubseteq (\multimap \text{nil} \cup (p \times 1)\text{cons})R^\circ$ (13.3.3d)	
shortening the tail and then taking the step lands inside taking the step and then shortening the result	
circuit — the cons branch of $F(R^\circ) \subseteq SR^\circ$	reason
$(1 \times R^\circ)(\multimap \text{nil} \cup (p \times 1)\text{cons})$	
	each operand is reached on its own (1.1a) — and $(1 \times R^\circ)$ $(p \times 1)$ is p and R° on the pair's two strands at once
$(1 \times R^\circ)\multimap \text{nil} \cup (p \times R^\circ)\text{cons}$	
	$\multimap \text{nil} \cup (p \times R^\circ)\text{cons}$
	$\multimap \text{nil} \cup (p \times 1)\text{cons} R^\circ$
	$\multimap \text{nil} R^\circ \cup (p \times 1)\text{cons} R^\circ$
	$(\multimap \text{nil} \cup (p \times 1)\text{cons})R^\circ$

the nil branch, which no row above draws, is $\text{nil} \sqsubseteq \text{nil} R^\circ$.

$\frac{S}{S} \text{ est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ (13.3.3e)	
the longest of the lists the algebra allows is the cons where the head passes p , and nil where it does not	
formula	reason
$ \begin{array}{c} A \\ [A] \quad \boxed{\frac{1}{S}} \rightarrow EF[A] \rightarrow E(S) \rightarrow E[A] \rightarrow \text{est}(R^\circ) \rightarrow [A] \end{array} $	
$ \begin{array}{c} = \\ F[A] \rightarrow \left\{ \begin{array}{l} \text{top branch: } \boxed{\frac{1}{S}} \rightarrow E1 \rightarrow E(\text{nil}) \rightarrow E[A] \rightarrow \text{est}(R^\circ) \\ \text{bottom branch: } \begin{array}{c} A \\ [A] \end{array} \rightarrow \boxed{\frac{1}{S}} \rightarrow E(A \times [A]) \rightarrow E(\neg \text{nil} \cup (p \times 1) \text{cons}) \rightarrow E[A] \rightarrow \text{est}(R^\circ) \end{array} \right\} \rightarrow [A] \end{array} $ $ \left[\frac{\text{nil}}{S} \text{ est}(R^\circ), \frac{\neg \text{nil} \cup (p \times 1) \text{cons}}{S} \text{ est}(R^\circ) \right] $	coproduct of maps
$ \begin{array}{c} = \\ F[A] \rightarrow \left\{ \begin{array}{l} \text{top branch: } \text{nil} \\ \text{bottom branch: } \begin{array}{c} A \\ [A] \end{array} \rightarrow \boxed{\frac{1}{S}} \rightarrow E(A \times [A]) \rightarrow E(\neg \text{nil} \cup (p \times 1) \text{cons}) \rightarrow E[A] \rightarrow \text{est}(R^\circ) \end{array} \right\} \rightarrow [A] \end{array} $ $ [\text{nil}, \frac{\neg \text{nil} \cup (p \times 1) \text{cons}}{S} \text{ est}(R^\circ)] $	singleton, R° reflexive
$ \begin{array}{c} = \\ F[A] \rightarrow \left\{ \begin{array}{l} \text{top branch: } \text{nil} \\ \text{bottom branch: } \begin{array}{c} A \\ [A] \end{array} \rightarrow \boxed{\pi_1 p \rightarrow \text{cons}, \neg \text{nil}} \end{array} \right\} \rightarrow [A] \end{array} $	$\text{nil } R = T$

the set is $\{\text{nil}\}$ where p fails on the head and $\{\text{nil}, \text{cons}(a, xs)\}$ where it holds, and nil loses the second — $\triangleq \text{est}$ at a two-element set.

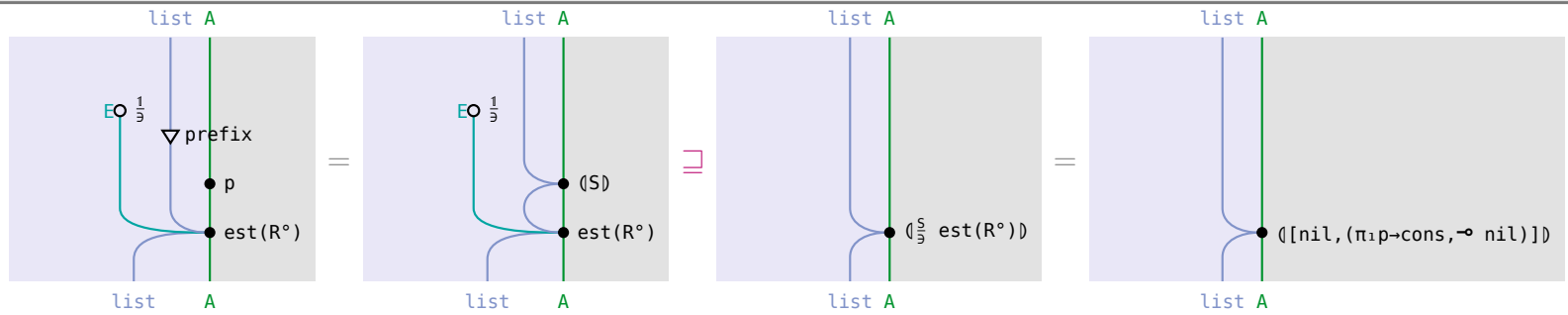
(13.3.3f)

name	definition	type	example	in words
S	$[\text{nil}, \neg \text{nil} \cup (p \times 1) \text{cons}] \triangleq \text{takewhile}$	$F[A] \rightarrow [A]$	$(4, [2]) \ S \ [4, 2]$, and $(3, [2]) \ S \ \text{nil only}$	prefix's algebra with one extra p — stop, or keep a head that passes p

$$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ) \quad (13.3.3g)$$

$$\text{takewhile}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})])$$

takewhile: $\text{takewhile}(p)(x)$ returns the longest prefix of x with the property that all its elements satisfy p ; the catamorphism is the standard implementation.



$$\begin{aligned} & [A] \xrightarrow{\frac{1}{\exists}} E(\text{prefix list}(p)) \xrightarrow{E[A]} \text{est}(R^\circ) \xrightarrow{[A]} [A] \\ & = [A] \xrightarrow{\frac{1}{\exists}} E(S) \xrightarrow{E[A]} \text{est}(R^\circ) \xrightarrow{[A]} [A] \end{aligned} \quad \text{the specification} \triangleq \text{est's } \text{est}(R^\circ) \quad (13.3.3c)$$

$$\begin{aligned} & [A] \xrightarrow{\frac{1}{\exists}} E(F[A]) \xrightarrow{E[A]} E(S) \xrightarrow{E[A]} \text{est}(R^\circ) \xrightarrow{[A]} [A] \\ & \text{greedy theorem at } R^\circ, \text{ with } F(R^\circ) S \subseteq SR^\circ \text{ — (13.3.3d) — for its hypothesis: one longest } p\text{-prefix kept at each } \text{cons}, \text{ instead of every } p\text{-prefix} \\ & \text{collected and one chosen at the end.} \end{aligned}$$

$$= [A] \xrightarrow{F[A]} \left(\begin{array}{c} \text{nil} \\ \text{cons } A \text{ } [A] \end{array} \right) \xrightarrow{[A]} [A] \quad (13.3.3e)$$

$\text{takewhile}(p)^\circ \text{takewhile}(p) \subseteq \text{prefix}^\circ \text{prefix} \cap R \cap R^\circ \subseteq 1$
 $\text{takewhile}(p) \subseteq \text{prefix list}(p)$ and $(\text{prefix list}(p))^\circ \text{takewhile}(p) \subseteq R$ — (13.1.5a) at $\text{est}(R^\circ)$ — and two prefixes of one list of equal length are equal, so $\text{takewhile}(p)$ is simple: the longest, not a longest.

$\frac{\text{prefix list}(p)}{\exists} \text{est}(R^\circ)$ entire
 nil is always a p -prefix and R is connected on the prefixes of one list, so the longest exists.

$$X \subseteq Y, X \text{ entire}, Y \text{ simple} \implies X = Y$$

$([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})])$ is a reduce of maps, hence entire — what turns the $\supseteq$ above into the heading's $=$.

$$\S 13.3.4. \text{mss} = ([\text{zero}(1, \frac{1}{\exists}), ((1 \times \pi_1) \oplus, ((1 \times \pi_1) \oplus \frac{1}{\exists}, \pi_2 \pi_2) \text{cup})]) \pi_2 \text{est}(\geq) \quad (13.3.4a)$$

$FX = 1 + A \times X$, $\alpha \triangleq [\text{nil}, \text{cons}]$, $\text{sum} = ([\text{zero}, \text{plus}])$ and $\text{segment} = \text{suffix prefix}$ from (7a) and $\triangleq$ subseq.

$\text{head} \triangleq \text{cons}^\circ \pi_1$, $\text{wrap} \triangleq (1, \neg \text{nil}) \text{cons}$ the head of a list and the one-element list, beside $\triangleq$ subseq's $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\neg \text{zero} \cup \text{plus}}{\exists} \text{est}(\geq)$ the set at (a, b) is $\{0, a+b\}$, so $\oplus$ is the larger of the two,

$\text{mss} = \frac{\text{suffix}}{\exists} E\left(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)\right) \text{est}(\geq) \quad (13.3.4b)$	
maximum segment sum problem: using segment=suffix prefix , the specification is expressed in this form	
formula — one wire from [A] to A, its type written along it	reason
$[A] \xrightarrow{\frac{1}{\exists}} E[A] \xrightarrow{E(\text{segment sum})} EA \xrightarrow{\text{est}(\geq)} A$	
$= [A] \xrightarrow{\frac{1}{\exists}} E[A] \xrightarrow{E(\text{suffix prefix sum})} EA \xrightarrow{\text{est}(\geq)} A$	segment=suffix prefix $\triangleq \text{subseq}, \triangleq \text{mss}$
$= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) \xrightarrow{E[A]} E(\text{prefix sum}) \xrightarrow{EA} \text{est}(\geq) \xrightarrow{EA} A$	absorption (10b) — $\text{frac}(S, \exists)$ $E(R) = \text{frac}(SR, \exists)$ at $S := \text{suffix}, R := \text{prefix}$ sum
$= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) \xrightarrow{E[A]} E\left(\frac{\text{prefix sum}}{\exists}\right) \xrightarrow{EEA} \text{union} \xrightarrow{EA} \text{est}(\geq) \xrightarrow{EA} A$	$\frac{R}{\exists} \exists = R, \text{union} = E(\exists)$ (10b)'s $\text{frac}(R, \exists) \exists = R$ at $R := \text{prefix sum}$ and $E(R) \triangleq \text{frac}(\exists R, \exists)$; (13.1b)'s $\text{union} \triangleq \text{frac}(\exists \exists, \exists)$; the middle equality is $\triangleq$ relator's $F(RS) = F(R)F(S)$ at $F := E$
$= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) \xrightarrow{E[A]} E\left(\frac{\text{prefix sum}}{\exists}\right) \xrightarrow{EEA} E(\text{est}(\geq)) \xrightarrow{EA} \text{est}(\geq) \xrightarrow{EA} A$	(13.1b), the sets non- empty
$= [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) \xrightarrow{E[A]} E\left(\frac{\text{prefix sum}}{\exists} \text{est}(\geq)\right) \xrightarrow{EA} \text{est}(\geq) \xrightarrow{EA} A$	relator $\triangleq$ relator — $F(RS) = F(R)F(S)$ at $F := E$

The **union** step is (13.1b)'s $P(\text{est}(R)) \text{est}(R) = P(\text{dom}(\text{est}(R))) \text{union} \text{est}(R)$ — every suffix has the empty prefix, so **Dom** is $\mathbb{1}$ here — and $P(f) = E(f)$ at the map it is applied to ((11.3d)).

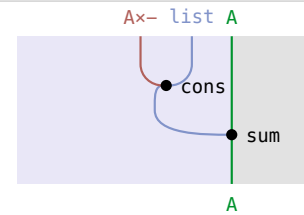
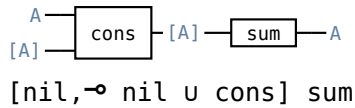
$$[\text{nil}, \multimap \text{nil} \text{ u } \text{cons}] \text{sum} = F(\text{sum})S$$

(13.3.4c)

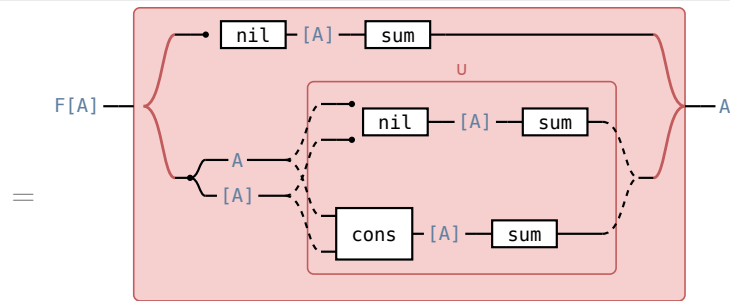
fusion: **prefix** expressed as a catamorphism on cons-lists, $([\text{nil}, \multimap \text{nil} \text{ u } \text{cons}])$, and this is the fusion condition that expresses **prefix sum** as a catamorphism

circuit — the fork is the bracket's case split $F([A]) = 1 + A \times [A]$: nil above, the pair and its u below

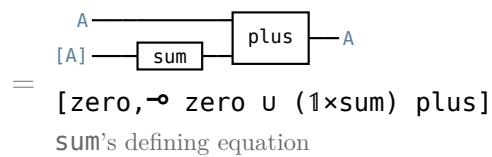
Hinze–Marsden



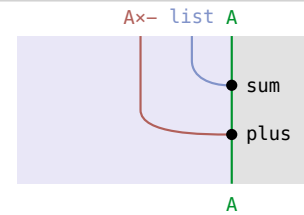
the **cons** operand of $\multimap \text{nil} \text{ u } \text{cons}$



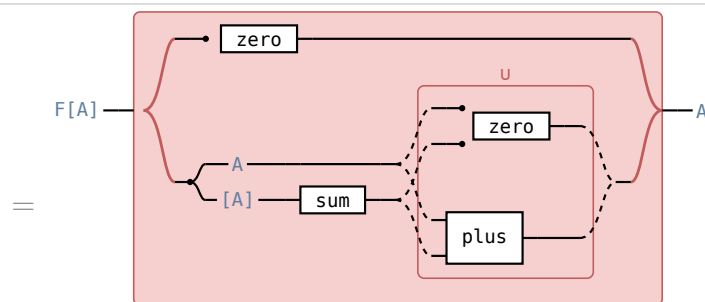
coproduct of maps, composition over U



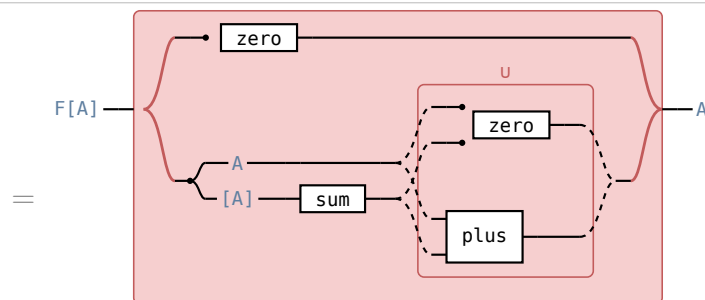
sum's defining equation



the $(1 \times \text{sum})$ **plus** operand of $\multimap \text{zero} \text{ u } (1 \times \text{sum})$ **plus**



$(1 \times \text{sum}) \multimap = \multimap$, sum entire



relator

fusion at $R := [\text{nil}, \multimap \text{nil} \text{ u } \text{cons}]$, $S := \text{sum}$: the side condition, so **prefix sum** $= ([\text{zero}, \multimap \text{zero} \text{ u } \text{plus}])$. **prefix** is the reduce, **sum** the map fused into it — the intermediate list is gone.

$(1 \times \geq)(\neg \text{zero} \cup \text{plus})E(\neg \text{zero} \cup \text{plus})\geq$				(13.3.4d)
monotonic algebra: an F -algebra S is monotonic on a relation R if $F(R)S \leq SR$ — the plus branch of $F(\geq)S \leq S\geq$; the zero branch is $\text{zero}E\text{zero}\geq$				
circuit — the head above, the running sum below; the tape is the u				
$(1 \times \geq)(\neg \text{zero} \cup \text{plus})$	$(1 \times \geq)\neg \text{zero} \cup$ $(1 \times \geq)\text{plus}$	$\neg \text{zero} \cup \text{plus}\geq$	$(\neg \text{zero} \cup \text{plus})\geq$	
				= relator, composition over U
$(5.1a), (\geq \times \geq)$ $\text{plus} \leq \text{plus} \geq; (\leq \times \leq)$ $\text{plus} \leq \text{plus} \leq$ is $\triangleq$ monotonic, written $+$ there, and plus is a map, so it is monotonic on an order and on its opposite together, which carries it to $\geq$.				

$\frac{S}{\geq} \text{est}(\geq) = [\text{zero}, \oplus]$		(13.3.4e)
the largest sum the algebra offers is zero from nothing and, from a head and a running sum, the larger of zero and the head added to it		
circuit — the tape is the coproduct: zero's branch above, plus's below	Hinze–Marsden	
the zero arm of the bracket		
the plus operand of the lower arm's $\neg \text{zero} \cup \text{plus}$, under its $1\% \exists E(\dots)$ and $\text{est}(\geq)$		
singleton, $\geq$ reflexive — (13.1b)'s $\frac{1}{\geq}$ $\text{est}(R) = 1 \cap R$ at $R := \geq$, zero a map; the lower branch is $\oplus$'s definition, $\triangleq$ mss , and no law		

with (13.3.4d) the greedy theorem gives $([\text{zero}, \oplus]) \leq \frac{\text{prefix sum}}{\geq} \text{est}(\geq)$ — and (13.3.4g) makes it an equality.

$[\text{nil}, \text{cons}] (\langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle \langle [\text{zero}, \oplus] \rangle) \rangle) = F(\langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle \langle [\text{zero}, \oplus] \rangle) \rangle) k \quad (13.3.4f)$ $k \triangleq [\text{zero} \langle 1, \frac{1}{\exists} \rangle, \langle w, \langle w \frac{1}{\exists}, \pi_2 \pi_2 \rangle \text{ cup} \rangle], w \triangleq (1 \times \pi_1) \oplus: \text{the value at the whole list, paired with the set of the values at its suffixes, runs } k \text{'s recursion.}$	
the equation at that branch	why
$\text{nil}(g, \frac{\text{suffix}}{\exists} E(g)) = \text{zero} \langle 1, \frac{1}{\exists} \rangle$	nil has one suffix, itself, and $g(\text{nil}) = \text{zero}$, so the set is the singleton $\{\text{zero}\}$
$\text{cons}(g, \frac{\text{suffix}}{\exists} E(g)) = (1 \times \langle g, \frac{\text{suffix}}{\exists} E(g) \rangle) \langle w, \langle w \frac{1}{\exists}, \pi_2 \pi_2 \rangle \text{ cup} \rangle$	$g(\text{cons}(a, x)) = a \oplus (g(x))$, which is w reading $g(x)$ off π_1 ; and the suffixes of $\text{cons}(a, x)$ are $\text{cons}(a, x)$ itself, whose value is that same w , together with those of x , which $\pi_2 \pi_2$ carries — so the two sets meet at cup

(13.3.4g)

$\text{mss} = (k) \pi_2 \text{ est}(\geq)$
 maximum segment sum problem: $\frac{\text{suffix}}{\exists} E(\langle \langle [\text{zero}, \oplus] \rangle) \rangle$ expressed as the catamorphism $\langle k \rangle$, hence mss
 implemented by a linear-time algorithm, $\oplus \triangleq \text{zero} \cup \text{plus}$ $\text{est}(\geq) \triangleq \text{mss}; k \text{ and } w \triangleq (13.3.4f)$.

list A

A

list A

A

list A

A

list A

A

mss is the greatest of the segment sums — $\triangleq \text{mss}$

$$\begin{aligned}
 & [A] \xrightarrow{\frac{1}{\exists}} E[A] \xrightarrow{E(\text{segment sum})} EA \xrightarrow{\text{est}(\geq)} A \\
 = & [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) E[A] \xrightarrow{E(\frac{\text{prefix sum}}{\exists} \text{est}(\geq))} EA \xrightarrow{\text{est}(\geq)} A \\
 & [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) E[A] \xrightarrow{E(\langle \langle [\text{zero}, \oplus] \rangle)} EA \xrightarrow{\text{est}(\geq)} A
 \end{aligned}$$

(13.3.4b)

$\frac{\text{suffix}}{\exists} E(\langle \langle [\text{zero}, \oplus] \rangle) \text{ est}(\geq)$

the greedy theorem greedy theorem at $R := \geq$, $S := [\text{zero}, \text{zero} \cup \text{plus}]$ — (13.3.4d) is its condition and (13.3.4e) its $\frac{S}{\exists} \text{ est}(\geq)$; its Ξ is an $=$ because $\langle \langle [\text{zero}, \oplus] \rangle$ is entire and $\frac{\text{prefix sum}}{\exists} \text{ est}(\geq)$ simple (13.3.3g)'s last row

$$\begin{aligned}
 & [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) E[A] \xrightarrow{E(\langle \langle [\text{zero}, \oplus] \rangle)} EA \xrightarrow{\text{est}(\geq)} A \\
 = & [A] \xrightarrow{\frac{1}{\exists}} E(\text{suffix}) E[A] \xrightarrow{E(\langle \langle [\text{zero}, \oplus] \rangle)} EA \xrightarrow{\text{est}(\geq)} A
 \end{aligned}$$

$\langle k \rangle \pi_2 \text{ est}(\geq)$
 $\triangleq \text{reduce at (13.3.4f)'s equation, so } \langle k \rangle = \langle \langle [\text{zero}, \oplus] \rangle, \frac{\text{suffix}}{\exists} E(\langle \langle [\text{zero}, \oplus] \rangle) \rangle$, of which π_2 is the row above

one fold builds the $n+1$ running maxima and the final $\text{est}(\geq)$ reads them in one more pass, so mss is linear.

§13.3.5. $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$

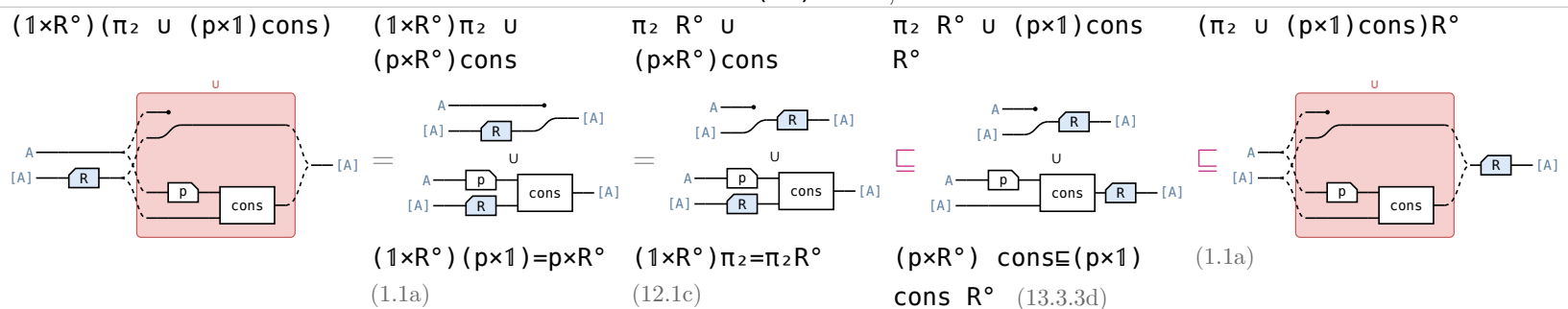
(13.3.5a)

name	definition	type	example	in words
F, α, p, R	as in $\triangleq \text{takewhile}$			
π_2	π_2 where $\triangleq \text{takewhile}$ has $\rightarrow \text{nil}$	$A \times [A] \rightarrow [A]$	$\pi_2(3, [1, 2]) = [1, 2]$	drop the head, keep the tail
subseq	$([\text{nil}, \pi_2 \cup \text{cons}]) \triangleq \text{subseq}$	$[A] \rightarrow [A]$	$[3, 1, 2] \text{ subseq } [3, 2]$	$xs \text{ subseq } ys \iff ys \text{ is } xs \text{ with elements dropped}$ — at each cons , drop the head or keep it
S	$[\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$	$F[A] \rightarrow [A]$	$(4, [2]) S [2]$ and $(4, [2]) S [4, 2]$, but $(3, [2]) S [2]$ only	subseq's algebra with one extra p — drop the head, or keep a head that passes p
$1 \sqsubseteq \pi_2 R \text{ cons}^\circ$				the tail is one shorter than the cons , so π_2 loses the $\text{est}(R^\circ)$ at every step — where $\triangleq \text{takewhile}$'s loser is nil

$$(1 \times R^\circ)(\pi_2 \cup (p \times 1) \text{ cons}) \sqsubseteq (\pi_2 \cup (p \times 1) \text{ cons}) R^\circ$$

(13.3.5b)

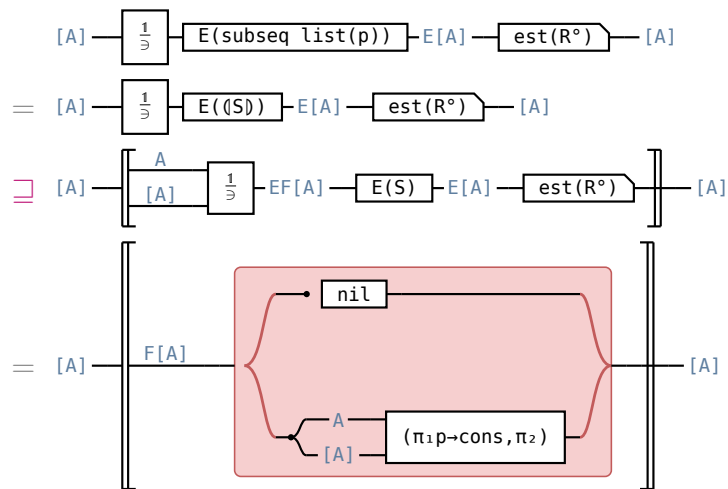
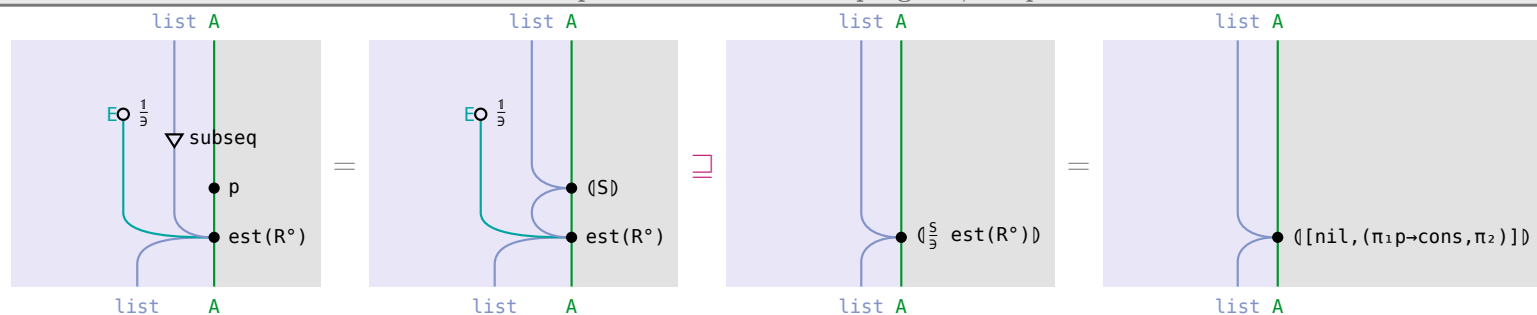
shortening the tail and then taking the step lands inside taking the step and then shortening the result

formula — the cons branch of $F(R^\circ) S \sqsubseteq SR^\circ$; **reason** under each circuit

 the nil branch: $\text{nil} \sqsubseteq \text{nil } R^\circ$.

$$\mathbf{filter}(p) = ([\mathbf{nil}, (\pi_1 p \rightarrow \mathbf{cons}, \pi_2)])$$

(13.3.5e)

filter: $\mathbf{filter}(p)$ x returns the longest subsequence of x with the property that all its elements satisfy p ; the catamorphism is the standard program; R a preorder.


 $\triangleq$ subseq

subseq list(p) = (S) — (13.3.3c)'s header, subseq for prefix

greedy theorem at R° , whose hypothesis $F(R^\circ) S \subseteq SR^\circ$ is (13.3.5b)

(13.3.5c)

the catamorphism is entire and $\mathbf{filter}(p)$ simple, so $\supseteq$ is $=$.

$(\mathbf{subseq\ list}(p))^\circ (\mathbf{subseq\ list}(p)) \cap R \cap R^\circ \subseteq 1$ fails — two p -subsequences of one list can be of equal length and different — so §13.3.3's uniqueness argument does not transfer. What survives $\mathbf{est}(R^\circ)$ is the one keeping **exactly** the passing elements: a subsequence that drops a passing element is beaten by the one that keeps it.

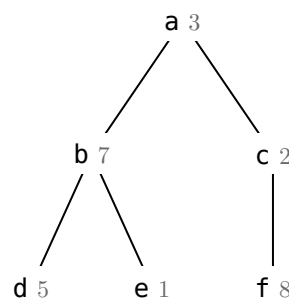
§13.4. Planning a company party

(13.4a)

definition	type	note
$\text{tree } A ::= \text{node } (A, [\text{tree } A])$	$\mathcal{A} \rightarrow \mathcal{A}$	The company hierarchy: an employee, and the list of subtrees under them.
$F(A, B) = A \times [B]$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor tree folds: an employee beside the recursive position, one layer deep.
rating	$A \rightarrow \text{Int}$	What one employee is worth as a guest.
$\text{cost} = \text{list}(\text{rating}) \text{sum}$	$[A] \rightarrow \text{Int}$	What a guest list is worth.
$R = \text{cost} \leq \text{cost}^\circ$	$[A] \rightarrow [A]$	The preorder the guest list is maximised over.
$\text{choose} \triangleq \pi_1 \cup \pi_2$	$[A] \times [A] \rightarrow [A]$	Takes one of the two parties a subtree returns.
$\text{include} = (1 \times (\text{list}(\pi_2) \text{concat})) \text{cons}$	$F(A, [A] \times [A]) \rightarrow [A]$	The party that invites the root, which puts every immediate subtree's root out. A map.
$\text{exclude} = (1 \times (\text{list}(\text{choose}) \text{concat})) \pi_2$	$F(A, [A] \times [A]) \rightarrow [A]$	The party that leaves the root out, so each subtree is free to choose. Not a map.
$S \triangleq (\text{include}, \text{exclude})$	$F(A, [A] \times [A]) \rightarrow [A] \times [A]$	The algebra: one step returns both parties of a subtree at once.
$\text{party} = (S) \text{choose}$	$\text{tree}(A) \rightarrow [A]$	Every guest list the president's ruling allows.
the specification $\frac{\text{party}}{\exists} \text{est}(R^\circ)$	$\text{tree}(A) \rightarrow [A]$	A guest list of greatest total conviviality.

§13.4.1. include and exclude

(13.4.1a)



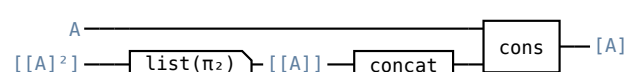
the small grey number is the employee's rating

(13.4.1b)

node	the algebra's input $A \times [[A] \times [A]]$	include	exclude
d, e, f	$(d, [])$	$[d]$	$[]$
b	$(b, ([[d], []], ([e], []]))$	$[b]$	$[d, e], [d], [e], []$
c	$(c, ([[f], []]))$	$[c]$	$[f], []$
a	$(a, ([[b], \dots], ([c], \dots)))$	$[a] \# \text{exclude}(b) \# \text{exclude}(c)$	$\text{choose}(b) \# \text{choose}(c)$

with those ratings, $\text{est}(R^\circ)$ keeps $([b]=7, [d, e]=6)$ at **b**, where **include** wins, and $([c]=2, [f]=8)$ at **c**, where **exclude** wins, so at the root $\text{include}=[a, d, e, f]=17$ beats $\text{exclude}=[b, f]=15$, and **choose** takes 17.

(13.4.1c)





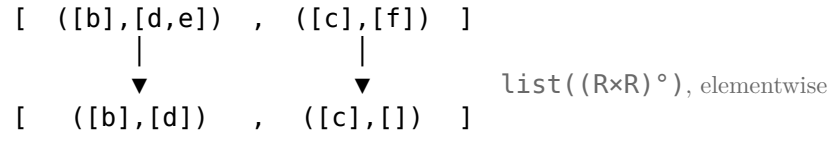
(13.4.1d)

list(choose) [([d],[]), ([e],[])]

1st item	([d],[])	choose→[d]	or	[]
2nd item	([e],[])	choose→[e]	or	[]
2×2=4 combinations:	[[d],[e]]	[[d],[]]	[[],[e]]	[[],[]]
concat flattens:	[d,e]	[d]	[e]	[]

(13.4.1e)

§13.4.2. list((R×R)°)



(13.4.2a)

([b],[d,e])

- └─ exclude: the best party in b's subtree when b does not come
- └─ include: the best party in b's subtree when b comes

(13.4.2b)

element	1st component include	2nd component exclude
1st, ([b], [d,e])	[b] R° [b] 7≥7, reflexivity	[d,e] R° [d] 6≥5, and ([b],[d,e]) is the pair est(R°) keeps at b
2nd, ([c],[f])	[c] R° [c] 2≥2, reflexivity	[f] R° [] 8≥0, and ([c],[f]) is the pair est(R°) keeps at c

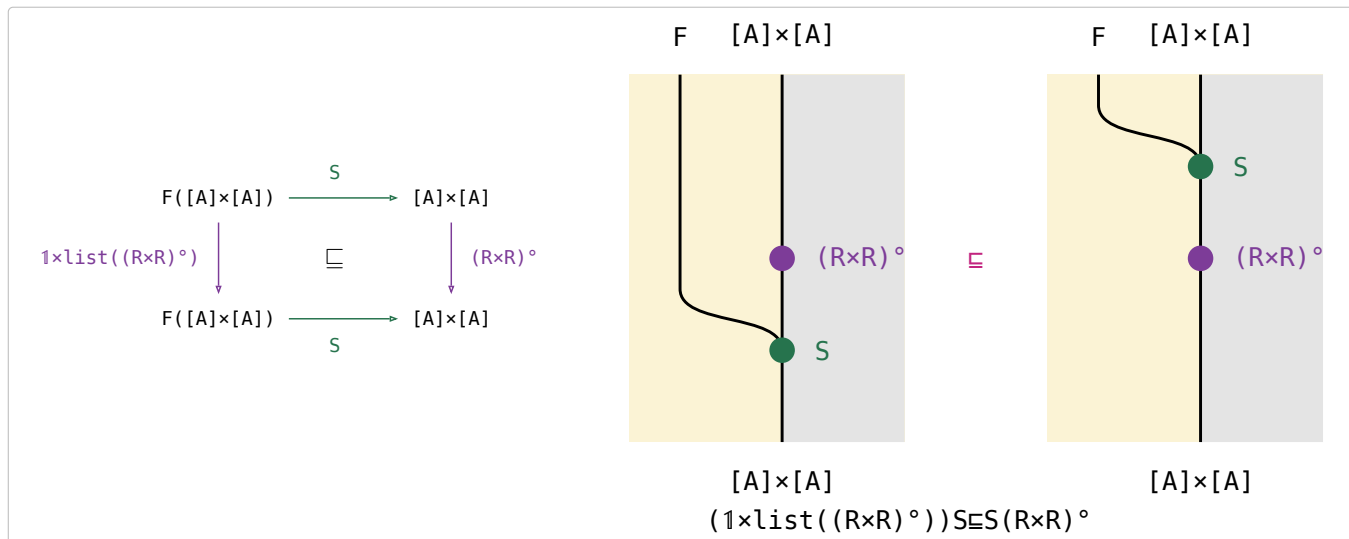
the first component of $1 \times \text{list}((R \times R)^\circ)$ is the root employee.

$R \triangleq \text{cost} \leq \text{cost}^\circ$ is a preorder, so the components that do not move are related by reflexivity — $(R \times R)^\circ$ demands a relation in **both** components, it cannot skip one.

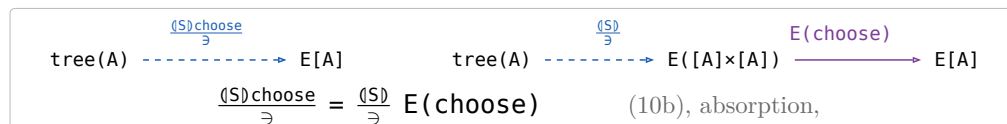
R compares cost only, not contents: $[f] R^\circ []$ is legal though $[f]$ has an element $[]$ does not. That is why the three leaves of §13.4.3's proof all reduce to “cost is a sum”.

§13.4.3. $(1 \times \text{list}((R \times R)^\circ)) S \subseteq S(R \times R)^\circ \text{ — } S : F([A] \times [A]) \rightarrow [A] \times [A] \text{ monotonic on } (R \times R)^\circ$

(13.4.3a)

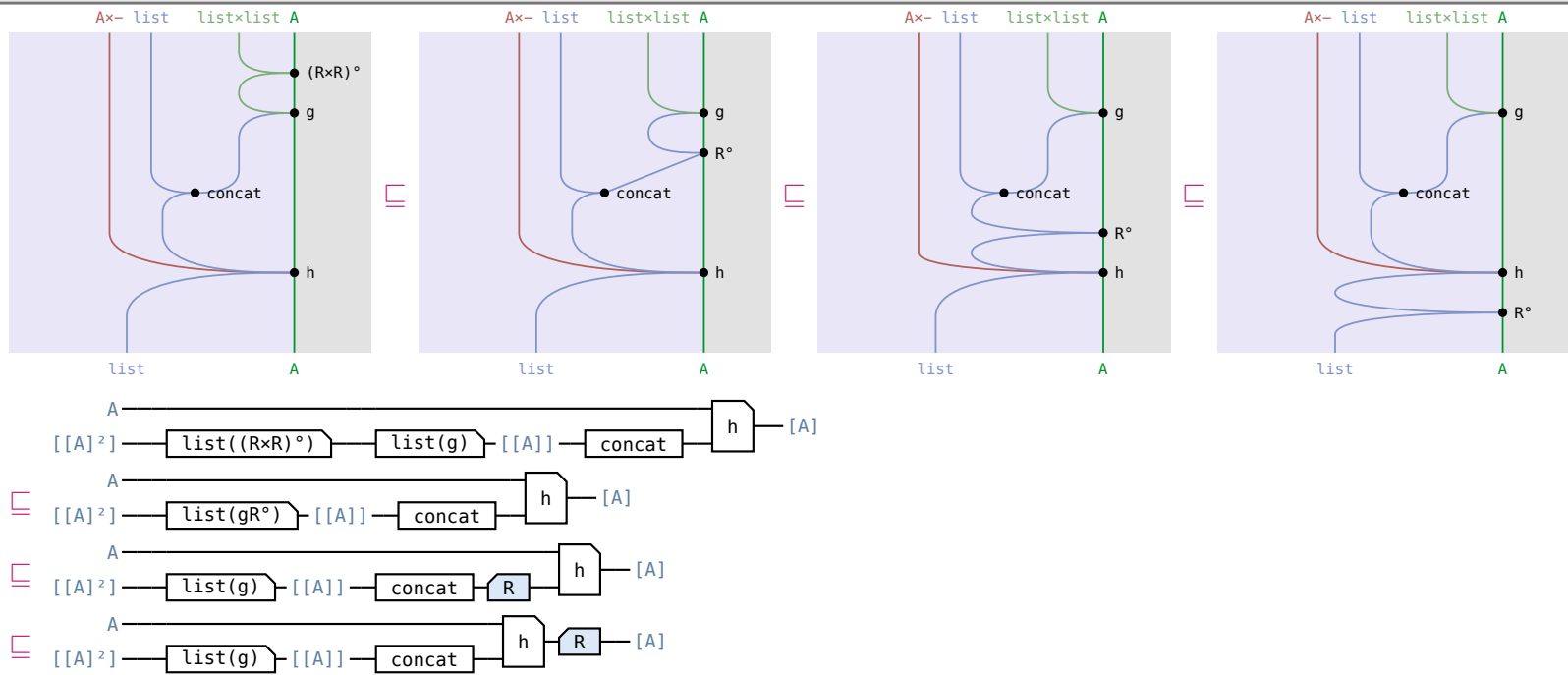


(13.4.3b)



$$\begin{aligned}
 S &\triangleq (\text{include}, \text{exclude}) \\
 (1 \times \text{list}((R \times R)^\circ)) S &\sqsubseteq S(R \times R)^\circ \\
 (1 \times \text{list}((R \times R)^\circ)) \text{include} &\sqsubseteq \text{include } R^\circ \\
 (1 \times \text{list}((R \times R)^\circ)) \text{exclude} &\sqsubseteq \text{exclude } R^\circ
 \end{aligned}
 \tag{13.4.3c}$$

bettering both parties of every subtree before the node's algebra runs gets no further than running it first and bettering the two parties it returns,



- g $(R \times R)^\circ g \sqsubseteq g R^\circ$
 $g := \pi_2$ is $(R \times R)^\circ \pi_2 = (\text{dom}(\pi_1 R^\circ)) \pi_2 R^\circ \sqsubseteq \pi_2 R^\circ$, $g := \pi_1$ its mirror $(\text{dom}(\pi_2 R^\circ)) \pi_1 R^\circ \sqsubseteq \pi_1 R^\circ$ — 1 and 4 of (11.1.2b), then $\text{dom} \sqsubseteq 1$; $g := \text{choose} \triangleq \pi_1 \cup \pi_2$ is the union of the two (13.2c). list monotonic, $\triangleq$ relator
 concat $\text{list}(R^\circ) \text{concat} \sqsubseteq \text{concat } R^\circ$
 a LEAF: no law above it. cost is a sum, so $\text{cost}(\text{concat}(xss)) = \text{sum}(\text{list}(\text{cost})(xss))$ and a cheaper part makes a cheaper whole.
 h $(1 \times R^\circ) h \sqsubseteq h R^\circ$
 a LEAF: $\text{cost}(\text{cons}(a, xs)) = \text{rating}(a) + \text{cost}(xs)$, so a cheaper tail makes a cheaper list. For $h := \pi_2$ it is an EQUALITY $(1 \times R^\circ) \pi_2 = (\text{dom}(\pi_1)) \pi_2 R^\circ = \pi_2 R^\circ$: π_1 is a map, hence entire, so $\text{dom}(\pi_1) = 1$ — (5c)

$(1 \times (\text{list}(g) \text{ concat})) h$	g	h
include	π_2	cons
exclude	choose	π_2

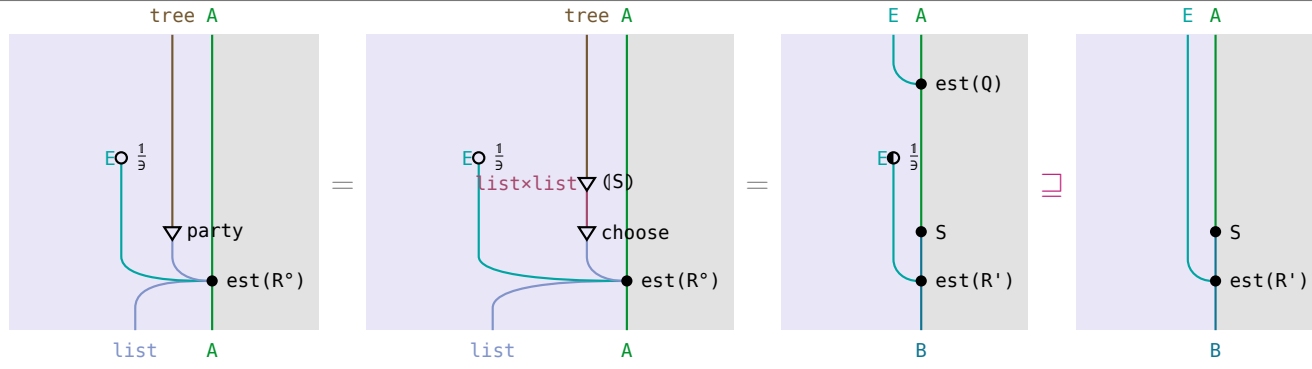
(13.4.3d)

$R \triangleq \{(0,0)\} : \{0,1\} \rightarrow \{0,1\}$	not entire — undefined at 1
$(R \times R) \text{ choose}$ at $(0,1)$	nothing — $R \times R$ needs BOTH components related, and R is undefined at 1
$\text{choose } R$ at $(0,1)$	0 — choose picks the first component 0, and $R \ 0 \ 0$ holds
$((0,1),0)$	in $\text{choose } R$, not in $(R \times R) \text{ choose}$
$(R \times R) \text{ choose} = \text{choose } R$	iff R is entire — $\text{dom} \sqsubseteq 1$ is the g row's only inequality step, and it is an equality iff $\text{dom}(R) = 1$

§13.4.4. The derivation

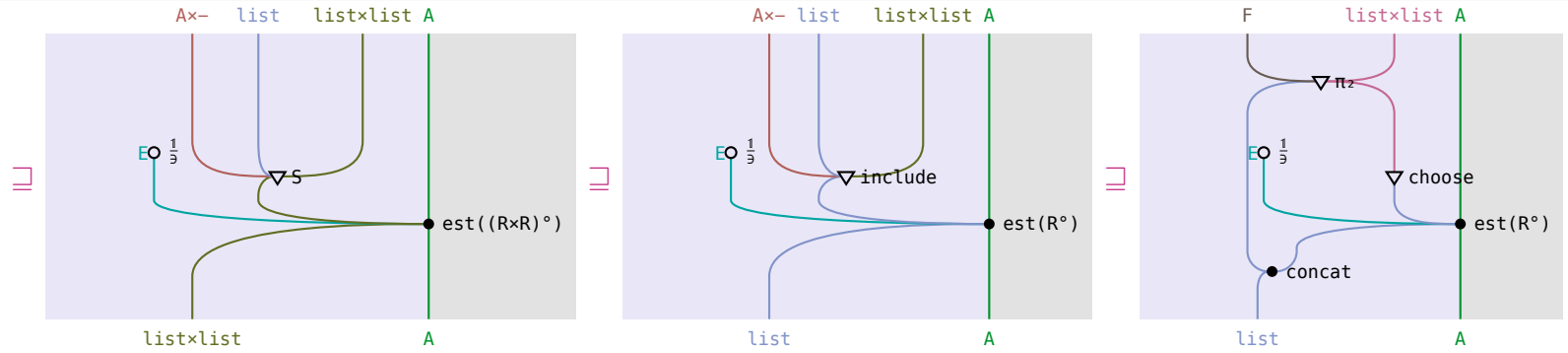
$$(\langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{\exists} \text{ est}(R^\circ)) \text{ concat} \rangle) \frac{\text{choose}}{\exists} \text{ est}(R^\circ) \sqsubseteq \frac{\text{party}}{\exists} \text{ est}(R^\circ) \quad (13.4.4a)$$

the best of every guest list the president allows is one pass up the tree, each subtree handing up its best party with its boss in and its best with the boss out, and **choose** taking the better of the two at the root



$$\begin{aligned} & \text{tree}(A) \xrightarrow{\frac{1}{\exists}} E(\text{tree}(A)) \xrightarrow{E(\text{party})} E[A] \xrightarrow{\text{est}(R^\circ)} [A] && \text{the specification} \triangleq \text{party} \\ = & \text{tree}(A) \xrightarrow{\frac{1}{\exists}} E(\text{tree}(A)) \xrightarrow{E(\langle S \rangle)} E[A]^2 \xrightarrow{E(\text{choose})} E[A] \xrightarrow{\text{est}(R^\circ)} [A] && \text{party} \triangleq \langle S \rangle \text{ choose} \triangleq \text{party} \\ = & EA \xrightarrow{\text{est}(Q)} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(S)} EB \xrightarrow{\text{est}(R')} B && \frac{\langle S \rangle \text{ choose}}{\exists} = \frac{\langle S \rangle}{\exists} E(\text{choose}) \quad (13.4.3b) \\ \sqsubseteq & EA \xrightarrow{E(S)} EB \xrightarrow{\text{est}(R')} B && (R \times R)^\circ \text{ choose} \sqsubseteq \text{choose } R^\circ \quad (13.4.3c) \text{'s g row} \end{aligned}$$

(13.4.4b)



$$\begin{aligned} & A \xrightarrow{\frac{1}{\exists}} E(A \times [A]^2) \xrightarrow{E(S)} E[A]^2 \xrightarrow{\text{est}((R \times R)^\circ)} [A] && \text{from here } \langle \rangle \text{ is drawn open} \text{ — the two bars, with the algebra's own circuit between them; } (1 \times \text{list}((R \times R)^\circ)) S \sqsubseteq S(R \times R)^\circ \text{ at } (R \times R)^\circ, \\ & [A]^2 \xrightarrow{\frac{1}{\exists}} E(A \times [A]^2) \xrightarrow{E(\text{include})} E[A] \xrightarrow{\text{est}(R^\circ)} [A] && (\frac{\text{include}}{\exists} \text{ est}(R^\circ), \frac{\text{exclude}}{\exists} \text{ est}(R^\circ)) \sqsubseteq \frac{S}{\exists} \text{ est}((R \times R)^\circ) \\ & A \xrightarrow{\frac{1}{\exists}} E(A \times [A]^2) \xrightarrow{\text{list}(\frac{\text{choose}}{\exists} \text{ est}(R^\circ))} [A] \xrightarrow{\text{concat}} [A] && \text{include a map, est}(R^\circ) \text{ into each branch} \end{aligned}$$

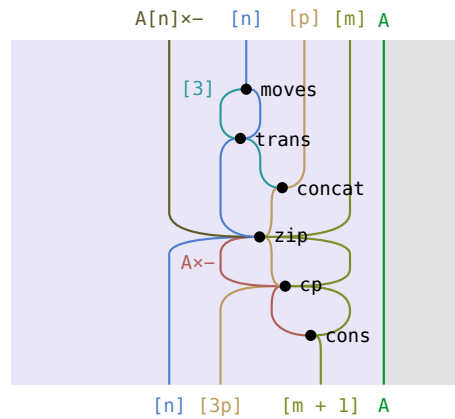
§13.5. Shortest paths on a cylinder

(13.5a)

	on $L = \text{list}^+$, 17.3.3	on Vec	note
F	$F(A, X) = A + A \times X$	$F(A, A[m]) = A[1] + A[m+1]$	A square alone, or in front of the path so far.
the input	$L(N\ A)$	$A[m][n] \triangleq (A[n])[m]$	m columns of n squares, outermost index first — the matrix itself.
a path	$L\ A$, a non-empty list	$A[1] \triangleq A$, $A[m+1] \triangleq A \times A[m]$	Through m columns, m squares: the count is in the type. A one-square path is the square; $\text{Vec}(m)$ is the functor $X \mapsto X[m]$.
the paths into a row	$E(L\ A)$	$A[p][m]$, $p = 3^{m-1}$	Three moves per column crossed, so this count is in the type too; a set forgets it.
α	$[\text{wrap}, \text{cons}]$ $F(A, LA) \rightarrow LA$	$\text{wrap} = 1 : A \rightarrow A[1]$ $\text{cons} : A \times A[m] \rightarrow A[m+1]$	wrap is an identity; cons is only an iso, so it stays a bead.
moves	$NA \rightarrow E(NA)$	$X[n] \rightarrow X[3][n]$	$\text{moves}(x) = (\text{up}(x), x, \text{down}(x))$, a $3 \times n$ matrix.
trans	$E(NX) \rightarrow N(EX)$	$X[3][n] \rightarrow X[n][3]$	The transpose.
union	$E(EX) \rightarrow EX$	$\text{concat} : X[j][k] \rightarrow X[jk]$	The j rows laid end to end; j is 3 inside gen and n inside paths .
zip	$F(NA, NB) \rightarrow NF(A, B)$	$F(A[n], B[n]) \rightarrow F(A, B)[n]$	Unchanged.
cp	$F(A, EB) \rightarrow E(F(A, B))$	$A \times B[p] \rightarrow (A \times B)[p]$	The square paired with each of the p candidates.
$\text{est}(R) \triangleq \exists n (\in \backslash \mathbb{R}^p) X \rightarrow X$		$X[k] \rightarrow X$, $\exists : X[k] \rightarrow X$	A component that is R -below every component; $\exists$ is “is some component” and $\in$ its converse.
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$L\ \text{Nat} \rightarrow L\ \text{Nat}$	$\text{Nat}[m] \rightarrow \text{Nat}[m]$	The cost of a path, which the cheapest minimises — one order per length m .
gen	$F(1, \text{moves trans } N(\text{union}))$ $\text{zip } N(\text{cp } P(\alpha))$ $F(NA, N(E(LA)))$ $\rightarrow N(E(LA))$	$F(1, \text{moves trans } \text{Vec}(n)(\text{concat}))$ $\text{zip } \text{Vec}(n)(\text{cp } \text{Vec}(3p)(\text{cons}))$ $F(A[n], A[n][p][m]) \rightarrow A[n][3p][m+1]$	The same composite without α ; the type shown is the cons side, the wrap side is $1 : A[n] \rightarrow A[n]$.
$\langle \text{gen} \rangle$	$L(N\ A) \rightarrow N(E(L\ A))$	$A[m][n] \rightarrow A[n][p][m]$	Columns in, rows out.
$\text{paths} \triangleq \langle \text{gen} \rangle \text{ concat}$	$\langle \text{gen} \rangle \text{ setify union}$ $L\ N\ \text{Nat} \rightarrow E(L\ \text{Nat})$	$A[m][n] \rightarrow A[np][m]$	The n rows of p paths laid end to end: every path of the cylinder, in one row.
Q , 13.5.3	$F(1, \text{moves trans } N(\text{est}(R)))$ $\text{zip } N(\alpha)$	$F(1, \text{moves trans } \text{Vec}(n)(\text{est}(R)))$ $\text{zip}, F(A[n], A[n][m]) \rightarrow A[n][m+1]$	The cheapest of three; no p anywhere. $\langle Q \rangle : A[m][n] \rightarrow A[n][m]$ has the type of a transpose.
the specification paths $\text{est}(R)$	$L\ N\ \text{Nat} \rightarrow L\ \text{Nat}$	$A[m][n] \rightarrow A[m]$, $\text{est}(R) : X[np] \rightarrow X$	A cheapest path from the entry side to the exit side.

§13.5.1. $\text{gen} \triangleq (1 \times (\text{moves trans } \text{Vec}(n)(\text{concat}))) \text{ zip } \text{Vec}(n)(\text{cp } \text{Vec}(3p)(\text{cons}))$

(13.5.1a)



(13.5.1b)

a row is the outer index, a cell one path

$u = ((1,2,3,4), C)$

$C = 5$

6

7

8

$(1 \times (\text{moves trans Vec}(n)(\text{concat})))$

$\text{moves}(C)$

$= 6 \ 7 \ 8 \ 5$

5 6 7 8

8 5 6 7

$\text{trans}(\text{that})$

$= 6 \ 5 \ 8$

7 6 5

8 7 6

5 8 7

$\text{Vec}(n)(\text{concat})(\text{that}) = \text{that}$

$\text{zip}(\text{that})$

$= (1, 6 \ 5 \ 8)$

(2, 7 6 5)

(3, 8 7 6)

(4, 5 8 7)

$\text{Vec}(n)(\text{cp})(\text{that})$

$= 1 \ 6 \ 2 \ 7 \ 3 \ 8 \ 4 \ 5$

1 5 2 6 3 7 4 8

1 8 2 5 3 6 4 7

$: A[n] \times A[n]$

one candidate per row, one square long:

a column of squares

1 keeps the column, the candidates move down, unmoved, up: a $3 \times n$ matrix

$: A[3][n]$

the transpose: row k gets rows $k-1, k, k+1$

$: A[n][3]$

$\text{concat}: X[3][1] \rightarrow X[3]$ is 1

each row: its square, and its candidates

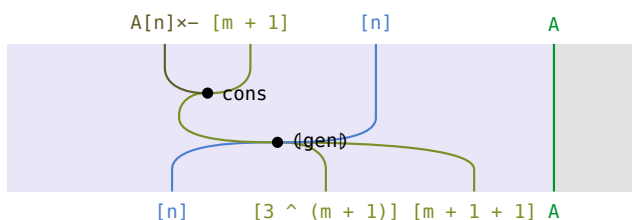
$: (A \times A[3])[n]$

cp pairs the square with each candidate:

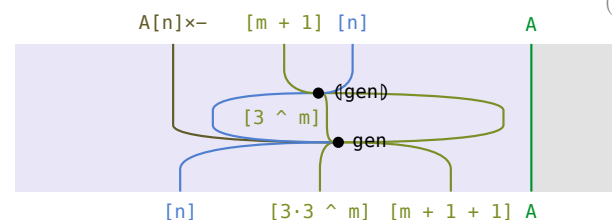
row k of that is the k -th 3×2 block, left to right; each line a two-square path

$: A[n][3][2]$

§13.5.2. gen is an F-algebra; $(\text{gen}): \text{cons } (\text{gen}) = (1 \times (\text{gen})) \text{gen}$



the $[m+1]$ bracket puts the column back on the matrix, then the fold reads all of it



the fold reads the rest under F , then one gen puts the column in front

(13.5.2b)

$xs = 1 \ 2 \ 3 \ 4$

5 6 7 8

$: A[m][n], m=2$

$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})(5,6,7,8))$

$(\text{gen})(5,6,7,8)$

$= 5$

6

7

8

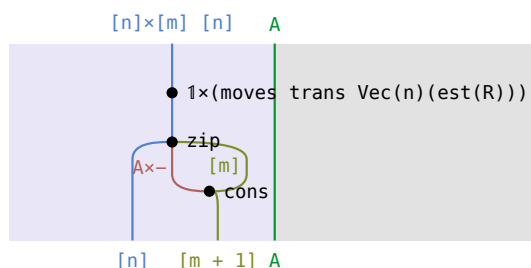
one column: each square its own one-square path

$: A[n][1][1]$

both halves are in (13.5.1b): the tail folds to the column 5 6 7 8, one path per row and each of them one square long, and gen on it is that walk.

§13.5.3. $Q \triangleq (1 \times (\text{moves trans Vec}(n)(\text{est}(R)))) \text{ zip}, A[n] \times A[n][m] \rightarrow A[n][m+1]$

(13.5.3a)



(13.5.3b)

$u = ((1,2,3,4), C)$
 $C = 5$
 6
 7
 8
 $(1 \times (\text{moves trans Vec}(n)(\text{est}(R))))$
 $\text{moves}(C)$
 $= 6 \ 7 \ 8 \ 5$
 $5 \ 6 \ 7 \ 8$
 $8 \ 5 \ 6 \ 7$
 $\text{trans}(\text{that})$
 $= 6 \ 5 \ 8$
 $7 \ 6 \ 5$
 $8 \ 7 \ 6$
 $5 \ 8 \ 7$
 $\text{Vec}(n)(\text{est}(R))(\text{that})$
 $= 5 \ 5 \ 6 \ 5$
 $\text{zip}(\text{that})$
 $= 1 \ 5$
 $2 \ 5$
 $3 \ 6$
 $4 \ 5$

$: A[n] \times A[n][m]$
 one path per row, one square long

 1 keeps the column, the paths move down, unmoved, up: a $3 \times n$ matrix

 $: A[3][n][m]$
 the transpose: row k gets rows $k-1, k, k+1$

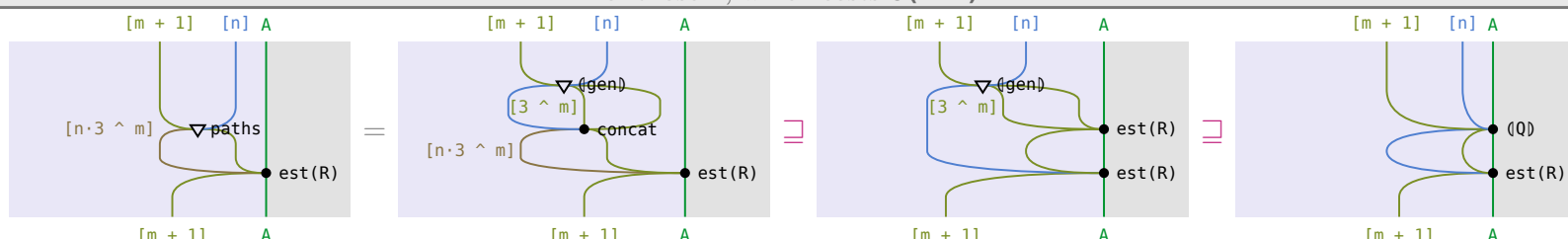
 $: A[n][3][m]$
 the cheapest into each row – chosen BEFORE
 $: A[n][m]$ the new square is put in front
 each row: its square, and the one predecessor that survived; the $[m+1]$ bracket is cons

 $: A[n][m+1]$

§13.5.4. paths $\text{est}(R) \sqsubseteq (\mathcal{Q}) \text{est}(R)$

$(\forall i. 1 \sqsubseteq R) \wedge (\forall i. R R \sqsubseteq R) \wedge (\forall i. (1 \times R) \text{cons} \sqsubseteq \text{cons } R) \Rightarrow (\mathcal{Q}) \text{est}(R) \sqsubseteq \text{paths est}(R)$ (13.5.4a)

a cheapest of all np paths of the cylinder is beaten by the greedy fold's one path per row and then a cheapest of those n , which costs $\mathcal{O}(n \times m)$.



$A[m+1][n] \text{---} \text{paths} \text{---} A[n-3 \wedge m][m+1] \text{---} \text{est}(R) \text{---} A[m+1]$
 $= A[m+1][n] \text{---} (\mathcal{Q}) \text{---} A[n][3 \wedge m][m+1] \text{---} \text{concat} \text{---} A[n-3 \wedge m][m+1] \text{---} \text{est}(R) \text{---} A[m+1]$ (13.5a) at paths
 $\sqsubseteq A[m+1][n] \text{---} (\mathcal{Q}) \text{---} A[n][3 \wedge m][m+1] \text{---} \text{Vec}(n)(\text{est}(R)) \text{---} A[n][m+1] \text{---} \text{est}(R) \text{---} A[m+1]$
 $\sqsubseteq A[m+1][n] \text{---} (\mathcal{Q}) \text{---} A[n][m+1] \text{---} \text{est}(R) \text{---} A[m+1]$

a cheapest of each of the n rows, then a cheapest of those; R transitive fusion at (13.5.4b)

$S \sqsubseteq S \Rightarrow \text{Vec}(j)(\text{est}(S)) \text{est}(S) \sqsubseteq \text{concat est}(S)$, R transitive

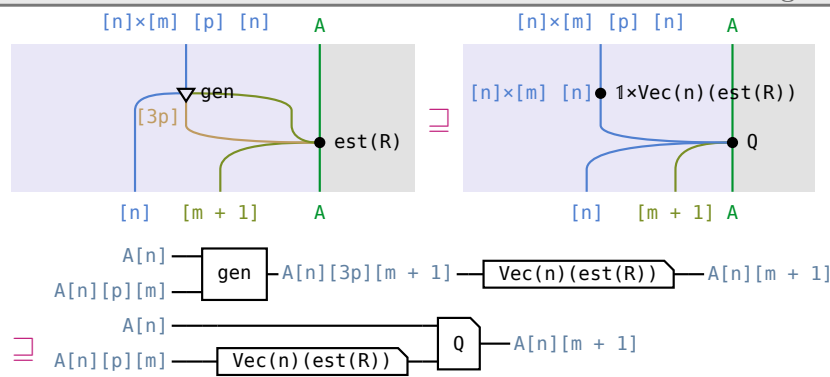
a cheapest of each of the j rows and then a cheapest of those j is a cheapest of all $j \times k$ entries laid end to end.

$(\forall i. 1 \sqsubseteq R) \wedge (\forall i. R R \sqsubseteq R) \wedge (\forall i. (1 \times R) \text{cons} \sqsubseteq \text{cons } R) \Rightarrow (\mathcal{Q}) \sqsubseteq (\mathcal{Q}) \text{Vec}(n)(\text{est}(R))$, R reflexive, transitive and monotonic

the one path per row the greedy fold keeps is one of the p that $(\mathcal{Q})$ generates for that row, and a cheapest of them.

$$RR \sqsubseteq R \wedge (1 \times R) \text{cons} \sqsubseteq \text{cons } R \Rightarrow (1 \times \text{Vec}(n)(\text{est}(R))) Q \sqsubseteq \text{gen } \text{Vec}(n)(\text{est}(R)) \quad (13.5.4b)$$

choosing a cheapest of each square's p paths before the column is extended is no better than extending first and choosing among the $3p$.



(7.13), then zip, trans, moves lax natural

$$(7.13) \text{ on } \text{Vec}: (1 \times R) \text{cons} \sqsubseteq \text{cons } R \Rightarrow (1 \times \text{est}(R)) \text{cons} \sqsubseteq \text{cp } \text{Vec}(k)(\text{cons}) \text{est}(R), R \text{ monotonic}$$

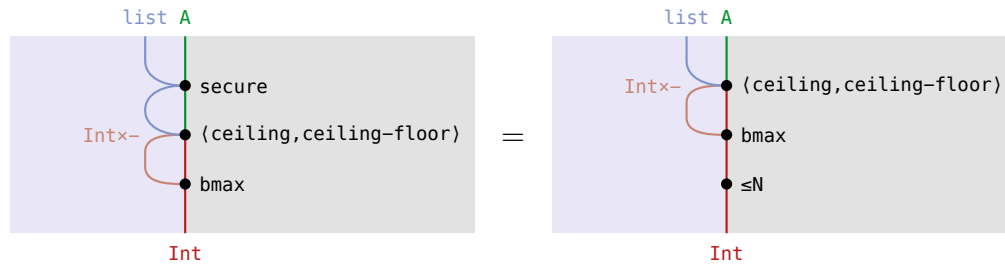
putting the new square in front of every one of the p candidates and then choosing a cheapest is beaten by choosing a cheapest first and putting the square in front of that one.

§13.6. The security van problem

definition	type	example
$R \triangleq \text{length} \leq \text{length}^\circ$	$[[A]] \rightarrow [[A]]$	$[[a,b,c]]$ to $[[a],[b,c]]$, and not back: $1 \leq 2$.
$\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\geq)$	$[A] \rightarrow \text{Int}$	$\text{ceiling}[a,b]$ is the largest of $0, a, a+b$.
$\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\leq)$	$[A] \rightarrow \text{Int}$	$\text{floor}[a,b]$ is the smallest of $0, a, a+b$, so $[a,b]$ carries $\text{ceiling} - \text{floor}$ cash.
secure the coreflexive on x with $\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$	$[A] \rightarrow [A]$	secure keeps $[a]$ exactly when $-N \leq a \leq N$.
ok the coreflexive on (a, xs) with xs non-empty and $[a] \# \text{head}(xs)$ secure	$A \times [[A]] \rightarrow A \times [[A]]$	ok keeps $(a, [[b],[c]])$ exactly when $[a,b]$ is secure.
$\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$	$A \times [[A]] \rightarrow [[A]]$	$\text{new}(a, [[b],[c]]) = [[a],[b],[c]]$.
$\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$	$A \times [[A]] \rightarrow [[A]]$	$\text{glue}(a, [[b],[c]]) = [[a,b],[c]]$.
$\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons secure}) \times 1) \text{ cons}$	$A \times [[A]] \rightarrow [[A]]$	$\text{old}(a, [[b],[c]]) = [[a,b],[c]]$ when $[a,b]$ is secure, and nothing otherwise.
partition $\triangleq ([\text{nil}, \text{new} \cup \text{glue}])$ cutting the transactions into non-empty segments is one pass along them that either opens a segment for the transaction in hand or puts it on the front of the segment already open	$[A] \rightarrow [[A]]$	partition $[a,b]$ gives $[[a],[b]]$ and $[[a,b]]$.
$S \triangleq [\text{nil}, \text{new} \cup \text{old}]$	$F[[A]] \rightarrow [[A]]$	$S(a, [[b],[c]])$ gives $[[a],[b],[c]]$, and $[[a,b],[c]]$ when $[a,b]$ is secure.
$(\forall a. \text{secureP}(\text{amount}, N, [a])) \implies$ partition list $(\text{secure}) = (S)$ cutting the transactions into non-empty segments every way and keeping the cuts whose every segment is secure is the one fold whose algebra is $[\text{nil}, \text{new} \cup \text{old}]$	$[A] \rightarrow [[A]]$	Both take $[a,b]$ to $[[a],[b]]$ — each $[a]$ is secure — and to $[[a,b]]$ when $[a,b]$ is.
$H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$	$[[A]] \rightarrow [[A]]$	$[[a],[b,c]]$ to $[[a,b],[c]]$: $[a]$ is a prefix of $[a,b]$.
$R; H \triangleq R \cap (R^\circ \Rightarrow H)$	$[[A]] \rightarrow [[A]]$	$[[a,b,c]]$ to $[[a],[b,c]]$ by $ R $, and $[[a],[b,c]]$ to $[[a,b],[c]]$ by $R \cap H$.
$ R \triangleq R \cap \neg R^\circ$	$[[A]] \rightarrow [[A]]$	$[[a,b,c]]$ to $[[a],[b,c]]$: $1 < 2$. The strict part R splits into: $R; H = R \cup (R \cap H)$.
the specification $\frac{\text{partition list}(\text{secure})}{\ni} \text{est}(R)$	$[A] \rightarrow [[A]]$	$[a,b]$ gives $[[a,b]]$ when $[a,b]$ is secure, and $[[a],[b]]$ otherwise.

§13.6.1. secure prefix $\sqsubseteq$ prefix secure

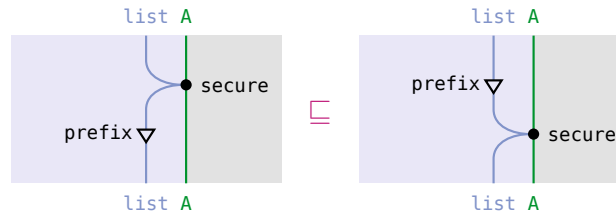
(13.6.1a)



$$\text{secure}(\text{ceiling}, \text{ceiling-floor}) \text{bmax} = (\text{ceiling}, \text{ceiling-floor}) \text{bmax} (\leq N)$$

a stretch passes **secure** before the test exactly where the test's own value passes $\leq N$ after it, which is $\triangleq$ **secure**'s **bmax**(**ceiling** **x**, **ceiling** **x** - **floor** **x**) $\leq N$

(13.6.1b)



$$\text{secure prefix} \sqsubseteq \text{prefix secure}$$

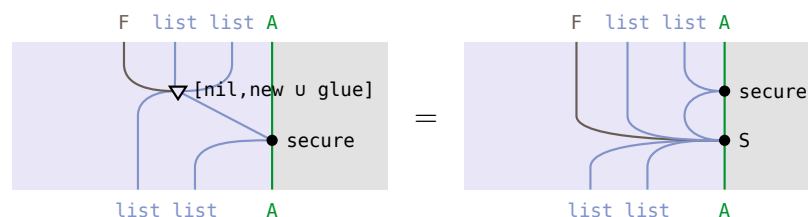
every pair **secure** then **prefix** gives, **prefix** then **secure** gives too — a prefix of a secure stretch is itself secure, which is the prefix-closure
B&dM p.185 names

§13.6.2. partition $\text{list}(\text{secure}) = ([\text{nil}, \text{new } u \text{ old}])$

(13.6.2a)

definition	Hinze–Marsden
new $\triangleq$ (wrap $\times 1$) cons wrap makes the transaction a segment on its own, and cons puts that segment at the front of the schedule: the van is called	
glue $\triangleq$ (1 $\times$ cons $^\circ$) assocl (cons $\times 1$) cons cons $^\circ$ splits the schedule into its first segment and the rest, the first cons puts the transaction at the front of that segment, and the second puts the segment back	
old $\triangleq$ (1 $\times$ cons $^\circ$) assocl ((cons secure) $\times 1$) cons glue with secure on the segment the transaction has just joined: the van stays away only where the longer segment still passes	

(13.6.2b)



$$(\forall a. \text{secureP}(\text{amount}, N, [a])) \Rightarrow [\text{nil}, \text{new } u \text{ glue}] \text{list}(\text{secure}) = F(\text{list}(\text{secure})) S$$

testing every segment for security after the algebra has run is testing it before it ran: a segment **glue** has extended is **secure** exactly where **old** would have let it be, and the one transaction **new** sets out on its own is **secure** whatever it is

$(\forall a. \text{secureP}(\text{amount}, N, [a])) \Rightarrow \text{partition list}(\text{secure}) = \langle S \rangle$ (13.6.2c)	
cutting the transactions into non-empty segments every way and keeping the cuts whose every segment is secure is the one fold whose algebra is $[\text{nil}, \text{new} \cup \text{old}]$	
step	Hinze–Marsden
$\text{partition list}(\text{secure})$ the specification: cut the transactions every way, then keep the cuts whose every segment is secure	
$= \langle [\text{nil}, \text{new} \cup \text{glue}] \rangle \text{list}(\text{secure})$ partition is the fold that either opens a segment for the transaction in hand or puts it on the front of the segment already open	
$= \langle [\text{nil}, \text{new} \cup \text{old}] \rangle$ fusion: the condition above moves $\text{list}(\text{secure})$ inside that fold, which is what turns glue into old	

(13.6.2d)

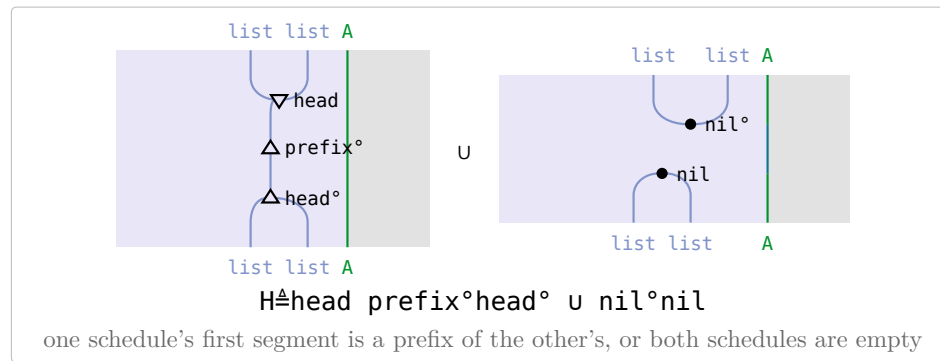
$(\forall a. \text{secureP}(\text{amount}, N, [a])) \Rightarrow \text{partition list}(\text{secure}) = \langle S \rangle$ cutting the transactions every way and then keeping the cuts whose every segment is secure is one pass along them that either calls the van or extends the open segment while it stays secure	

§13.6.3. $(1 \times R) \text{new} \sqsubseteq (\text{new} \cup \text{old}) R$

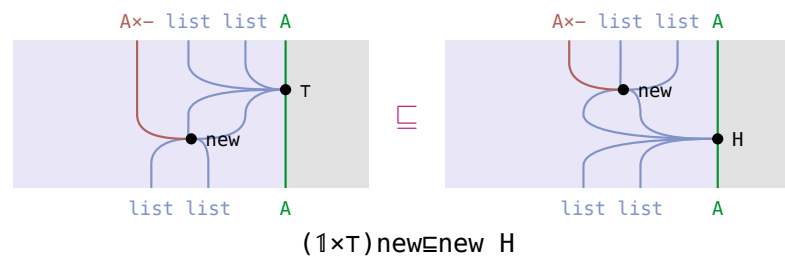
$(1 \times R) \text{new} \sqsubseteq (\text{new} \cup \text{old}) R$ (13.6.3a)	
calling the van for the transaction on a no-longer schedule gets no further than calling it on this one and shortening the schedule afterwards	
step	Hinze–Marsden
$(1 \times R) \text{new}$ the left side of (7.14) — $\triangleq$ secure	
$= (\text{wrap} \times R) \text{cons}$ $\text{new} \triangleq (\text{wrap} \times 1) \text{cons}$ — (13.6.2a) — and $(\text{wrap} \times 1) (1 \times R) = (\text{wrap} \times R)$	
$\sqsubseteq \text{new } R$ cons is monotonic on R — $(1 \times R) \text{cons} \sqsubseteq \text{cons } R$, consing onto a no-longer schedule leaves it no longer — and $\text{new} \sqsubseteq \text{new} \cup \text{old}$	

§13.6.4. $H \triangleq (\text{head prefix} \circ \text{head}) \cup (\text{nil} \circ \text{nil})$

(13.6.4a)

§13.6.5. $(1 \times (R;H)) \text{new} \in (\text{new} \cup \text{old})(R;H)$

(13.6.5a)



whatever schedule the van is called on, the result's first segment is the one transaction $[a]$, so any two results of **new** on that transaction stand in H

 $(1 \times (R;H)) \text{new} \in (\text{new} \cup \text{old})(R;H)$

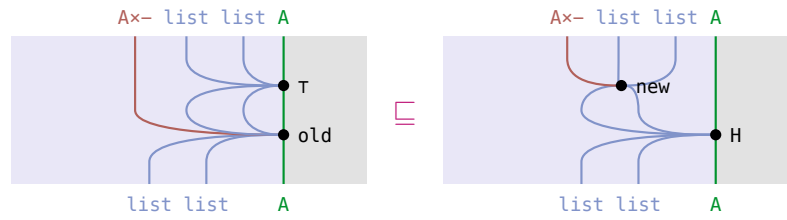
(13.6.5b)

calling the van for the transaction on a $R;H$ -better schedule gets no further than calling it on this one and bettering the whole schedule afterwards

step	Hinze–Marsden
$(1 \times (R;H)) \text{new}$ the left side of (7.16)	
$\sqsubseteq (1 \times R) \text{new}$ $R;H \in R \triangleq \text{secure}$	
$\sqsubseteq \text{new } R \cap \text{new } H$ (7.14) as far as its new R line — (13.6.3a) — and (7.18)	
$= \text{new } (R \cap H)$ new is a map, and a map distributes over $\cap$	
$\sqsubseteq \text{new } (R;H)$ $X \cap Y \in X;Y$, and $\text{new} \in \text{new} \cup \text{old}$	

§13.6.6. $(1 \times (R;H)) \text{old} \in (\text{new} \cup \text{old})(R;H)$

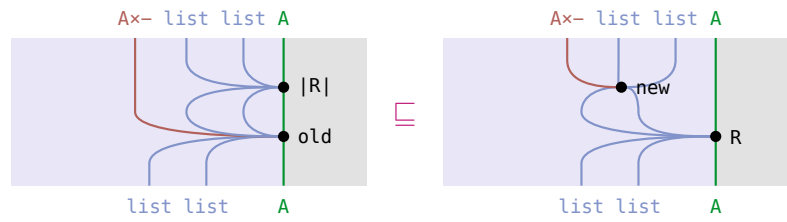
(13.6.6a)



$$(1 \times \tau) \text{old} \sqsubseteq \text{new H}$$

old leaves the transaction **[a]** at the front of the first segment, and **[a]** is what **new** makes that segment, so the two first segments are **prefix**-related whatever the schedules were

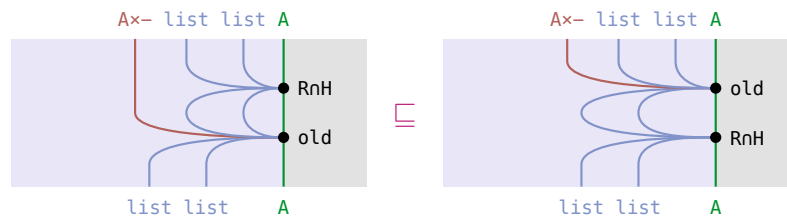
(13.6.6b)



$$(1 \times |R|) \text{old} \sqsubseteq \text{new R}$$

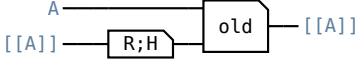
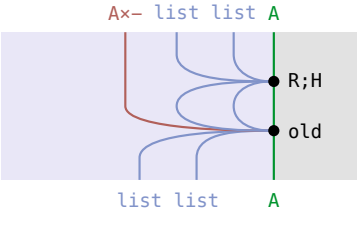
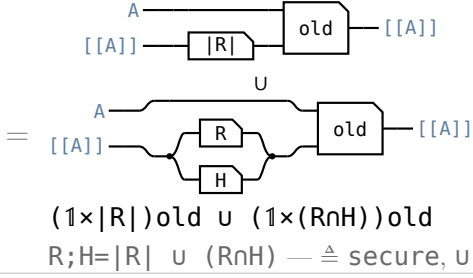
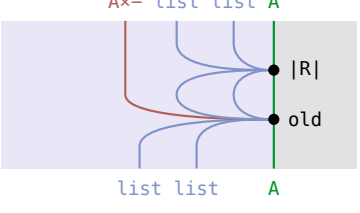
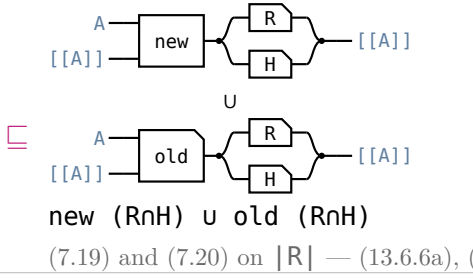
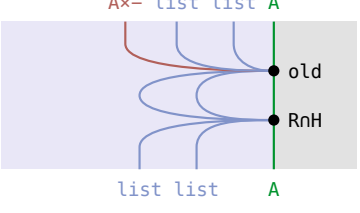
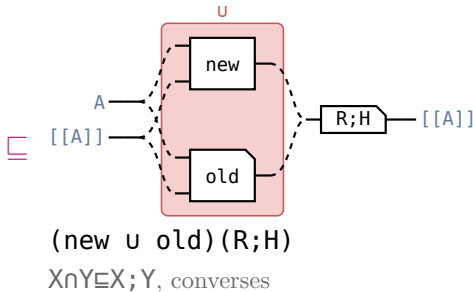
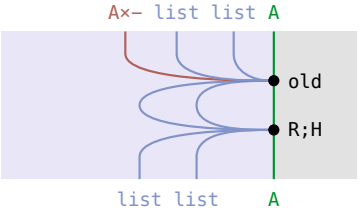
old keeps the schedule's length, so a strictly shorter one still comes out no longer than the one the van's own segment lengthens

(13.6.6c)



$$(1 \times (R \cap H)) \text{old} \sqsubseteq \text{old (RnH)}$$

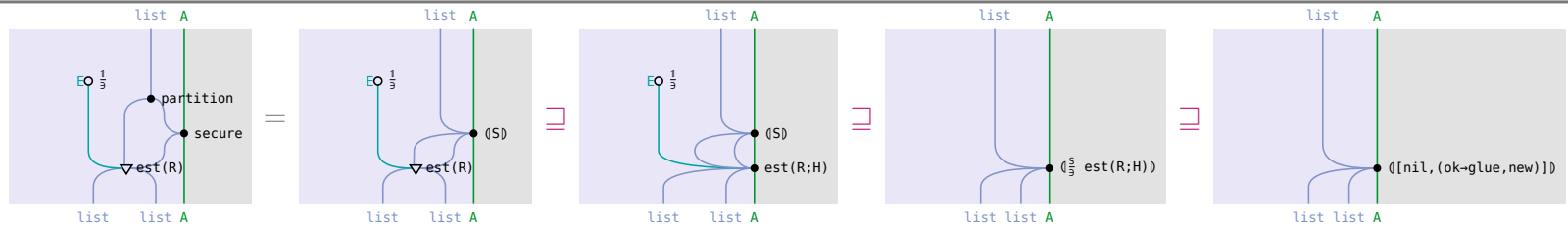
on a tie the one first segment is a prefix of the other, so prefix-closure of **secure** — (13.6.1b) — lets **old** fire on this side too, and it keeps both the length and the prefix

$(1 \times (R;H)) \text{old} \sqsubseteq (\text{new} \cup \text{old})(R;H)$ (13.6.6d)	
gluing the transaction onto a better schedule for the rest gets no further than gluing it on, or calling the van, and bettering the whole schedule after,	
circuit — the old branch of each union	Hinze–Marsden
	
the old operand of new $\cup$ old , in every row	
 $(1 \times R) \text{old} \cup (1 \times (R \text{ n } H)) \text{old}$ $R;H = R \cup (R \text{ n } H) \text{ — } \triangleq \text{ secure, } U \text{ distributes,}$	
 $\sqsubseteq$ $\text{new } (R \text{ n } H) \cup \text{old } (R \text{ n } H)$ (7.19) and (7.20) on R — (13.6.6a), (13.6.6b) — and (7.21) on $R \text{ n } H$ — (13.6.6c)	
 $\sqsubseteq$ $(\text{new} \cup \text{old})(R;H)$ $X \text{ n } Y \sqsubseteq X;Y, \text{ converses}$	

§13.6.7. The derivation

$$(\forall a. \text{secureP}(\text{amount}, N, [a])) \Rightarrow ([\text{nil}, (\text{ok-glue}, \text{new})]) \sqsubseteq \frac{\text{partition list}(\text{secure})}{\supset} \text{est}(\text{R}) \quad (13.6.7a)$$

the fewest secure segments the transactions can be cut into are one pass along them, the next transaction glued onto the open segment wherever that segment stays secure and the van called where it does not



$$[A] \xrightarrow{\frac{1}{\supset}} E[A] \xrightarrow{E(\text{partition list}(\text{secure}))} E[[A]] \xrightarrow{\text{est}(\text{R})} [[A]]$$

$$[A] \xrightarrow{\frac{1}{\supset}} E[A] \xrightarrow{E(\{S\})} E[[A]] \xrightarrow{\text{est}(\text{R})} [[A]]$$

=

$$\text{partition list}(\text{secure}) = \{S\} \xrightarrow{\supset} \text{est}(\text{R}) \quad \text{the specification} \xrightarrow{\supset} \text{secure}$$

$$[A] \xrightarrow{\frac{1}{\supset}} E[A] \xrightarrow{E(\{S\})} E[[A]] \xrightarrow{\text{est}(\text{R};H)} [[A]]$$

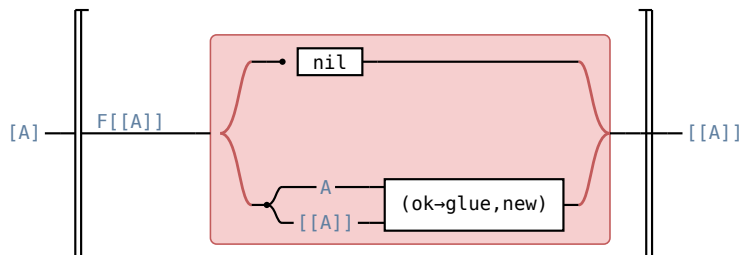
⊆

$$R;H \sqsubseteq R \xrightarrow{\supset} \text{secure}; (7.15) (1 \times R) \text{old} \sqsubseteq (\text{new} \cup \text{old})R \text{ is FALSE, the shorter partition need not stay secure, where (7.14) } (1 \times R) \text{new} \sqsubseteq (\text{new} \cup \text{old})R \text{ holds}$$

$$[A] \xrightarrow{\frac{1}{\supset}} E[A] \xrightarrow{E(\{S\})} E[[A]] \xrightarrow{\text{est}(\text{R};H)} [[A]]$$

⊆

$$\text{greedy theorem at } R;H, \text{ its hypothesis } F(R;H) S \sqsubseteq S(R;H) \text{ the old half (7.17) — (13.6.6d) — and the new half (7.16) — (13.6.5b) — which rests on (7.18) } (1 \times T) \text{new} \sqsubseteq \text{new } H \text{ — (13.6.5a)}$$



⊆

$$([\text{nil}, (\text{ok-glue}, \text{new})])$$

$$\text{old} \sqsubseteq \text{new } (R;H)^\circ: \text{old returns the shorter result wherever it returns one, and ok is where it does}$$

§14. Thinning Algorithms

§14.1. Thinning

(14.1a)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

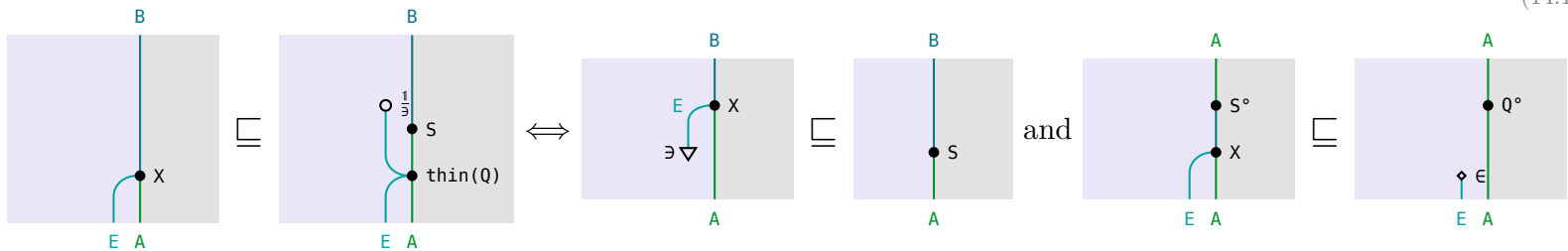
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(14.1b)

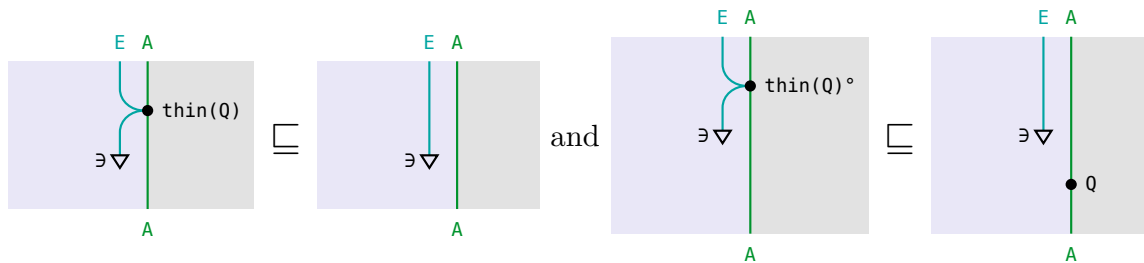
the law	what it says
$X \subseteq \frac{S}{\exists} \text{ thin}(Q) \iff X \ni S$ and $S^\circ X \subseteq Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \ni \exists \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \ni \exists \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{ thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin}(Q) \ni \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{ thin}(Q) \ni \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§14.1.1. $X \subseteq (S^\circ \exists) \text{ thin}(Q) \iff X \ni S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \ni \exists = \wedge(\exists) = 1$

§14.1.2. $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

(14.1.2a)

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

$EA \text{ est}(R) A$
 $\sqsubseteq EA \text{ thin}(Q) \text{ est}(R) A$
 $1 \sqsubseteq \text{thin}(Q) \text{ — (14.1b), } Q \text{ reflexive}$

(14.1.2b)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangle \text{thin}(Q) \text{ est}(R)$ — (13.1.1a)

$EA \text{ thin}(Q) \text{ est}(R) A$
 $\sqsubseteq EA \text{ thin}(Q) \exists A$
 $\text{est}(R) \sqsubseteq \exists \text{ — (13.1b) first row at } X \triangle \text{est}(R)$
 $EA \exists A$
 $\sqsubseteq \text{thin}(Q) \exists \exists \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q): \exists \exists = 1, \text{ so}$
 $\text{the left side is } \text{thin}(Q) \sqsubseteq \text{thin}(Q)$

(14.1.2c)

$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R, R$ transitive

$A \epsilon \text{ thin}(Q) EA \text{ est}(R) A$
 $\sqsubseteq A \text{ Q } \epsilon EA \text{ est}(R) A$
 $\sqsubseteq A \text{ Q } R A$
 $\sqsubseteq A \text{ R } R A$
 $\sqsubseteq A \text{ R } A$

$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon \text{ — (14.1.1a) at } S \triangle \exists, X \triangle \text{thin}(Q), \text{ second half}$
 $\epsilon \text{ est}(R) \sqsubseteq R^\circ \text{ — (13.1b) first row at } X \triangle \text{est}(R), \text{ second half, conversed}$
 $Q \sqsubseteq R, \text{ conversed}$
 $R \text{ transitive, conversed}$

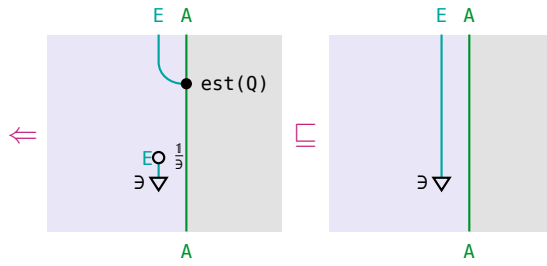
$$\text{est}(Q) \stackrel{1}{\exists} \text{thin}(Q)$$

(14.1.2d)

 the singleton holding a Q -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \exists \exists$$

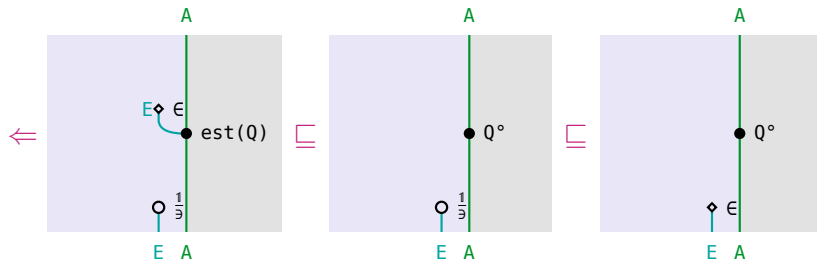
every member of the singleton is a member of the set



$$\begin{aligned} \Leftarrow & EA \text{---} \boxed{\text{est}(Q)} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \text{---} \boxed{\exists} \text{---} A \\ \sqsubseteq & EA \text{---} \boxed{\exists} \text{---} A \end{aligned}$$

 the $\exists/\exists$ half of $\triangle$ thin at $X \triangle \text{est}(Q) \stackrel{1}{\exists}$
 $\frac{1}{\exists} \exists = 1$ — (10b) — then $\text{est}(Q) \sqsubseteq \exists$ — (13.1b)

$$\epsilon \text{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

 the singleton's member stands in Q to every member of the set


$$\begin{aligned} \Leftarrow & A \text{---} \boxed{\epsilon} \text{---} EA \text{---} \boxed{\text{est}(Q)} \text{---} A \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \\ \sqsubseteq & A \text{---} \boxed{Q} \text{---} \boxed{\frac{1}{\exists}} \text{---} EA \\ \sqsubseteq & A \text{---} \boxed{Q} \text{---} \boxed{\epsilon} \text{---} EA \end{aligned}$$

 the $\epsilon \setminus (Q^\circ \epsilon)$ half of $\triangle$ thin, $\neg \sqsubseteq Q^\circ \epsilon$
 $\epsilon \text{est}(Q) \sqsubseteq Q^\circ$ — (13.1.1a) at $X \triangle \text{est}(Q)$, conversed

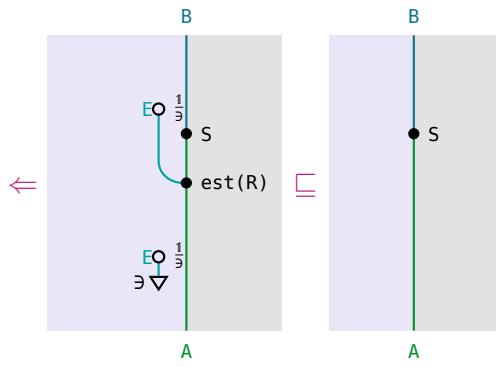
 $\frac{1}{\exists} \sqsubseteq \epsilon$, since $\frac{1}{\exists} \exists = 1$ with $\frac{1}{\exists}$ a map — (10b)

$$\text{Rn}S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q) \quad (14.1.2e)$$

given $\text{Rn}(S^\circ S) \sqsubseteq Q$, Q a preorder

$$\frac{S}{S} \text{est}(R) \frac{1}{S} \ni S$$

every member of the singleton is a value S returns



$$\Leftarrow B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{S}} EA \xrightarrow{\ni} A$$

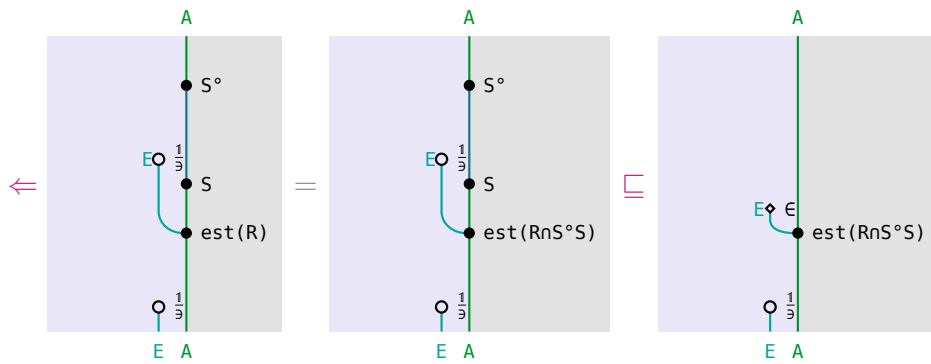
the first of (14.1b)' two conditions at $X \stackrel{S}{=} \text{est}(R) \frac{1}{S}$

$$\sqsubseteq B \xrightarrow{S} A$$

$\frac{1}{S} \ni 1$ — (10b) — then $\frac{S}{S} \text{est}(R) = S n(S^\circ \backslash R^\circ) \sqsubseteq S$ — (13.1b)

$$\text{Rn}S^\circ S \sqsubseteq Q \Rightarrow S^\circ \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument



$$\Leftarrow A \xrightarrow{S} B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{S}} EA$$

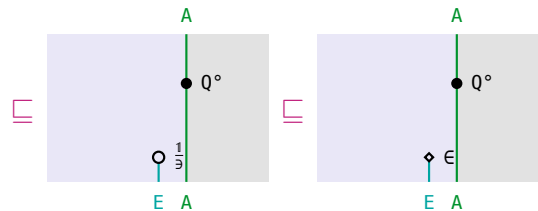
the second condition, $\neg Q^\circ \in$

$$= A \xrightarrow{S} B \xrightarrow{\frac{1}{S}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(RnS^\circ S)} A \xrightarrow{\frac{1}{S}} EA$$

$$\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(RnS^\circ S) \text{ — (13.1b)}$$

$$\sqsubseteq A \xrightarrow{\in} EA \xrightarrow{\text{est}(RnS^\circ S)} A \xrightarrow{\frac{1}{S}} EA$$

$$S^\circ \frac{S}{S} \sqsubseteq \in, \text{ since } \frac{S}{S} \ni S \text{ with } \frac{S}{S} \text{ a map — (10b)}$$



$$\sqsubseteq A \xrightarrow{Q} \frac{1}{S} \xrightarrow{EA}$$

$$\in \text{est}(RnS^\circ S) \sqsubseteq (RnS^\circ S)^\circ \text{ — (13.1.1a) at } X \stackrel{S}{=} \text{est}(RnS^\circ S), \text{ conversed — then } \text{Rn}(S^\circ S) \sqsubseteq Q$$

$$\sqsubseteq A \xrightarrow{Q} \in \xrightarrow{EA}$$

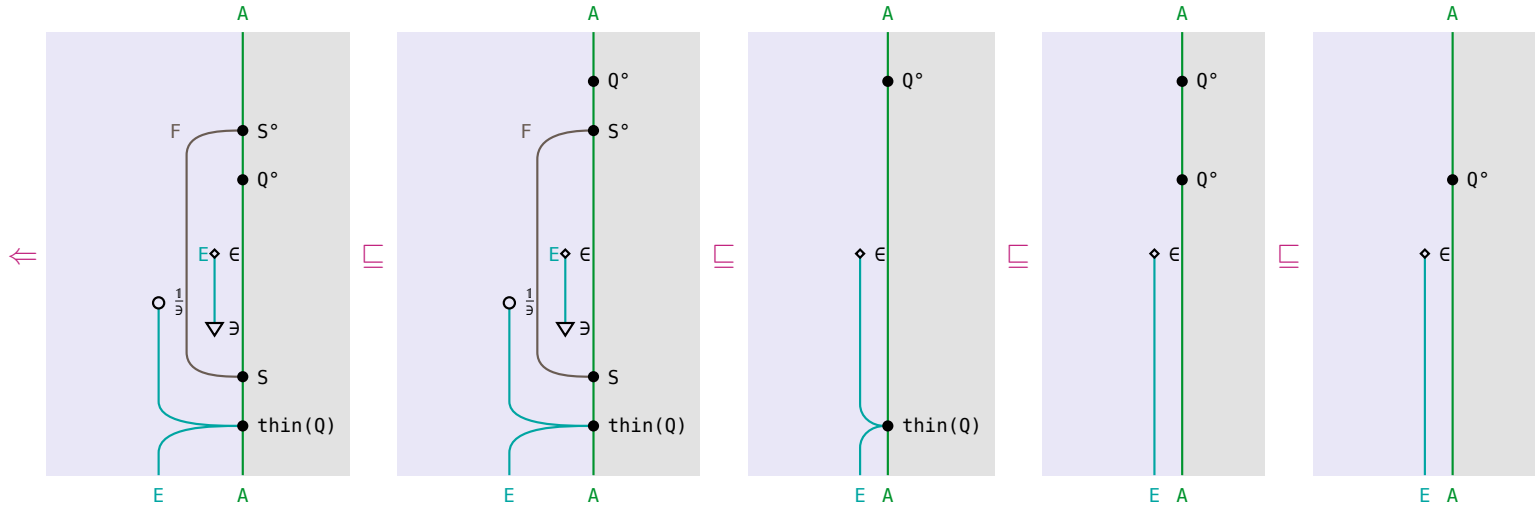
$$\frac{1}{S} \sqsubseteq \in, \text{ since } \frac{1}{S} \ni 1 \text{ with } \frac{1}{S} \text{ a map — (10b)}$$

$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2f)$$

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$\langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad \text{and} \quad \langle \frac{(S)}{\exists} \rangle \circ \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq Q^\circ \epsilon$$

(14.1b) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle \frac{(S)}{\exists} \rangle$ for its S



$$\leftarrow A \rightarrow \boxed{S} \rightarrow FA \rightarrow \boxed{F(Q^\circ \epsilon)} \rightarrow FEA \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

the first by fusion; (11.6.4b) at the bound $Q^\circ \epsilon$ reduces the second to $\neg \epsilon Q^\circ \epsilon$

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{S} \rightarrow FA \rightarrow \boxed{F(\epsilon)} \rightarrow FEA \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

$S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — (13.3b) at S , conversed; $F(R)^\circ = F(R^\circ)$ — (11b)

$S^\circ F(\epsilon) \sqsubseteq \frac{F(\exists)S}{\exists} \epsilon$, since $\frac{F(\exists)S}{\exists} \sqsubseteq F(\exists)S$ with $\frac{F(\exists)S}{\exists}$ a map — (10b)

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA$$

$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$, the $\epsilon \setminus (Q^\circ \epsilon)$ half of $\triangle \text{thin}$ — (1.1a)

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow EA$$

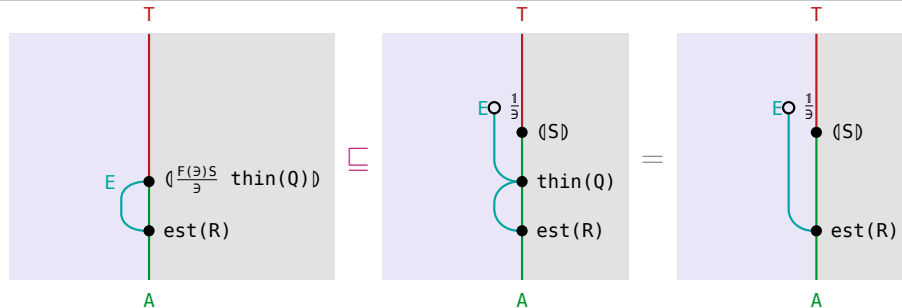
$Q^\circ Q^\circ = Q^\circ$, Q a preorder

$$\sqsubseteq A \rightarrow \boxed{Q} \rightarrow \boxed{\epsilon} \rightarrow EA$$

$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \quad (14.1.2g)$$

$\text{est}(R)$

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders



$$T \rightarrow \boxed{FEA} \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEA \rightarrow \boxed{E(F(\exists)S)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

$$\sqsubseteq T \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T) \rightarrow \boxed{E(\langle \frac{(S)}{\exists} \rangle)} \rightarrow \boxed{\text{thin}(Q)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

thinning theorem

$$= T \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T) \rightarrow \boxed{E(\langle \frac{(S)}{\exists} \rangle)} \rightarrow EA \rightarrow \boxed{\text{est}(R)} \rightarrow A$$

$\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (14.1b), $Q \sqsubseteq R$

§14.2. Paths in a layered network

(14.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head} \text{head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

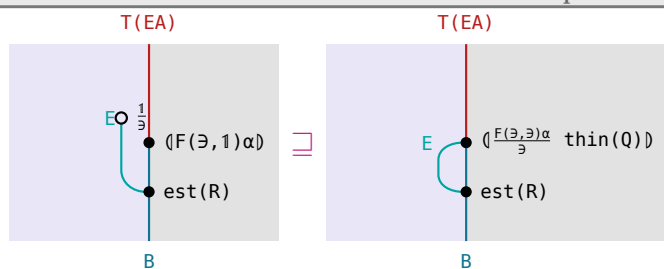
(14.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $Rn(S \circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \circ R \circ \sqsubseteq R \circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge Rn(F(1, \exists) \alpha) \circ F(1, \exists) \alpha \sqsubseteq Q \quad (14.2c)$$

$$\Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



$$\begin{aligned} & T(EA) \rightarrow \boxed{\frac{1}{\exists}} \rightarrow E(T(EA)) \rightarrow E(F(\exists, 1) \alpha) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \\ & \sqsubseteq T(EA) \rightarrow \boxed{FEB} \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFEB \rightarrow E(F(\exists, \exists) \alpha) \rightarrow \text{thin}(Q) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \end{aligned}$$

 (14.1.2g), at $F(\exists, 1) \alpha$ monotonic on Q — (14.2b).

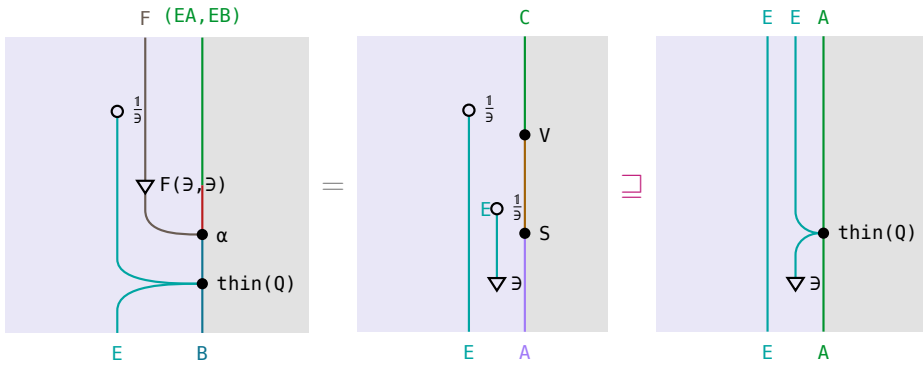
 $\sqsubseteq$ (14.2d) under $\langle \rangle$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (14.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



$$\begin{aligned} & F(EA, EB) \xrightarrow{\frac{1}{\exists}} E(F(EA, EB)) \xrightarrow{E(F(\exists, \exists)\alpha)} \text{thin}(Q) \xrightarrow{} EB \\ & = C \xrightarrow{\frac{1}{\exists}} EC \xrightarrow{E(V)} EW \xrightarrow{P(\frac{S}{\exists})} EEA \xrightarrow{\text{union}} EA \\ & \sqsubseteq EEA \xrightarrow{P(\text{thin}(Q))} \text{union} \xrightarrow{} EA \end{aligned}$$

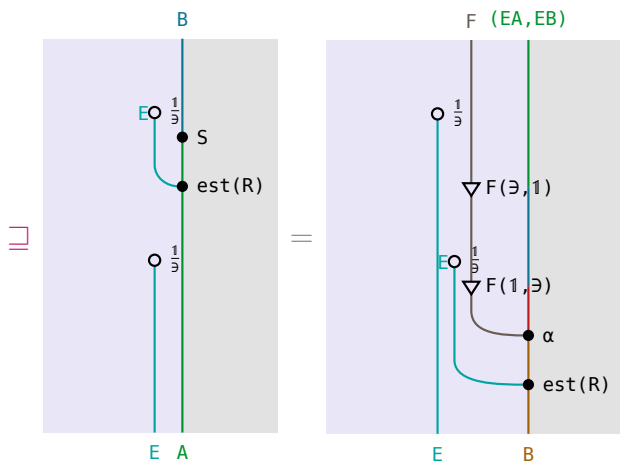
the algebra of (14.2c) row 2

$$F(\exists, \exists) = F(\exists, 1)F(1, \exists); \quad \frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}.$$

$$\text{union thin}(Q) \sqsupseteq P(\text{thin}(Q)) \text{union} \quad (14.1b).$$

$$Rn(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



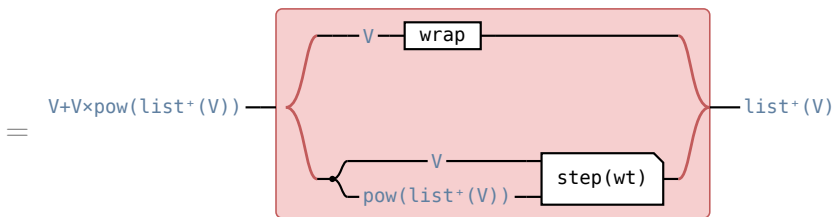
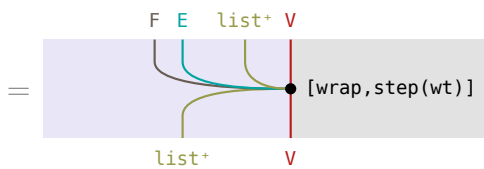
$$\begin{aligned} & B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{\exists}} EA \\ & = F(EA, EB) \xrightarrow{\frac{1}{\exists}} E(F(EA, EB)) \xrightarrow{E(F(\exists, 1))} E(FA, EB) \xrightarrow{P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R))} EB \end{aligned}$$

$$\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \xrightarrow{\frac{1}{\exists}} \quad (14.1b) \text{ at } S \triangleq F(1, \exists)\alpha, Rn(S \circ S) \sqsubseteq Q \quad (14.2b).$$

$$P(\frac{1}{\exists}) \text{union} = 1.$$

$$cp(F, \text{list}^+(V)) P(\alpha) \text{est}(\text{path}(\text{wt})) = [\text{wrap}, \text{step}(\text{wt})]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



$$\alpha \text{ a map, so } \frac{F(1, \exists)\alpha}{\exists} = \frac{F(1, \exists)}{\exists} P(\alpha); \quad \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}.$$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

§14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

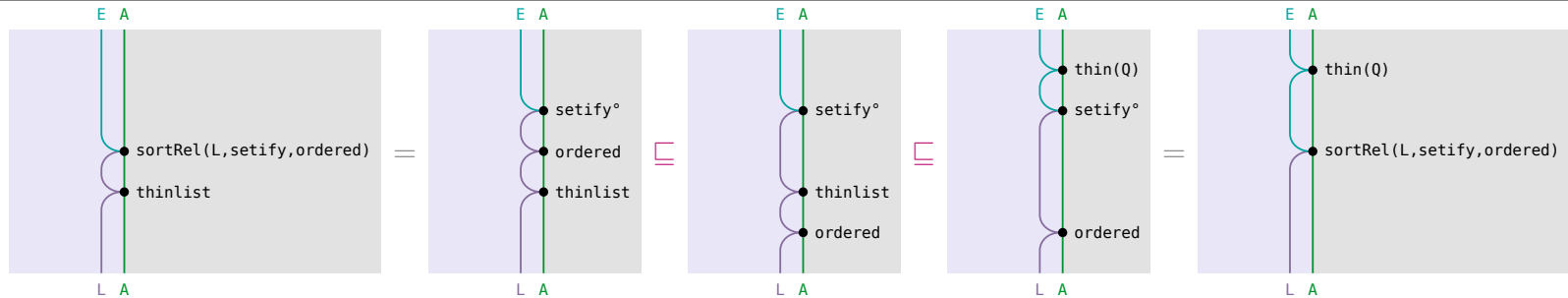
(14.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered subseq} \sqsubseteq \text{subseq ordered} \wedge$ (14.3c)

$\text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel}(L, \text{setify}, \text{ordered}) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}(L, \text{setify}, \text{ordered})$

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



$EA \xrightarrow{\text{setify}} \text{ordered} \xrightarrow{\text{thinlist}} LA$

$= EA \xrightarrow{\text{setify}} \text{ordered} \xrightarrow{\text{thinlist}} LA$

$EA \xrightarrow{\text{setify}} \text{thinlist} \xrightarrow{\text{ordered}} LA$

$\sqsubseteq \text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P$, since $\text{thinlist}(Q) \sqsubseteq \text{subseq} \triangleq \text{thinlist}$ — and a subsequence of a P -ordered list is P -ordered

$EA \xrightarrow{\text{thin}(Q)} \text{setify} \xrightarrow{\text{ordered}} LA$

$\triangleq \text{thinlist's thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$ after setify° , at $\text{setify}^\circ \text{ setify} \sqsubseteq 1$ for setify simple — (5c) — then $\cdot \text{setify} \dashv \cdot \text{setify}^\circ$ — (1.3a)

$= EA \xrightarrow{\text{thin}(Q)} \text{setify} \xrightarrow{\text{ordered}} LA$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$

$\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \triangleq \text{thinlist}$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \quad (14.3d)$$

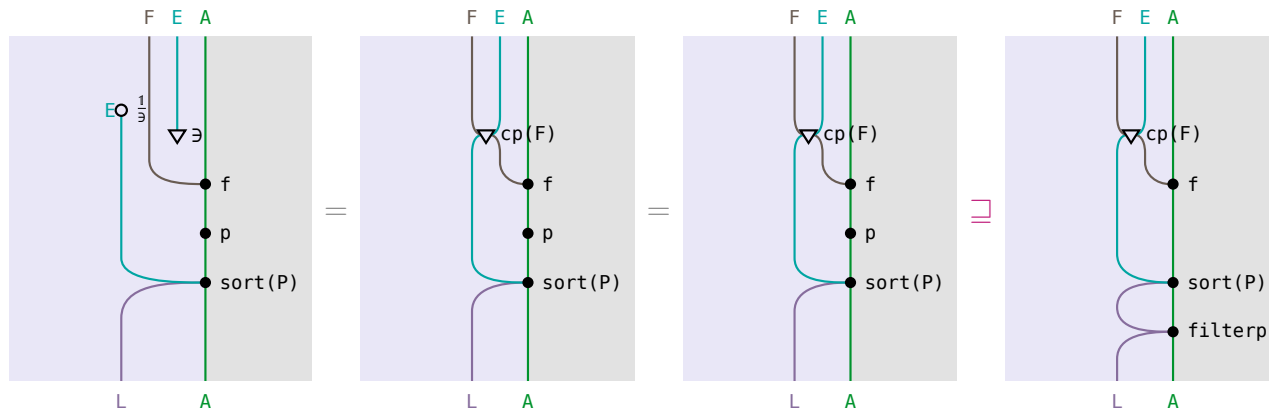
$$\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf}$$

$$\text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic on P , p coreflexive, F linear.

$$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$$

sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



$$\begin{aligned} & \text{FEA} \rightarrow \boxed{\frac{1}{\exists}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)fp)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ = & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{E(fp)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ = & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{P(f)} \rightarrow \boxed{E(p)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{P(f)} \rightarrow \text{EA} \rightarrow \boxed{\text{sort}(P)} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \end{aligned}$$

$$\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp) \text{ — (10b); } \text{cp}(F) \triangleq \frac{F(\exists)}{\exists} \text{ — } \triangleq \text{thinlist}$$

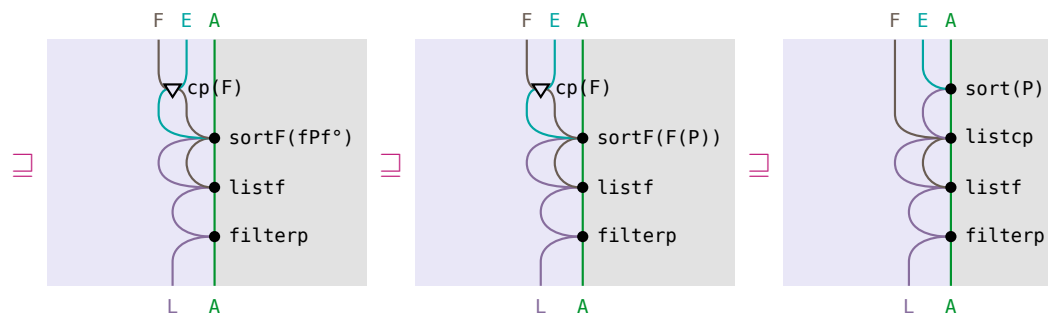
$$E(fp) = E(f)E(p); E(f) = P(f) \text{ for } f \text{ a map — (11.3d)}$$

$$\text{sort}(P) \text{filter}(p) \sqsubseteq E(p) \text{sort}(P) \text{ — (14.3b)}$$

$$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$$

$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F, A) \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp}(F, A) P(f) \text{sort}(P) \text{filterp}$$

combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



$$\begin{aligned} \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{\text{sortF}(fP f^\circ)} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{\text{cp}(F, A)} \rightarrow \text{EFA} \rightarrow \boxed{\text{sortF}(F(P))} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \\ \sqsubseteq & \text{FEA} \rightarrow \boxed{F(\text{sort}(P))} \rightarrow \text{FLA} \rightarrow \boxed{\text{listcp}} \rightarrow \text{LFA} \rightarrow \boxed{\text{listf}} \rightarrow \boxed{\text{filterp}} \rightarrow \text{LA} \end{aligned}$$

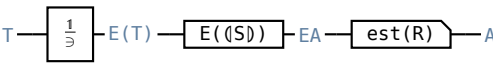
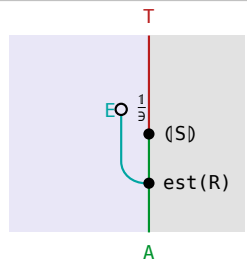
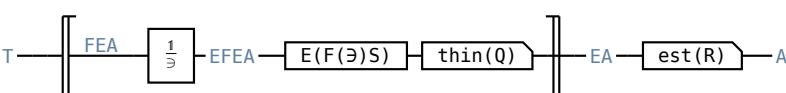
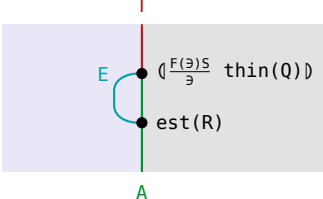
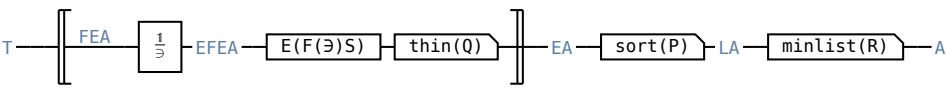
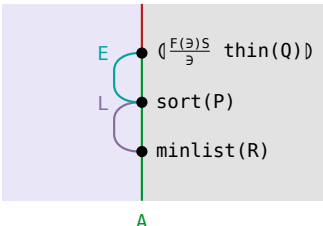
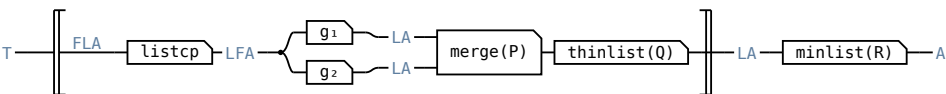
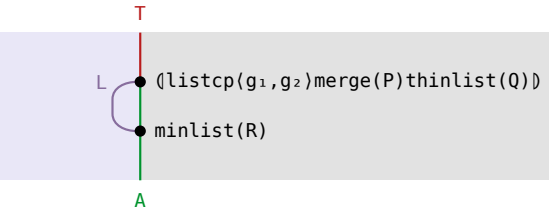
$$\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P) \text{ — (14.3b)}$$

$$FP \sqsubseteq fP f^\circ \text{ — (13.3b) at } f \text{ a map; } \text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ grows with } P \text{ — } \triangleq \text{thinlist}$$

$$F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP) \text{ — (14.3b), } F \text{ linear}$$

$$\begin{aligned}
& \text{PreservesRecip}(F) \wedge \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1, p_1, Q) \wedge \\
& \text{MonoAlg}(f_2, p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sort}F(X) \sqsubseteq \text{sort}F(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \\
& \text{sort}F(f_1 \text{ P } f_1^\circ) \text{list}f_1 \sqsubseteq P(f_1) \text{sort}(P) \wedge \text{sort}F(f_2 \text{ P } f_2^\circ) \text{list}f_2 \sqsubseteq P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filter}p_1 \sqsubseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p_2 \sqsubseteq E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F, A) \text{sort}F(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{list}f_1 \text{ filter}p_1 \wedge \\
& g_2 = \text{list}f_2 \text{ filter}p_2 \wedge S = f_1 \text{ p}_1 \cup f_2 \text{ p}_2 \implies (\text{list}cp(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \\
& \text{est}(R)
\end{aligned} \tag{14.3e}$$

a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

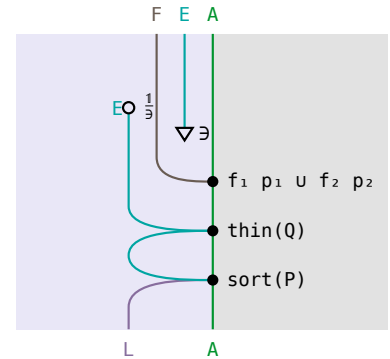
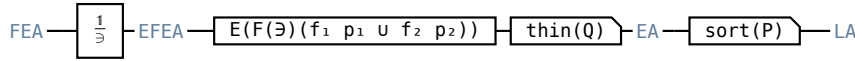
circuit	Hinze–Marsden
	
 (14.1.2g) at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$	
 $\text{sort}(P) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ — (14.3b) at its $Q \triangleq R$	
 fusion at (14.3f)	

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (14.3f) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \subseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \subseteq P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \subseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \subseteq E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{listcp} \subseteq \text{cp}(F, A) \text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \Rightarrow F(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \subseteq \frac{F(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

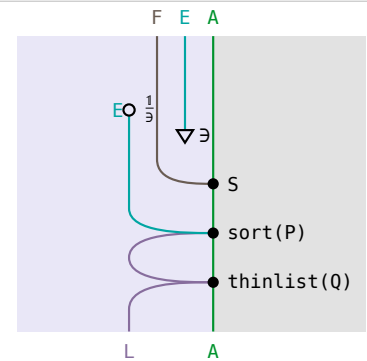
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)S)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \quad \text{--- (14.3b)}
\end{aligned}$$



$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{cup} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \\
& \left\langle \frac{F(\exists)f_1 \text{ p}_1}{\exists}, \frac{F(\exists)f_2 \text{ p}_2}{\exists} \right\rangle \text{cup} \text{sort}(P) \text{thinlist}(Q) \\
& S = (f_1 \text{ p}_1) \cup (f_2 \text{ p}_2) \quad \text{--- } \triangleq \text{thinlist, then } \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)f_2 \text{ p}_2)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \left\langle \frac{F(\exists)f_1 \text{ p}_1}{\exists}, \frac{F(\exists)f_2 \text{ p}_2}{\exists} \right\rangle (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q) \\
& (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup} \text{sort}(P) \quad \text{--- (14.3b)}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)f_1 \text{ p}_1)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{FEA} \rightarrow \boxed{\frac{1}{3}} \rightarrow \text{EFEA} \rightarrow \boxed{E(F(\exists)f_2 \text{ p}_2)} \rightarrow \text{EA} \rightarrow \text{sort}(P) \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \left\langle \frac{F(\exists)f_1 \text{ p}_1}{\exists} \text{sort}(P), \frac{F(\exists)f_2 \text{ p}_2}{\exists} \text{sort}(P) \right\rangle \text{merge}(P) \text{thinlist}(Q) \\
& (X, Y) (\text{sort}(P) \times \text{sort}(P)) = (X \text{sort}(P), Y \text{sort}(P)) \quad \text{--- (11.1.2b)}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \rightarrow \boxed{F(\text{sort}(P))} \rightarrow \text{FLA} \rightarrow \text{listcp} \rightarrow \text{LFA} \rightarrow \left\langle \text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2 \right\rangle \rightarrow \text{merge}(P) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{F}(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q) \\
& (14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \subseteq Xg_1, Xg_2 \quad \text{--- (11.1.2b); } g_i \triangleq \text{list}(f_i) \\
& \text{filter}(p_i) \quad \text{--- } \triangleq \text{thinlist}
\end{aligned}$$

§14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$, $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$, $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$.

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])$, within w the coreflexive on xs with weight $xs \leq w$, $0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R$.

$\text{FA} = 1 + \text{Item} \times A$, $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$ $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$.

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2)$.

the law	what it says
$(1 \times R) \text{ (cons (within w))} \sqsubseteq \text{cons (within w)R}$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) \text{ (cons (within w))} \sqsubseteq \text{cons (within w)Q}$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq wt(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
& \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq_P ([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
& \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq_P ([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP} \text{ filterp}_1 \sqsubseteq_E (\text{within}(w)) \text{sortP} \wedge \\
& F(\text{sortP}) \text{listcp} \sqsubseteq_{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq_{cup} \text{sortP} \wedge \text{sortP} \\
& \text{thinlist} \sqsubseteq_{\text{thin}} (Q(\text{vol}, wt)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq_{\text{est}} (R) \implies (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
& \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq_{\substack{\text{subseq} \text{ within}(w) \\ \exists}} \text{est}(R)
\end{aligned}
\tag{14.4c}$$

circuit subseq (within w) $\}$ est(R)	Hinze–Marsden
$= \frac{E([nil, cons] \text{ within } w \cup [nil, \pi_2])}{\supset} \text{est}(R)$ fusion, weights non-negative.	
$\sqsubseteq \text{listcp } \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist}(Q) \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq R$, F linear, Q from (14.4b)	
$= \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ — } \triangleq \text{within};$ minlist(R) is head. packings coming out in descending value	

§14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(14.5b)

the law	what it says
(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°.	both halves are monotonic on Q once ties in waste are broken by the first line
merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

$$\begin{aligned}
 & (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
 & \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \\
 & \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{filterp}_2 \sqsubseteq \text{E}(\text{ok}(w)) \text{sortP} \wedge \\
 & \text{F}(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{list}^+((\text{list}^+ \text{Word}))) \text{sortF}(\text{F}(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cp} \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(\text{Q}(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \\
 & \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(\text{R}) \\
 & \text{a paragraph laid out as a fold that thins the layouts kept at each word}
 \end{aligned} \tag{14.5c}$$

circuit	Hinze–Marsden
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & \frac{\text{partition list}^+(\text{fits } w)}{\supset} \text{est}(\text{R}) \end{aligned} $	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))}{\supset} \text{est}(\text{R}) \\ & \text{fusion, every word fits on a line by itself.} \end{aligned} $	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(\text{R}) \\ & \text{the algebra as } (f_1 p_1) \cup (f_2 p_2), p_1 \triangleq 1 \text{ — } \triangleq \\ & \text{thinlist.} \end{aligned} $	
$ \begin{aligned} & \text{listcp}(\text{g}_1, \text{g}_2) \text{cat thinlist}(\text{Q}) \\ & \sqsubseteq \text{minlist}(\text{R}) \\ & \text{binary thinning theorem, at } \text{P} \triangleq \text{T} \text{ with merge } \tau = \text{cat}, \text{Q} \\ & \text{from (14.5b)} \end{aligned} $	
$ \begin{aligned} & ([\text{start}, \text{cpr}(\text{h}_1, \text{h}_2) \text{cat thinlist}(\text{Q})]) \\ & = \text{minlist}(\text{R}) \\ & \text{listcp} = \text{wrap} + \text{cpr}, \text{g}_i \text{ along the coproduct — } \triangleq \\ & \text{partition} \end{aligned} $	

§14.6. Bitonic tours

FA = (City × City) + (City × A), the base functor of cons-lists of length at least two; listcp = wrap + cpr; tc : City × City → Real, neither positive nor symmetric.

start (a,b) = ([a,b], [a,b]), drop_l (a, ([b]#xs, ys)) = ([a]#xs, [a]#ys), drop_r (a, (xs, [b]#ys)) = ([a]#xs, [a]#ys), tour_Δ = ([start, drop_l u drop_r]).

cost (xs, ys) = outcost xs + incost ys, outcost [a₀, ..., a_n] = tc (a₀, a₁) + ... + tc (a_{n-1}, a_n), incost [a₀, ..., a_n] = tc (a₁, a₀) + ... + tc (a_n, a_{n-1}).

next_Δ = tail head, next₂ = next × next, head₂ = head × head, R_Δ = cost ≤ cost[°], Q_Δ = R_n(next₂ next₂[°])_n(head₂ head₂[°]), P_Δ = T_Δ, g₁ = list([start, drop_l]), g₂ = list([start, drop_r]).

the law	what it says
(1 × R) drop _l ⊆ drop _l R FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
(1 × R) drop _r ⊆ drop _r R FALSE	
(1 × Q) drop _l ⊆ drop _l Q	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
(1 × Q) drop _r ⊆ drop _r Q	

$ \begin{aligned} & (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{start}, \text{droplFn}] \top [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\ & \text{sortF}([\text{start}, \text{droprFn}] \top [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{dTour}(\text{City})) \text{sortF}(F(\top)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{tc})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \\ & \quad \underbrace{(\text{tourAlg})}_{\exists} \text{est}(R) \\ & \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city} \end{aligned} $ (14.6c)	
circuit	Hinze–Marsden
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \frac{\text{tour}}{\exists} \text{est}(R) \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & = \frac{([\text{start}, \text{dropl} \cup \text{dropr}])}{\exists} \text{est}(R) \\ & \text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}]) \triangleq \text{tour} \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \sqsubseteq (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q)) \text{minlist}(R) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat}, Q \text{ from (14.6b)} \end{aligned} $	
$ \begin{aligned} & ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\ & = \text{thinlist}(Q)] \text{minlist}(R) \\ & \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\ & \text{quadratic, two tours added per step} \end{aligned} $	

§15. Dynamic Programming

§15.1. Example of Hylo $(T)^\circ(h)$: segmenting a list

- $(T)=\text{concat}$
 $T=[\text{nil}, \text{cat}]$ takes a non-empty first list, so T° cuts a non-empty prefix off a list every way, and folding with T flattens a list of segments
- $(h)=\text{list}(\text{sum})$
 $h=[\text{nil}, \text{cons}(\text{sum} \times 1)]$ puts a segment's sum in front, so folding with h sums every segment
- $H=\text{partition list}(\text{sum})$
 H segments a list every way, then sums each segment; with R comparing the largest entry, M cuts the list so that the largest segment sum is as small as possible

(15.1b)

x	$h(x)$
$\text{inl}(())$	$[\]$
$\text{inr}([c], [])$	$[c]$
$\text{inr}([a, b], [c])$	$[a + b, c]$
$\text{inr}([a], [b + c])$	$[a, b + c]$

- $H([a, b, c], ys) \iff ys=[a + b + c] \vee ys=[a, b + c] \vee ys=[a + b, c] \vee ys=[a, b, c]$
 H sends $[a, b, c]$ to the sums of its four segmentations and nothing else
- $T^\circ F(H)h=H$
cut once, solve the rest, put the pieces back together

(15.1c)

§15.2. Theory

(15.2a)

$h : FB \rightarrow B$ a map, $T : FA \rightarrow A$ an F -algebra, $R : B \rightarrow B$.

$H \triangleq (T)^\circ(h) : A \rightarrow B$, $M \triangleq \frac{H}{\exists} \text{est}(R)$ the problem to be solved, $(\mu X : G(X))$ as in $\triangleq \mu$.

$\text{DPSetting}(h, R) \triangleq \text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \sqsubseteq R^\circ$

h is a map, h is monotonic on R° , and R° is transitive — assumed by every theorem below

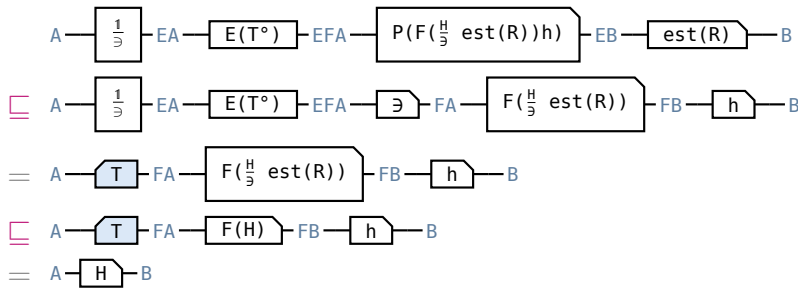
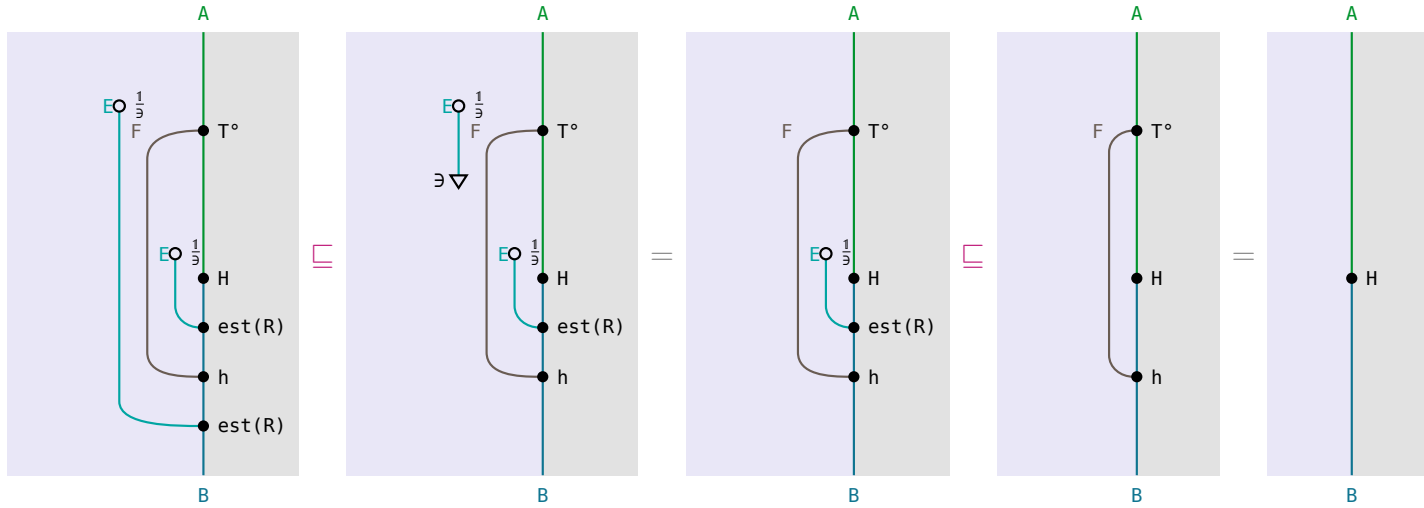
$(\mu X : \frac{T^\circ}{\exists} P(F(X)h) \text{est}(R)) \sqsubseteq \frac{H}{\exists} \text{est}(R)$
every answer the recursion returns is an optimal one

(15.2b)

- $\frac{T^\circ}{\exists}$ takes the input apart one step every way; $F(X)$ solves each part by the recursion X ; h assembles each candidate; $\text{est}(R)$ keeps a best one
- $\sqsubseteq M$: that never misses the optimum, though all candidates are never listed as H would
- h monotonic on R : a better part never makes the whole worse
- $\sqsubseteq$, not $=$: some optima may not be returned; that the recursion returns anything at all needs T° to stop taking apart — dynamic programming theorem

$$T^\circ F(H)h=H \Rightarrow \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq H$$

(15.2c)

 taking the input apart every way T allows ($\frac{T^\circ}{\exists}$), solving each part by M and keeping an optimum ($P(F(M)h)$
 $\text{est}(R)$) returns only what H returns


$$(9.4) P(X)\text{est}(R) \sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ --- (13.1.8a)}$$

$$\frac{T^\circ}{\exists} \exists = T^\circ \text{ --- (10b)}$$

$$M \triangleq \frac{H}{\exists} \text{est}(R) \sqsubseteq \frac{H}{\exists} \exists = H \text{ --- (13.1.1a)}$$

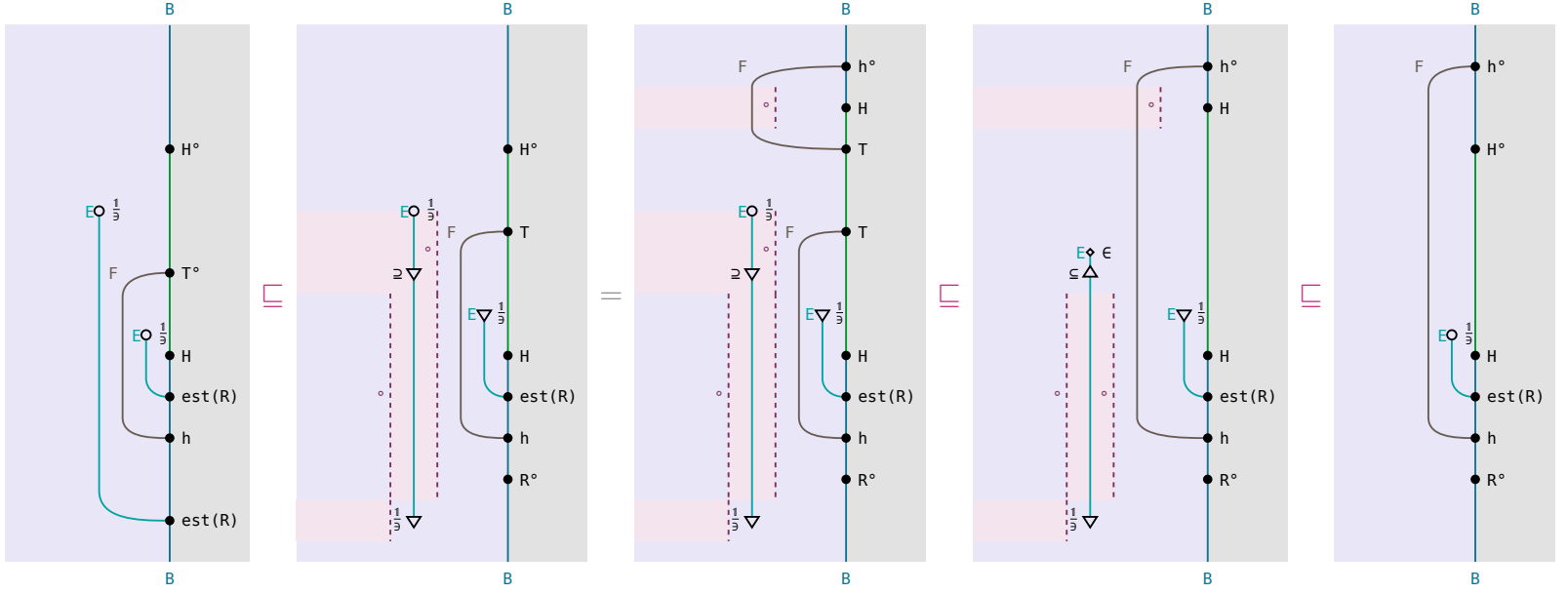
$$T^\circ F(H)h=H: H \triangleq (T)^\circ \circ (h) \text{ and (11.6.4a)}$$

$$T^\circ F(H)h=H \Rightarrow H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq R^\circ \quad (15.2d)$$

for every $\mathbf{b}$ that $\mathbf{H}$ returns from an input, the step $\frac{T^\circ}{\exists} P(F(\mathbf{M})h) \text{est}(R)$ returns from that input only $\mathbf{b}'$ with $\mathbf{R}$ relating $\mathbf{b}'$ to $\mathbf{b}$

$$T^\circ F(H)h=H \Rightarrow H^\circ \frac{T^\circ}{\exists} P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq h^\circ F(H^\circ)F(\frac{H}{\exists} \text{est}(R))hR^\circ$$

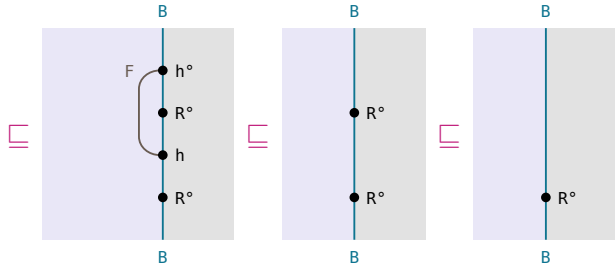
from an answer of $\mathbf{H}$, the step's answer is reached by taking $\mathbf{H}$'s parts back, rebuilding them by $\mathbf{F}(\mathbf{M})h$, then one $\mathbf{R}^\circ$



$$\begin{aligned}
 & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{P(F(\frac{H}{\exists} \text{est}(R))h)} EB \xrightarrow{\text{est}(R)} B \\
 \sqsubseteq & B \xrightarrow{H} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))hR^\circ} B \quad (9.4) \quad P(X)\text{est}(R) \sqsubseteq \epsilon \setminus (XR^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ --- (13.1.8a)} \\
 = & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{T} A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))hR^\circ} B \quad H^\circ = h^\circ F(H^\circ)T: H \triangleq (T)^\circ(h) \text{ and (11.6.4a)} \\
 \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{\epsilon} EFA \xrightarrow{\epsilon \setminus F(\frac{H}{\exists} \text{est}(R))hR^\circ} B \quad T \frac{T^\circ}{\exists} \sqsubseteq \epsilon: \frac{T^\circ}{\exists} \text{ a map, } \frac{T^\circ}{\exists} \ni T^\circ \text{ --- (10b)} \\
 \sqsubseteq & B \xrightarrow{h} FB \xrightarrow{F(H^\circ)} FA \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} R \xrightarrow{} B \quad \epsilon(\epsilon \setminus Y) \sqsubseteq Y \text{ at } Y \triangleq F(\mathbf{M})hR^\circ \text{ --- division}
 \end{aligned}$$

$$h^\circ F(H^\circ)F(\frac{H}{\exists} \text{est}(R))hR^\circ \sqsubseteq R^\circ$$

rebuilding an answer of $\mathbf{H}$ from its parts, each replaced by $\mathbf{M}$'s, and one $\mathbf{R}^\circ$ stays within $\mathbf{R}^\circ$



$$\begin{aligned}
 \sqsubseteq & B \xrightarrow{h} F(R^\circ) \xrightarrow{FB} h \xrightarrow{R} B \\
 \sqsubseteq & B \xrightarrow{R} R \xrightarrow{} B \\
 \sqsubseteq & B \xrightarrow{R} B
 \end{aligned}$$

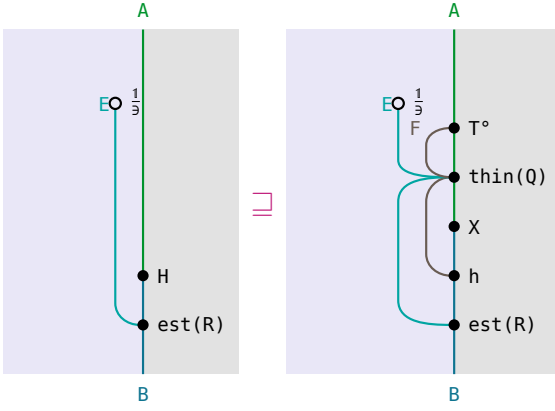
$$H^\circ M \sqsubseteq R^\circ: M \triangleq \frac{H}{\exists} \text{est}(R) \text{ --- (13.1.1a)}$$

$$h^\circ F(R^\circ)h \sqsubseteq R^\circ: h \text{ monotonic on } R^\circ$$

$$R^\circ R^\circ \sqsubseteq R^\circ: R \text{ transitive}$$

$$\text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{thin}(Q)P(F(X)h)\text{est}(R)) \sqsubseteq_{\exists}^H \text{est}(R) \quad (15.2e)$$

an optimum over everything H returns is reached by taking the input apart every way T allows, dropping the parts that can never win, solving each of the rest and keeping one optimum

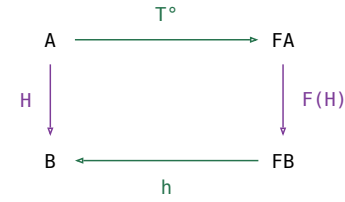
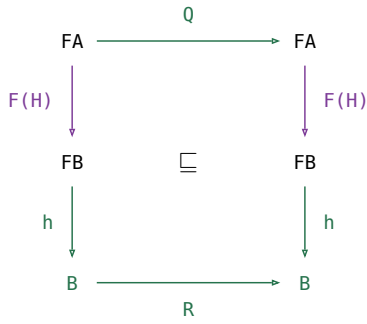


$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(H)} EB \xrightarrow{\text{est}(R)} B$$

the problem to be solved, $H \triangleq (T)^\circ \circ (h) \triangleq H$

$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(X)h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B$$

$\sqsubseteq$ h monotonic on R and Q a preorder with $QF(H)h \sqsubseteq F(H)hR$; $\text{thin}(Q)$ as in (14.1b). Theorem 9.1 is this with the thinning step dropped — $1 \sqsubseteq \text{thin}(Q)$. Knaster–Tarski leaves (9.1): the body at M is $\sqsubseteq M$; $M = H \cap (H^\circ \setminus R^\circ)$ splits that into (9.2) and (9.3) below — (13.1.1a). The fixed point is unique and entire when T° followed by F 's membership relation is inductive, $\frac{T^\circ}{\exists}$ finite and non-empty, R connected

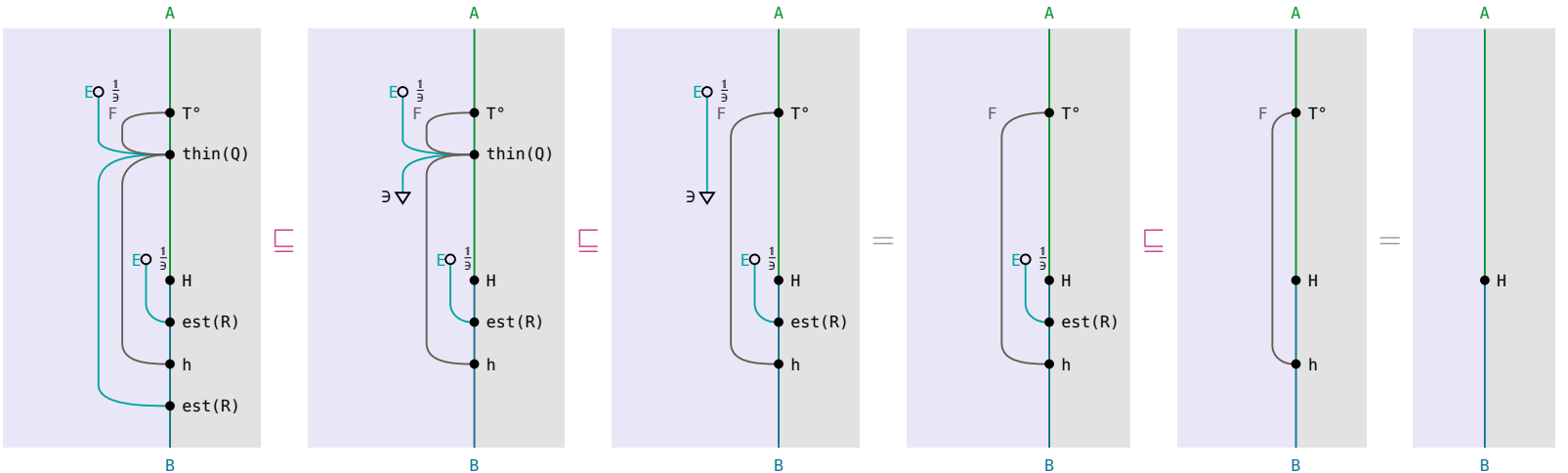


$T^\circ F(H)h = H$: taking the input apart once by T° , solving the parts by $F(H)$ and assembling by h returns exactly what H returns

$QF(H)h \sqsubseteq F(H)hR$: every result of thinning by Q , solving the parts by $F(H)$ and assembling by h is reached by R from a result of solving by $F(H)$ and assembling by h without Q

$$T^\circ F(H)h = H \Rightarrow \frac{T^\circ}{\exists} \text{thin}(Q)P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq_{\exists}^H H \quad (15.2f)$$

taking the input apart every way T allows ($\frac{T^\circ}{\exists}$), dropping the parts $\text{thin}(Q)$ rejects, solving each rest by M and keeping an optimum returns only what H returns



$$A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} P(F(\frac{H}{\exists} \text{est}(R))h) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \quad (9.2)$$

(9.2)

$$\sqsubseteq A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} \text{thin}(Q) \xrightarrow{EFA} \exists \text{FA} \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \quad (9.4) \quad P(X)\text{est}(R) \sqsubseteq \exists X \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ — (13.1.8a)}$$

$$\sqsubseteq A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E(T^\circ)} EFA \xrightarrow{\exists} \text{FA} \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \quad \text{thin}(Q) \exists \sqsubseteq \exists \text{ — (14.1b)}$$

$$= A \xrightarrow{T} \text{FA} \xrightarrow{F(\frac{H}{\exists} \text{est}(R))} FB \xrightarrow{h} B \quad \frac{T^\circ}{\exists} \exists = T^\circ \text{ — (10b)}$$

$$\sqsubseteq A \xrightarrow{T} \text{FA} \xrightarrow{F(H)} FB \xrightarrow{h} B \quad M \sqsubseteq \frac{H}{\exists} \exists = H \text{ — (13.1.1a)}$$

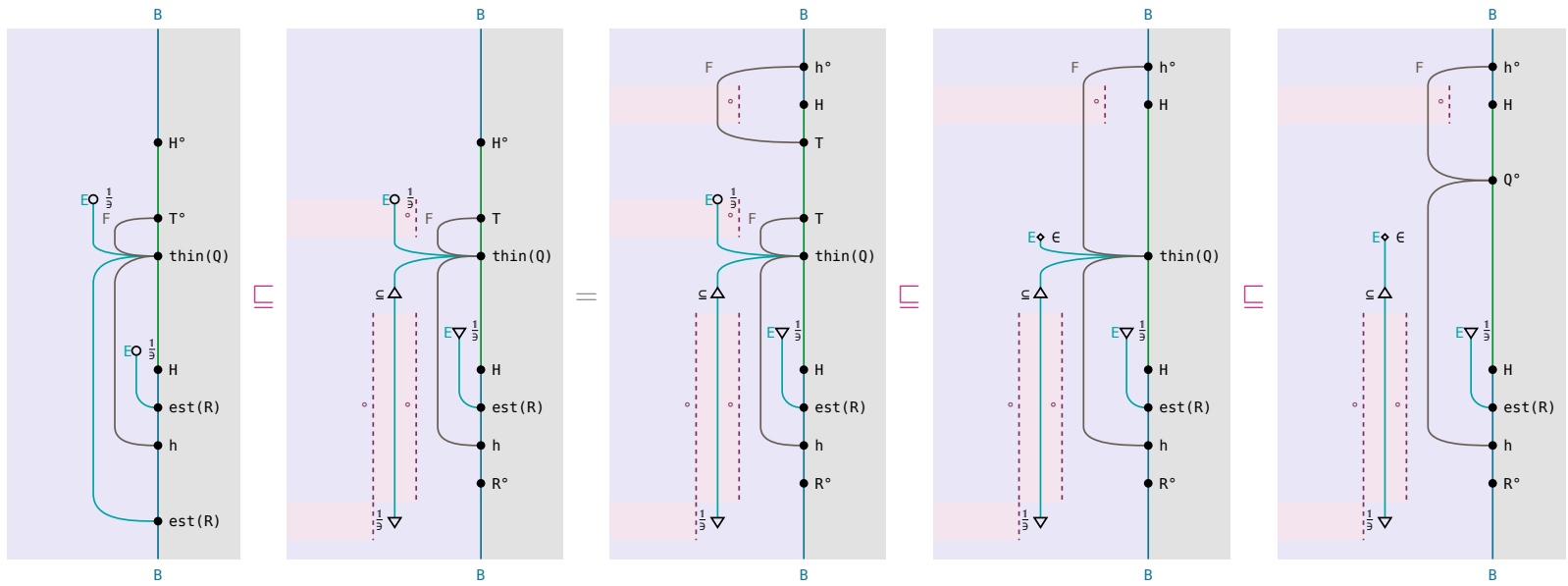
$$= A \xrightarrow{H} B \quad T^\circ F(H)h = H: H \triangleq (T)^\circ \circ (h) \text{ and (11.6.4a)}$$

$$T^\circ F(H)h=H \wedge QF(H)h \sqsubseteq F(H)hR \Rightarrow H^\circ \frac{T^\circ}{\exists} \text{thin}(Q)P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq R^\circ \quad (15.2g)$$

H° followed by the body at M is ER° : an answer of the body is never worse than an answer of H to the same input

$$T^\circ F(H)h=H \Rightarrow H^\circ \frac{T^\circ}{\exists} \text{thin}(Q)P(F(\frac{H}{\exists} \text{est}(R))h)\text{est}(R) \sqsubseteq h^\circ F(H^\circ)Q^\circ \in (\in \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ)$$

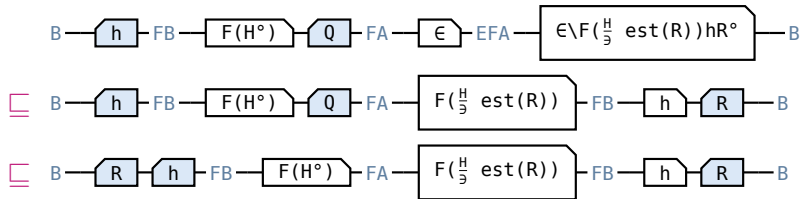
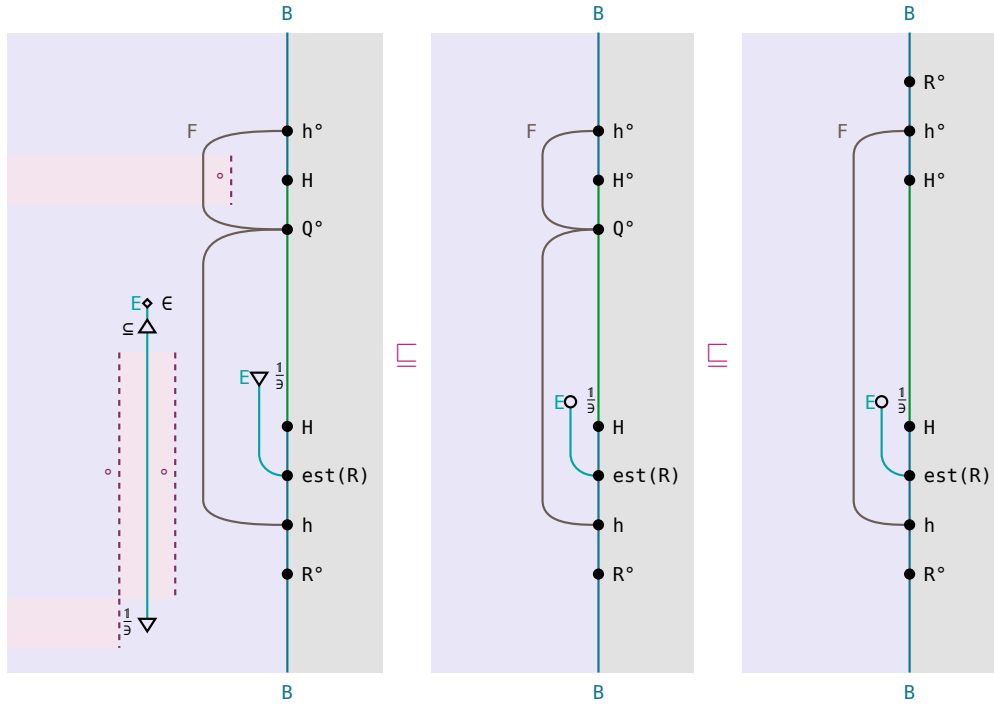
from an answer of H , the thinned step's answer is reached through H 's parts, a Q° step to a candidate, and that candidate's answer



$$\begin{aligned}
 & B \rightarrow [H] \rightarrow A \rightarrow \left[\frac{1}{\exists} \right] \rightarrow EA \rightarrow E(T^\circ) \rightarrow \text{thin}(Q) \rightarrow EFA \rightarrow P(F(\frac{H}{\exists} \text{est}(R))h) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B \\
 & \sqsubseteq B \rightarrow [H] \rightarrow A \rightarrow \left[\frac{1}{\exists} \right] \rightarrow EA \rightarrow E(T^\circ) \rightarrow \text{thin}(Q) \rightarrow EFA \rightarrow \in \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \rightarrow B \\
 & \quad (9.4) P(X)\text{est}(R) \sqsubseteq \in \backslash (XR^\circ) \text{ at } X \triangleq F(\frac{H}{\exists} \text{est}(R))h \text{ --- (13.1.8a)} \\
 & = B \rightarrow [h] \rightarrow FB \rightarrow F(H^\circ) \rightarrow FA \rightarrow T \rightarrow A \rightarrow \left[\frac{1}{\exists} \right] \rightarrow EA \rightarrow E(T^\circ) \rightarrow \text{thin}(Q) \rightarrow EFA \rightarrow \in \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \rightarrow B \\
 & \quad H^\circ = h^\circ F(H^\circ)T, \text{ the converse of } T^\circ F(H)h=H \text{ --- (11.6.4a)} \\
 & \sqsubseteq B \rightarrow [h] \rightarrow FB \rightarrow F(H^\circ) \rightarrow FA \rightarrow \in \rightarrow \text{thin}(Q) \rightarrow EFA \rightarrow \in \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \rightarrow B \\
 & \quad T \frac{T^\circ}{\exists} \sqsubseteq \in, \text{ not a tabulated row: } \frac{T^\circ}{\exists} \ni = T^\circ \text{ conversed} \\
 & \sqsubseteq B \rightarrow [h] \rightarrow FB \rightarrow F(H^\circ) \rightarrow Q \rightarrow FA \rightarrow \in \rightarrow EFA \rightarrow \in \backslash F(\frac{H}{\exists} \text{est}(R))hR^\circ \rightarrow B \quad \in \text{thin}(Q) \sqsubseteq Q^\circ \in: \text{ a dropped candidate is } Q\text{-below a kept one --- (14.1b)}
 \end{aligned}$$

$$QF(H)h \sqsubseteq F(H)hR \implies h^\circ F(H^\circ)Q^\circ \in (\epsilon \setminus F(\frac{H}{3} \text{ est}(R)))hR^\circ \sqsubseteq R^\circ h^\circ F(H^\circ)F(\frac{H}{3} \text{ est}(R))hR^\circ$$

the hypothesis on Q turns the Q° step into an R° step in front of rebuilding by $F(M)h$

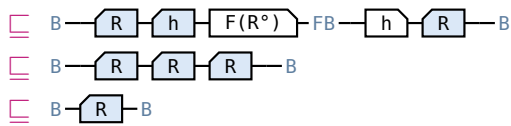
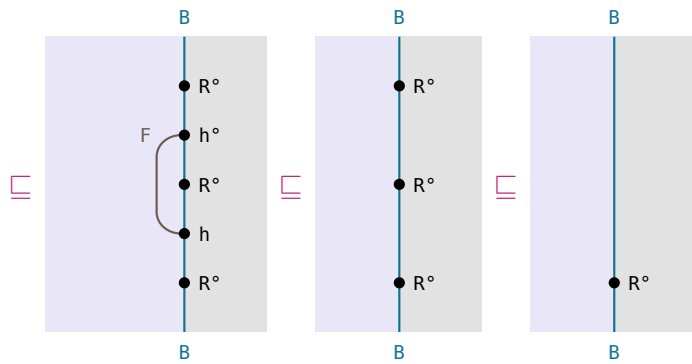


$\epsilon \in (\epsilon \setminus Y) \sqsubseteq Y$, not a tabulated row: division cancels

$QF(H)h \sqsubseteq F(H)hR$ conversed — the hypothesis on Q

$$R^\circ h^\circ F(H^\circ)F(\frac{H}{3} \text{ est}(R))hR^\circ \sqsubseteq R^\circ$$

rebuilding from parts replaced by M 's, between two R° steps, stays within R°



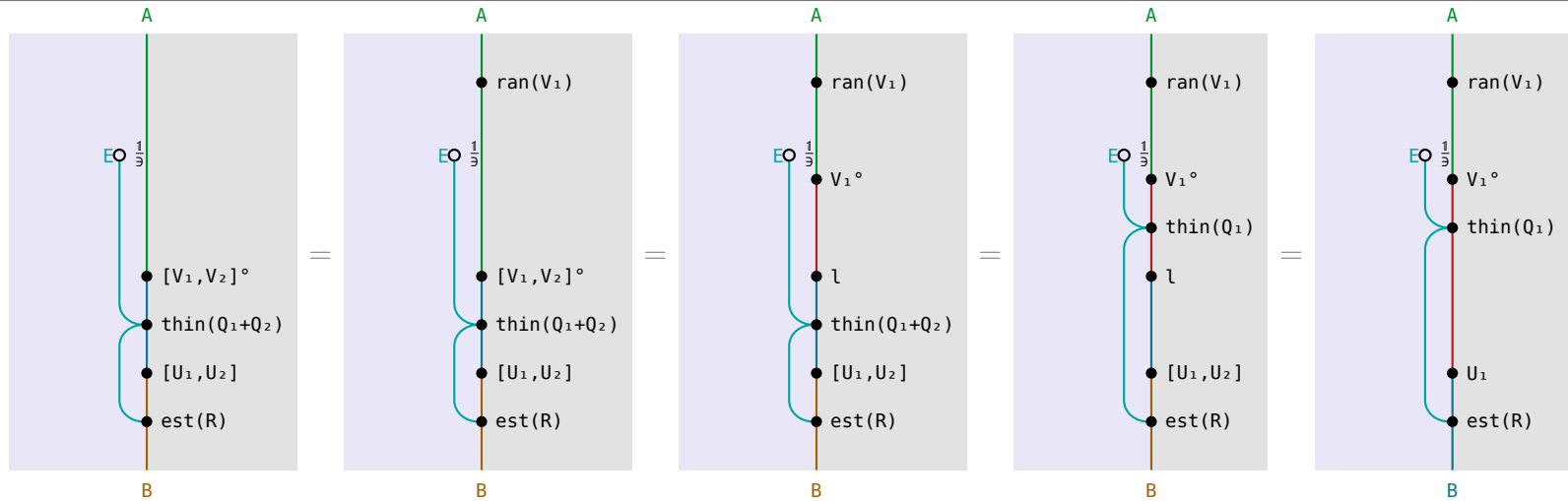
$H^\circ M \sqsubseteq R^\circ$ under F — (13.1.1a)

$h^\circ F(R^\circ)h \sqsubseteq R^\circ$: h monotonic on R , shunted — the hypothesis on h

R transitive, twice

$$V_2 \quad V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{thin}(Q_1+Q_2)P([U_1, U_2])\text{est}(R) = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{thin}(Q_1)P(U_1)\text{est}(R) \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{thin}(Q_2)P(U_2)\text{est}(R) \quad (15.2i)$$

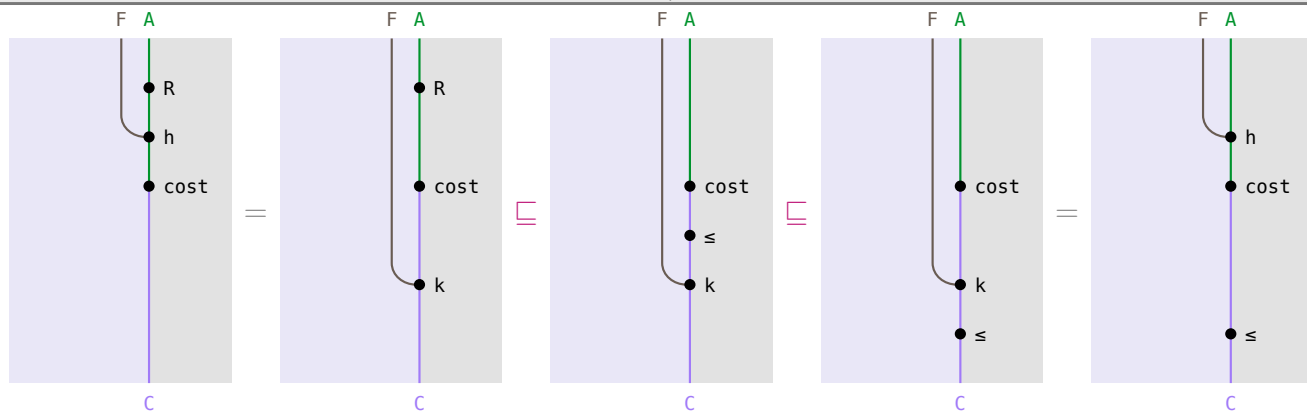
when V_1 and V_2 have disjoint ranges, the body over $[V_1, V_2]$ runs the V_1 problem on inputs V_1 reaches and the V_2 problem on inputs V_2 reaches



$$\begin{aligned} & A \xrightarrow{\frac{1}{\exists}} EA \xrightarrow{E([V_1, V_2]^\circ)} \text{thin}(Q_1+Q_2) \xrightarrow{E(\alpha+\beta)} P([U_1, U_2]) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ &= A \xrightarrow{\text{ran}(V_1)} \frac{1}{\exists} EA \xrightarrow{E([V_1, V_2]^\circ)} \text{thin}(Q_1+Q_2) \xrightarrow{E(\alpha+\beta)} P([U_1, U_2]) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ &\quad \text{off } \text{ran } V_1 \cup \text{ran } V_2 \text{ the set is empty and } \text{est}(R) \text{ of it is nothing; not a tabulated row — the } V_2 \text{ branch is the same} \\ &= A \xrightarrow{\text{ran}(V_1)} \frac{1}{\exists} EA \xrightarrow{E(V_1^\circ)} E\alpha \xrightarrow{P(l)} \text{thin}(Q_1+Q_2) \xrightarrow{E(\alpha+\beta)} P([U_1, U_2]) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ &\quad \text{Exercise 9.5: on } \text{ran } V_1, \frac{[V_1, V_2]^\circ}{\exists} = \frac{V_1^\circ}{\exists} P(\text{inl}), \text{ as } V_1, V_2 \text{ have disjoint ranges} \\ &= A \xrightarrow{\text{ran}(V_1)} \frac{1}{\exists} EA \xrightarrow{E(V_1^\circ)} \text{thin}(Q_1) \xrightarrow{E\alpha} P(l) \xrightarrow{E(\alpha+\beta)} P([U_1, U_2]) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ &\quad \text{Exercise 9.5: } P(\text{inl})\text{thin}(Q_1+Q_2) = \text{thin}(Q_1)P(\text{inl}) \\ &= A \xrightarrow{\text{ran}(V_1)} \frac{1}{\exists} EA \xrightarrow{E(V_1^\circ)} \text{thin}(Q_1) \xrightarrow{E\alpha} P(U_1) \xrightarrow{EB} \text{est}(R) \xrightarrow{} B \\ &\quad P \text{ a relator and } \text{inl}[U_1, U_2] = U_1 \end{aligned}$$

$$\text{Map}(\text{cost}) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge h \text{cost} = F(\text{cost})k \wedge F(\leq)k \leq k \implies \text{MonoAlg}(h, R) \quad (15.2j)$$

if R compares two values by comparing their **costs** under $\leq$, and h then **cost** equals $F(\text{cost})$ then a k monotonic on $\leq$, then h is monotonic on R



$$\begin{aligned} & FA \xrightarrow{F(R)} h \xrightarrow{} A \xrightarrow{\text{cost}} C \\ &= FA \xrightarrow{F(R)} F(\text{cost}) \xrightarrow{FC} k \xrightarrow{} C \\ &\sqsubseteq FA \xrightarrow{F(\text{cost})} F(\leq) \xrightarrow{FC} k \xrightarrow{} C \\ &\sqsubseteq FA \xrightarrow{F(\text{cost})} FC \xrightarrow{k} \leq \xrightarrow{} C \\ &= FA \xrightarrow{h} A \xrightarrow{\text{cost}} \leq \xrightarrow{} C \end{aligned}$$

$F(R)h \sqsubseteq hR$ iff $F(R)h \text{ cost} \sqsubseteq h \text{ cost} \leq$: definition of R and shunting
 assumption $h \text{ cost} = F(\text{cost})k$
 $R \text{ cost} \sqsubseteq \text{cost} \leq$, as cost is a map; functors
 assumption $F(\leq)k \leq k$: k monotonic on $\leq$
 assumption $h \text{ cost} = F(\text{cost})k$

(15.2k)

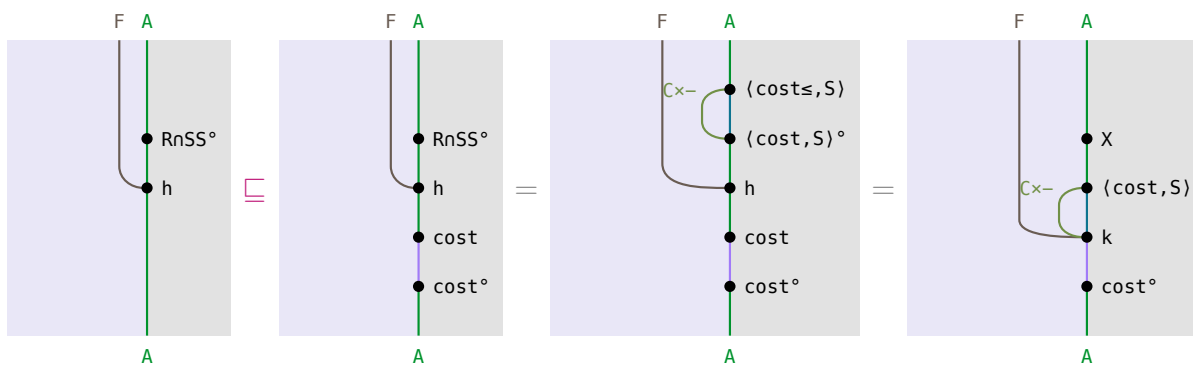
the condition	how it is discharged
$F(R)h \sqsubseteq hR$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost})k$, $F(\leq)k \leq k$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$
$F(Rn(H^\circ H))h \sqsubseteq hR$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$, $F(\leq \times 1)k \leq k$, H° simple;	monotonicity in context : k may also read the input the part was built from
$QF(H)h \sqsubseteq F(H)hR$ at $Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R)h \sqsubseteq hR$, $VH \sqsubseteq HR$	both conditions at once, split along the two arguments of a bifunctor

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge \text{hcost} = F(\langle \text{cost}, S \rangle)k \wedge F(\leq \times 1)k \leq k \implies F(RnSS^\circ)h \leq hR \quad (15.2l)$$

improving each part by R within its S -context, then assembling by h , is below hR

$$\text{Map}(\text{cost}) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge \text{hcost} = F(\langle \text{cost}, S \rangle)k \implies F(RnSS^\circ)h \leq F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) F(\langle \text{cost}, S \rangle)k \text{cost}^\circ$$

improving the parts within their S -context and assembling by h is below the same improvement on the $(\text{cost}, \text{context})$ bundles, assembled by k



$$\begin{aligned} & FA \text{---} F(RnSS^\circ) \text{---} h \text{---} A \\ \sqsubseteq & FA \text{---} F(RnSS^\circ) \text{---} h \text{---} A \text{---} \text{cost} \text{---} C \text{---} \text{cost} \text{---} A \\ = & FA \text{---} F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) \text{---} h \text{---} A \text{---} \text{cost} \text{---} C \text{---} \text{cost} \text{---} A \\ = & FA \text{---} F(X) \text{---} F(\langle \text{cost}, S \rangle) \text{---} F(C \times B) \text{---} k \text{---} C \text{---} \text{cost} \text{---} A \end{aligned}$$

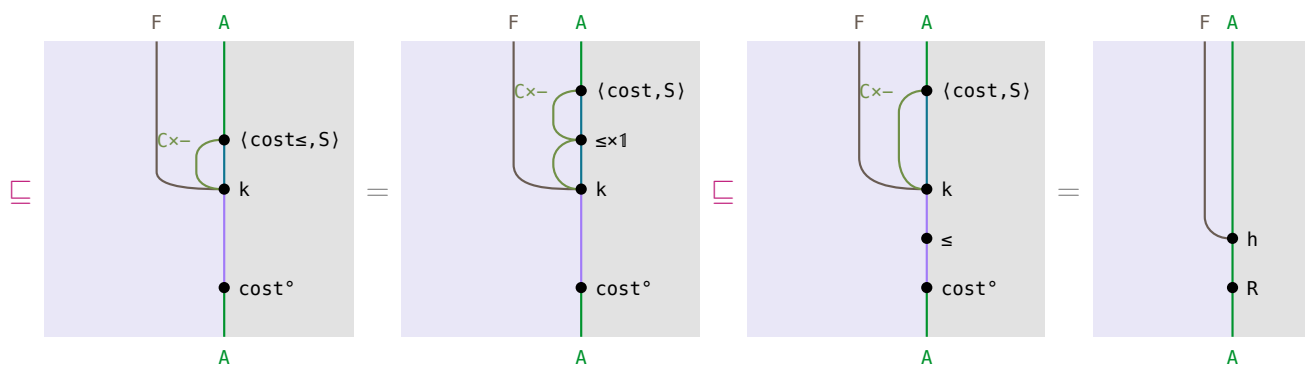
shunting: cost a map, so $1 \leq \text{cost} \text{cost}^\circ$ — (1.3a)

products: $RnSS^\circ = \langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ$ at $R = \text{cost} \leq \text{cost}^\circ$ — $\triangleq \times$

assumption on cost : $h \text{cost} = F(\langle \text{cost}, S \rangle)k$

$$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge R = \text{cost} \leq \text{cost}^\circ \wedge \text{hcost} = F(\langle \text{cost}, S \rangle)k \wedge F(\leq \times 1)k \leq k \implies F(\langle \text{cost} \leq, S \rangle \langle \text{cost}, S \rangle^\circ) F(\langle \text{cost}, S \rangle)k \text{cost}^\circ \leq hR$$

the improvement on the bundles, assembled by k , is below assembling by h and improving by R



$$\begin{aligned} \sqsubseteq & FA \text{---} F(\langle \text{cost} \leq, S \rangle) \text{---} F(C \times B) \text{---} k \text{---} C \text{---} \text{cost} \text{---} A \\ = & FA \text{---} F(\langle \text{cost}, S \rangle) \text{---} F(\leq \times 1) \text{---} F(C \times B) \text{---} k \text{---} C \text{---} \text{cost} \text{---} A \\ \sqsubseteq & FA \text{---} F(\langle \text{cost}, S \rangle) \text{---} F(C \times B) \text{---} k \text{---} \leq \text{---} C \text{---} \text{cost} \text{---} A \\ = & FA \text{---} h \text{---} R \text{---} A \end{aligned}$$

S simple, so $\langle \text{cost}, S \rangle$ simple: $\langle \text{cost}, S \rangle^\circ \langle \text{cost}, S \rangle \leq 1$

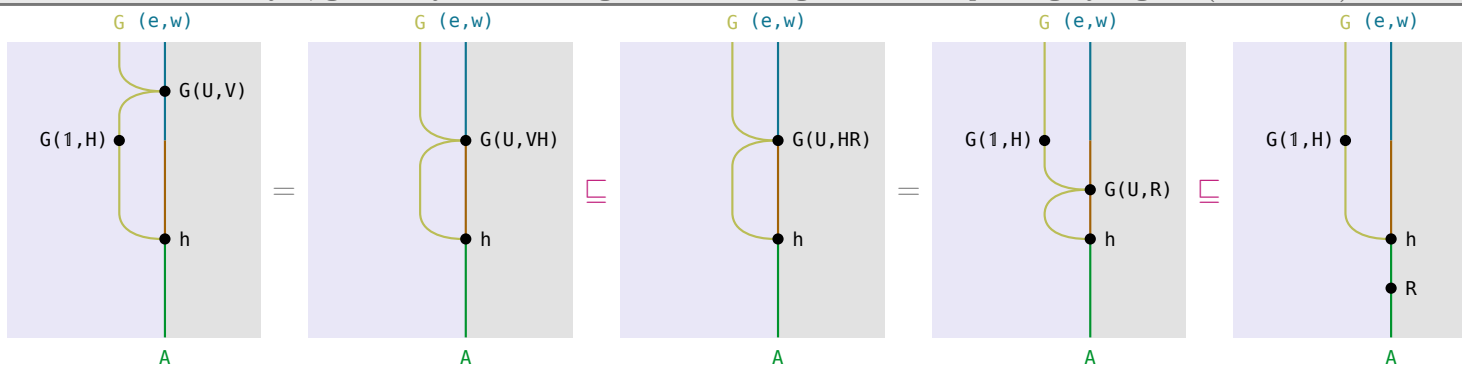
products; functors: $\langle \text{cost} \leq, S \rangle = \langle \text{cost}, S \rangle (\leq \times 1)$ — (11.1.2b), (11b)

assumption on k : $F(\leq \times 1)k \leq k$

assumption on cost read backwards, then $R = \text{cost} \leq \text{cost}^\circ$

$$G(U, R)h \leq hR \wedge \forall H \in HR \implies G(U, V)G(1, H)h \leq G(1, H)hR \quad (15.2m)$$

thinning the parts by U in the first argument and by V in the second ($G(U, V)$), then solving by H and assembling by h , gives only what solving and assembling and then improving by R gives ($G(1, H)hR$)



$$\begin{aligned} & G e w \text{---} G(U, V) \text{---} G(1, H) \text{---} G e A \text{---} h \text{---} A \\ = & G e w \text{---} G(U, VH) \text{---} G e A \text{---} h \text{---} A \\ \sqsubseteq & G e w \text{---} G(U, HR) \text{---} G e A \text{---} h \text{---} A \\ = & G e w \text{---} G(1, H) \text{---} G(U, R) \text{---} G e A \text{---} h \text{---} A \\ \sqsubseteq & G e w \text{---} G(1, H) \text{---} G e A \text{---} h \text{---} R \text{---} A \end{aligned}$$

taking $Q \triangleq G(U, V)$; bifunctors: $G(U, V)G(1, H) = G(U, VH)$ — (11b)

assumption on V : $\forall H \in HR$

bifunctors: $G(U, HR) = G(1, H)G(U, R)$ — (11b)

assumption on h : $G(U, R)h \leq hR$

§15.3. The string edit problem

(15.3a)

- $(\exists a. o = \text{cpy}(a)) \vee (\exists a. o = \text{del}(a)) \vee (\exists a. o = \text{ins}(a))$
an edit operation copies, deletes or inserts one character
- $\text{FX} = 1 + (0p \times X)$
 $F(0p, X)$ is the empty sequence or one operation paired with an X
- $\text{edit} = ([\text{base}, \text{step}])$
 edit is the fold of $[\text{base}, \text{step}]$ over an edit sequence, returning the two strings it reconstitutes
- $\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$
the fold of $[\text{zero}, \pi_2 \text{ succ}]$ counts the operations of a sequence
- $R = \text{length} \leq \text{length}^\circ$
 R relates es to fs when es is no longer than fs
- $V = \text{suffix}^\circ \times \text{suffix}^\circ$
 V relates two pairs of strings when each string of the first is a suffix of the matching string of the second
- $Q(\text{inl}(u), \text{inl}(u))$
 Q is the identity on the **base** summand
- $Q(\text{inr}(p), \text{inr}(q)) \iff V(\pi_2(p), \pi_2(q))$
on the **step** summand Q compares any two operations ($U \triangleq T$) and the two pairs of strings by V
- $\text{empty}(p, q) \iff p = q \wedge p = ([], [])$
 empty is the coreflexive holding only at the pair of two empty strings
- $q \in \text{unstep}(p) \implies \text{step}(\text{inr}(q)) = p$
every decomposition unstep returns is sent back to its input by step

(15.3b)

x	$\alpha(x)$
$\text{inl}(u)$	$\text{wrap}(u)$
$\text{inr}((o, \text{es}))$	$[o] \# \text{es}$

(15.3c)

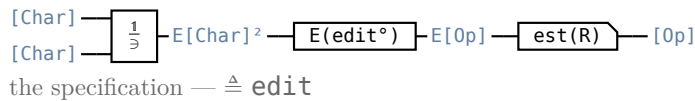
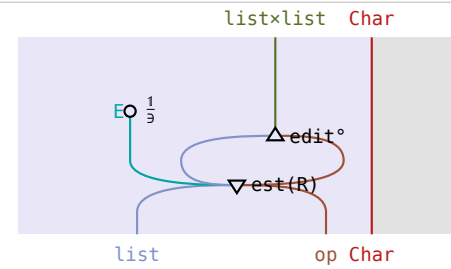
x	$[\text{base}, \text{step}](x)$
$\text{inl}(u)$	$([], [])$
$\text{inr}((\text{cpy}(a), (xs, ys)))$	$([a] \# xs, [a] \# ys)$
$\text{inr}((\text{del}(a), (xs, ys)))$	$([a] \# xs, ys)$
$\text{inr}((\text{ins}(a), (xs, ys)))$	$(xs, [a] \# ys)$

(15.3d)

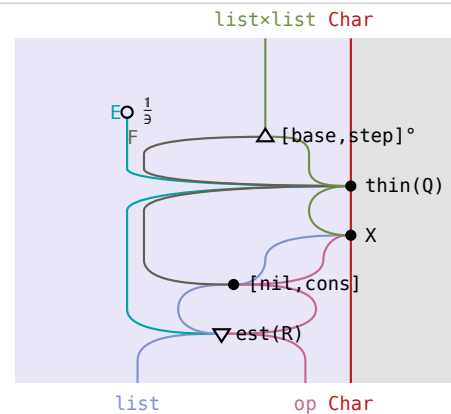
p	$\text{unstep}(p)$
$([], [])$	$[]$
$([a] \# xs, [])$	$[(\text{del}(a), (xs, []))]$
$([], [b] \# ys)$	$[(\text{ins}(b), ([], ys))]$
$([a] \# xs, [b] \# ys)$	if $a = b$ then $[(\text{cpy } a, xs, ys)]$ else $[(\text{del } a, xs, [b] \# ys), (\text{ins } b, [a] \# xs, ys)]$

$$(\mu X : \frac{[base, step]^{\circ}}{3} \text{thin}(Q)P(F(X)[nil, cons])est(R)) \sqsubseteq \frac{edit^{\circ}}{3} est(R), \text{ mle}=(empty \rightarrow nil, unstep \text{list}((1 \times mle)cons) \text{minlist}(R)) \quad (15.3e)$$

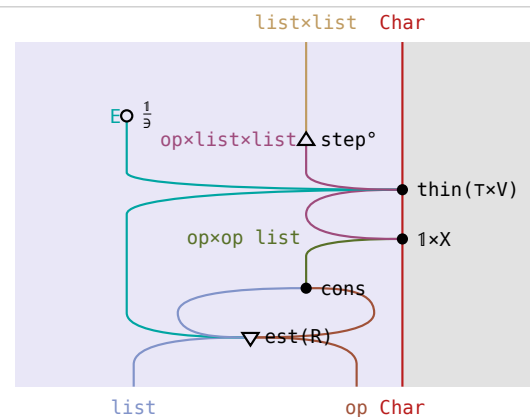
a shortest edit sequence from which both strings can be reconstituted is one pass over the two of them, each step copying, deleting or inserting one character and the best sequence for what is left taken from the entries already computed

circuit

Hinze–Marsden


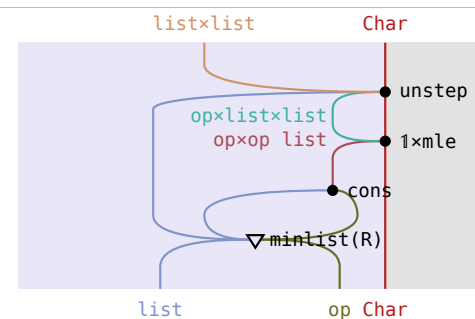
at $Q \triangleq 1 + (U \times V)$. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is Proposition 9.2 at $length \triangleq ([zero, \pi_2 succ])$ with $succ$ monotonic on $\leq$, so $cons$ is monotonic on R . The thinning condition $QF(1, edit^{\circ}) \alpha \sqsubseteq F(1, edit^{\circ}) \alpha R$ over $F(0p, [Char] \times [Char])$ is Proposition 9.4 at $U \triangleq T - F(T, R) \alpha \sqsubseteq \alpha R$, left as an exercise, so any two operations may be compared — and $V \triangleq suffix^{\circ} \times suffix^{\circ}$, — $edit(1 \times suffix) \sqsubseteq R^{\circ} edit, edit(suffix \times 1) \sqsubseteq R^{\circ} edit$. $suffix = tail^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ cut those to one step each: drop the operation that produced the head, or weaken its cpy to a del , never lengthening the sequence



Proposition 9.1: **base** and **step** have disjoint ranges, so the fixed point splits into one branch per summand — **empty**, the coreflexive on (xs, ys) with both lists empty, is where **base** returns

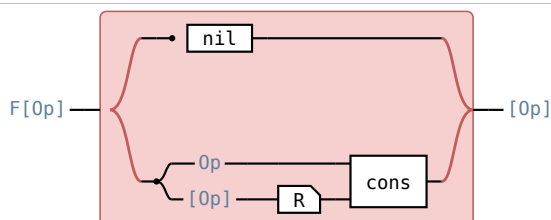
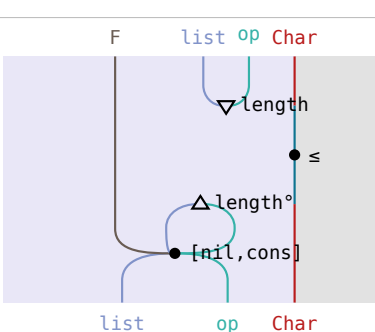
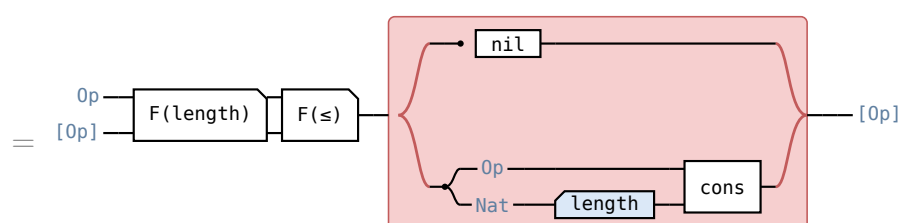
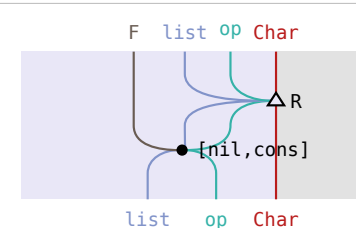


unstep implements $\frac{step^{\circ}}{3} \text{thin}(U \times V)$ — at most two decompositions survive, a copy beating a delete or an insert wherever it is available — and **minlist**(R) implements $est(R)$. The same subproblem is solved many times over, so the running time is exponential in the two lengths



$$F(R)[nil, cons] \sqsubseteq [nil, cons]R \quad (15.3f)$$

comparing the tails by R and then putting an operation in front ($F(R) \alpha$) relates only sequences that R also relates once the operation is in front (αR)

circuit

Hinze–Marsden


$R \triangleq length \leq length^{\circ} \triangleq edit$; F preserves composition

circuit	Hinze–Marsden
length is a map, so entire: $1 \sqsubseteq \text{length } \text{length}^\circ$	
$\alpha \text{ length} = F(\text{length})[\text{zero}, \pi_2 \text{ succ}] : \text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}]) \text{ — edit}$	
length is a map, so simple, and F is monotonic: $F(\text{length}^\circ)F(\text{length}) \sqsubseteq F(1) = 1$	
succ is monotonic on $\leq$: $F(\leq)[\text{zero}, \pi_2 \text{ succ}] \sqsubseteq [\text{zero}, \pi_2 \text{ succ}] \leq$	
$F(\text{length})[\text{zero}, \pi_2 \text{ succ}] = \alpha \text{ length}$ again	
$R \triangleq \text{length} \leq \text{length}^\circ \text{ — edit}$	

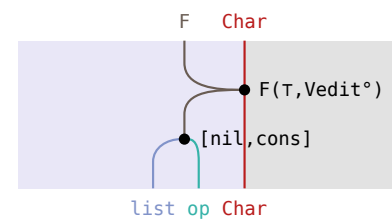
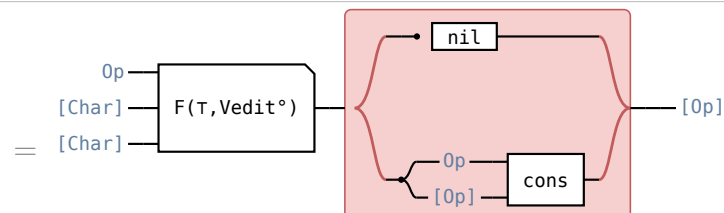
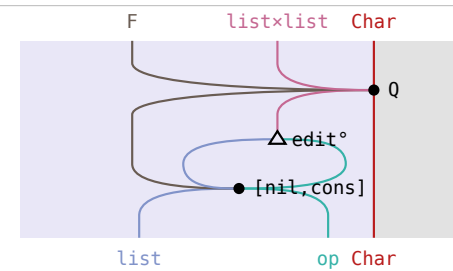
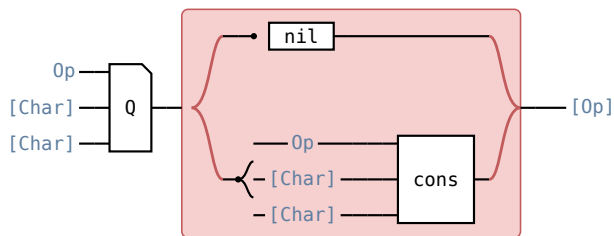
$$QF(\text{edit}^\circ)[\text{nil}, \text{cons}] \sqsubseteq F(\text{edit}^\circ)[\text{nil}, \text{cons}]R$$

(15.3g)

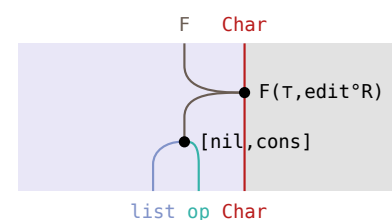
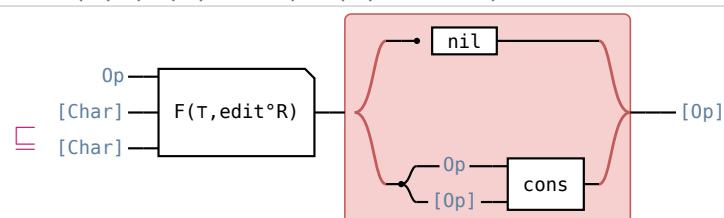
every edit sequence built from a decomposition Q puts above a given one — any operation, each string lengthened at the front — is at least as long as one built from the given one

circuit

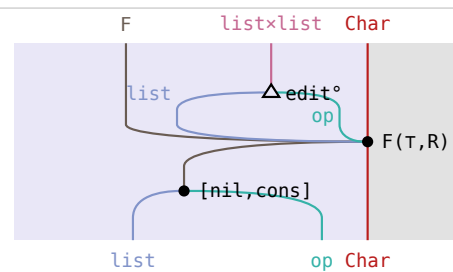
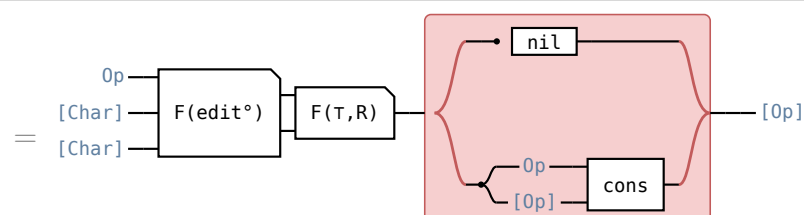
Hinze–Marsden



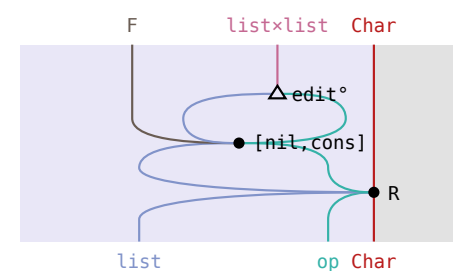
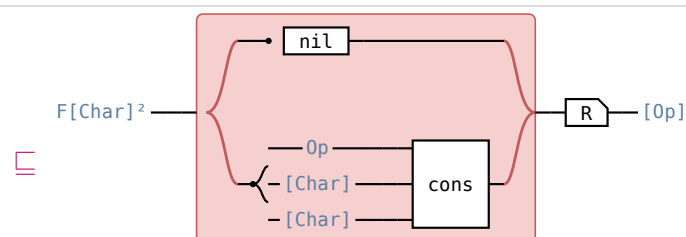
$Q \triangleq 1 + (U \times V)$ is $F(U, V)$ at $U \triangleq \tau$ — $\triangleq \text{edit}$; and the bifunctor F preserves composition: $F(U, V)F(1, \text{edit}^\circ) = F(U, V \text{ edit}^\circ)$



Proposition 9.4's second condition $V \text{ edit}^\circ \sqsubseteq \text{edit}^\circ R$ — (15.3h)

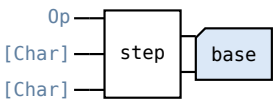
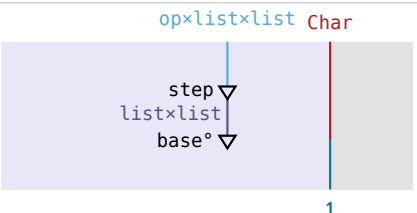
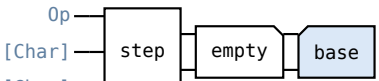
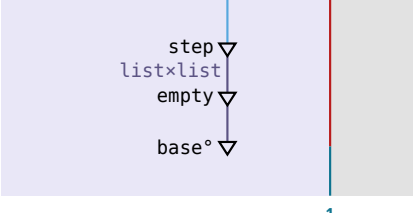
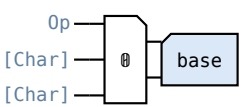
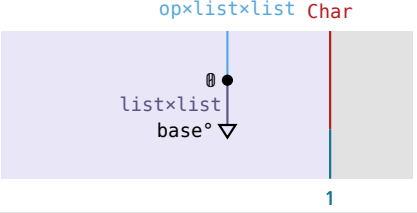
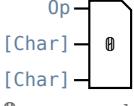
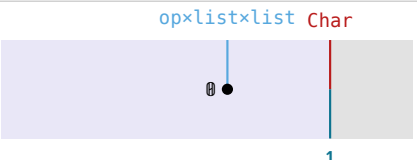


F preserves composition: $F(U, \text{edit}^\circ R) = F(1, \text{edit}^\circ)F(U, R)$



Proposition 9.4's first condition $F(\tau, R) \alpha \sqsubseteq \alpha R$, left as an exercise in the book: CONS adds one to both lengths whatever the two operations are

$\text{edit } V^\circ \sqsubseteq R^\circ \text{ edit}$ (15.3h) cutting a front off either string an edit sequence produces ($\text{edit } V^\circ$) leaves a pair that a sequence no longer produces ($R^\circ \text{ edit}$)	
circuit	Hinze–Marsden
$=$ $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, and $\times$ preserves composition	
 $\sqsubseteq \text{edit } (\text{suffix} \times 1) \sqsubseteq R^\circ \text{ edit}$: drop the operation that produced the head, or weaken its cpy to an ins . Proved by induction at suffix directly, not through $\text{suffix} = \text{tail}^*$ and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$ as the book does	
 $\sqsubseteq \text{edit } (1 \times \text{suffix}) \sqsubseteq R^\circ \text{ edit}$, the mirror image: cpy weakens to a del	
 R° is transitive, $\leq$ being so	

$\text{step } \text{base}^\circ = \emptyset$ (15.3i) no pair of strings is both a result of step and the result $([], [])$ of base	
circuit	Hinze–Marsden
	
 base returns only $([], [])$, so $\text{base}^\circ = \text{empty } \text{base}^\circ$	
 $\text{step } \text{empty} = \emptyset$: cpy and del put a character on the left string, ins one on the right	
 $\emptyset$ composed with anything is $\emptyset$	

(15.3j)

$\text{mle}(xs, ys) = \text{head}(\text{column}(xs, ys))$, $\text{column}(xs, ys) = [\text{mle}(u, ys) \mid u \leftarrow \text{tails}(xs)]$ $\text{column}(xs) = ([\text{fstcol}(xs), \text{nextcol}(xs)])$, $\text{fstcol} = \text{list}(\text{del}) \text{ tails}$ the tabulation: $\text{mle}(xs, ys)$ needs $\text{mle}(u, v)$ for every tail u of xs and v of ys , so the columns are built right to left	
$\text{column}(xs, [b] \# ys) = \text{nextcol}(xs, (b, \text{column}(xs, ys)))$ $\text{nextcol}(xs)(b, us) = ([\text{base}(b, \text{last}(us)), \text{step}(b)])(xus)$, $xus = \text{zip}(xs, \text{zip}(\text{init}(us), \text{tail}(us)))$ each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right	
$\text{base}(b, u) = [[\text{ins}(b)] \# u]$ $\text{step}(b)((a, (u, v)), ws) = (a \rightarrow b \rightarrow [[\text{cpy}(a)] \# v] \# ws, [\text{bmin}(R)([\text{del}(a)] \# w, [\text{ins}(b)] \# u)] \# ws)$ $w = \text{head}(ws)$; these base , step are not edit 's. An entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	

§15.4. Optimal bracketing

(15.4a)

tree $A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, $FX = A + X^2$, so $F(R) = 1 + R^2$; $h \triangleq [\text{tip}, \text{bin}]$, $\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A$, $H = \text{flatten}^\circ$.

$\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])$, $\text{opt} \triangleq (\text{zero}, \text{st})$, $\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))$.

sb associative, so $\text{size} = \text{flatten } \text{sz}$ for a map sz $R \triangleq \text{cost} \leq \text{cost}^\circ$, $g \triangleq [\text{zero}, (1 \times \text{sz})^2 \text{ opb } \pi_1]$, single the coreflexive on singleton lists.

$\text{splits} \triangleq (\text{inits}^+, \text{tails}^+)$ zip , an implementation of $\frac{\text{cat}^\circ}{\text{sz}}$; $\text{array} \triangleq \text{inits } \text{list}(\text{row})$, $\text{row} \triangleq \text{tails } \text{list}(\text{mct})$, $\text{col} \triangleq \text{inits } \text{list}(\text{mct})$.

$\text{mix} \triangleq \text{zip } \text{list}(\text{bin}) \text{ minlist}(R)$, $\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}$, $\text{process} \triangleq ((\text{tip } \text{wrap}) \times 1) \text{ loop}(\text{next})$.

$\text{Assoc}(\text{sb}) \Rightarrow F(\langle \text{cost}, \text{flatten} \rangle)g = [\text{tip}, \text{bin}] \text{cost}$ (15.4b)	
pairing each subtree with its cost and its flattening and then applying g gives what building the tree by $[\text{tip}, \text{bin}]$ and taking its cost gives	
circuit	Hinze–Marsden
$F(\text{tree}(A)) \rightarrow F(\langle \text{cost}, \text{flatten} \rangle) \rightarrow F(\text{Int} \times \text{list}^+(A)) \rightarrow g \rightarrow \text{Int}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{cost} \rightarrow \text{flatten} \rightarrow \text{list}^+(A) \rightarrow \text{sz} \rightarrow \text{opb} \rightarrow \text{Int}$ definition of g; coproducts and products $\triangleq \text{flatten}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{cost} \rightarrow \text{size} \rightarrow \text{opb} \rightarrow \text{Int}$ $\text{flatten sz} = \text{size}$, since sb is associative $\triangleq \text{flatten}$	
$A + \text{tree}(A)^2 \rightarrow \text{zero} \rightarrow \text{bin} \rightarrow \text{cost} \rightarrow \text{size} \rightarrow \text{opb} \rightarrow \text{Int}$ $\langle \text{cost}, \text{size} \rangle \triangleq ([\text{opt}, \text{opb}])$, at a node $\text{bin}(\text{cost}, \text{size}) = (\text{cost}, \text{size})^2 \text{ opb}$	
$A + \text{tree}(A)^2 \rightarrow \text{tip} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ $\text{tip cost} = \text{zero}$; products, $\langle \text{cost}, \text{size} \rangle \pi_1 = \text{cost}$	
$A + \text{tree}(A)^2 \rightarrow \text{tip} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{bin} \rightarrow \text{tree}(A) \rightarrow \text{cost} \rightarrow \text{Int}$ coproducts, $[\text{tip cost}, \text{bin cost}] = [\text{tip}, \text{bin}] \text{cost}$	

$F(\leq \times 1)g \sqsubseteq g \leq$ (15.4c)	
raising the cost of either subtree ($\leq \times 1$ in both slots) and then applying g returns a value at least what g returns on the costs before raising ($g \leq$)	
circuit	Hinze–Marsden
$F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{F(\leq \times 1)} \rightarrow \boxed{g} \rightarrow \text{Int}$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $=$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $\sqsubseteq$	
$A + \text{Int} \times \text{list}^+(A) \times \text{Int} \times \text{list}^+(A) \rightarrow \text{Int}$ $\sqsubseteq$	
$\leq$ reflexive, so $\text{zero} \sqsubseteq \text{zero} \leq$; coproducts	
$F(\text{Int} \times \text{list}^+(A)) \rightarrow \boxed{g} \rightarrow \boxed{\leq} \rightarrow \text{Int}$ $=$ definition of $g \triangleq \text{flatten}$	

$$\text{Assoc}(\text{sb}) \Rightarrow (\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{3} P([\text{tip}, (X \times X) \text{bin}]) \text{est}(R)) \sqsubseteq \frac{\text{flatten}^\circ}{3} \text{est}(R), \text{ mct} = (\text{single} \rightarrow \text{head tip}, \text{ (init col, tail row) mix}) \quad (15.4d)$$

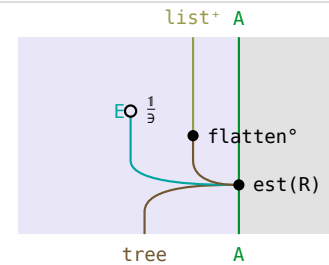
a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ is read off an array holding one best tree per non-empty segment, each entry built from the column to its left and the row below it

circuit

$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E(\text{flatten}^\circ)} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

the specification $\text{---} \triangleq \text{flatten}$

Hinze–Marsden



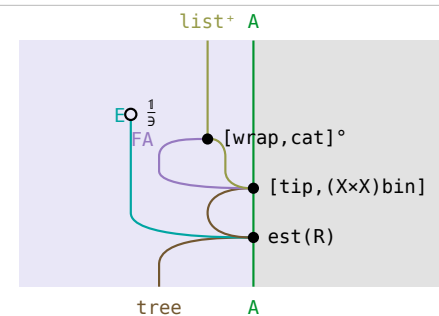
$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E([\text{wrap}, \text{cat}]^\circ)} E(F(\text{list}^+(A))) \xrightarrow{P([\text{tip}, (X \times X) \text{bin}])} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

split the list in every way, bracket both halves, join. The condition is monotonicity **in context**,

$\sqsubseteq F(Rn(\text{flatten} \text{ flatten}^\circ)) h \sqsubseteq hR$ — only trees with the same flattening are compared —

which is Proposition 9.3 at $H^\circ = \text{flatten}$, a map, with (9.5) $[\text{tip}, \text{bin}]$

$\text{cost} = (1 + (\text{cost}, \text{flatten})^2)g$ (the cost of a node reads only the cost and the flattening of its two subtrees) and (9.6) $(1 + (\leq 1)^2)g \sqsubseteq g$ (g monotonic on $\leq$ in its two cost arguments)

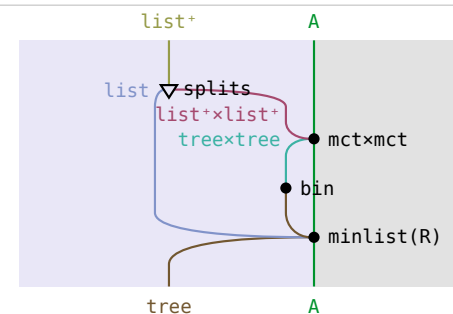


$$\text{list}^+(A) \xrightarrow{\frac{1}{3}} E(\text{list}^+(A)) \xrightarrow{E(\text{cat}^\circ)} E(\text{list}^+(A)^2) \xrightarrow{P((X \times X) \text{bin})} E(\text{tree}(A)) \xrightarrow{\text{est}(R)} \text{tree}(A)$$

$\equiv$ Proposition 9.1: **wrap** and **cat** have disjoint ranges, and **single** is the coreflexive on singleton lists, where **wrap** returns

$$\text{list}^+(A) \xrightarrow{\text{splits}} [\text{list}^+(A)^2] \xrightarrow{\text{list}((\text{mct} \times \text{mct}) \text{bin})} [\text{tree}(A)] \xrightarrow{\text{minlist}(R)} \text{tree}(A)$$

$\sqsubseteq$ $\text{splits} \triangleq \langle \text{inits}^+, \text{tails}^+ \rangle$ zip implements $\frac{\text{cat}^\circ}{3}$ and $\text{minlist}(R)$ implements $\text{est}(R)$. Exponential, since the segments of one list overlap



$$\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{mix}) \quad (9.7)$$

$\equiv$ the tabulation: **mct** $\times S$ is needed for every non-empty segment $\times S$, so the values are held as an array of rows, $\text{array} \triangleq \text{inits list}(\text{row}), \text{row} \triangleq \text{tails list}(\text{mct}), \text{col} \triangleq \text{inits list}(\text{mct}), \text{mix} \triangleq \text{zip list}(\text{bin}) \text{minlist}(R)$

$$\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{next}) \quad (9.8)$$

$$\text{cons col} = (1 \times \text{array}) \text{process} \quad (9.9)$$

$$\equiv \text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{cons}) \quad (9.10)$$

a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop, $\text{next} \triangleq (\pi_1, \text{mix}) \text{snoc}, \text{process} \triangleq ((\text{tip wrap}) \times 1) \text{loop}(\text{next})$

$$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$$

$\equiv \text{addcol} \triangleq (\pi_1 \text{tip wrap}, \text{step}) \text{cons}, \text{step} \triangleq (\text{process tail}, \pi_2) \text{zip list}(\text{cons})$
the program: one fold building the array column by column, cubic in the length of the input

cons col=(1×array)process (15.4h) the column of a consed onto x is what process makes of a and the array of x	
circuit	Hinze–Marsden
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{cons} \\ \hline \end{array} \text{list}^+(A) \begin{array}{ c} \hline \text{col} \\ \hline \end{array} [\text{tree}(A)]$	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{tip} \\ \hline \end{array} \text{tree}(A) \begin{array}{ c} \hline \text{wrap} \\ \hline \end{array} \begin{array}{ c} \hline \text{loop(next)} \\ \hline \end{array} [\text{tree}(A)]$ Exercise 9.13 with $k \triangleq \text{cons}$, $g \triangleq \text{tip}$, $h \triangleq \text{wrap}$, $f \triangleq \text{next}$: its two equations are (9.8) — (15.4f) — at $a:[b]$ and at $a:(u++[b])$, with $[a]$ $\text{col}=[\text{tip}(a)]$	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{array} \\ \hline \end{array} \begin{array}{ c} \hline \text{tip} \\ \hline \end{array} \text{tree}(A) \begin{array}{ c} \hline \text{wrap} \\ \hline \end{array} \begin{array}{ c} \hline \text{loop(next)} \\ \hline \end{array} [\text{tree}(A)]$ products	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{array} \\ \hline \end{array} \begin{array}{ c} \hline \text{process} \\ \hline \end{array} [\text{tree}(A)]$ definition of process	

tops=(1×array)process tail (15.4i) the trees at the top of the new rows are what process makes of a and the array of x , less its first	
circuit	Hinze–Marsden
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{tops} \\ \hline \end{array} [\text{tree}(A)]$	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{cons} \\ \hline \end{array} \text{list}^+(A) \begin{array}{ c} \hline \text{inits} \\ \hline \end{array} \begin{array}{ c} \hline \text{tail} \\ \hline \end{array} [\text{list}^+(A)] \begin{array}{ c} \hline \text{list(mct)} \\ \hline \end{array} [\text{tree}(A)]$ definition of $\text{tops} \triangleq \text{tic list(mct)}$, our name	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{cons} \\ \hline \end{array} \text{list}^+(A) \begin{array}{ c} \hline \text{col} \\ \hline \end{array} \begin{array}{ c} \hline \text{tail} \\ \hline \end{array} [\text{tree}(A)]$ $\text{tail list}(f)=\text{list}(f) \text{ tail}$; definition of col	
$\begin{array}{c} A \\ \text{list}^+(A) \end{array} \begin{array}{ c} \hline \text{array} \\ \hline \end{array} \begin{array}{ c} \hline \text{process} \\ \hline \end{array} \begin{array}{ c} \hline \text{tail} \\ \hline \end{array} [\text{tree}(A)]$ (9.9) — (15.4h)	

rests=π_2 array (15.4j) the rest of each new row is the array of x	
circuit	Hinze–Marsden
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} \text{tail} \xrightarrow{[\text{list}^+(A)]} \text{list}(\text{tail row}) \xrightarrow{[[\text{tree}(A)]]}$ definition of $\text{rests} \triangleq \text{tic list}(\text{tail row})$, our name	
$= \text{list}^+(A) \xrightarrow{\text{list}^+(A)} \text{array} \xrightarrow{[[\text{tree}(A)]]}$ $\text{tic list}(\text{tail}) = \pi_2 \text{ inits}$; definition of array	

newrows=($1 \times \text{array}$)step (15.4k) the rows of the array of a:x below its first are what step makes of a and the array of x	
circuit	Hinze–Marsden
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} \text{tail} \xrightarrow{[\text{list}^+(A)]} \text{list}(\text{row}) \xrightarrow{[[\text{tree}(A)]]}$ definition of $\text{newrows} \triangleq \text{tic list}(\text{row})$, our name	
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} \text{tail} \xrightarrow{[\text{list}^+(A)]} \text{list}(\text{mct, tail row cons}) \xrightarrow{[[\text{tree}(A)]]}$ (9.10) — (15.4g) — on the non-singleton tic lists	
$= \text{list}^+(A) \xrightarrow{\text{tops}} \text{rests} \xrightarrow{[\text{tree}(A)]} \text{zip} \xrightarrow{[\text{tree}(A) \times [\text{tree}(A)]]} \text{list}(\text{cons}) \xrightarrow{[[\text{tree}(A)]]}$ $\text{list}(f, g) = (\text{list}(f), \text{list}(g))$ zip; products; definitions of tops and rests	
$= \text{list}^+(A) \xrightarrow{\text{array}} \text{step} \xrightarrow{[[\text{tree}(A)]]}$ (15.4i) and (15.4j); products; definition of step	

cons array=(1×array)addcol (15.4l) the array of a consed onto x is what addcol makes of a and the array of x	
circuit	Hinze–Marsden
$\text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{array}} [[\text{tree}(A)]]$	$\text{list} \text{ list } \text{tree } A$
$= \text{list}^+(A) \xrightarrow{\text{cons}} \text{list}^+(A) \xrightarrow{\text{inits}} [[\text{list}^+(A)]] \xrightarrow{\text{list}(\text{row})} [[\text{tree}(A)]]$ definition of array	$\text{list} \text{ list } \text{tree } A$
$= \text{cons } \text{inits} = (\pi_1 \text{ wrap, tic}) \text{ cons, abbreviating cons inits tail by tic}$	$\text{list} \text{ list } \text{tree } A$
$= \text{list}^+(A) \xrightarrow{\text{newrows}} \text{list}^+(A) \xrightarrow{\text{wrap}} \text{list}^+(A) \xrightarrow{\text{row}} [[\text{tree}(A)]] \xrightarrow{\text{cons}} [[\text{tree}(A)]]$ $\text{cons } \text{list}(f) = (f \times \text{list}(f)) \text{ cons; definition of newrows}$	$\text{list} \text{ list } \text{tree } A$
$= \text{list}^+(A) \xrightarrow{\text{newrows}} \text{list}^+(A) \xrightarrow{\text{tip}} \text{tree}(A) \xrightarrow{\text{wrap}} [[\text{tree}(A)]] \xrightarrow{\text{cons}} [[\text{tree}(A)]]$ $\text{wrap row} = \text{tip wrap}$	$\text{list} \text{ list } \text{tree } A$
$= \text{list}^+(A) \xrightarrow{\text{array}} \text{array} \xrightarrow{\text{addcol}} [[\text{tree}(A)]]$ (15.4k); products; definition of addcol	$\text{list} \text{ list } \text{tree } A$

§15.5. Data compression

(15.5a)

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym } \text{Char} \mid \text{ptr } (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$

$\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{ extend } (xs, \text{sym } a) = xs \# [a], \text{ extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$

$\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}]) \text{ with } c, p \text{ the constant costs of a symbol and a pointer; } R \triangleq \text{size} \leq \text{size}^\circ, Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau)), \text{ the two } \tau \text{ on symbols and on pointers.}$

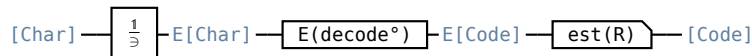
$\text{lrt } ws = \text{est}(\text{prefix}^\circ \times (\tau + \tau)) \{(xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws\}, \text{ the longest repeated tail; reduce } (ws \# [a]) = (zs \# [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)]) \text{ with } (xs, (ys, zs)) = \text{lrt } (ws \# [a]).$

$$\theta \leq c \wedge \theta \leq p \Rightarrow (\mu X : \frac{[nil, extend]^\circ}{\exists} \text{thin}(Q)P([nil, (X \times 1)snoc])est(R)) \sqsubseteq \frac{decode^\circ}{\exists} est(R), \quad (15.5b)$$

$$\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1)snoc) \text{minlist}(R))$$

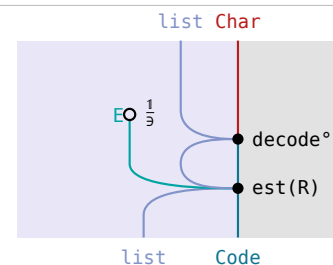
a smallest code sequence decoding to the given string is built from the right, each step emitting the last character as a symbol or ending with a pointer back into what has already been decoded

circuit



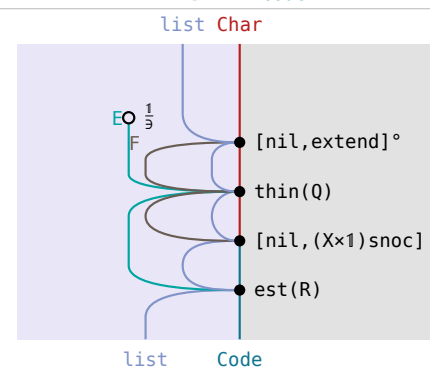
the specification $\dashv\vdash decode$

Hinze–Marsden

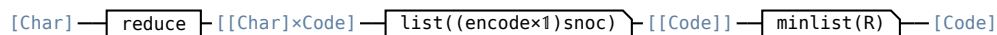
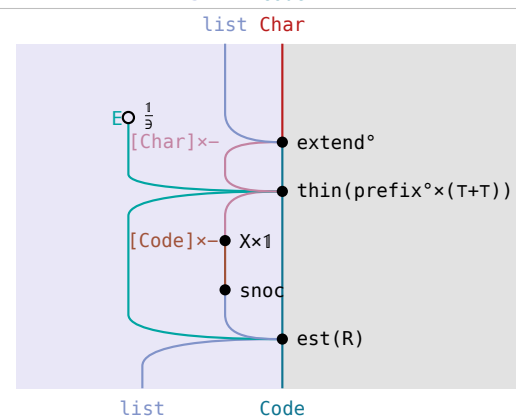


at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers. Monotonicity $F(R) \alpha \sqsubseteq \alpha R$ is routine, the two costs being constants: $(\tau + \tau)[c, p] = [c, p]$.

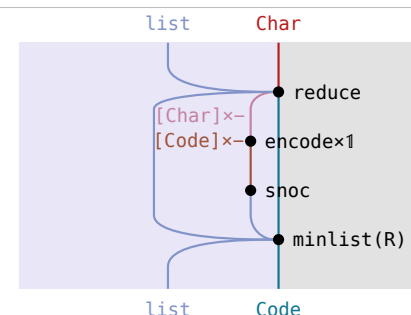
Proposition 9.4 splits the thinning condition in two — $F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$, and $\text{decode prefix} \sqsubseteq R^\circ \text{decode}$, for which $\text{decode init} \sqsubseteq R^\circ \text{decode}$ is enough: dropping the last character of the output shortens or removes the last code element and never raises the cost. So between a symbol and a pointer nothing can be decided in advance, and between two pointers the longer match wins



Proposition 9.1: nil and extend have disjoint ranges. The decompositions of one string are $\frac{\text{extend}^\circ}{\exists} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr}(ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$ — take the last character as a symbol, or end with a pointer



reduce implements $\frac{\text{extend}^\circ}{\exists} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$: thinning leaves at most two, the symbol and the pointer of the longest repeated tail lrt . Again exponential; the book gives no tabulation for it

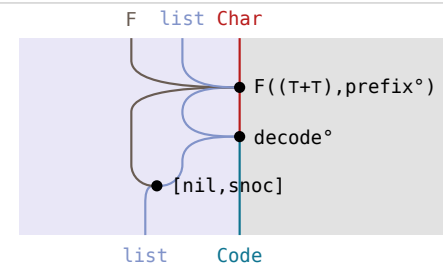
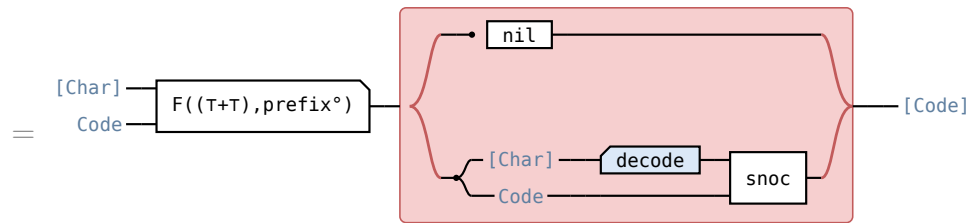
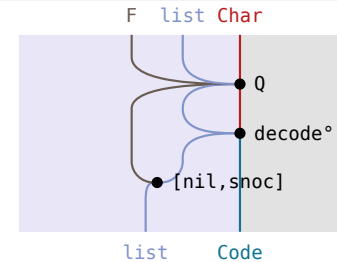
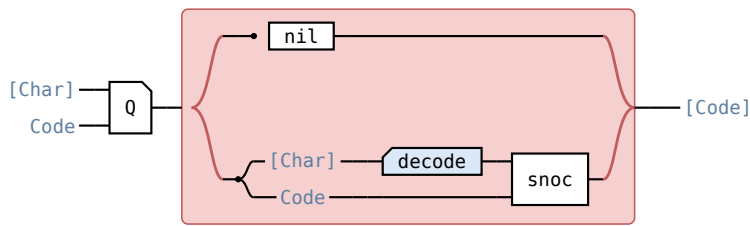


$$\theta \leq c \wedge \theta \leq p \Rightarrow QF(\text{decode}^\circ)[\text{nil}, \text{snoc}] \sqsubseteq F(\text{decode}^\circ)[\text{nil}, \text{snoc}]R \quad (15.5c)$$

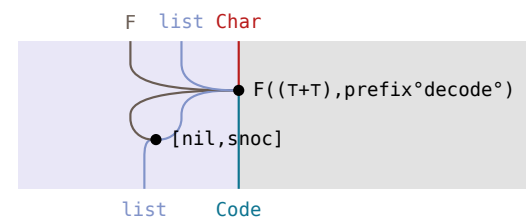
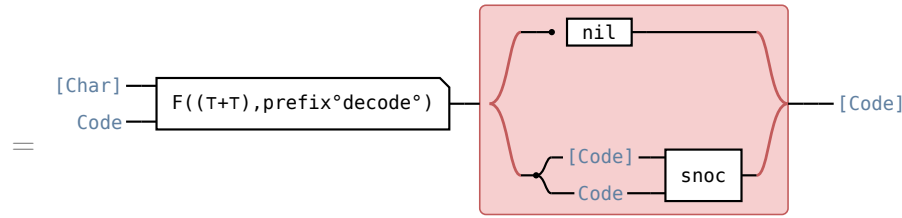
every code sequence built from a decomposition Q puts above a given one — a code element of the same kind, a longer front string — costs at least as much as one built from the given one

circuit

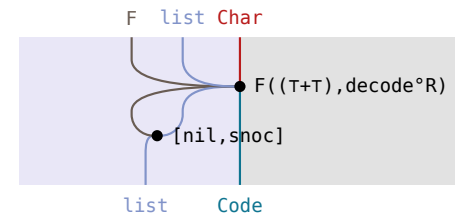
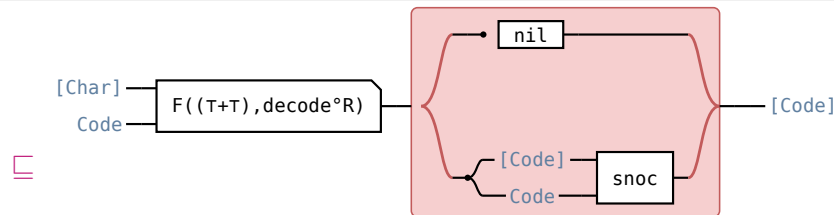
Hinze–Marsden



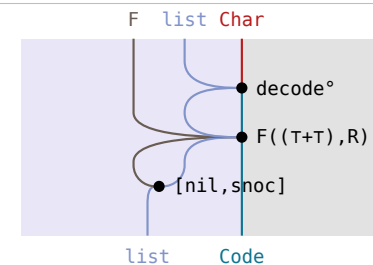
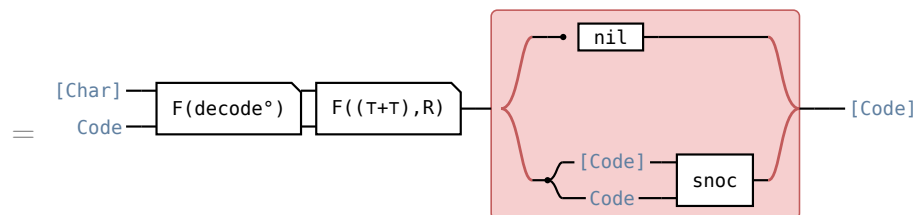
$Q \triangleq 1 + (\text{prefix}^\circ \times (\tau + \tau))$ is $F(\tau + \tau, \text{prefix}^\circ) \triangleq \text{decode}$



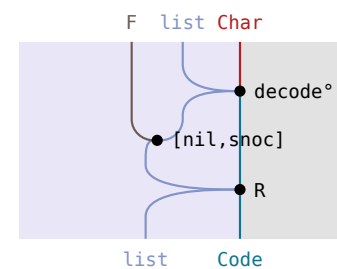
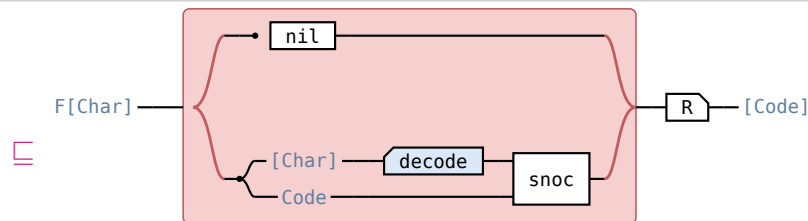
the bifunctor F preserves composition: $F(U, \text{prefix}^\circ)F(1, \text{decode}^\circ) = F(U, \text{prefix}^\circ \text{decode}^\circ)$



Proposition 9.4's second condition $\text{prefix}^\circ \text{decode}^\circ \sqsubseteq \text{decode}^\circ R$: dropping the last **sym**, or shortens or drops the last pointer. Proved at **prefix** directly by induction, not through **init**



F preserves composition: $F(U, \text{decode}^\circ R) = F(1, \text{decode}^\circ)F(U, R)$



Proposition 9.4's first condition $F(\tau + \tau, R) \alpha \sqsubseteq R$, left as an exercise in the book: **snoc** adds the cost of the last element to both sides, and $[c, p](\tau + \tau) = [c, p]$

extend nil°=⊘ (15.5d) no string is both a result of extend and the result [] of nil	
circuit	Hinze–Marsden
[Char] Code extend [Char] nil	[Char]×- Code list nil°▽ extend 1
[Char] Code extend null [Char] nil nil returns only [], so nil°=null nil°	[Char]×- Code list nil°▽ extend null 1
[Char] Code ⊘ [Char] nil extend null=⊘: a symbol ends the string with a character, a pointer with its non-empty ZS	[Char]×- Code ⊘ list nil°▽ 1
[Char] Code ⊘ ⊘ composed with anything is ⊘	[Char]×- Code ⊘ 1

§16. Greedy Algorithms

§16.1. Theory

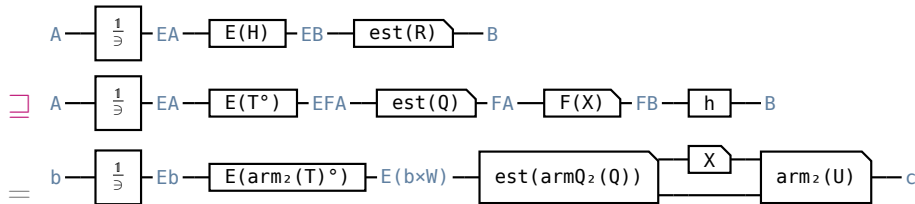
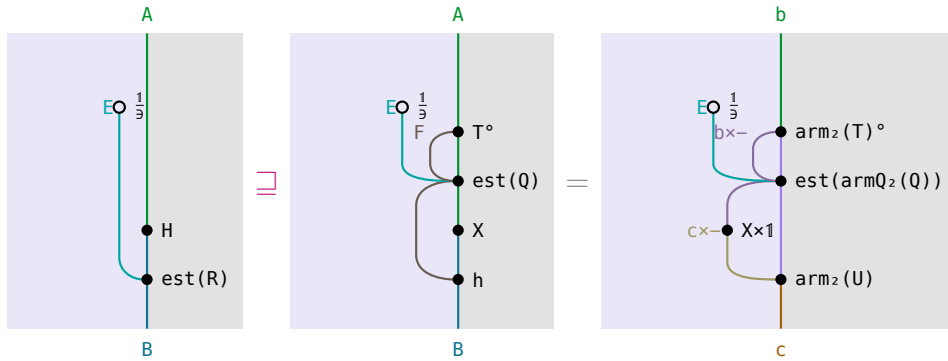
(16.1a)

h, T, R, H, M as in $\triangleq H$; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\frac{1}{3}}$ returns, so that $\frac{T^\circ}{\frac{1}{3}} \text{est}(Q)$ is entire.

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R) \wedge \text{RR} \in R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\frac{1}{3}} \quad (16.1b)$$

$$\text{est}(Q)F(X)h \sqsubseteq_{\frac{H}{\frac{1}{3}}} \text{est}(R)$$

the same optimum reached by keeping **ONE** decomposition at each step, so that no set is ever carried and the recursion runs on values alone

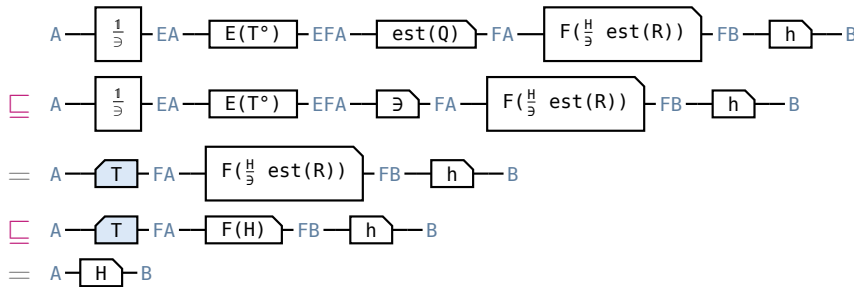
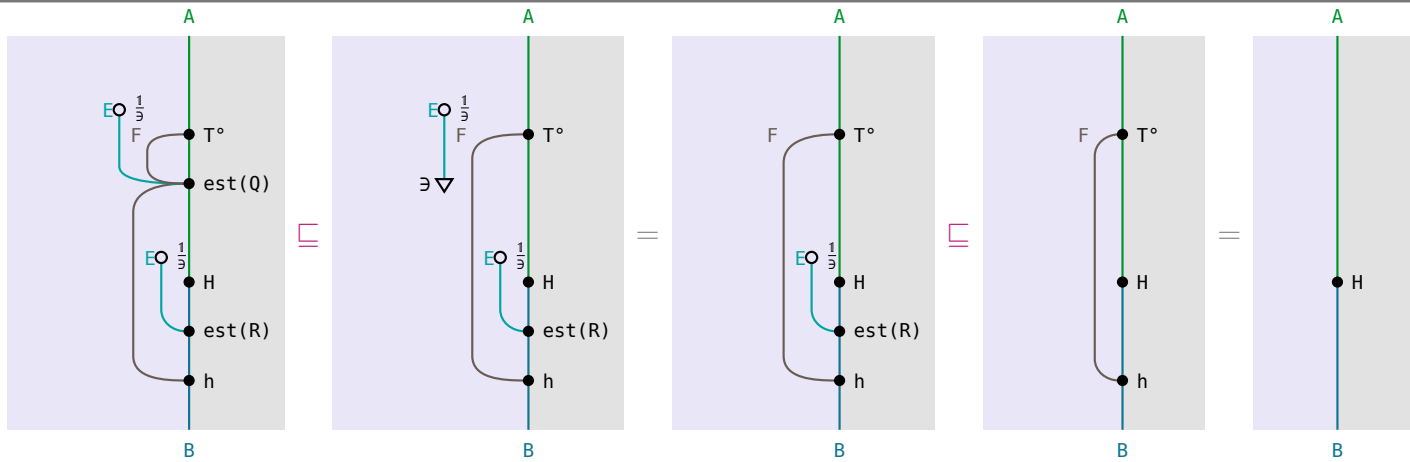


the problem to be solved, $H \triangleq (T)^\circ(h) \triangleq H, Q$

Proposition 10.1 at $T=[V_1, V_2]$, $h=[U_1, U_2]$, $Q=Q_1+Q_2$, $V_2 V_1^\circ = \emptyset$

$$T^\circ F(H)h=H \Rightarrow \frac{T^\circ}{\frac{1}{3}} \text{est}(Q)F(\frac{H}{\frac{1}{3}} \text{est}(R))h \sqsubseteq H \quad (16.1c)$$

taking the input apart in one Q -extreme way, solving the parts by M and assembling by h returns only what H returns



$\text{est}(Q) \sqsubseteq \exists \triangleq \text{est}$

$$\frac{T^\circ}{\frac{1}{3}} \exists = T^\circ \quad (10b)$$

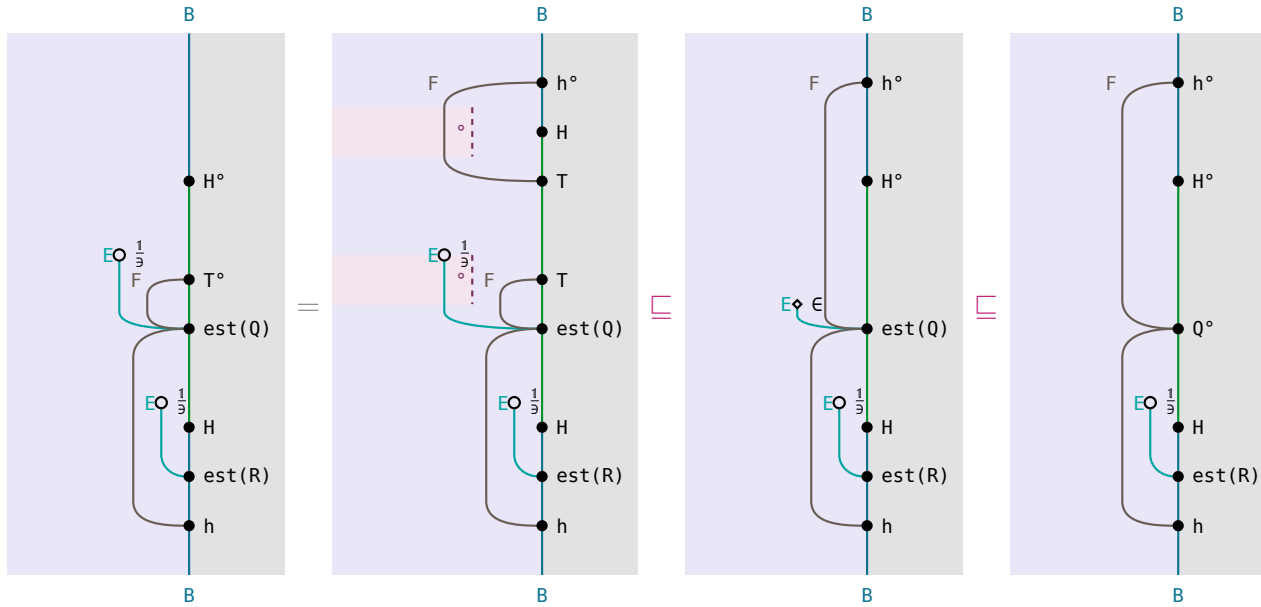
$$M \sqsubseteq \frac{H}{\frac{1}{3}} \exists = H \quad (13.1.1a)$$

$T^\circ F(H)h=H$: $H \triangleq (T)^\circ(h)$ and (11.6.4a)

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H)h=H \wedge QF(H)h \subseteq F(H)hR \Rightarrow H^\circ \frac{T^\circ}{\exists} \quad (16.1d)$$

$$\text{est}(Q)F(\frac{H}{\exists} \text{est}(R))h \subseteq R^\circ$$

H° followed by the greedy body at M is $\subseteq R^\circ$: an answer of the body is never worse than an answer of H to the same input

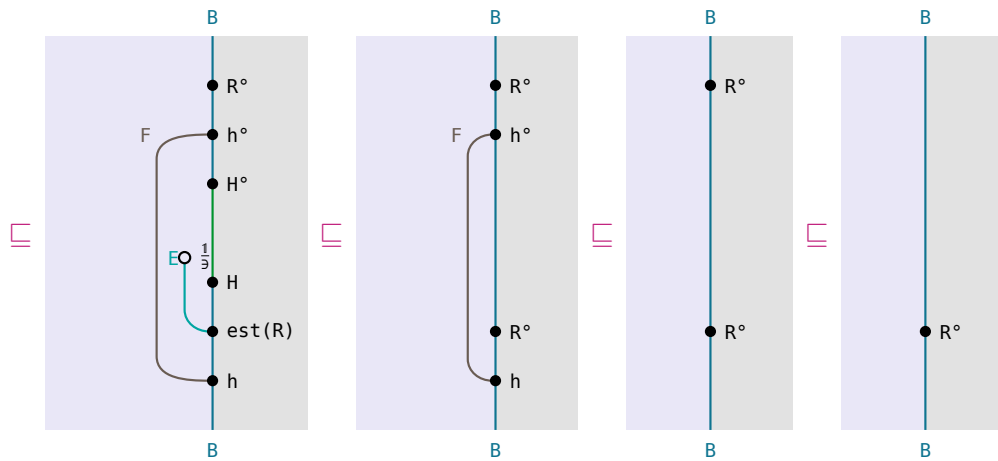


$$\begin{aligned} & B \rightarrow [H] \rightarrow A \rightarrow \left[\frac{1}{\exists} \right] \rightarrow EA \rightarrow [E(T^\circ)] \rightarrow EFA \rightarrow [\text{est}(Q)] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \\ = & B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [T] \rightarrow A \rightarrow \left[\frac{1}{\exists} \right] \rightarrow EA \rightarrow [E(T^\circ)] \rightarrow EFA \rightarrow [\text{est}(Q)] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \end{aligned}$$

$H^\circ = h^\circ F(H^\circ)T$, the converse of $T^\circ F(H)h=H$ — (11.6.4a)

$$\subseteq B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [\epsilon] \rightarrow EFA \rightarrow [\text{est}(Q)] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \quad T \frac{T^\circ}{\exists} \subseteq \epsilon, \text{ not a tabulated row: } \frac{T^\circ}{\exists} \ni T^\circ \text{ conversed}$$

$$\subseteq B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow [Q] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \quad \epsilon \text{est}(Q) \subseteq Q^\circ \text{ — (13.1.1a)}$$



$$\subseteq B \rightarrow [R] \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B$$

$QF(H)h \subseteq F(H)hR$ conversed — the hypothesis on Q

$$\subseteq B \rightarrow [R] \rightarrow [h] \rightarrow [F(R^\circ)] \rightarrow FB \rightarrow [h] \rightarrow B$$

$H^\circ M \subseteq R^\circ$ under F — (13.1.1a)

$$\subseteq B \rightarrow [R] \rightarrow [R] \rightarrow B$$

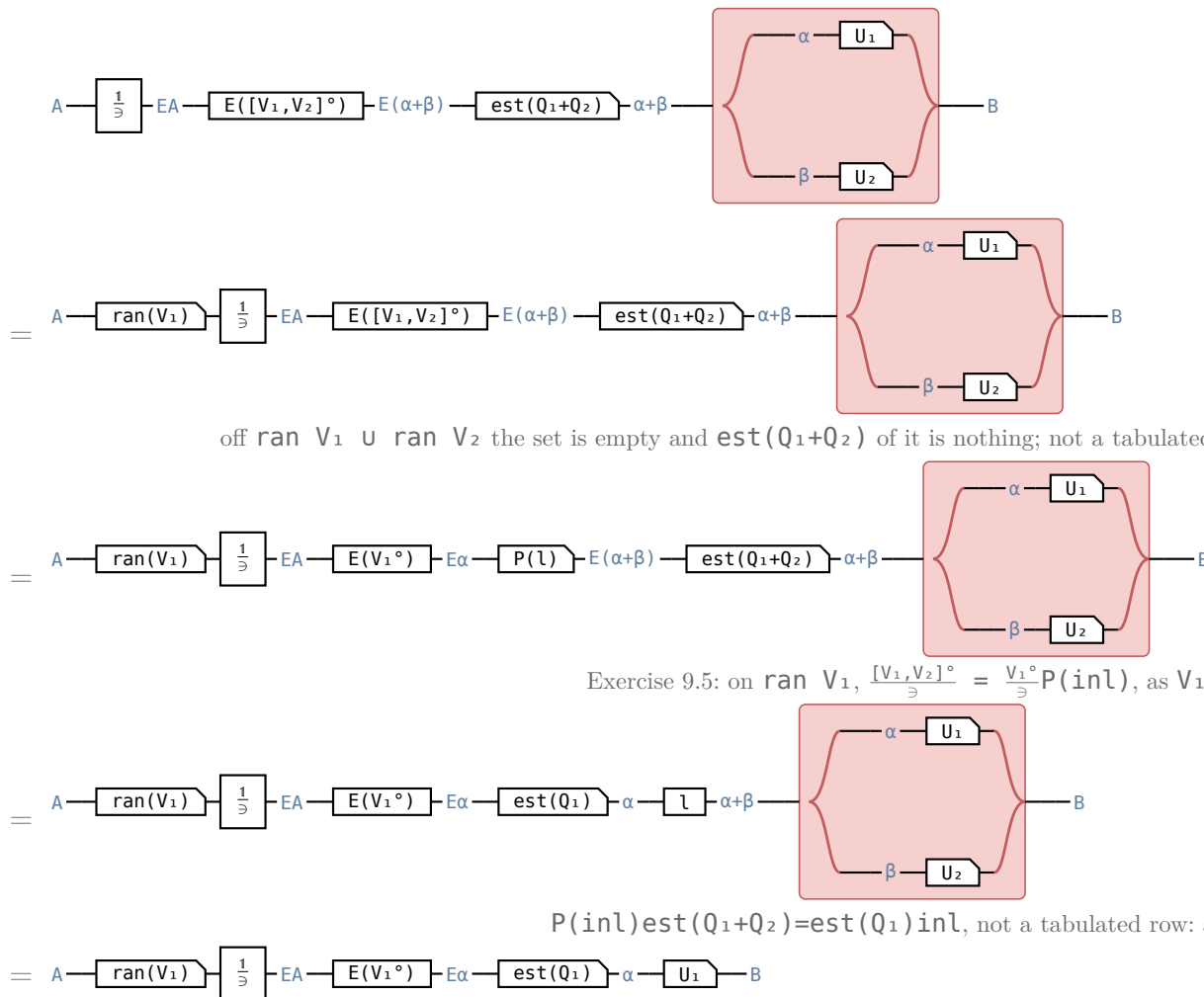
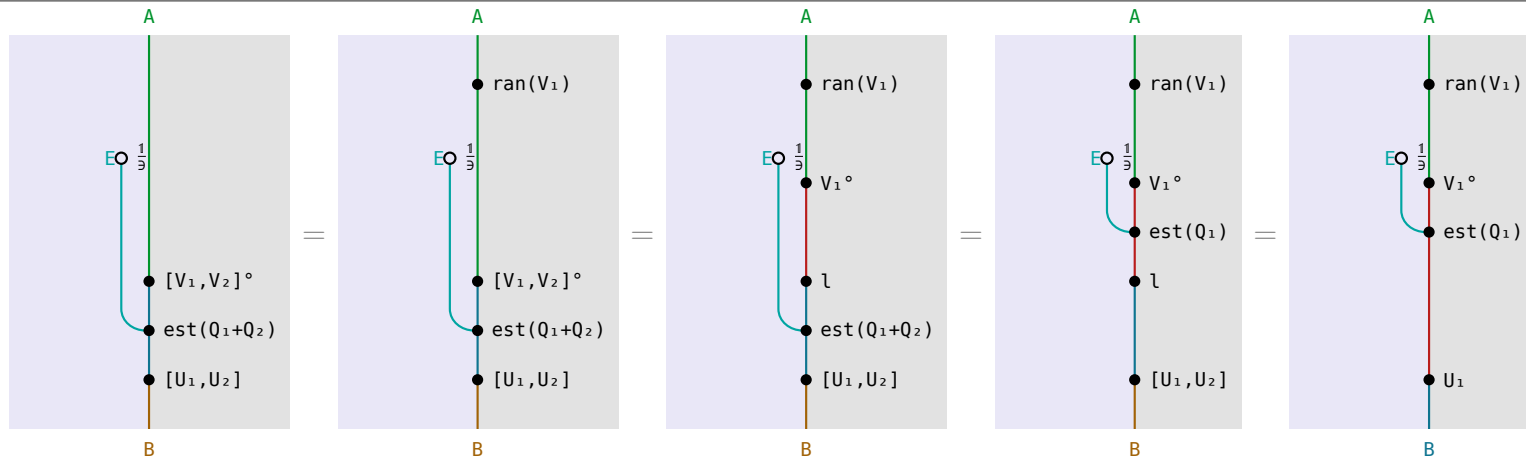
$h^\circ F(R^\circ)h \subseteq R^\circ$: h monotonic on R , shunted — the hypothesis on h

$$\subseteq B \rightarrow [R] \rightarrow B$$

R transitive

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1+Q_2)[U_1, U_2] = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{est}(Q_1)U_1 \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{est}(Q_2)U_2 \quad (16.1e)$$

when V_1 and V_2 have disjoint ranges, the greedy step over $[V_1, V_2]$ runs the V_1 step on inputs V_1 reaches and the V_2 step on inputs V_2 reaches



§16.2. The detab-entab problem

(16.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$ over snoc -lists, $\alpha \triangleq [\text{nil}, \text{snoc}]$, $H = \text{detab}^\circ$; $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$, $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$.

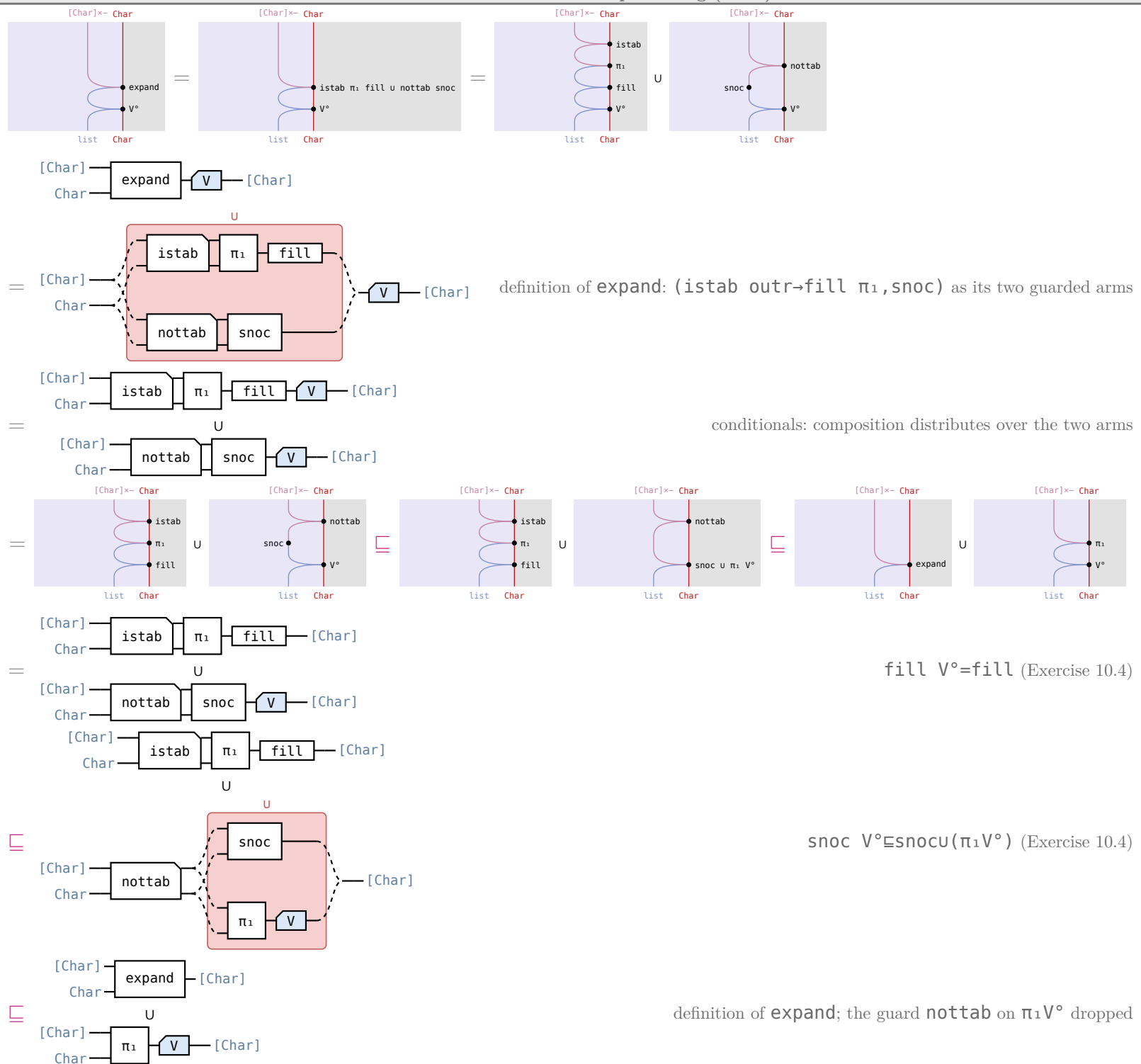
$\text{col} \triangleq ([\text{zero}, \text{count}])$, $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c+1)$; TB the tab, BL the blank, NL the newline, tab stops every n columns.

$R \triangleq \text{length} \leq \text{length}^\circ$, U the preorder with $a \cup b \iff a = \text{TB} \vee a = b$, $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$, $Q \triangleq 1 + (V \times U)$.

$\text{unfill } xs$ the shortest prefix of xs with $\text{fill } (\text{unfill } xs) = \text{fill } xs$; tbc the trailing blank count, $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$.

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand} \cup \pi_1 V^\circ \quad (16.2b)$$

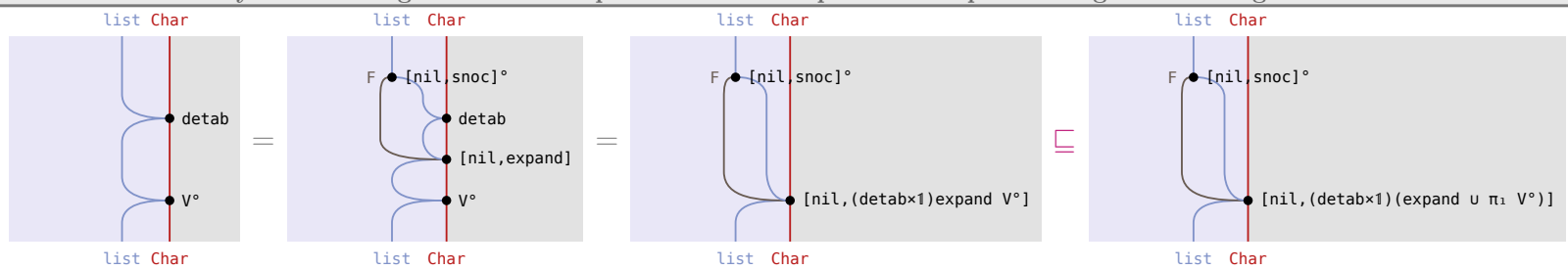
shortening the output of one **expand** step to a V -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ($\pi_1 V^\circ$)



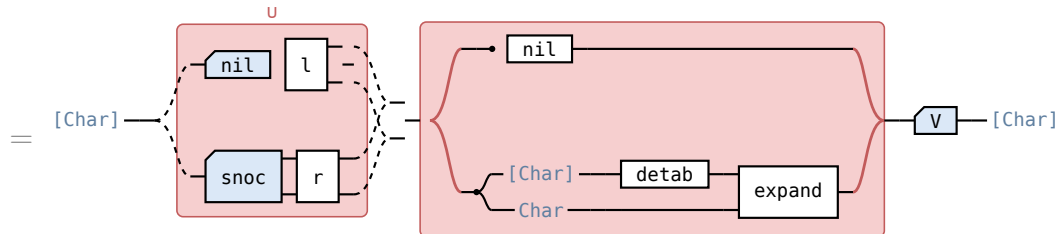
$$0 < n \wedge \text{blank} \neq n1 \Rightarrow \text{detab } V^\circ \in R^\circ \text{detab}$$

(16.2c)

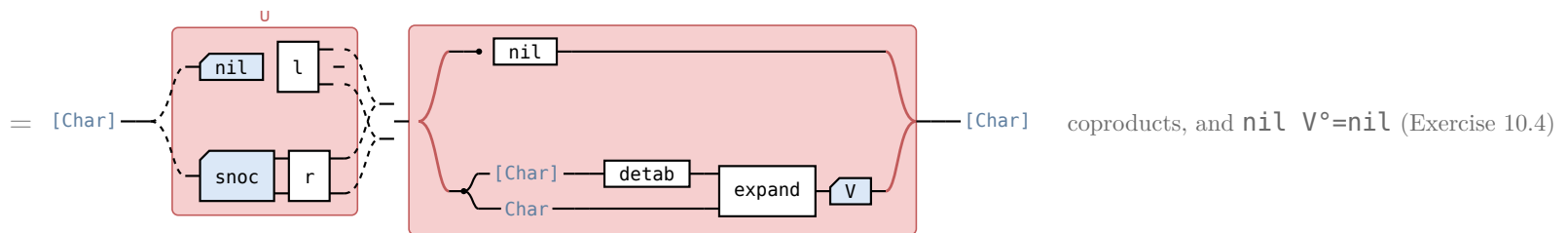
any V -shortening of detab 's output is detab 's output on an input no longer than the given one



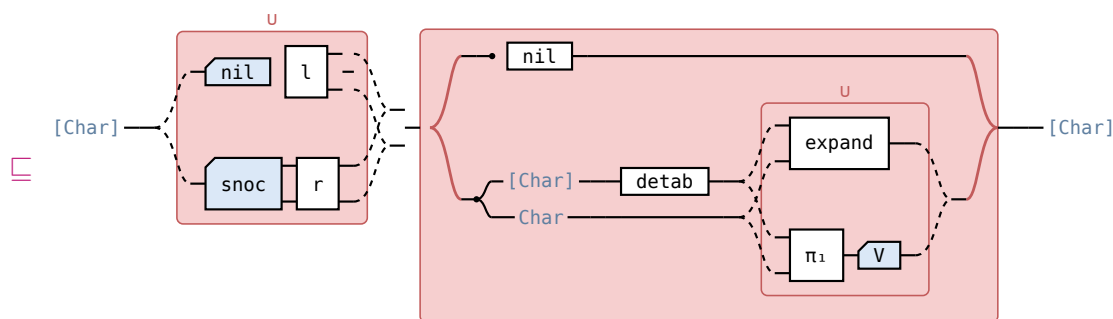
$$[\text{Char}] \xrightarrow{\text{detab}} V \xrightarrow{[Char]}$$



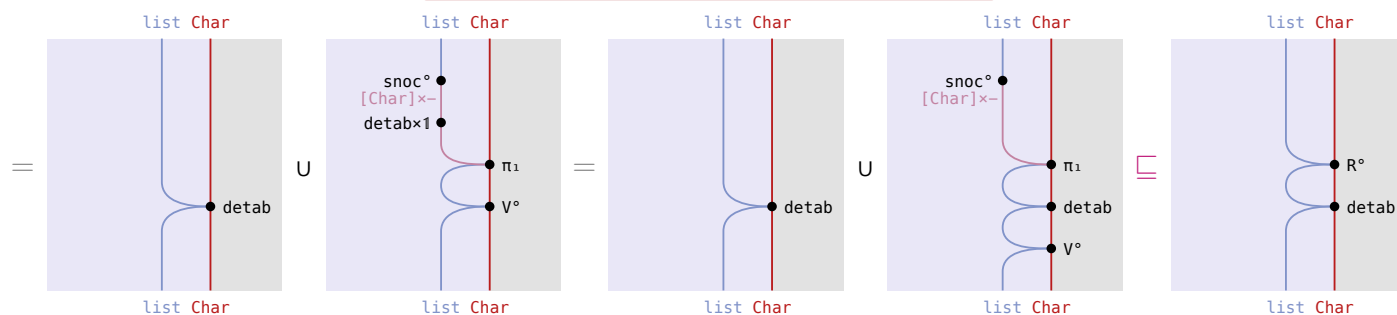
detab is a fold: $\text{detab} = \alpha^\circ F(\text{detab}) [\text{nil}, \text{expand}] \triangleq \text{detab}$



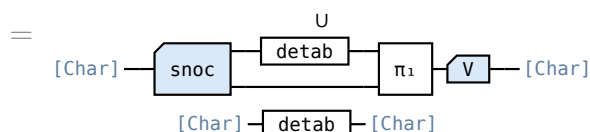
coproducts, and $\text{nil } V^\circ = \text{nil}$ (Exercise 10.4)



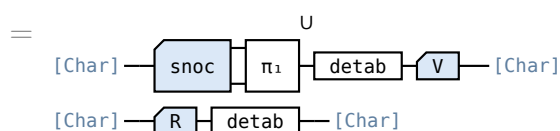
the claim — (16.2b)



$$[\text{Char}] \xrightarrow{\text{detab}} [\text{Char}]$$



distributing U ; the fold again, and the definition of F

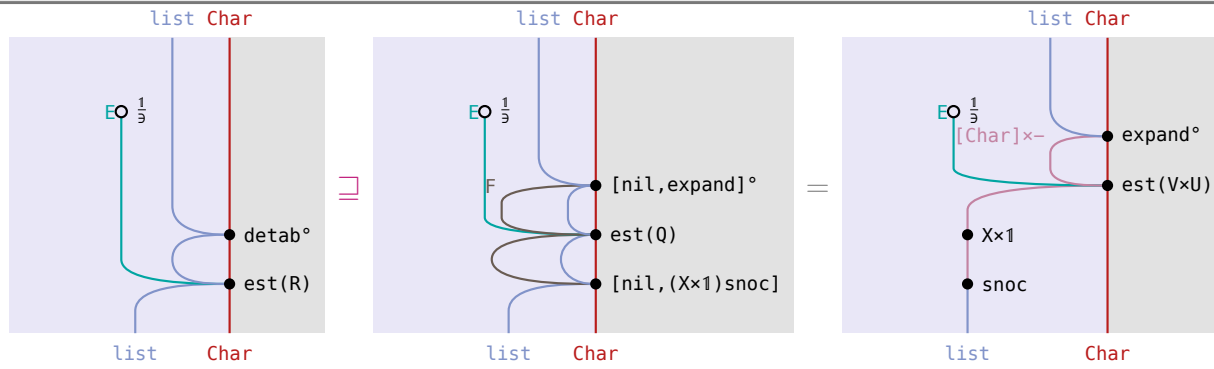


naturality of π_1 : $(\text{detab} \times 1) \pi_1 = \pi_1 \text{ detab}$; $\text{snoc}^\circ \pi_1$ is init

init is inductive, so $X \triangleq \text{detab } V^\circ$, a solution of $X \sqsubseteq \text{detab} u (\text{init } X)$, lies below the greatest one, prefix detab (induction on the input); $\text{prefix} \in R^\circ$

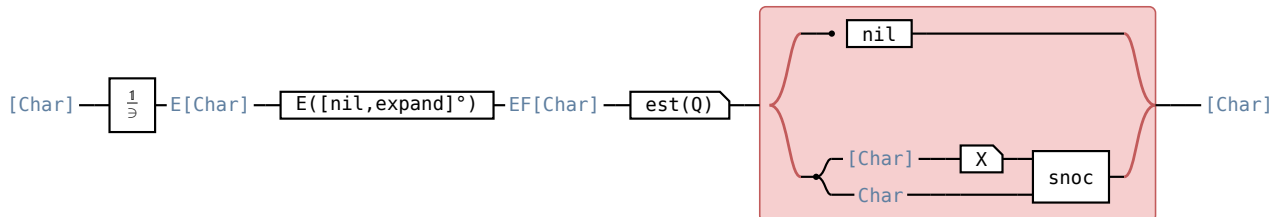
$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R) \quad (16.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop



$$[\text{Char}] \xrightarrow{\frac{1}{3}} E(\text{detab}^\circ) E[\text{Char}] \xrightarrow{\text{est}(R)} [\text{Char}]$$

the specification $\triangleq \text{detab}; \text{detab} \text{ entab}=1$ and nothing shorter does



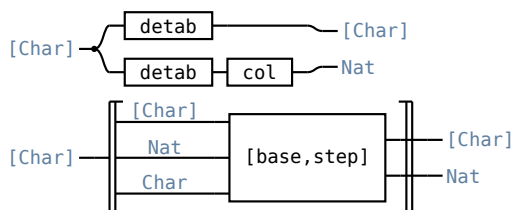
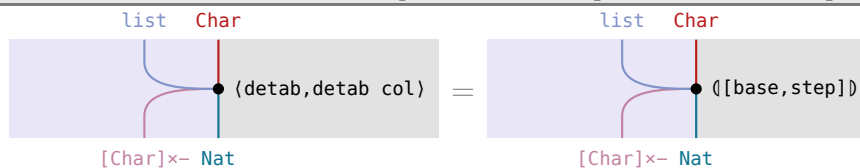
at $Q \triangleq 1 + (V \times U)$: one character of input is decided at each step. $F(T, R) \alpha \sqsubseteq R$. **detab** prefix $\sqsubseteq R^\circ$ **detab** is FALSE, — at $n=8$, **detab** $[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$, whose prefix $[a, b, c, d, e, \text{BL}, \text{BL}]$ is longer than any input giving it, and **detab** $V^\circ \sqsubseteq R^\circ$ **detab** holds. $\text{expand } V^\circ \sqsubseteq \text{expand } U$ ($\pi_1 V^\circ$) — shortening the output either leaves the last step alone or discards it

$$[\text{Char}] \xrightarrow{\frac{1}{3}} E[\text{Char}] \xrightarrow{E(\text{expand}^\circ)} E([\text{Char}] \times \text{Char}) \xrightarrow{\text{est}(V \times U)} X \xrightarrow{\text{snoc}} [\text{Char}]$$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle \text{detab}, \text{detab col} \rangle = \langle [\text{base}, \text{step}] \rangle \quad (16.2e)$$

one pass over the input carries the output so far together with its column



both components are folds over the same input; **step** on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row

$$\text{blank} \neq n1 \wedge \text{tb} \neq n1 \Rightarrow \langle [\text{base}, \text{step}] \rangle (\text{convert } l) r \Leftrightarrow r = \text{loop}(\text{step}, (([]), 0), l) \quad (16.2f)$$

the fold of **[base, step]** over the input converted to a snoc-list is the left loop of **step** from **base**

$$\pi_1(\text{loop}(\text{step}, (([]), c_0), x)) = \text{loop}'((f, g), (c_0, x))$$

Exercise 10.1: when **step** only appends $f(c, a)$ to the output and moves the state by g , the output of the loop is $\text{loop}'(f, g)$, which never carries the output

$$a \neq \text{tb} \Rightarrow \text{expand}(y, b) = \text{snoc}(x, a) \Leftrightarrow \text{fill}(y) = \text{snoc}(x, a) \wedge b = \text{tb} \vee (y, b) = (x, a) \quad (16.2g)$$

expand produces $x \# [a]$ from a tab after a string that fills to it, or from x and a

$$0 < n \wedge \text{blank} \neq n1 \Rightarrow (\exists y. \text{fill}(y) = \text{snoc}(x, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

$$\text{entab}(x) = \text{entab}(\text{unfill } x) ++ \text{blanks}(\text{tbc } x) \quad (16.2h)$$

(10.1): the output of **entab** is that of **unfill** x followed by the trailing blanks held back

$$\text{tbc}[] = 0$$

the empty string has no trailing blanks

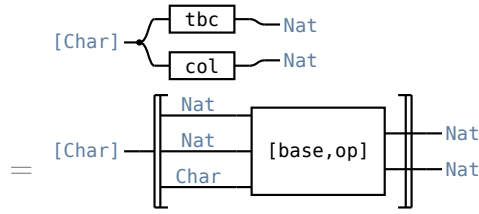
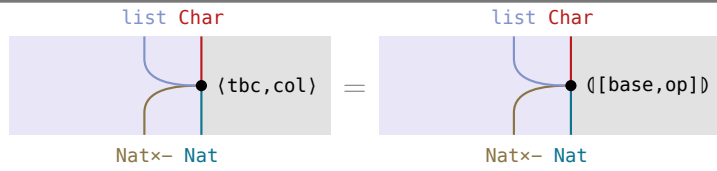
$$\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } x \text{ } a) \% n \neq 0 \text{ then } \text{tbc } x + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

$$\text{blank} \neq \text{nl} \Rightarrow (\text{tbc}, \text{col}) = ([\text{base}, \text{op}])$$

(16.2i)

one pass counts the held blanks and the column together

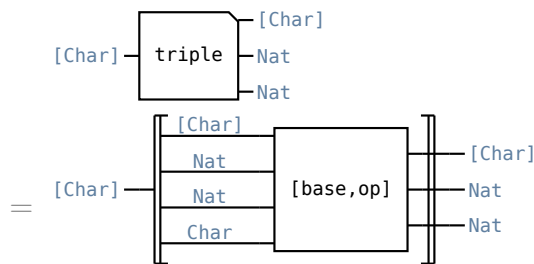
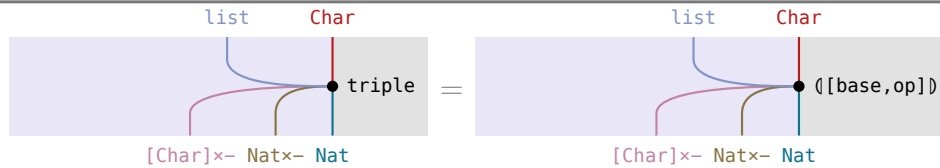


op on (tbc x, col x) is (tbc, col) one character further, from the equations for tbc — (16.2h)

$$\text{blank} \neq \text{nl} \Rightarrow \text{triple} = ([\text{base}, \text{op}])$$

(16.2j)

one pass carries the output up to the held blanks, their count and the column

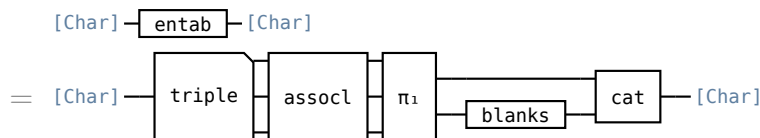
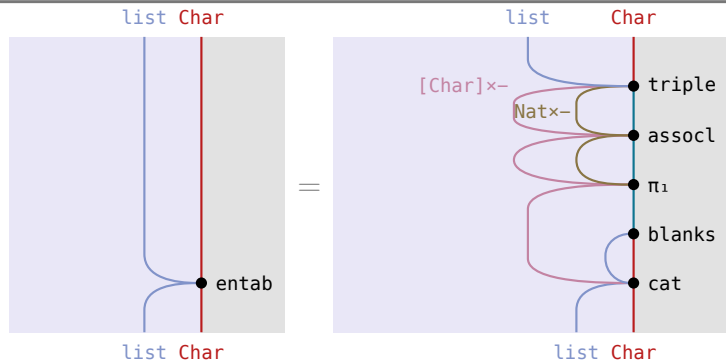


op on triple x is triple one character further: (10.1) where the blanks are released — (16.2h)

$$\text{entab} = \text{triple} \text{ assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

(16.2k)

entab is triple with the held blanks appended to its output



(10.1) — (16.2h)

§16.3. The minimum tardiness problem

(16.3a)

$\text{FX} = 1 + (\text{X} \times \text{Job})$, $\alpha \triangleq [\text{nil}, \text{snoc}]$ on schedules, $\beta \triangleq [\text{nil}, \text{snag}]$ on bags, snag putting a job into a bag; $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$, $\text{H} = \text{bagify}^\circ$.

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$ the completion, due and weighting quantities of a job; $\text{penalty}(\text{xs}, j) = (\text{sum}(\text{list}(\text{ct})(\text{xs})) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j)$.

$\text{cost} \triangleq \frac{\text{prefix}}{\sup} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{ est}(\geq)$, $\text{cost} [] = 0$, $\text{cost} (\text{xs} \# [j]) = \text{bmax} (\text{cost xs}, \text{penalty} (\text{xs}, j))$, $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$.

$\text{perm} \triangleq \text{bagify} \text{ bagify}^\circ = ([\text{nil}, \text{add}])$, $\text{add} (\text{xs}, j) = \text{ys} \# [j] \# \text{zs}$ for some $\text{xs} = \text{ys} \# \text{zs}$.

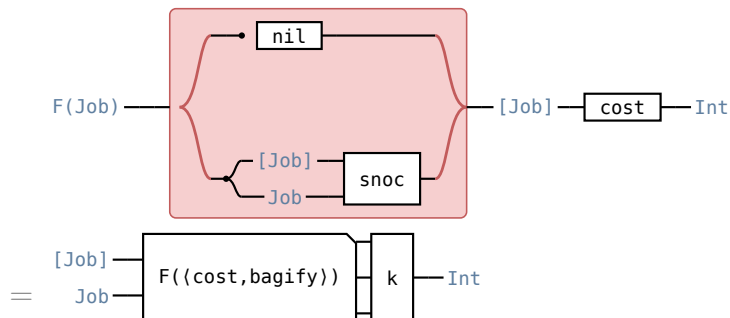
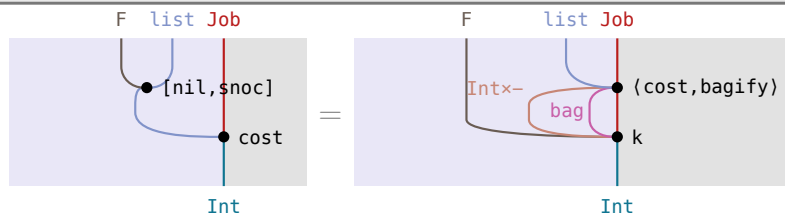
$\text{k} \triangleq [\text{zero}, \text{assocr} (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$, $\text{f} \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$, $\text{Q} \triangleq \text{f} \leq \text{f}^\circ$, $\text{Q}' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$.

$\text{g} \triangleq [\text{zero}, \text{penalty}]$, $\text{m} \triangleq [\text{zero}, \pi_1 \text{ cost}]$ (B&dM's h, renamed as in (16.3k)).

$$[\text{nil}, \text{snoc}] \text{cost} = F(\langle \text{cost}, \text{bagify} \rangle) k$$

(16.3b)

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

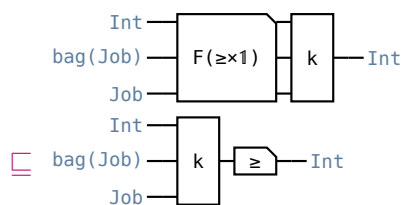
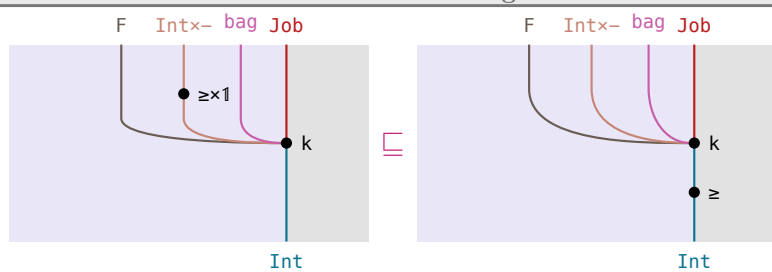


$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$, and $\text{penalty}(xs, j)$ reads only the bag of xs — $\triangleq \text{bagify}$

$$F(\geq \times 1) k \leq k \geq$$

(16.3c)

a larger cost before the last job gives a larger cost after it

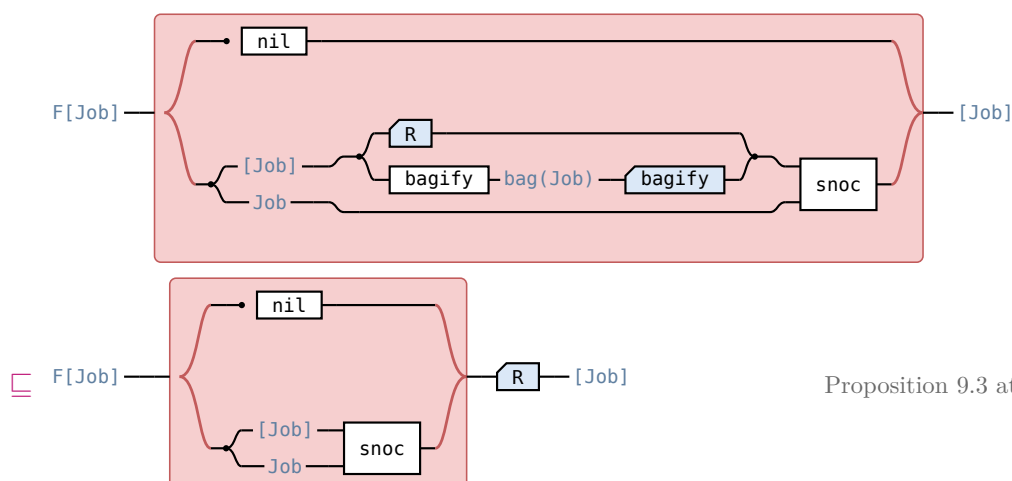
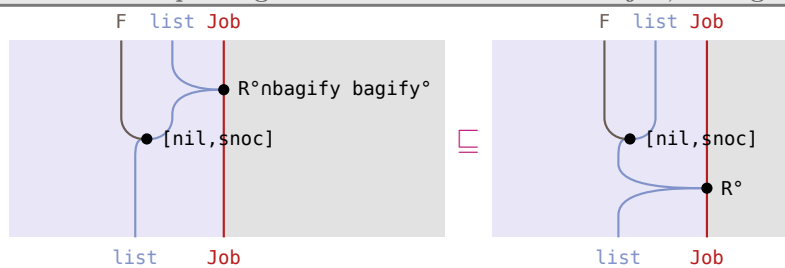


bmax is monotone in its first argument

$$F(R^\circ \text{bagify } \text{bagify}^\circ) [\text{nil}, \text{snoc}] \leq [\text{nil}, \text{snoc}] R^\circ$$

(16.3d)

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule

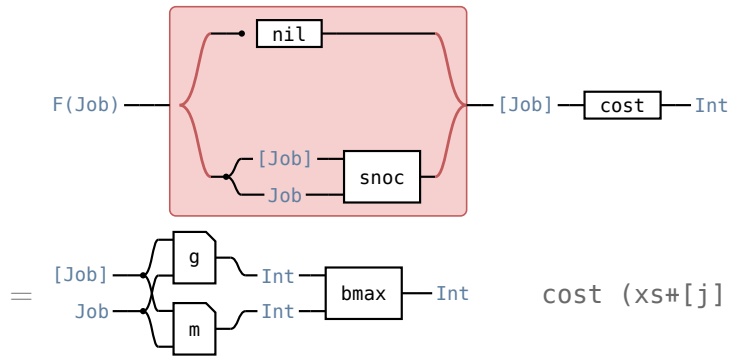
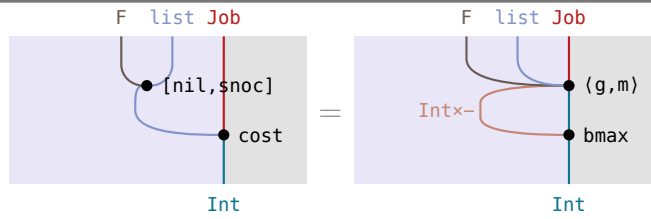


Proposition 9.3 at $S \triangleq \text{bagify}$, $\leq \triangleq$ — (15.2l), with (16.3b) and (16.3c)

$$[\text{nil}, \text{snoc}] \text{cost} = \langle g, m \rangle \text{bmax}$$

(16.3e)

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

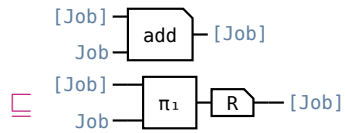
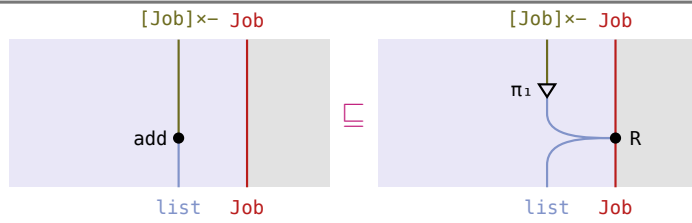


$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ and bmax commutes $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 R$$

(16.3f)

putting one more job anywhere into a schedule never lowers its cost

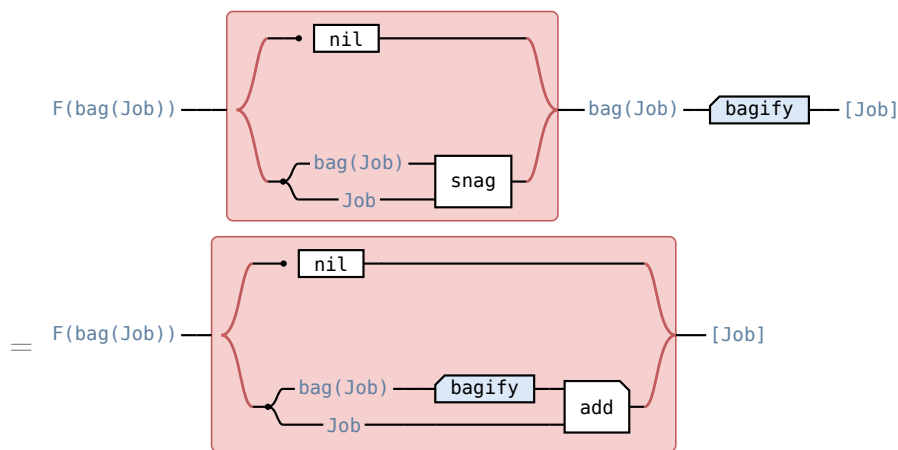
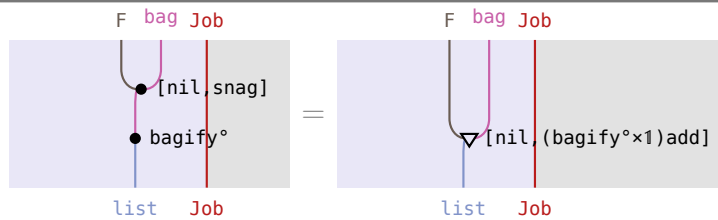


induction on where j goes: every job after it starts later, so no penalty falls

$$[\text{nil}, \text{snag}] \text{bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{add}]$$

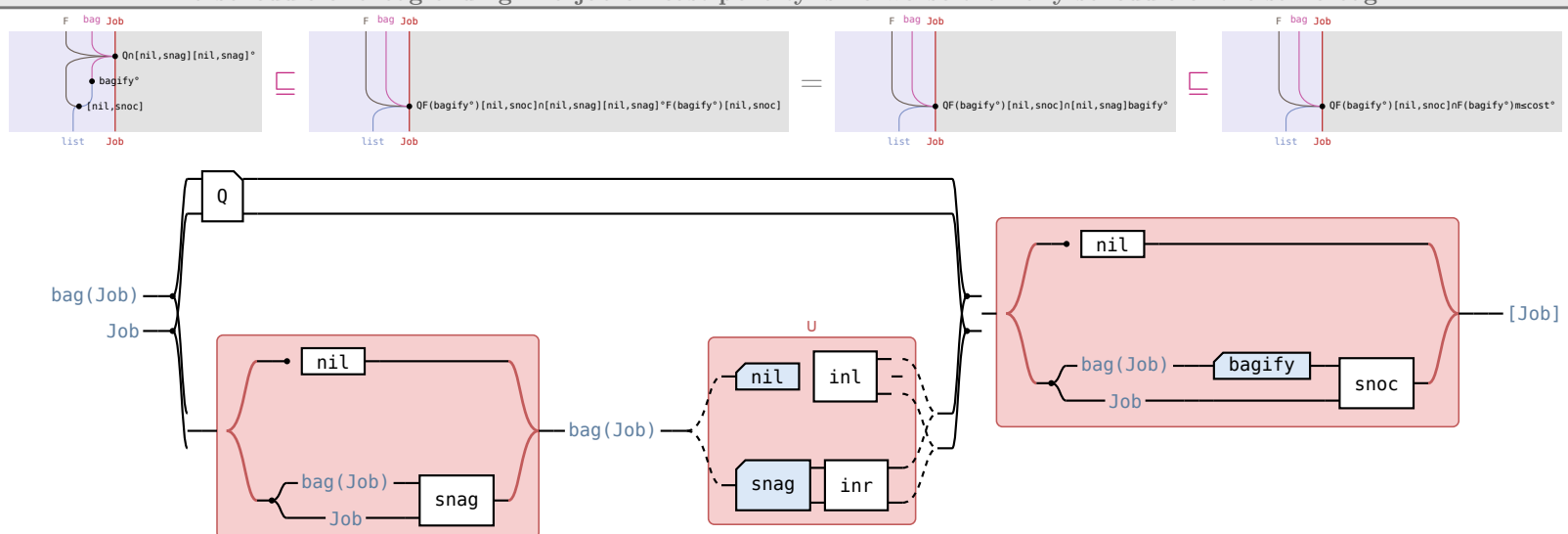
(16.3g)

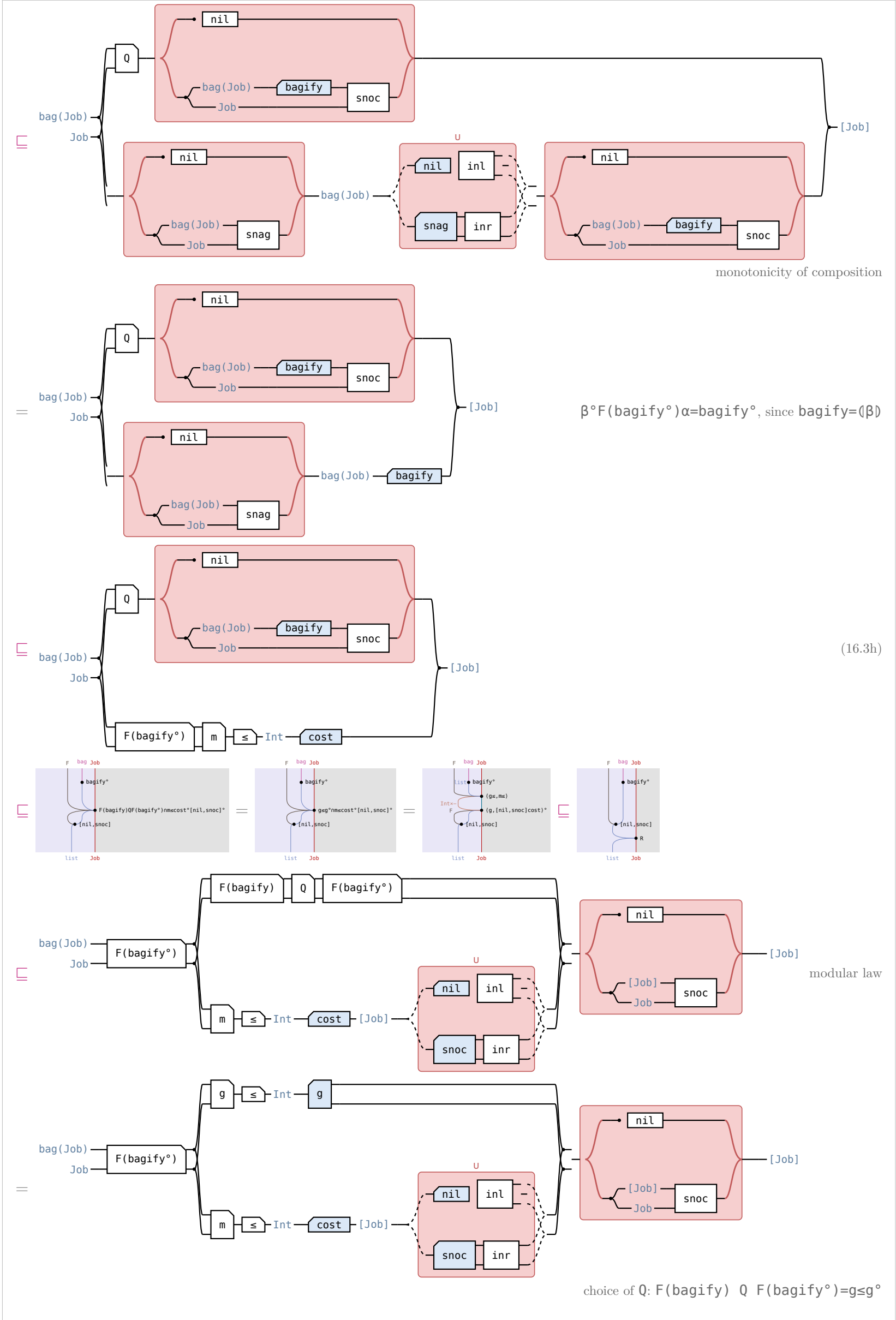
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere

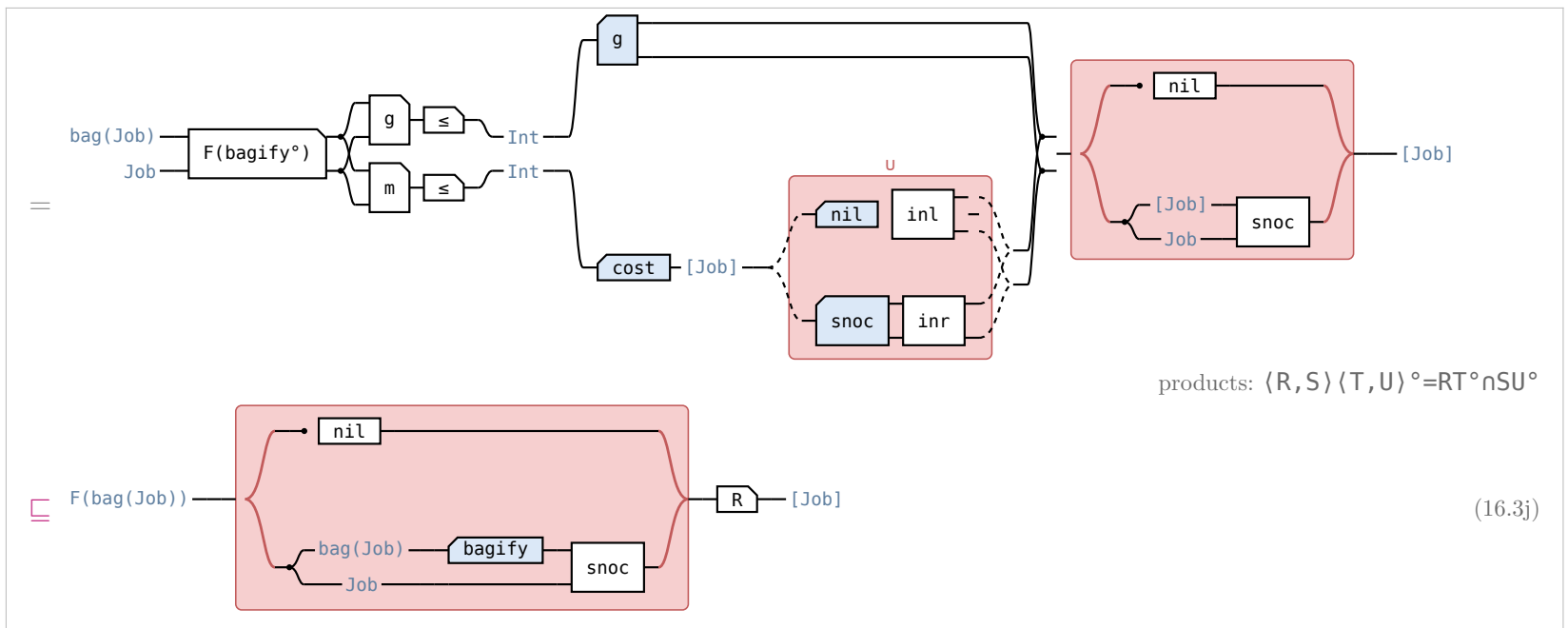


take out the last copy of j and put it back $\triangleq \text{bagify}$

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (Qn[\text{nil}, \text{snag}][\text{nil}, \text{snag}]^\circ F(\text{bagify}^\circ)[\text{nil}, \text{snoc}] \models F(\text{bagify}^\circ)[\text{nil}, \text{snoc}]R) \quad (16.3i)$$



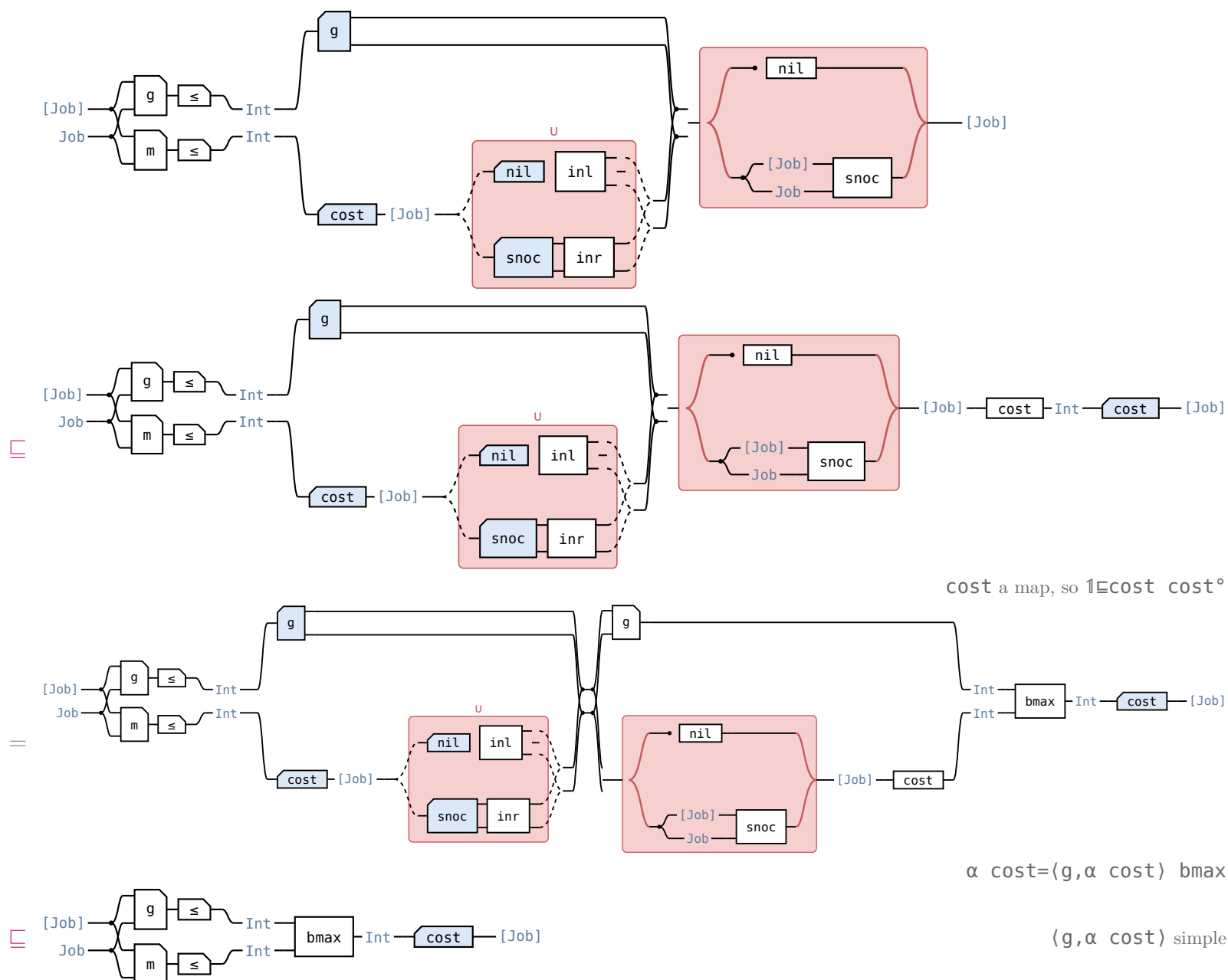
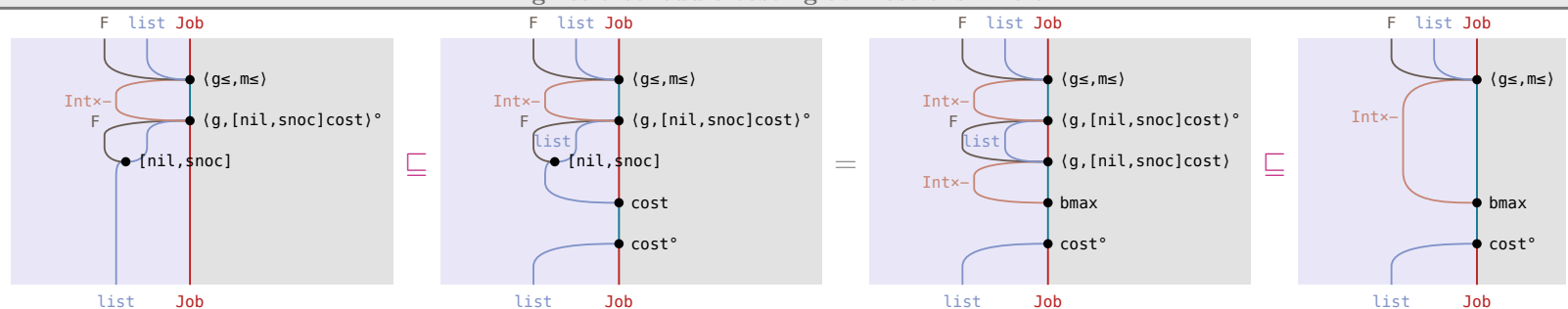


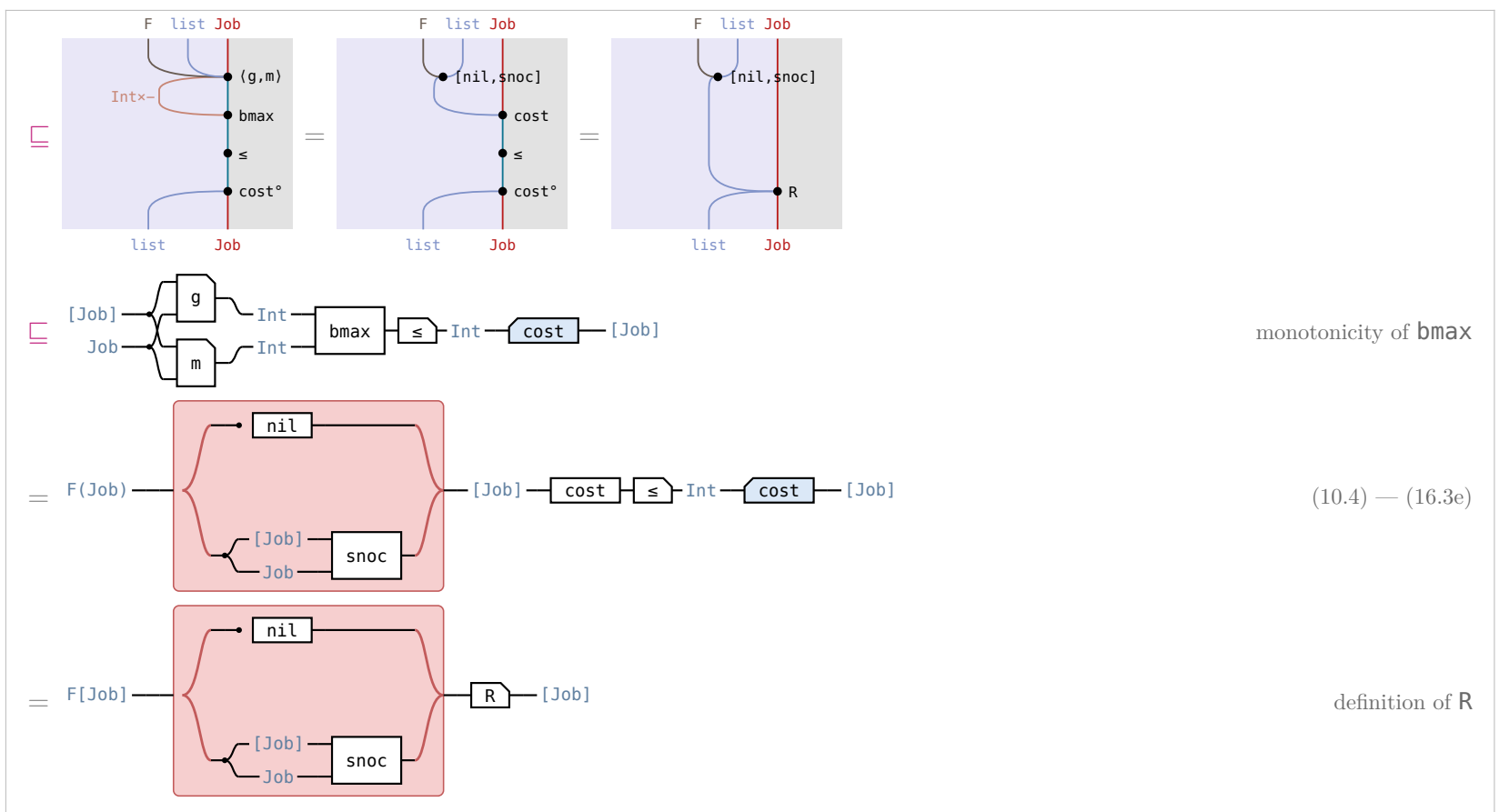


$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R$$

(16.3j)

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job,
 gives a schedule costing at most the whole

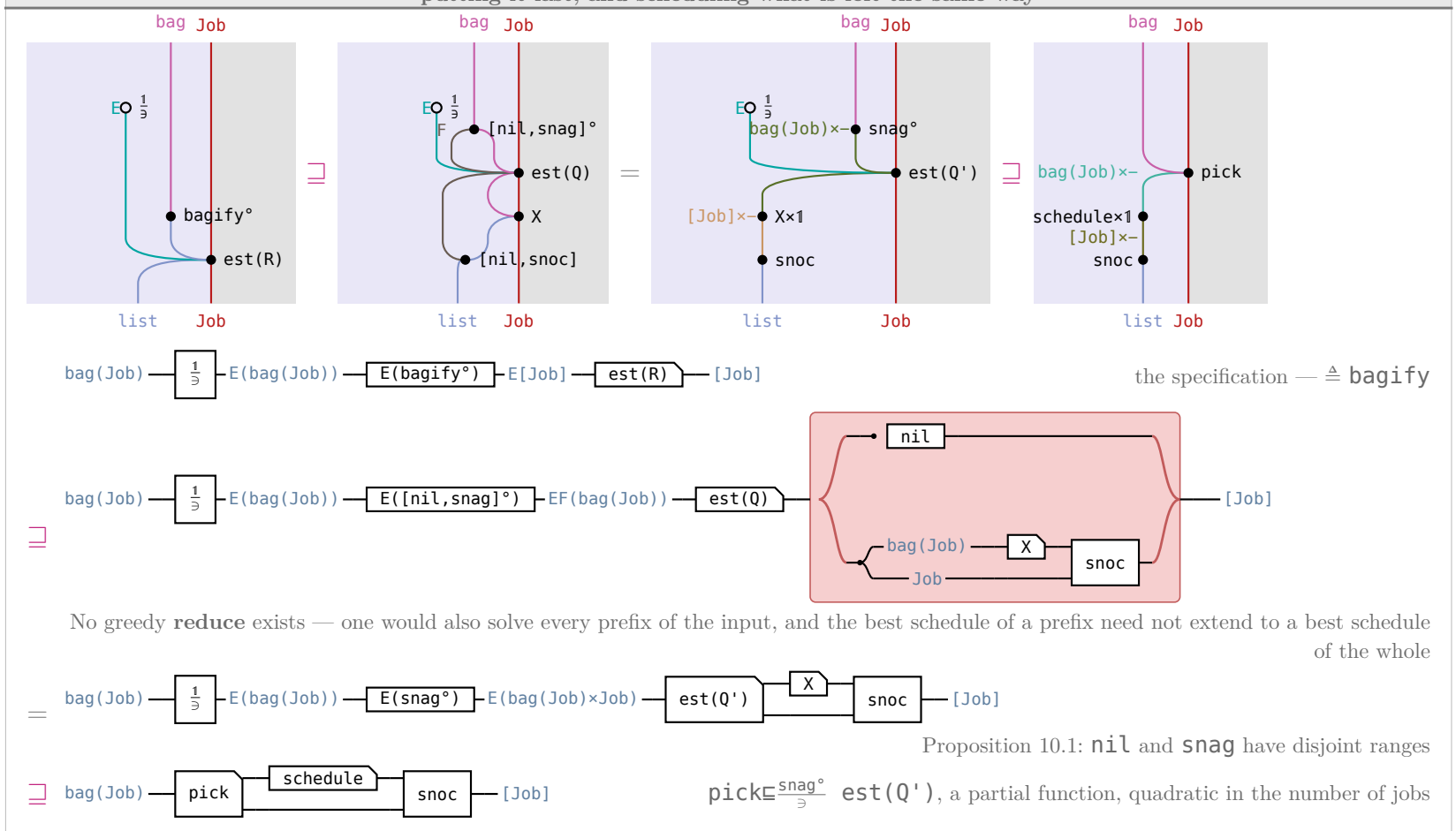




$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq_{\frac{\text{bagify}^\circ}{3}} \text{est}(R), \quad (16.3k)$$

schedule = (null-nil, pick (schedule × 1) snoc)

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way



§16.4. The TeX problem

(16.4a)

intern $\triangleq$ val round : Decimal $\rightarrow$ [0,2¹⁶), val $\triangleq$ Q[zero,shift], shift (d,r)=(d+r)/10, round r rounds 2¹⁶r to the nearest integer: round r=n $\Leftrightarrow$ 2n-1<2¹⁷r<2n+1.

interval n=((2n-1)/2¹⁷,(2n+1)/2¹⁷), r inrange (a,b) $\Leftrightarrow$ a<r<b, round $\circ$ =interval inrange, R $\triangleq$ length $\leq$ length $\circ$.

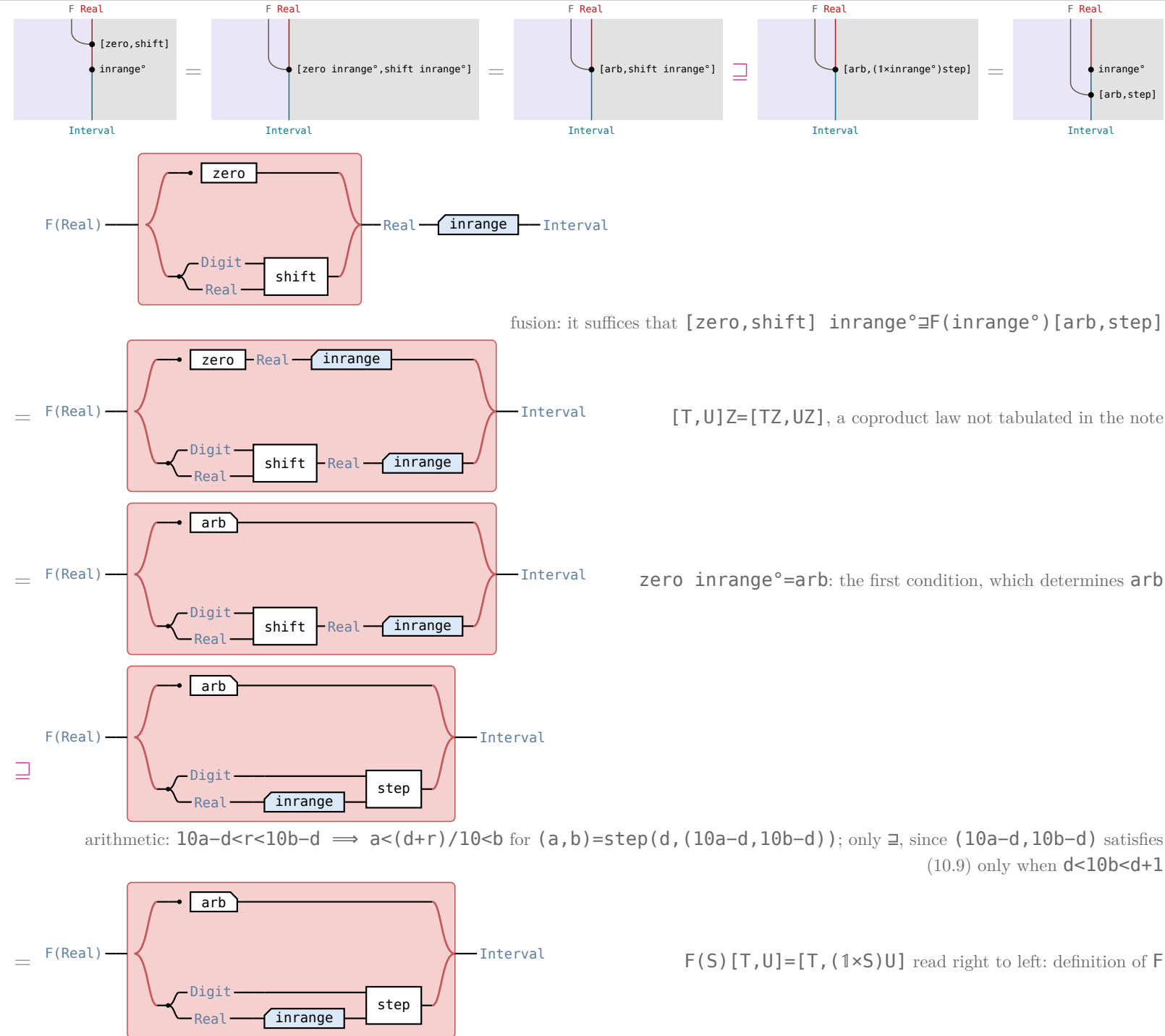
Interval the pairs (a,b) with Legal(a,b) $\Leftrightarrow$ 0<b<1 and a<b (10.9); step_legal is what types step [arb,step] : 1+(Digit $\times$ Interval) $\rightarrow$ Interval, step (d,(a,b))=((d+a)/10,(d+b)/10).

FX=1+(Digit $\times$ X), $\alpha\triangleq$ [nil,cons], H $\triangleq$ Q[arb,step] $\circ$, ! : Digit $\times$ Interval $\rightarrow$ 1, Q $\triangleq$ (l! $\circ$ r) $\cup$ 1 l, r are (11.2a)'s injections into FX=1+(Digit $\times$ X), so l : 1 $\rightarrow$ FX and r : Digit $\times$ X $\rightarrow$ FX, w $\triangleq$ 2¹⁷.

Q[arb,step]Eval inrange $\circ$

(16.4b)

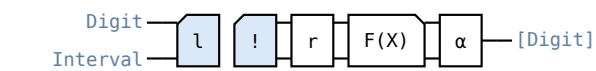
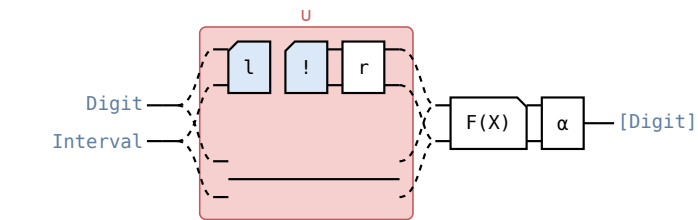
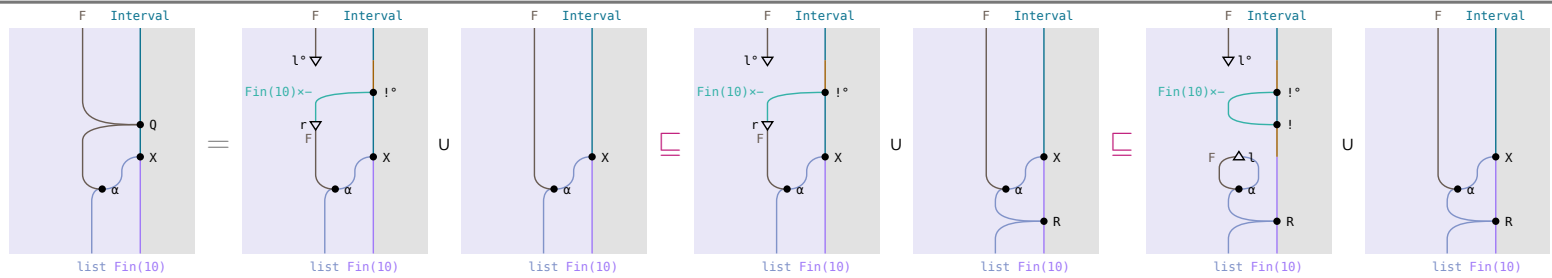
every interval got by folding a decimal's digits with [arb,step] has the decimal's value strictly inside it



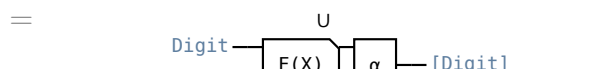
$$QF(X)\alpha \sqsubseteq F(X)\alpha R$$

(16.4c)

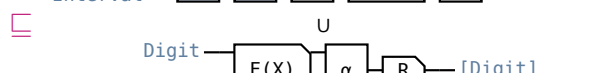
choosing by Q among the one-step decompositions before building with $F(X)$ and α yields only decimals that are R -related to one built without choosing



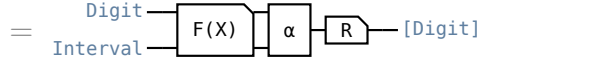
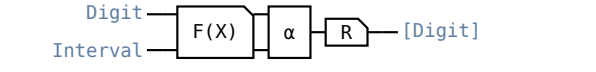
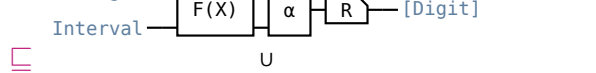
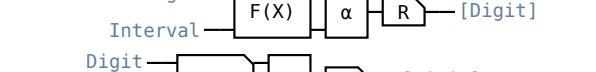
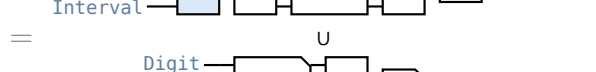
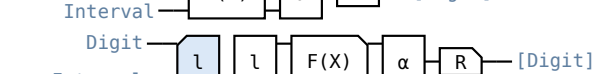
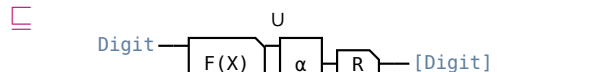
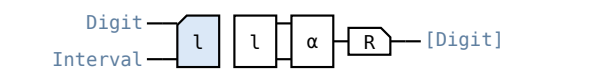
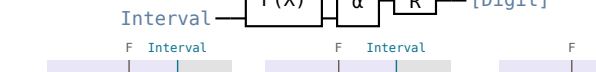
definition of Q ; composition distributes over U



R is reflexive



$r F(X) \alpha \sqsubseteq ! l \alpha R: l \alpha = \text{nil}, \text{ and } \text{length}(\text{nil}) = 0 \text{ is at most any length}$



$! \circ ! \sqsubseteq 1$ on 1 : universal property of $!$

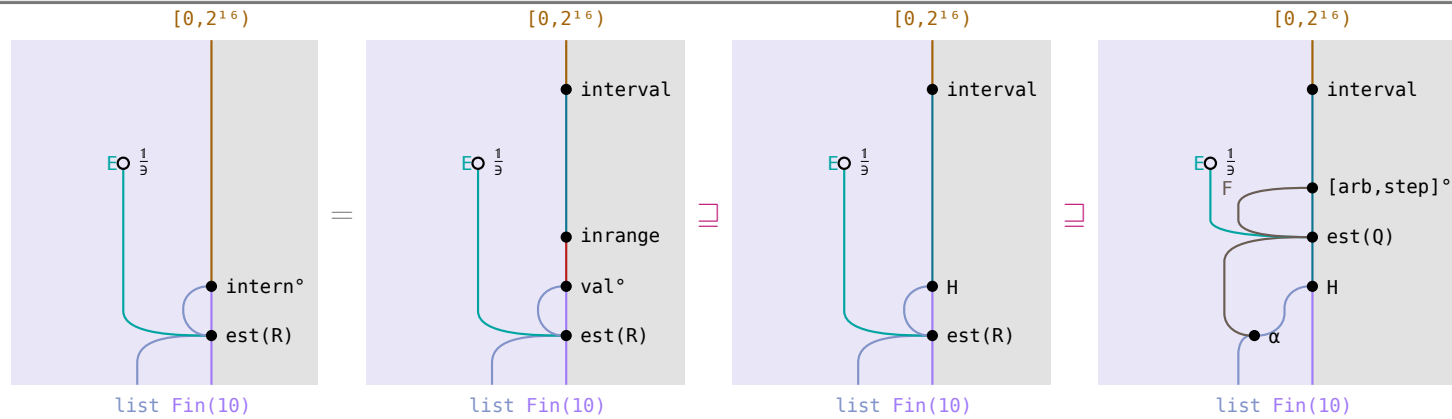
$l F(X) = l$: definition of F

l is simple: $l \circ l \sqsubseteq 1$

U is idempotent

$$\text{interval}(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) , \text{extern}(n)=f(2n-1, 2n+1) \quad (16.4d)$$

a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



$$\begin{aligned}
 & [0, 2^{16}) \xrightarrow{\frac{1}{3}} E([0, 2^{16})) \xrightarrow{E(\text{intern}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] && \text{the specification } \triangleq \text{intern} \\
 = & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(\text{inrange val}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 \sqsubseteq & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(H)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 & \text{the type restriction (10.9): a shortest decimal H gives an interval is a shortest one among all decimals inside it, and is inside it by (16.4b)} \\
 \sqsubseteq & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E([\text{arb}, \text{step}]^\circ)} EF(\text{Interval}) \xrightarrow{\text{est}(Q)} F(H) \xrightarrow{\alpha} [\text{Digit}] \\
 & \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ returns at most two elements — stop, or take one more digit — and ! nil} \sqsubseteq \text{cons } R^\circ \text{ makes it stop whenever stopping is legal}
 \end{aligned}$$

- $f(a, b) = x \iff a < 0 \wedge x = [] \vee \neg a < 0 \wedge (\exists d. \exists y. d = \lfloor 10b \rfloor \wedge f(10a-d, 10b-d) = y \wedge x = [d] \# y)$
 f gives (a, b) the empty decimal when $a < 0$, and otherwise the digit $d = \lfloor 10b \rfloor$ followed by what it gives $(10a-d, 10b-d)$
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) = f$
the least solution of the greedy recursion of (16.4d) is f
- $d = \lfloor 10b \rfloor \wedge e = \lfloor 10b \rfloor \implies d = e$
Exercise 10.12: two digits d and e with $0 < 10b-d < 1$ and $0 < 10b-e < 1$ are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p, q) = []$
the program's first clause: $f(p, q) = []$ when $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \wedge d = \lfloor 10q/w \rfloor \implies f(p, q) = [d] \# f(10p-wd, 10q-wd)$
the program's second clause: with $d = \lfloor 10q \rfloor \text{ div } w$, $f(p, q)$ is d followed by $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p, q) \implies f(p/w, q/w) = f(p, q)$
on the pairs the program reaches from $\text{interval } n$, the integer f at (p, q) is a value of the rational f at $(p/w, q/w)$
- $\text{interval } f = \text{extern}$
 f after interval is the program extern , whose value at n is $f(2n-1, 2n+1)$

(16.4e)

§17. Appendix

§17.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(17.1a)

(p, q) tabulates $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$, a map.

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

§17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \\ \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \quad (13.1.9a) \end{array}$$

(17.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \\ \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus (\epsilon \setminus R^\circ)))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \quad \text{counit of } T \cdot \neg T \setminus \\ \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \\ \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est} \end{array}$$

§17.3. Shortest paths on a cylinder, on lists

(17.3a)

definition	type	note
$F(A, X) = A + A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
$L = \text{list}^+$	$\mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
α the initial algebra	$F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$	$\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$: a path is started by one square, or extended by one.
N the n-tuple relator	$\mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$[\text{Nat}] \rightarrow [\text{Nat}]$	The cost of a path, which the cheapest minimises.
setify	$A[n] \rightarrow EA$	$\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.
moves	$A[n] \rightarrow E(A[n])$	$\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.
trans	$E(A[n]) \rightarrow (E A)[n]$	$\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components.
zip	$F(a[n]) \rightarrow (F a)[n]$	$\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$.
$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$F(EA) \rightarrow EFA$	$\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$.
$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{zip}$ $N(\text{cp}(F(A, -), T)E(\alpha))$	$F(NA, N(E(TA))) \rightarrow N(E(TA))$	$\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (17.3.1b).
$\text{paths} \triangleq (\text{gen}) \text{setify union}$	$T(NA) \rightarrow E(TA)$	$\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (17.3.1b)'s four sets: 12 paths, 3 from each entry row.
the specification $\text{paths est}(R)$	$[\text{Nat}[n]] \rightarrow [\text{Nat}]$	A cheapest path from the entry side to the exit side.

§17.3.1. $N(E(L A))$

$$\begin{array}{l} 1 \quad 5 \\ 2 \quad 6 \\ 3 \quad 7 \\ 4 \quad 8 \end{array} \quad [(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$$

(17.3.1a)

$$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$$

(17.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ (\{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \}) \end{aligned}$$

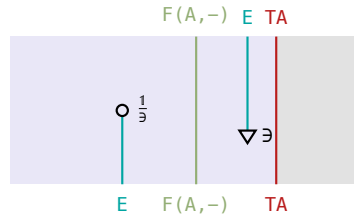
(17.3.1c)

N	the 4-tuple, one component per row of the entry column
E	the set of paths that can start in that row
L	a path is the non-empty list of the squares it crosses

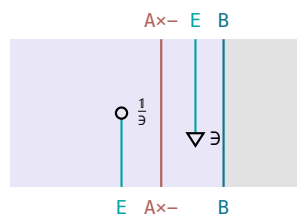
row 1's set has no $[1,7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1,8]$, because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of $\triangleq$ **paths**: component k collects the paths of rows $k-1, k, k+1$.

§17.3.2. The cross product $cp \triangleq \frac{F(1, \exists)}{\exists}$

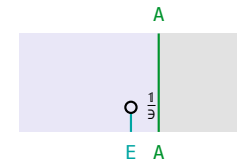
(17.3.2a)



cp on all of $F(A, E(L A))$



the $A \times -$ summand: $\exists$ picks one path, $1\% \exists$ collects the results



the A summand, no E to distribute: $1\% \exists$ alone, $a \mapsto \{a\}$

(17.3.2b)

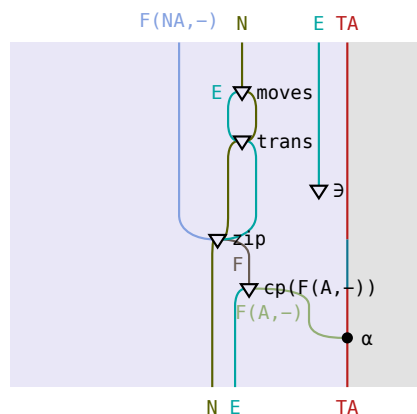
type	note
$F(A, -) = A + A \times -$	one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$
$\exists : E(L A) \rightarrow L A$	the $\exists$ inside $F(1, \exists)$: a set of paths, one of them — $L A$ is one path, A one square
$F(1, \exists) = 1 + 1 \times \exists$	$: A + A \times E(L A) \rightarrow A + A \times L A$, the relator acting on each summand
$\exists : E(A + A \times L A) \rightarrow A + A \times L A$	the $\exists$ under the bar: a set of cells, one of them — a different $\exists$
$cp : A + A \times E(L A) \rightarrow E(A + A \times L A)$	$\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$
$cp = \frac{1}{\exists} E(F(1, \exists))$	what the picture draws: the unit makes the outer E , leaving one $\exists$ — (10b)

(17.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$ the right summand
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$ 1 keeps the square 1, $\exists$ picks one path
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$ one output per path in w
 $cp(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$ the three of them, collected

§17.3.3. $gen = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(cp P(\alpha))$

(17.3.3a)

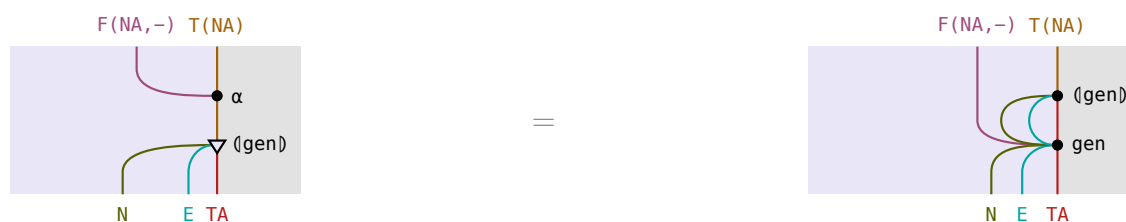


(17.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$: $F(N\ A, N(E(L\ A)))$
 $F(1, \text{moves trans } N(\text{union}))$: 1 keeps the column, the path SETS move
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$
 $= (\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})$: $E(N(E(L\ A)))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$
 $\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$: $N(E(E(L\ A)))$ row k gets rows k-1, k, k+1
 $N(\text{union})(\text{that})$
 $= (\{[5],[6],[8]\},\{[5],[6],[7]\},$
 $\{[6],[7],[8]\},\{[5],[7],[8]\})$: $N(E(L\ A))$ every path into row k, none dropped
 $\text{zip}(\text{that})$
 $= ((1,\{[5],[6],[8]\}), (2,\{[5],[6],[7]\}),$
 $(3,\{[6],[7],[8]\}), (4,\{[5],[7],[8]\}))$: $N(F(A, E(L\ A)))$ each row: its square, and the paths it may be put in front of
 $N(\text{cp } P(\alpha))(\text{that})$
 $= (\{[1,5],[1,6],[1,8]\},\{[2,5],[2,6],[2,7]\},$
 $\{[3,6],[3,7],[3,8]\},\{[4,5],[4,7],[4,8]\})$: $N(E(L\ A))$ cp pairs the square with each path, α prefixes it: $\alpha(1,[5])=[1,5]$

§17.3.4. gen is an F-algebra; $(\text{gen}): \alpha(\text{gen})=F(1, (\text{gen}))\text{gen}$

(17.3.4a)



α puts the column back on the list, then the fold reads all of it

the fold reads the rest under F , then one gen puts the column in front

(17.3.4b)

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N\ A)$$

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (17.3.1b): the tail folds to $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$, one path per row and each of them one square long, and gen on it is (17.3.3b)'s walk.

§17.3.5. $Q : F(N\ A, N(L\ A)) \rightarrow N(L\ A)$, an $F(N\ A, -)$ -algebra: $Q=F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(17.3.5a)

$Q : F(N\ A, N(L\ A)) \rightarrow N(L\ A)$: the fold's algebra: a new column, one path per row
 $u = ((1,2,3,4),([5],[6],[7],[8]))$: $F(N\ A, N(L\ A))$
 $F(1, \text{moves trans } N(\text{est}(R)))$: 1 keeps the column, the paths move
 $\text{moves}([5],[6],[7],[8])$
 $= (\{[6],[7],[8],[5]\},$
 $\{[5],[6],[7],[8]\},$
 $\{[8],[5],[6],[7]\})$: $E(N(L\ A))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6],[5],[8]\},\{[7],[6],[5]\},$
 $\{[8],[7],[6]\},\{[5],[8],[7]\})$: $N(E(L\ A))$ row k gets rows k-1, k, k+1
 $N(\text{est}(R))(\text{that})$
 $= ([5],[5],[6],[5])$: $N(L\ A)$ the cheapest into each row – chosen BEFORE the new square
 $\text{zip}(\text{that})$
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$: $N(F(A, L\ A))$ each row: its square, and the one predecessor that survived
 $N(\alpha)(\text{that})$
 $= ([1,5], [2,5], [3,6], [4,5])$: $N(L\ A)$ $\alpha(1,[5])=[1,5]$

$Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (17.3.5b)	
at the last column Q starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (17.3.5c)'s last step folds	
circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$	Hinze–Marsden
$F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$	
$FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$ the fusion condition (17.3.5d) read as a definition,	
$FN(N A) (N (T A)) \text{---} \left\{ \begin{array}{l} N(p_{\circ}) \text{---} \boxed{N(\text{wrapA})} \\ q \text{---} \boxed{\text{pm}} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_{\circ}) \text{---} \boxed{N(\text{consA})} \end{array} \right\} \text{---} N(T A)$ $\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$	

paths $\text{est}(R) \exists (Q)$ setify $\text{est}(R)$ (17.3.5c)	
shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; Q is (17.3.5b)'s algebra.	
circuit	Hinze–Marsden
$T(N\ A) \text{---} \boxed{\text{paths}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$	
$= T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{\text{union}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\triangleq \text{paths at paths}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{P(\text{est}(R))} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $P(\text{est}(R))\ \text{est}(R) \sqsubseteq \text{union}\ \text{est}(R) \text{--- a minimum in each set, then a minimum of those; } R \text{ transitive}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{setify lax natural}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (T\ A))} \text{---} \boxed{Q} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{fusion at (17.3.5d)}$	

gen $N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q$ (17.3.5d)	
fusion: the condition for fusion in (17.3.5c)'s last step, used to derive a definition of Q .	
circuit	Hinze–Marsden
$F(N\ A)\ (N\ (E\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)$	
$\sqsubseteq FNE(T\ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T\ A) \text{---} \boxed{Q} \text{---} N(T\ A)$ $(7.13), \text{ then zip, trans, moves lax natural}$	

(7.13) is $F(1, \text{est}(R)) \alpha \sqsubseteq_{cp} P(\alpha)\ \text{est}(R)$, (13.3.1c) at the map α with $\frac{F(1, \exists)\alpha}{\exists} =_{cp} P(\alpha)$: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.